

Project Data Page:

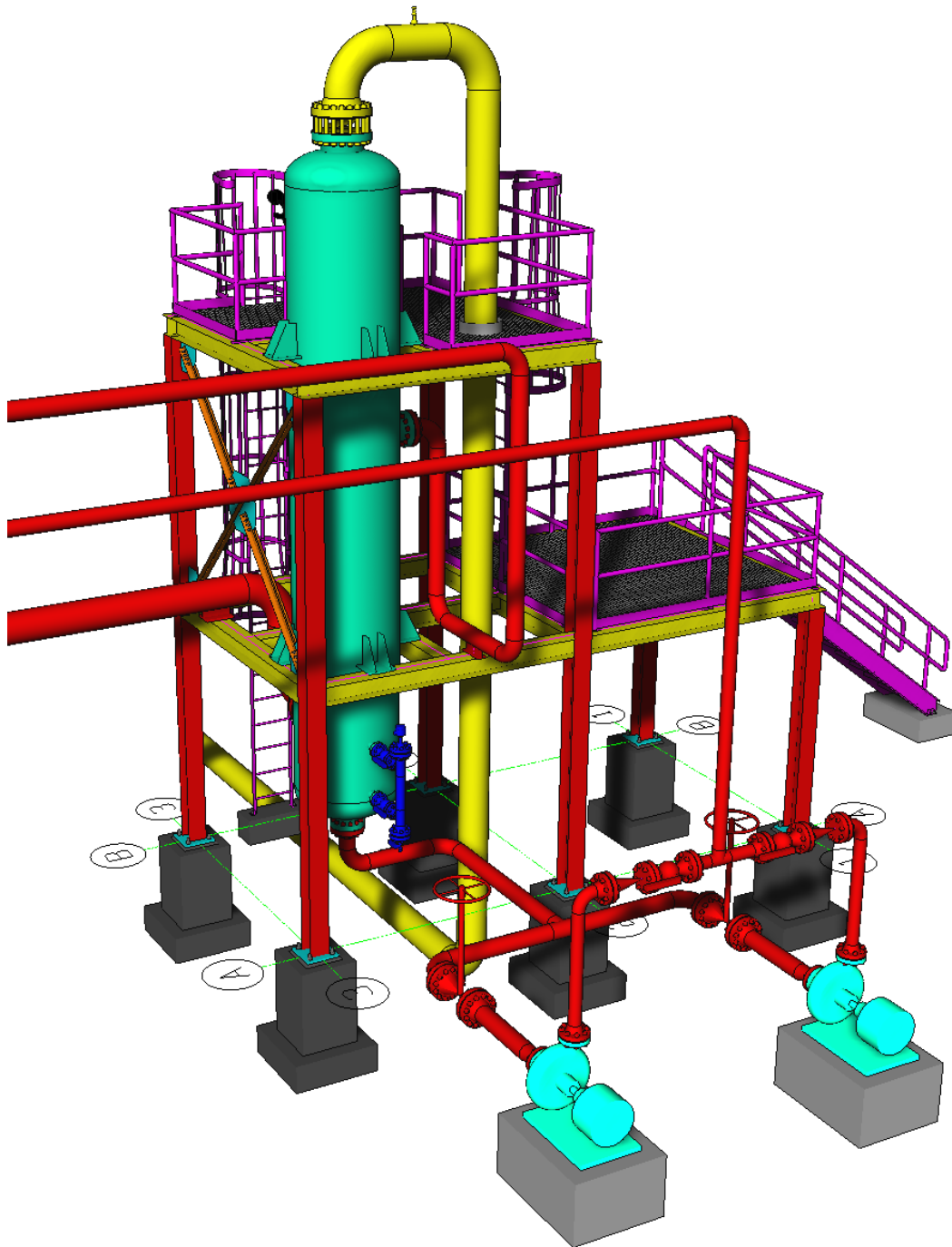




Table of Contents

Cover Page	1	Nozzle Calcs.: K4a 2"	230
Vessel Design Summary:	2	Nozzle Calcs.: K1b 2"	233
Nozzle Summary:	8	Nozzle Calcs.: K2b 2"	236
Nozzle Schedule:	11	Nozzle Calcs.: K3b 2"	239
Bill of Material:	15	Nozzle Calcs.: K4b 2"	242
MDMT Summary:	17	Nozzle Calcs.: N10 2"	245
Warnings and Errors:	20	Nozzle Calcs.: N14 4"	248
Input Echo:	21	Nozzle Calcs.: N13A 8"	261
XY Coordinate Calculations:	61	Nozzle Calcs.: MH1 30"	274
Internal Pressure Calculations:	62	Nozzle Calcs.: N12 2"	280
Element: Btm Dish End	64	Nozzle Calcs.: N13B 8"	283
Element: Shell-1	67	Nozzle Calcs.: N13C 8"	296
Element: Shell-2	70	Nozzle Calcs.: N2A 84"	309
Element: Shell-3	73	Nozzle Calcs.: MH2 30"	321
Element: Shell-4	76	Nozzle Calcs.: K9a 2"	327
Element: Shell-5	79	Nozzle Calcs.: N2B 84"	330
Element: Shell-6	82	Nozzle Calcs.: K9b 2"	342
Element: Shell-7	85	Nozzle Calcs.: MH3 30"	345
Element: Shell-9	88	Nozzle Calcs.: N4 20"	351
Element: Shell-10	91	Nozzle Calcs.: N3a 24"	364
Element: Shell-11	94	Nozzle Calcs.: N3B 24"	377
Element: Shell-12	97	Nozzle Calcs.: K5a 2"	390
Element: Shell-13	100	Nozzle Calcs.: K6a 2"	393
Element: Shell-14	103	Nozzle Calcs.: K14a 2"	396
Element: Shell-15	106	Nozzle Calcs.: K14b 2"	399
Element: Cone-1	109	Nozzle Calcs.: K6b	402
Element: Cone-2	111	Nozzle Calcs.: K5b 2"	405
Element: Cone-3	113	Nozzle Calcs.: MH4 30"	408
Element: Shell-16	115	Nozzle Calcs.: N15 38"	414
Element: Top head	117	Nozzle Calcs.: K10B	427
Shop Pressure Test Results:	120	Nozzle Calcs.: N5 14"	430
Field Pressure Test Results:	124	Nozzle Calcs.: MH5 30"	444
External Pressure Calculations:	128	Nozzle Calcs.: N6a 18"	450
Element and Detail Weights:	149	Nozzle Calcs.: N6b 18"	464
Nozzle Flange MAWP:	156	Nozzle Calcs.: K7a	478
Natural Frequency Calculation:	160	Nozzle Calcs.: K8a	481
Wind Load Calculation:	161	Nozzle Calcs.: K15a	484
Earthquake Load Calculation:	166	Nozzle Calcs.: K7b	487
Wind/Earthquake Shear, Bending:	168	Nozzle Calcs.: K8b	490
Wind/Earthquake Deflection:	169	Nozzle Calcs.: K15b	493
Longitudinal Stress Constants:	171	Nozzle Calcs.: MH6 30"	496
Longitudinal Allowable Stresses:	172	Nozzle Calcs.: K11a 2"	502
Longitudinal Stresses due to:	173	Nozzle Calcs.: N7 16"	505
Stress due to Combined Loads:	177	Nozzle Calcs.: MH7	518
Skirt Stress Calculations:	188	Nozzle Calcs.: K16a	524
Center of Gravity Calculation:	193	Nozzle Calcs.: K16b	527
Basering Calculations:	195	Nozzle Calcs.: K11b	530
Conical Section: Cone-1	199	Nozzle Calcs.: K12	533
Conical Section: Cone-2	204	Nozzle Calcs.: K13b	536
Conical Section: Cone-3	206	Nozzle Calcs.: P1(frm V-1403)	539
Nozzle Calcs.: N1 22"	211	Nozzle Calcs.: P2(frm ME-1702)	545
Nozzle Calcs.: K1a 2"	221	Nozzle Calcs.: P3 (from C-1402)	551
Nozzle Calcs.: K2a 2"	224	Problems/Failures Summary:	564
Nozzle Calcs.: K3a 2"	227		



DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\C-14010peRev 0.pvdb

Date of Analysis : Jul 21, 2026 4:01pm

PV Elite 28 U2, April 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 6762.000 x 6706.000 x 6738.000 x 3422.400 mm ID
 Vessel Design Length, Tangent to Tangent 46211.00 mm

Distance of Bottom Tangent above Grade 14600.00 mm
 Distance of Base above Grade 300.00 mm
 Specified Datum Line Distance 14300.00 mm

Internal Design Temperature 160 °C
 Internal Design Pressure 0.347 MPa

External Design Temperature 150 °C
 External Design Pressure 0.049 MPa

Maximum Allowable Working Pressure 0 MPa
 External Max. Allowable Working Pressure 0 MPa
 Shop Test Pressure 0.616 MPa
 Field Test Pressure 0.451 MPa

Required Minimum Design Metal Temperature -1.0 °C
 Warmest Computed Minimum Design Metal Temperature -28.0 °C

Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 60	...	...	K02100	
Head	SA-516 60	...	...	K02100	
Head	SA-516 60	...	...	K02100	
Cone	SA-516 60	...	...	K02100	
Skirt	E250BR	...	20 <= t <= 40	SA/IS 20	
Skirt	SA-516 60	...	...	K02100	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-516 60	...	...	K02100	+IT
Nozzle	SA-266 2	...	...	K03506	
Nozzle	SA-516 70	...	...	K02700	+IT
Nozzle	SA-105	...	...	K03504	+IT
Re-Pad	SA-516 70	...	...	K02700	
Re-Pad	SA-516 60	...	...	K02100	
Nozzle Flg	SA-105	...	...	K03504	
Rings	SA-516 70	...	...	K02700	
Basing	E250BR	...	20 <= t <= 40	SA/IS 20	
Base Bolting	SA-36	...	...	K02600	+IT

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Skirt-1	...	...	...	1.0000	N/A	N/S
Skirt-2	...	...	...	1.0000	N/A	N/S
Skirt-3	...	...	...	1.0000	N/A	N/S
Skirt-4	...	...	...	1.0000	N/A	N/S
Skirt-5	...	...	...	1.0000	N/A	N/S
Skirt-6	...	...	...	1.0000	N/A	No
Btm Dish End	0.377	0.05	1.000	1.0000	No	No



Shell-1	0.366	0.05	1.000	1.0000	N/A	No
Shell-2	0.347	0.05	1.000	1.0000	N/A	No
Shell-3	0.347	0.05	1.000	1.0000	N/A	No
Shell-4	0.347	0.05	1.000	1.0000	N/A	No
Shell-5	0.347	0.05	1.000	1.0000	N/A	No
Shell-6	0.347	0.05	1.000	1.0000	N/A	No
Shell-7	0.347	0.05	1.000	1.0000	N/A	No
Shell-9	0.347	0.05	1.000	1.0000	N/A	No
Shell-10	0.347	0.05	1.000	1.0000	N/A	No
Shell-11	0.347	0.05	1.000	1.0000	N/A	No
Shell-12	0.347	0.05	1.000	1.0000	N/A	No
Shell-13	0.347	0.05	1.000	1.0000	N/A	No
Shell-14	0.343	0.05	1.000	1.0000	N/A	No
Shell-15	0.347	0.05	1.000	1.0000	N/A	No
Cone-1	0.347	0.05	1.000	1.0000	N/A	No
Cone-2	0.347	0.05	0.000	1.0000	N/A	No
Cone-3	0.347	0.05	1.000	1.0000	N/A	No
Shell-16	0.347	0.05	1.000	1.0000	N/A	No
Top head	0.347	0.05	1.000	1.0000	No	No

N/S in the Creep Range column specifies that the Creep Range Temperature was not specified in the ASME Code Section II Part D for this material.

Liquid Level: 2843.97 mm Dens.: 0.001 kg/cm³ Sp. Gr.: 1.054

Stiffener Ring Specifications:

Elevation mm	Selected Type	Ring Description
850.00	W10X77	R1
5950.00	W10X88	R2
8400.00	W10X77	R3
10400.00	W10X49	R4
12400.00	W10X45	R5
14400.00	W10X88	R6
16400.00	W10X45	R7
18400.00	W10X60	R8
21400.00	W10X60	R9
23400.00	W10X49	R10
25400.00	W10X45	R11
27400.00	W10X45	R12
29400.00	W10X45	R13
31400.00	W10X45	R14
33400.00	W10X45	R15
35400.00	W10X45	R16
37400.00	W10X45	R17
38899.99	W10X39	R18
40136.74	Bar 150.0 x 20.0	R19
43196.99	Bar 250.0 x 14.0	R20

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Skirt	-13000.0	1300.0	56.0	56.0	...	...	0.70	0.70
Skirt	-10000.0	3000.0	34.0	34.0	...	...	0.70	0.70
Skirt	-7000.0	3000.0	34.0	34.0	...	...	0.70	0.70
Skirt	-4000.0	3000.0	32.0	32.0	...	...	0.70	0.70
Skirt	-1000.0	3000.0	32.0	32.0	...	...	0.70	0.70
Skirt	-0.0	1000.0	36.0	36.0	...	...	0.70	0.50
Torisph	50.0	50.0	46.0	38.0	20.2	...	1.00	1.00
Cylinder	3050.0	3000.0	28.0	28.0	11.5	...	1.00	1.00
Cylinder	6050.0	3000.0	26.0	26.0	10.9	...	1.00	1.00
Cylinder	9050.0	3000.0	26.0	26.0	10.9	...	1.00	1.00
Cylinder	12050.0	3000.0	24.0	24.0	10.9	...	1.00	1.00
Cylinder	15050.0	3000.0	22.0	22.0	10.9	...	1.00	1.00
Cylinder	18050.0	3000.0	20.0	20.0	10.9	...	1.00	1.00
Cylinder	21050.0	3000.0	20.0	20.0	10.9	...	1.00	1.00
Cylinder	24050.0	3000.0	18.0	18.0	10.9	...	1.00	1.00



Cylinder	27050.0	3000.0	18.0	18.0	10.9	...	1.00	1.00
Cylinder	30050.0	3000.0	16.0	16.0	10.9	...	1.00	1.00
Cylinder	33050.0	3000.0	16.0	16.0	10.9	...	1.00	1.00
Cylinder	36050.0	3000.0	16.0	16.0	10.9	...	1.00	1.00
Cylinder	38550.0	2500.0	16.0	16.0	10.8	...	1.00	1.00
Cylinder	40200.0	1650.0	20.0	20.0	10.9	...	1.00	1.00
Conical	41161.0	961.0	22.0	18.0	12.3	...	1.00	1.00
Conical	42161.0	1000.0	18.0	14.0	10.5	...	1.00	1.00
Conical	43161.0	1000.0	18.0	14.0	8.7	...	1.00	1.00
Cylinder	46161.0	3000.0	16.0	16.0	6.0	...	1.00	1.00
Ellipse	46211.0	50.0	18.0	14.0	6.0	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	...	No Calc	...	No Calc
50	60	...	No Calc	...	No Calc
60	70	...	No Calc	...	No Calc
70	80	38	14.512	0.049033	0.36769
80	Ring	28	8.89236	0.049033	0.55917
Ring	90	28	15.3431	0.049033	0.23832
90	Ring	26	15.3379	0.049033	0.19812
Ring	100	26	11.5602	0.049033	0.39989
100	Ring	26	11.5157	0.049033	0.40347
Ring	110	26	10.6187	0.049033	0.43795
110	Ring	24	10.6582	0.049033	0.38126
Ring	120	24	10.6582	0.049033	0.38126
120	Ring	22	10.6549	0.049033	0.33107
Ring	Ring	22	10.6549	0.049033	0.33107
Ring	130	22	10.6549	0.049033	0.33107
130	Ring	20	10.6516	0.049033	0.27364
Ring	140	20	10.6516	0.049033	0.27364
140	Ring	20	10.6516	0.049033	0.27364
Ring	150	20	12.5237	0.049033	0.17615
150	Ring	18	12.5197	0.049033	0.13087
Ring	Ring	18	10.6484	0.049033	0.2029
Ring	160	18	10.6484	0.049033	0.2029
160	Ring	18	10.6484	0.049033	0.2029
Ring	170	18	10.6484	0.049033	0.2029
170	Ring	16	10.6451	0.049033	0.14911
Ring	Ring	16	10.645	0.049033	0.14911
Ring	180	16	10.6451	0.049033	0.14911
180	Ring	16	10.6451	0.049033	0.14911
Ring	190	16	10.6451	0.049033	0.14911
190	Ring	16	10.6451	0.049033	0.14911
Ring	Ring	16	10.6451	0.049033	0.14911
Ring	200	16	10.645	0.049033	0.14911
200	Ring	16	10.645	0.049033	0.14911
Ring	210	16	9.59101	0.049033	0.20381
210	Ring	20	9.59096	0.049033	0.32014
Ring	Ring	20	8.90705	0.049033	0.34417
Ring	220	20	No Calc	0.049033	No Calc
220	230	18	8.83629	0.049033	0.26887
230	240	14	8.14833	0.049033	0.21272
240	250	14	7.25469	0.049033	0.27766
250	Ring	16	No Calc	0.049033	No Calc
Ring	260	16	9.06906	0.049033	0.23267
260	270	14	7.15341	0.049033	0.21887

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners	Allowable Length Bet. Stiffeners	Ring Inertia Required	Ring Inertia Available
------	----	-------------------------------	----------------------------------	-----------------------	------------------------

		mm	mm	mm^4	mm^4
10	20	14300	No Calc	No Calc	No Calc
20	30	14300	No Calc	No Calc	No Calc
30	40	14300	No Calc	No Calc	No Calc
40	50	14300	No Calc	No Calc	No Calc
50	60	14300	No Calc	No Calc	No Calc
60	70	14300	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc
80	Ring	1226.19	23247.3	No Calc	No Calc
Ring	90	5100	23248.6	33134616	339015552
90	Ring	5100	19498.4	No Calc	No Calc
Ring	100	2450	19502.2	39345020	374900512
100	Ring	2450	19705.9	No Calc	No Calc
Ring	110	2000	19703.9	22592618	327487040
110	Ring	2000	15933.5	No Calc	No Calc
Ring	120	2000	15933.5	20825078	205572512
120	Ring	2000	12940.2	No Calc	No Calc
Ring	Ring	2000	12940.2	20720058	186720480
Ring	130	2000	12940.2	20859732	348160704
130	Ring	2000	10238.7	No Calc	No Calc
Ring	140	2000	10238.7	20809194	179315616
140	Ring	2000	10238.7	No Calc	No Calc
Ring	150	3000	10235.5	25801040	229672928
150	Ring	3000	7703.47	No Calc	No Calc
Ring	Ring	2000	7705.12	25826942	219824128
Ring	160	2000	7705.12	20760554	182236864
160	Ring	2000	7705.12	No Calc	No Calc
Ring	170	2000	7705.12	20597792	171523552
170	Ring	2000	5725.19	No Calc	No Calc
Ring	Ring	2000	5725.19	20569066	163332544
Ring	180	2000	5725.19	20569066	163332544
180	Ring	2000	5725.19	No Calc	No Calc
Ring	190	2000	5725.19	20569074	163332544
190	Ring	2000	5725.19	No Calc	No Calc
Ring	Ring	2000	5725.19	20569066	163332544
Ring	200	1999.99	5725.19	20569038	163332544
200	Ring	1999.99	5725.19	No Calc	No Calc
Ring	210	1500	5726.93	17989436	163332544
210	Ring	1500	10254.6	No Calc	No Calc
Ring	Ring	1236.75	10253.5	14197833	155579952
Ring	220	No Calc	No Calc	No Calc	No Calc
220	230	884.94	884.94	No Calc	No Calc
230	240	902.207	902.207	No Calc	No Calc
240	250	878.462	878.462	No Calc	No Calc
250	Ring	No Calc	No Calc	No Calc	No Calc
Ring	260	3299.2	14627.7	No Calc	No Calc
260	270	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1 22"	WRC-107/537	0.391	n/a	Passed
N14 4"	WRC-297	0.517	Shell	Passed
N13A 8"	WRC-297	0.517	Shell	Passed
N13B 8"	WRC-297	0.517	Shell	Passed
N13C 8"	WRC-297	0.517	Shell	Passed
N2A 84"	WRC-297	0.946	Shell	Passed
N2B 84"	WRC-297	0.946	Shell	Passed
N4 20"	WRC-297	1.467	Nozzle	Failed
N3a 24"	WRC-297	0.832	Nozzle	Passed
N3B 24"	WRC-297	0.957	Nozzle	Passed
N15 38"	WRC-297	1.429	Shell	Failed
N5 14"	WRC-297	1.391	Nozzle	Failed
N6a 18"	WRC-297	1.764	Nozzle	Failed
N6b 18"	WRC-297	1.764	Nozzle	Failed
N7 16"	WRC-297	1.391	Nozzle	Failed



P3 (from C-140) WRC-297 1.918 Shell Failed

Wind/Earthquake Shear, Bending:

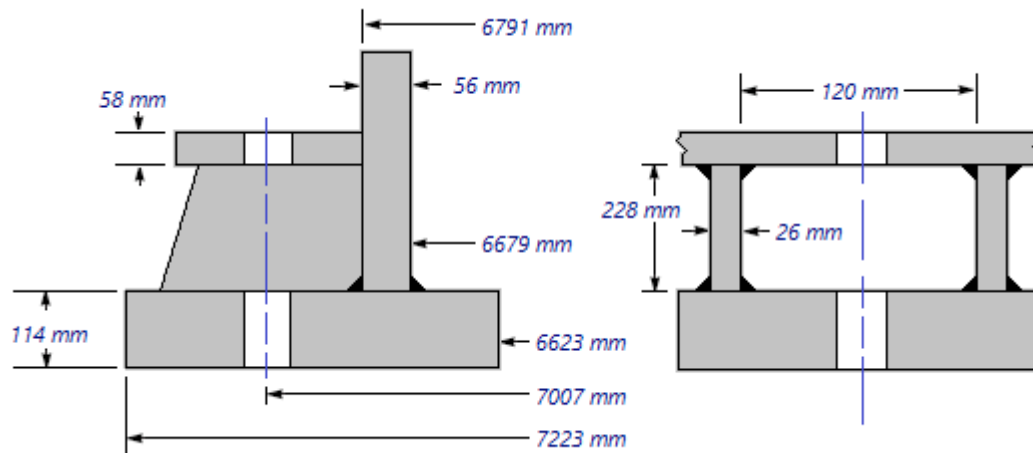
From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	650	284220	973525	8.545E+09	41.25E+09
20	30	2800	278425	973382	8.179E+09	39.98E+09
30	40	5800	264911	972738	7.364E+09	37.06E+09
40	50	8800	251582	971220	6.589E+09	34.14E+09
50	60	11800	238135	968468	5.854E+09	31.23E+09
60	70	13800	224350	964428	5.160E+09	28.33E+09
70	80	14325	219906	962752	4.938E+09	27.37E+09
80	90	15850	219680	949978	4.927E+09	27.32E+09
90	100	18850	204397	917398	4.291E+09	24.52E+09
100	110	21850	189325	902721	3.700E+09	21.79E+09
110	120	24850	174274	884871	3.154E+09	19.10E+09
120	130	27850	160817	848586	2.651E+09	16.50E+09
130	140	30850	145805	804946	2.191E+09	14.02E+09
140	150	33850	130740	758491	1776219648	11.67E+09
150	160	36850	115624	673355	1406522752	9.526E+09
160	170	39850	102191	616611	1079667712	7.591E+09
170	180	42850	87127.4	552099	795575424	5.837E+09
180	190	45850	72032.4	462109	556738944	4.315E+09
190	200	48850	58606.5	393958	360701312	3.030E+09
200	210	51600	43581	345078	207358128	1921259520
210	220	53675	30750.7	251705	114405792	1174978176
220	230	54980.5	23366.4	202928	69741024	799754048
230	240	55961	17760.1	179918	49971684	615721920
240	250	56961	14350.8	175019	33909712	438181920
250	260	58961	11669.4	171093	20894344	265056272
260	270	60486	1771.38	4353	725072	1781799

Abs Max of the all of the Stress Ratio's : 0.5013

Basing Data : Continuous Top Ring W/Gussets

Thickness of Basing	114.0000	mm
Inside Diameter of Basing	6623.0005	mm
Outside Diameter of Basing	7223.0000	mm
Nominal Diameter of Bolts	64.0000	mm
Diameter of Bolt Circle	7007.0000	mm
Number of Bolts	56	
Thickness of Gusset Plates	26.0000	mm
Average Width of Gusset Plates	206.0000	mm
Height of Gussets	228.0000	mm
Distance between Gussets	120.0000	mm
Thickness of Top Plate or Ring	58.0000	mm
Radial Width of the Top Plate	216.0000	mm
Number of Gussets	112	

Basing Sketch for Cylindrical Skirts:



Basing Cross Section View

Loads for Foundation/Support Design:

Total Wind Shear on Support	284220.	N
Total Earthquake Shear on Support	973525.	N
Wind Moment on Support	9575769088.	N-mm
Earthquake Moment on Support	42276552704.	N-mm
Maximum Initial Bolt Load	247546.	N

Wind and Earthquake moments include the effects of user defined forces and moments if any exist in the job and were specified to act (compute loads and stresses) during these cases. Also included are moment effects due to eccentric weights if any are present in the input.

Weights:

Fabricated - Bare w/o removable internals	389665.7	kgm
Shop Test - Fabricated + water (full)	1925227.9	kgm
Shipping - Fab. + rem. intls.+ shipping app.	397585.6	kgm
Erected - Fab. + rem. intls.+ insul. (etc)	451178.7	kgm
Empty - Fab. + intls. + details + wghts.	636377.8	kgm
Operating - Empty + operating liquid (no ca)	1080336.6	kgm
Field Test - Empty weight + test medium (full)	1985240.6	kgm

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Nozzle Calculation Summary (MPa & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1 22"	1	OK	...	OK	8.33	OK	Passed
K1a 2"	1	...	...	OK	3.42	OK	No Calc[*]
K2a 2"	1	...	...	OK	3.42	OK	No Calc[*]
K3a 2"	1	...	...	OK	3.42	OK	No Calc[*]
K4a 2"	1	...	...	OK	3.42	OK	No Calc[*]
K1b 2"	1	...	...	OK	3.42	OK	No Calc[*]
K2b 2"	1	...	...	OK	3.42	OK	No Calc[*]
K3b 2"	1	...	...	OK	3.42	OK	No Calc[*]
K4b 2"	1	...	...	OK	3.42	OK	No Calc[*]
N10 2"	1	...	...	OK	3.42	OK	No Calc[*]
N14 4"	1	OK	...	OK	5.26	OK	Passed
N13A 8"	1	OK	...	OK	7.16	OK	Passed
MH1 30"	1	OK	...	...	...	OK	Passed
N12 2"	1	...	...	OK	3.42	OK	No Calc[*]
N13B 8"	1	OK	...	OK	7.16	OK	Passed
N13C 8"	1	OK	...	OK	7.16	OK	Passed
N2A 84"	1	OK	...	OK	8.33	OK	Passed
MH2 30"	1	OK	...	...	...	OK	Passed
K9a 2"	1	...	...	OK	3.42	OK	No Calc[*]
N2B 84"	1	OK	...	OK	8.33	OK	Passed
K9b 2"	1	...	...	OK	3.42	OK	No Calc[*]
MH3 30"	1	OK	...	...	...	OK	Passed
N4 20"	1	OK	...	OK	8.33	OK	Passed
N3a 24"	1	OK	...	OK	8.33	OK	Passed
N3B 24"	1	OK	...	OK	8.33	OK	Passed
K5a 2"	1	...	...	OK	3.42	OK	No Calc[*]
K6a 2"	1	...	...	OK	3.42	OK	No Calc[*]
K14a 2"	1	...	...	OK	3.42	OK	No Calc[*]
K14b 2"	1	...	...	OK	3.42	OK	No Calc[*]
K6b	1	...	...	OK	3.42	OK	No Calc[*]
K5b 2"	1	...	...	OK	3.42	OK	No Calc[*]
MH4 30"	1	OK	...	...	...	OK	Passed
N15 38"	1	OK	...	OK	8.33	OK	Passed
K10B	1	...	...	OK	3.42	OK	No Calc[*]
N5 14"	1	OK	...	OK	8.33	OK	Passed
MH5 30"	1	OK	...	...	...	OK	Passed
N6a 18"	1	OK	...	OK	8.33	OK	Passed
N6b 18"	1	OK	...	OK	8.33	OK	Passed
K7a	1	...	...	OK	3.42	OK	No Calc[*]
K8a	1	...	...	OK	3.42	OK	No Calc[*]
K15a	1	...	...	OK	3.42	OK	No Calc[*]
K7b	1	...	...	OK	3.42	OK	No Calc[*]
K8b	1	...	...	OK	3.42	OK	No Calc[*]
K15b	1	...	...	OK	3.42	OK	No Calc[*]
MH6 30"	1	OK	...	...	...	OK	Passed
K11a 2"	1	...	...	OK	3.42	OK	No Calc[*]
N7 16"	1	OK	...	OK	8.33	OK	Passed
MH7	1	OK	...	...	...	OK	Passed
K16a	1	...	...	OK	3.42	OK	No Calc[*]
K16b	1	...	...	OK	3.42	OK	No Calc[*]
K11b	1	...	...	OK	3.42	OK	No Calc[*]
K12	1	...	...	OK	3.42	OK	No Calc[*]
K13b	1	...	...	OK	3.42	OK	No Calc[*]
P1(frm V-1403)	1	OK	...	OK	5.26	OK	Passed
P2(frm ME-1702)	1	OK	...	OK	5.26	OK	Passed
P3 (from C-1402	1	OK	...	OK	7.89	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1 Nozzle : K11a 2"
 Minimum MAWP Shells/Flanges : 0 Element : Cone-2
 Minimum MAPnc Shells/Flanges : 1 Element : Cone-2

Computed Vessel M.A.W.P. : 0 MPa

[*] - This was a small opening and the areas were not computed or the MAWP of this connection could not be computed because the longitudinal bending stress was greater than the hoop stress.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
70	N1 22"	0.000	0.000	1077.600
80	K1a 2"	249.999	0.000	114.325
80	K2a 2"	249.999	90.000	114.325
80	K3a 2"	249.999	180.000	114.325
80	K4a 2"	249.999	270.000	114.325
80	K1b 2"	2050.000	0.000	114.325
80	K2b 2"	2050.000	90.000	114.325
80	K3b 2"	2050.000	180.000	114.325
80	K4b 2"	2050.000	270.000	114.325
80	N10 2"	249.999	300.000	114.325
80	N14 4"	750.000	300.000	204.521
80	N13A 8"	2550.000	240.000	405.435
80	MH1 30"	849.999	45.000	1468.000
80	N12 2"	249.999	330.000	114.325
80	N13B 8"	2550.000	270.000	405.435
80	N13C 8"	2550.000	300.000	405.435
90	N2A 84"	4050.000	200.000	4187.200
90	MH2 30"	5850.001	45.000	1452.000
90	K9a 2"	5850.001	90.000	110.325
90	N2B 84"	4050.000	110.000	4187.200
130	K9b 2"	17650.002	90.000	105.004
140	MH3 30"	18150.006	45.000	1444.000
140	N4 20"	18150.006	200.000	976.000
140	N3a 24"	19870.006	90.000	1171.200
140	N3B 24"	19870.006	270.000	1179.200
140	K5a 2"	20070.006	200.000	105.004
140	K6a 2"	20070.006	180.000	105.004
140	K14a 2"	20070.006	220.000	105.004
140	K14b 2"	20900.004	220.000	105.004
140	K6b	20900.004	180.000	105.004
140	K5b 2"	20900.004	200.000	105.004
140	MH4 30"	20550.006	200.000	1444.000
140	N15 38"	19550.004	245.000	1850.400
180	K10B	30300.004	200.000	105.004
180	N5 14"	32000.002	250.000	671.200
180	MH5 30"	32000.002	45.000	1444.000
180	N6a 18"	32070.002	90.000	874.400
180	N6b 18"	32070.002	210.000	874.400
180	K7a	30050.004	0.000	105.004
180	K8a	32870.004	30.000	105.004
180	K15a	32870.004	70.000	105.004
190	K7b	33800.004	0.000	105.004
190	K8b	33800.004	30.000	105.004
190	K15b	33800.004	70.000	105.004
190	MH6 30"	34220.008	45.000	1444.000
190	K11a 2"	34220.008	120.000	105.004
210	N7 16"	39549.996	240.000	772.800
210	MH7	39870.004	45.000	1444.000
210	K16a	39870.004	90.000	105.004
210	K16b	39870.004	90.000	105.004
210	K11b	39870.004	120.000	105.004
210	K12	39870.004	150.000	105.004
210	K13b	39870.004	200.000	105.004
250	P1(frm V-1403)	43360.988	285.000	204.521
250	P2(frm ME-1702)	43360.988	245.000	204.521
250	P3 (from C-1402)	44360.992	45.000	1553.600



The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

Diameter Limit clash between : N2B 84" and K9a 2".

The Centerline Dist. is 2147.25 while the sum of the limit Radii is 2148.76 mm.

Diameter Limit clash between : N15 38" and N3B 24".

The Centerline Dist. is 1498.04 while the sum of the limit Radii is 1514.80 mm.

Diameter Limit clash between : MH4 30" and K5a 2".

The Centerline Dist. is 480.00 while the sum of the limit Radii is 774.50 mm.

Diameter Limit clash between : MH4 30" and K5b 2".

The Centerline Dist. is 350.00 while the sum of the limit Radii is 774.50 mm.

Diameter Limit clash between : K16b and K16a.

The Centerline Dist. is 0.00 while the sum of the limit Radii is 105.00 mm.

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Thk mm	Cut Length mm	Flg Class
K1a 2"	2.000 in	40	WNF	2.375	3.912	...	...	228.14	300
K2a 2"	2.000 in	40	WNF	2.375	3.912	...	...	228.14	300
K3a 2"	2.000 in	40	WNF	2.375	3.912	...	...	228.14	300
K4a 2"	2.000 in	40	WNF	2.375	3.912	...	...	228.14	300
K1b 2"	2.000 in	40	WNF	2.375	3.912	...	...	228.14	300
K2b 2"	2.000 in	40	WNF	2.375	3.912	...	...	228.14	300
K3b 2"	2.000 in	40	WNF	2.375	3.912	...	...	228.14	300
K4b 2"	2.000 in	40	WNF	2.375	3.912	...	...	228.14	300
N10 2"	2.000 in	40	WNF	2.375	3.912	...	...	228.14	300
N12 2"	2.000 in	40	WNF	2.375	3.912	...	...	228.14	300
K9a 2"	2.000 in	STD	WNF	2.375	3.912	...	...	226.14	300
K9b 2"	2.000 in	STD	WNF	2.375	3.912	...	...	220.14	300
K5a 2"	2.000 in	STD	WNF	2.375	3.912	...	...	220.14	300
K6a 2"	2.000 in	STD	WNF	2.375	3.912	...	...	220.14	300
K14a 2"	2.000 in	STD	WNF	2.375	3.912	...	...	220.14	300
K14b 2"	2.000 in	STD	WNF	2.375	3.912	...	...	220.14	300
K6b	2.000 in	STD	WNF	2.375	3.912	...	...	220.14	300
K5b 2"	2.000 in	STD	WNF	2.375	3.912	...	...	220.14	300
K10B	2.000 in	STD	WNF	2.375	3.912	...	...	216.14	300
K7a	2.000 in	STD	WNF	2.375	3.912	...	...	216.14	300
K8a	2.000 in	STD	WNF	2.375	3.912	...	...	216.14	300
K15a	2.000 in	STD	WNF	2.375	3.912	...	...	216.14	300
K7b	2.000 in	STD	WNF	2.375	3.912	...	...	216.14	300
K8b	2.000 in	STD	WNF	2.375	3.912	...	...	216.14	300
K15b	2.000 in	STD	WNF	2.375	3.912	...	...	216.14	300
K11a 2"	2.000 in	STD	WNF	2.375	3.912	...	...	216.14	300
K16a	2.000 in	STD	WNF	2.375	3.912	...	...	220.14	300
K16b	2.000 in	STD	WNF	2.375	3.912	...	...	220.14	300
K11b	2.000 in	STD	WNF	2.375	3.912	...	...	220.14	300
K12	2.000 in	STD	WNF	2.375	3.912	...	...	220.14	300
K13b	2.000 in	STD	WNF	2.375	3.912	...	...	220.14	300
N14 4"	4.000 in	STD	WNF	4.500	6.020	354.30	28.00	228.49	300
P1(frm V-1403)	4.000 in	STD	WNF	4.500	6.020	250.00	18.00	428.05	150
P2(frm ME-1702)	4.000 in	STD	WNF	4.500	6.020	250.00	14.00	440.75	150
N13A 8"	8.000 in	STD	Socke	8.625	8.179	379.08	26.00	229.79	300
N13B 8"	8.000 in	STD	WNF	8.625	8.179	379.08	26.00	229.79	300
N13C 8"	8.000 in	STD	WNF	8.625	8.179	379.08	26.00	229.79	300
N5 14"	14.000 in	Actual	WNF	14.000	10.000	855.60	16.00	420.72	150
N7 16"	16.000 in	Actual	WNF	16.000	10.000	1066.40	16.00	426.17	150
N6a 18"	18.000 in	Actual	WNF	18.000	10.000	1257.20	16.00	423.80	150
N6b 18"	18.000 in	Actual	WNF	18.000	10.000	1257.20	16.00	423.80	150
N4 20"	20.000 in	Actual	WNF	20.000	10.000	1148.00	20.00	429.63	150
N1 22"	22.000 in	Actual	WNF	22.000	10.000	658.80	30.00	643.73	150
N3a 24"	24.000 in	Actual	WNF	24.000	12.000	1563.60	36.00	433.88	150
N3B 24"	24.000 in	Actual	WNF	24.000	10.000	1509.60	30.00	433.88	150
MH1 30"	30.000 in	Actual	WNF	30.000	14.000	1522.00	30.00	449.72	150
MH2 30"	30.000 in	Actual	WNF	30.000	18.000	1462.00	32.00	447.72	150
MH3 30"	30.000 in	Actual	WNF	30.000	20.000	1422.00	28.00	441.72	150
MH4 30"	30.000 in	Actual	WNF	30.000	20.000	1422.00	26.00	441.72	150
MH5 30"	30.000 in	Actual	WNF	30.000	20.000	1422.00	28.00	437.72	150
MH6 30"	30.000 in	Actual	WNF	30.000	20.000	1422.00	28.00	437.72	150
MH7	30.000 in	Actual	WNF	30.000	20.000	1422.00	28.00	441.74	150
P3 (from C-1402)	32.000 in	Actual	WNF	32.000	18.000	1212.80	16.00	314.96	150
N15 38"	38.000 in	Actual	WNF	38.000	20.000	1665.20	26.00	454.91	150
N2A 84"	84.000 in	Actual	WNF	84.000	20.000	...	...	1014.23	150
N2B 84"	84.000 in	Actual	WNF	84.000	20.000	...	...	1014.23	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width



around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
K1a 2"	SA-106 B	20.000	10.000	...	...	...
K2a 2"	SA-106 B	20.000	10.000	...	...	...
K3a 2"	SA-106 B	20.000	10.000	...	...	...
K4a 2"	SA-106 B	20.000	10.000	...	...	...
K1b 2"	SA-106 B	20.000	10.000	...	...	...
K2b 2"	SA-106 B	20.000	10.000	...	...	...
K3b 2"	SA-106 B	20.000	10.000	...	...	...
K4b 2"	SA-106 B	20.000	10.000	...	...	...
N10 2"	SA-106 B	20.000	10.000	...	...	...
N12 2"	SA-106 B	20.000	10.000	...	...	...
K9a 2"	SA-105	20.000	10.000	...	...	...
K9b 2"	SA-106 B	20.000	10.000	...	...	...
K5a 2"	SA-106 B	18.000	10.000	...	...	...
K6a 2"	SA-106 B	18.000	10.000	...	...	...
K14a 2"	SA-106 B	18.000	10.000	...	...	...
K14b 2"	SA-106 B	18.000	10.000	...	...	...
K6b	SA-106 B	18.000	10.000	...	...	...
K5b 2"	SA-106 B	18.000	10.000	...	...	...
K10B	SA-106 B	16.000	10.000	...	...	...
K7a	SA-106 B	16.000	10.000	...	...	...
K8a	SA-106 B	16.000	10.000	...	...	...
K15a	SA-106 B	16.000	10.000	...	...	...
K7b	SA-106 B	16.000	10.000	...	...	...
K8b	SA-106 B	16.000	10.000	...	...	...
K15b	SA-106 B	16.000	10.000	...	...	...
K11a 2"	SA-106 B	16.000	10.000	...	...	...
K16a	SA-106 B	18.000	10.000	...	...	...
K16b	SA-106 B	18.000	10.000	...	...	...
K11b	SA-106 B	18.000	10.000	...	...	...
K12	SA-106 B	18.000	10.000	...	...	...
K13b	SA-106 B	18.000	10.000	...	...	...
N14 4"	SA-106 B	26.000	10.000	24.000	28.000	...
P1(frm V-14	SA-106 B	14.000	10.000	14.000	18.000	...
P2(frm ME-1	SA-106 B	14.000	10.000	12.000	14.000	...
N13A 8"	SA-106 B	20.000	10.000	16.000	26.000	...
N13B 8"	SA-106 B	20.000	10.000	16.000	26.000	...
N13C 8"	SA-106 B	20.000	10.000	16.000	26.000	...
N5 14"	SA-106 B	16.000	10.000	16.000	16.000	...
N7 16"	SA-516 60	16.000	10.000	16.000	16.000	...
N6a 18"	SA-516 60	16.000	20.000	16.000	16.000	...
N6b 18"	SA-516 60	16.000	20.000	16.000	16.000	...
N4 20"	SA-516 60	18.000	10.000	16.000	20.000	...
N1 22"	SA-106 B	28.000	10.000	20.000	30.000	...
N3a 24"	SA-516 60	18.000	14.000	16.000	36.000	...
N3B 24"	SA-516 60	18.000	10.000	16.000	30.000	...
MH1 30"	SA-516 60	20.000	10.000	16.000	30.000	...
MH2 30"	SA-516 70	20.000	10.000	16.000	32.000	...
MH3 30"	SA-516 60	18.000	10.000	16.000	28.000	...
MH4 30"	SA-516 60	18.000	10.000	20.000	26.000	...
MH5 30"	SA-106 B	16.000	10.000	16.000	28.000	...
MH6 30"	SA-106 B	16.000	10.000	16.000	28.000	...
MH7	SA-106 B	18.000	10.000	16.000	28.000	...
P3 (from C-	SA-516 60	14.000	10.000	12.000	16.000	...
N15 38"	SA-516 60	18.000	20.000	20.000	24.000	...
N2A 84"	SA-266 2	20.000	10.000	...	...	...
N2B 84"	SA-266 2	20.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:



Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
K1a 2"	249.998	0.0	200.00	0.00	Shell-1
K2a 2"	249.998	90.0	200.00	0.00	Shell-1
K3a 2"	249.998	180.0	200.00	0.00	Shell-1
K4a 2"	249.998	270.0	200.00	0.00	Shell-1
K1b 2"	2049.999	0.0	200.00	0.00	Shell-1
K2b 2"	2049.999	90.0	200.00	0.00	Shell-1
K3b 2"	2049.999	180.0	200.00	0.00	Shell-1
K4b 2"	2049.999	270.0	200.00	0.00	Shell-1
N10 2"	249.998	300.0	200.00	0.00	Shell-1
N12 2"	249.998	330.0	200.00	0.00	Shell-1
K9a 2"	5850.000	90.0	200.00	0.00	Shell-2
K9b 2"	17650.000	90.0	200.00	0.00	Shell-6
K5a 2"	20070.002	200.0	200.00	0.00	Shell-7
K6a 2"	20070.002	180.0	200.00	0.00	Shell-7
K14a 2"	20070.002	220.0	200.00	0.00	Shell-7
K14b 2"	20900.002	220.0	200.00	0.00	Shell-7
K6b	20900.002	180.0	200.00	0.00	Shell-7
K5b 2"	20900.002	200.0	200.00	0.00	Shell-7
K10B	30300.000	200.0	200.00	0.00	Shell-12
K7a	30050.002	0.0	200.00	0.00	Shell-12
K8a	32870.000	30.0	200.00	0.00	Shell-12
K15a	32870.000	70.0	200.00	0.00	Shell-12
K7b	33800.000	0.0	200.00	0.00	Shell-13
K8b	33800.000	30.0	200.00	0.00	Shell-13
K15b	33800.000	70.0	200.00	0.00	Shell-13
K11a 2"	34220.004	120.0	200.00	0.00	Shell-13
K16a	39869.996	90.0	200.00	0.00	Shell-15
K16b	39869.996	90.0	200.00	0.00	Shell-15
K11b	39869.996	120.0	200.00	0.00	Shell-15
K12	39869.996	150.0	200.00	0.00	Shell-15
K13b	39869.996	200.0	200.00	0.00	Shell-15
N14 4"	749.998	300.0	200.00	0.00	Shell-1
P1(frm V-1403)	43360.988	285.0	411.10	0.00	Shell-16
P2(frm ME-1702)	43360.988	245.0	423.80	0.00	Shell-16
N13A 8"	2549.999	240.0	200.00	0.00	Shell-1
N13B 8"	2549.999	270.0	200.00	0.00	Shell-1
N13C 8"	2549.999	300.0	200.00	0.00	Shell-1
N5 14"	31999.998	250.0	400.00	0.00	Shell-12
N7 16"	39549.992	240.0	400.00	0.00	Shell-15
N6a 18"	32070.000	90.0	400.00	0.00	Shell-12
N6b 18"	32070.000	210.0	400.00	0.00	Shell-12
N4 20"	18150.004	200.0	400.00	0.00	Shell-7
N1 22"	...	0.0	600.00	0.00	Btm Dish End
N3a 24"	19870.006	90.0	400.00	0.00	Shell-7
N3B 24"	19870.006	270.0	400.00	0.00	Shell-7
MH1 30"	849.998	45.0	400.00	0.00	Shell-1
MH2 30"	5850.000	45.0	400.00	0.00	Shell-2
MH3 30"	18150.004	45.0	400.00	0.00	Shell-7
MH4 30"	20550.006	200.0	400.00	0.00	Shell-7
MH5 30"	31999.998	45.0	400.00	0.00	Shell-12
MH6 30"	34220.004	45.0	400.00	0.00	Shell-13
MH7	39869.996	45.0	400.00	0.00	Shell-15
P3 (from C-1402)	44360.992	45.0	250.00	0.00	Shell-16
N15 38"	19550.002	245.0	400.00	0.00	Shell-7
N2A 84"	4049.999	200.0	800.00	0.00	Shell-2
N2B 84"	4049.999	110.0	800.00	0.00	Shell-2

Weld Sizes for Slip On/Socket Weld Nozzle Flanges per UW-21:

Nozzle to Flange Fillet Weld Leg dimension [xmin]:
 = min(1.4 • tn, Hub Thickness)

The Nozzle Wall thicknesses shown below are in the corroded condition. Hubs are considered to be straight.



Nominal or Actual Description Size	Schd or FVC Type	Flg Type	Noz. O/Dia in	Wall Thk mm	Hub Thk mm	Throat Thk mm	xmin Thk mm
N13A 8" 8.000 in	STD	Socket	8.625	8.179	19.431	8.015	11.450

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	SKIRT: 56.0mm THK X 6679.0mm ID X 1300.0mm	E250BR
2	SKIRT: 34.0mm THK X 6701.0mm ID X 3000.0mm	E250BR
2	SKIRT: 32.0mm THK X 6703.0mm ID X 3000.0mm	E250BR
1	SKIRT: 36.0mm THK X 6699.0mm ID X 1000.0mm	SA-516 60
1	TORISPHERICAL HEAD: 46.0mm THK X 6706.0mm ID X 50.0mm	SA-516 60
1	CYLINDER: 26.0mm THK X 6706.0mm ID X 3000.0mm	SA-516 60
1	CYLINDER: 26.0mm THK X 6654.0mm ID X 3000.0mm	SA-516 60
1	CYLINDER: 24.0mm THK X 6706.0mm ID X 3000.0mm	SA-516 60
1	CYLINDER: 22.0mm THK X 6706.0mm ID X 3000.0mm	SA-516 60
2	CYLINDER: 20.0mm THK X 6706.0mm ID X 3000.0mm	SA-516 60
2	CYLINDER: 18.0mm THK X 6706.0mm ID X 3000.0mm	SA-516 60
3	CYLINDER: 16.0mm THK X 6706.0mm ID X 3000.0mm	SA-516 60
1	CYLINDER: 16.0mm THK X 6706.0mm ID X 2500.0mm	SA-516 60
1	CYLINDER: 20.0mm THK X 6698.0mm ID X 1650.0mm	SA-516 60
1	ELLIPTICAL HEAD: 2.0 X 1, 18.0mm THK X 3422.4mm ID X 50.0mm	SA-516 60
16	PLATFORM: 1000mm X 1500mm WIDE	
12	LADDER: 4000.0mm	...
2	LADDER: 3000.0mm	...
1	INSULATION: 1016mm X 40mm THK	Exp Perlite
1	LINING: 1180.410mm X 3.000mm THK	...
12	INSULATION: 3000mm X 40mm THK	Exp Perlite
12	LINING: 3000.000mm X 3.000mm THK	...
2	SECTION RING STIFFENER: W10X77	SA-516 70
2	SECTION RING STIFFENER: W10X88	SA-516 70
1	PACKING: 900.000mm	...
4	PACKING: 3000.000mm	...
2	SECTION RING STIFFENER: W10X49	SA-516 70
9	SECTION RING STIFFENER: W10X45	SA-516 70
1	PACKING: 2100.000mm	...
2	SECTION RING STIFFENER: W10X60	SA-516 70
1	PACKING: 2999.998mm	...
1	PACKING: 530.000mm	...
1	PACKING: 2470.000mm	...



1	INSULATION: 2500mm X 40mm THK	Exp Perlite
1	LINING: 2500.000mm X 3.000mm THK	...
1	INSULATION: 1650mm X 40mm THK	Exp Perlite
1	CLADDING: 1650.000mm X 3.000mm THK	...
1	SECTION RING STIFFENER: W10X39	SA-516 70
1	BAR RING STIFFENER: 6738.0mm ID X 7038.0mm OD X 20.0mm	SA-516 70
1	INSULATION: 961mm X 40mm THK	
2	INSULATION: 1000mm X 40mm THK	
2	LADDER: 4500.0mm	...
1	INSULATION: 3000mm X 40mm THK	
1	BAR RING STIFFENER: 3454.4mm ID X 3954.4mm OD X 14.0mm	SA-516 70
1	INSULATION: 905mm X 40mm THK	
7	CLASS 150 GR 1.1, 30.0" BLIND FLANGE(S)	SA-105
1	BASERING: 7223.0mm OD X 6623.0 ID X 114.0mm	E250BR
56	BASERING BOLTS: 64.0mm DIA	SA-36
56	BASERING BOLT NUTS: 64.0mm DIA	...
1	BASERING TOP RING: 7223.0mm OD X 6791.0 ID X 58.0mm	...
112	BASERING GUSSET PLATES: 206.0mm X 206.0mm X 228.0mm X 26.0mm	...
1	CYLINDRICAL SEGMENT 28.0mm THK X 8426.9mm X 3000.0mm	SA-516 60
1	CYLINDRICAL SEGMENT 28.0mm THK X 8368.3mm X 3000.0mm	SA-516 60
1	CYLINDRICAL SEGMENT 28.0mm THK X 4272.0mm X 3000.0mm	SA-516 60
6	CON. CONICAL SEGMENT 22.0mm THK X 3043.8mm X 961.0mm X 2559.4mm	SA-516 60
1	CON. CONICAL SEGMENT 22.0mm THK X 2775.7mm X 961.0mm X 2333.9mm	SA-516 60
6	CON. CONICAL SEGMENT 18.0mm THK X 2551.4mm X 1000.0mm X 2049.9mm	SA-516 60
1	CON. CONICAL SEGMENT 18.0mm THK X 2410.8mm X 1000.0mm X 1936.9mm	SA-516 60
5	CON. CONICAL SEGMENT 18.0mm THK X 2434.8mm X 1000.0mm X 1839.3mm	SA-516 60
1	CON. CONICAL SEGMENT 18.0mm THK X 2060.9mm X 1000.0mm X 1556.8mm	SA-516 60
4	CYLINDRICAL SEGMENT 16.0mm THK X 2285.9mm X 3000.0mm	SA-516 60
1	CYLINDRICAL SEGMENT 16.0mm THK X 1607.8mm X 3000.0mm	SA-516 60
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Btm Dish End	[10]	C	-11	-28		0.692	38.000	1.00	Yes
Btm Dish End	[7]	C	-6	-104		0.319	46.000	1.00	Yes
Shell-1	[8]	C	-18	-48		0.522	28.000	1.00	No
Shell-2	[8]	C	-20	-48		0.540	26.000	1.00	No
Shell-3	[8]	C	-20	-48		0.538	26.000	1.00	No
Shell-4	[8]	C	-22	-45	-29	0.587	24.000	1.00	No
Shell-5	[8]	C	-24	-44	-29	0.643	22.000	1.00	No
Shell-6	[8]	C	-27	-43	-29	0.711	20.000	1.00	No
Shell-7	[8]	C	-27	-43	-29	0.711	20.000	1.00	No
Shell-9	[8]	C	-29	-41	-29	0.794	18.000	1.00	No
Shell-10	[8]	C	-29	-41	-29	0.794	18.000	1.00	No
Shell-11	[8]	C	-32	-38	-29	0.900	16.000	1.00	No
Shell-12	[8]	C	-32	-38	-29	0.900	16.000	1.00	No
Shell-13	[8]	C	-32	-38	-29	0.900	16.000	1.00	No
Shell-14	[8]	C	-32	-38	-29	0.900	16.000	1.00	No
Shell-15	[8]	C	-27	-43	-29	0.711	20.000	1.00	No
Cone-1	[8]	C	-29	-34	-29	0.908	18.000	1.00	No
Cone-2	[8]	C	-36	-36	-29	1.000	14.000	1.00	No
Cone-3	[8]	C	-36	-46	-29	0.804	14.000	1.00	No
Shell-16	[8]	C	-32	-48	-29	0.459	16.000	1.00	No
Top head	[10]	C	-36	-48	-29	0.529	14.000	1.00	No
Top head	[7]	C	-29	-48	-29	0.405	18.000	1.00	No
R19	[23]	B	-7	-23	-29	0.711	20.000	1.00	No
R20	[23]	B	-18	-48	-29	0.459	14.000	1.00	No
N1 22"	[1]	B	-13	-48	-29	0.391	30.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1a 2"	[1]	C	-8	-104	-29	0.037	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2a 2"	[1]	B	-8	-104	-29	0.037	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K3a 2"	[1]	B	-8	-104	-29	0.037	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K4a 2"	[1]	B	-8	-104	-29	0.037	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K1b 2"	[1]	B	-8	-104	-29	0.037	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K2b 2"	[1]	B	-8	-104	-29	0.037	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K3b 2"	[1]	B	-8	-104	-29	0.037	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K4b 2"	[1]	B	-8	-104	-29	0.037	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N10 2"	[1]	B	-8	-104	-29	0.037	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N14 4"	[1]	B	2	-28		0.520	28.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N13A 8"	[1]	B	0	-30	-29	0.520	26.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
MH1 30"	[1]	C	-18	-48		0.520	28.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N12 2"	[1]	B	-8	-104	-29	0.037	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N13B 8"	[1]	B	0	-30	-29	0.520	26.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N13C 8"	[1]	B	0	-30	-29	0.520	26.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N2A 84"	[1]	B	12	-104		0.183	20.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
MH2 30"	[1]	B	0	-28		0.540	26.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K9a 2"	[1]	C	-8	-104	-29	0.030	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N2B 84"	[1]	B	12	-104		0.183	20.000	1.00	No



Nozzle Flg	[5]	A	-18	-104					
K9b 2"	[1]	B	-8	-104	-29	0.035	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
MH3 30"	[1]	C	-27	-43	-29	0.711	20.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N4 20"	[1]	C	-27	-43	-29	0.711	20.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N3a 24"	[1]	C	-27	-43	-29	0.711	20.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-104					
N3B 24"	[1]	C	-27	-43	-29	0.711	20.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K5a 2"	[1]	B	-8	-104	-29	0.035	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K6a 2"	[1]	C	-8	-104	-29	0.035	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K14a 2"	[1]	B	-8	-104	-29	0.035	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K14b 2"	[1]	B	-8	-104	-29	0.035	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K6b	[1]	B	-8	-104	-29	0.035	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K5b 2"	[1]	B	-8	-104	-29	0.035	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
MH4 30"	[1]	C	-27	-43	-29	0.711	20.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N15 38"	[1]	C	-27	-43	-29	0.711	20.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K10B	[1]	B	-8	-104	-29	0.035	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N5 14"	[1]	B	-14	-29	-29	0.900	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
MH5 30"	[1]	B	-14	-29	-29	0.900	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N6a 18"	[1]	B	-14	-29	-29	0.900	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N6b 18"	[1]	B	-14	-29	-29	0.900	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K7a	[1]	B	-8	-104	-29	0.035	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K8a	[1]	B	-8	-104	-29	0.035	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K15a	[1]	B	-8	-104	-29	0.035	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K7b	[1]	B	-8	-104	-29	0.035	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K8b	[1]	B	-8	-104	-29	0.035	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K15b	[1]	B	-8	-104	-29	0.035	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
MH6 30"	[1]	B	-14	-29	-29	0.900	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K11a 2"	[1]	B	-8	-104	-29	0.035	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
N7 16"	[1]	C	-32	-48	-29	0.710	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
MH7	[1]	B	-7	-29	-29	0.710	20.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K16a	[1]	B	-8	-104	-29	0.035	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K16b	[1]	B	-8	-104	-29	0.035	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K11b	[1]	B	-8	-104	-29	0.035	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K12	[1]	B	-8	-104	-29	0.035	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
K13b	[1]	B	-8	-104	-29	0.035	3.423	1.00	No
Nozzle Flg	[5]	A	-18	-104					
P1(frm V-1403)	[1]	B	-14	-48	-29	0.459	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104					
P2(frm ME-1702)	[1]	B	-18	-48	-29	0.459	14.000	1.00	No



Nozzle Flg	[5]	A	-18	-104				
P3 (from C-140	[1]	C	-1	-39	0.459	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-104				
Nozl. Bolting	[22]		-48		1.000	64.000	1.00	No

Warmest MDMT: 12 -23

Required Minimum Design Metal Temperature -1.0 °C

Warmest Computed Minimum Design Metal Temperature -28.0 °C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Note 60 80 There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (SIASME):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	4.4480	N
Mass	Pounds	0.4536	kgm
Area	sq. inches	645.1600	mm ²
Moment	ft. lbs.	1356.3000	N-mm
Stress	lbs./sq.in.	0.0069	MPa
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0069	MPa
Elast. Modulus	lbs./sq.in.	0.0069	MPa
Pipe Density	lbs./cu.in.	0.0277	kg/cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg/cm ³
Wind Speed	miles/hr	1.6093	km/hr
Tray Weight	lbs./sq.ft.	0.0005	kg/cm ²
Inertia	in.4	416231.0000	mm ⁴
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	0.0479	kPa
Elevation	Feet	304.8000	mm
Volume	in.3	0.0164	ltr
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	0.3472	MPa
Design Internal Temperature	160.0	°C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	-1.0	°C
Type of Construction	Welded	
Special Service	None	
Degree of Radiography	RT-1	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(b) Note [35]
Pressure Test Position	Vertical
Test Performed in Corroded Condition	Yes

Load Case 1	NP+EW+1.1WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+1.1WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+1.1WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+1.1WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW



Load Case 16		IP+VE+EW	
Load Case 17		NP+VO+OW	
Load Case 18		FS+BS+IP+OW	
Load Case 19		FS+BS+EP+OW	
Load Case 20		BL+IP+OW	
Wind Design Code		ASCE/SEI 7-16	
Wind Load Reduction Scale Factor		1.100	
Basic Wind Speed	[V]	97.2	km/hr
Surface Roughness Category		C: Open Terrain	
Type of Surface		Moderately Smooth	
Base Elevation		300	mm
Percent Wind for Hydrotest		50.0	
Using User defined Wind Press. Vs Elev.		N	
Height of Hill or Escarpment	H or Hh	0	mm
Distance Upwind of Crest	Lh	0	mm
Distance from Crest to the Vessel	x	0	mm
Type of Terrain (Hill, Escarpment)		Flat	
Damping Factor (Beta) for Wind (Ope)		0.0100	
Damping Factor (Beta) for Wind (Empty)		0.0000	
Damping Factor (Beta) for Wind (Filled)		0.0000	
Site Elevation above Sea Level		0	mm
Seismic Design Code		ASCE/SEI 7-16	
Seismic Load Reduction Scale Factor		1.000	
Importance Factor		1.250	
Table Value Fa		1.600	
Table Value Fv		2.400	
Max. Mapped Res. Acceleration	[Ss]	0.147	
Max. Eff. Ground Acceleration	[S]	0.065	
Force Modification Factor R		1.250	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	2.500	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Moment Reduction Factor Tau		1.000	
M.A.W.P. + Static Head to Element Bottom		Y	
Consider MAP New and Cold in Noz. Design		N	
Consider External Loads for Nozzle Des.		Y	
Use ASME VIII-1 Appendix 1-9		N	
Perform Blast Load Analysis		No	
Material Database Year	Current w/Addenda or Code Year		

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	No
Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/26
Element From Node	10



Element To Node	20
Element Type	Skirt Support
Description	Skirt-1
Distance "FROM" to "TO"	1300 mm
Skirt Outside Diameter	6791 mm
Diameter of Skirt at Base	6791 mm
Skirt Thickness	56 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	56 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	160 °C
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.38
Material Name	E250BR
Allowable Stress, Ambient	117 MPa
Allowable Stress, Operating	117 MPa
Allowable Stress, Hydrotest	216 MPa
Material Density	0.00775 kg/cm³
P Number Thickness	32.004 mm
Yield Stress, Operating	210.8 MPa
UCS-66 Chart Curve Designation	A
External Pressure Chart Name	CS-2
UNS Number	SA/IS 20
Class / Thickness / Grade	:: 20 <= t <= 40
Product Form	Plate; bar; shapes
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No
Element From Node	10
Detail Type	Weight
Detail ID	TAILING LUG WT
Dist. from "FROM" Node / Offset dist	300 mm
Miscellaneous Weight	24515 N
Offset from Element Centerline	3400 mm
Element From Node	10
Detail Type	Weight
Detail ID	BRACING WT
Dist. from "FROM" Node / Offset dist	300 mm
Miscellaneous Weight	49030 N
Offset from Element Centerline	0 mm

Element#	2/26
Element From Node	20
Element To Node	30
Element Type	Skirt Support
Description	Skirt-2
Distance "FROM" to "TO"	3000 mm
Skirt Outside Diameter	6769 mm
Diameter of Skirt at Base	6769 mm
Skirt Thickness	34 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	34 mm
External Corrosion Allowance	0 mm
Design Temperature Internal Pressure	160 °C
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.38
Material Name	E250BR
Efficiency, Longitudinal Seam	0.7
Efficiency, Head-to-Skirt or Circ. Seam	0.7
Weld is pre-Heated	No
Element From Node	20
Detail Type	Platform
Detail ID	SK PLT1
Dist. from "FROM" Node / Offset dist	2700 mm
Platform Start Angle (degrees)	270.0



Platform End Angle (degrees)	0.0	
Platform Wind Area	157982	mm ²
Platform Weight	18167	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	270.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.3628	N/mm
Platform Length (top head platform)	0	mm

Element#	3/26	
Element From Node	30	
Element To Node	40	
Element Type	Skirt Support	
Description	Skirt-3	
Distance "FROM" to "TO"	3000	mm
Skirt Outside Diameter	6769	mm
Diameter of Skirt at Base	6769	mm
Skirt Thickness	34	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	34	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	160	°C
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.38	
Material Name	E250BR	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element#	4/26	
Element From Node	40	
Element To Node	50	
Element Type	Skirt Support	
Description	Skirt-4	
Distance "FROM" to "TO"	3000	mm
Skirt Outside Diameter	6767	mm
Diameter of Skirt at Base	6767	mm
Skirt Thickness	32	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	32	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	160	°C
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.38	
Material Name	E250BR	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Platform	
Detail ID	PLAT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	700	mm
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	157982	mm ²
Platform Weight	18155	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm



Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.3628	N/mm
Platform Length (top head platform)	0	mm

Element#	5/26	
Element From Node	50	
Element To Node	60	
Element Type	Skirt Support	
Description	Skirt-5	
Distance "FROM" to "TO"	3000	mm
Skirt Outside Diameter	6767	mm
Diameter of Skirt at Base	6767	mm
Skirt Thickness	32	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	32	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	160	°C
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.38	
Material Name	E250BR	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.7	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Platform	
Detail ID	SKIRT PLTF-2	
Dist. from "FROM" Node / Offset dist	1000	mm
Platform Start Angle (degrees)	270.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	568486	mm ²
Platform Weight	17793	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	270.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	3000	mm
Unit Weight of Ladder	0.363	N/mm
Platform Length (top head platform)	0	mm

Element#	6/26	
Element From Node	60	
Element To Node	70	
Element Type	Skirt Support	
Description	Skirt-6	
Distance "FROM" to "TO"	1000	mm
Skirt Outside Diameter	6771	mm
Diameter of Skirt at Base	6771	mm
Skirt Thickness	36	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	36	mm
External Corrosion Allowance	0	mm
Design Temperature Internal Pressure	160	°C
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.38	
Material Name	SA-516 60	
Allowable Stress, Ambient	118	MPa



Allowable Stress, Operating	118	MPa
Allowable Stress, Hydrotest	198	MPa
Material Density	0.00775	kg/cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	193.8	MPa
UCS-66 Chart Curve Designation	C	
External Pressure Chart Name	CS-2	
UNS Number	K02100	
Product Form	Plate	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Head-to-Skirt or Circ. Seam	0.5	
Weld is pre-Heated	No	

Element#	7/26	
Element From Node	70	
Element To Node	80	
Element Type	Torispherical	
Description	Btm Dish End	
Distance "FROM" to "TO"	50	mm
Inside Diameter	6706	mm
Element Thickness	38	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	46	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3472	MPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	0.04903	MPa
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.38	
Material Name	SA-516 60	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Tori Head Crown Radius	6781	mm
Tori Head Knuckle Radius	406.86	mm
Weld is pre-Heated	No	

Element From Node	70	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	-966.97	mm
Height/Length of Liquid	1017	mm
Liquid Density	0.001053	kg/cm ³

Element From Node	70	
Detail Type	Insulation	
Detail ID	PP	
Dist. from "FROM" Node / Offset dist	-966.97	mm
Height/Length of Insulation	1017	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm ³

Element From Node	70	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	-1130.4	mm
Height/Length of Lining	1180.4	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³

Element From Node	70	
Detail Type	Nozzle	
Detail ID	N1 22"	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	22	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	



Weight of Nozzle (Used if > 0)	12423	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	70	
Detail Type	Weight	
Detail ID	N1 PIPE WT	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	40920	N
Offset from Element Centerline	4755	mm

Element#	8/26	
Element From Node	80	
Element To Node	90	
Element Type	Cylinder	
Description	Shell-1	
Distance "FROM" to "TO"	3000	mm
Element Outside Diameter	6762	mm
Element Thickness	28	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	28	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3472	MPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	0.04903	MPa
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.38	
Material Name	SA-516 60	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Platform	
Detail ID	BTL PLTF	
Dist. from "FROM" Node / Offset dist	0.001131	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	360.0	
Platform Wind Area	2368490	mm ²
Platform Weight	67802	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	3000	mm
Unit Weight of Ladder	0.363	N/mm
Platform Length (top head platform)	0	mm

Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid: 20	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1827	mm
Liquid Density	0.001053	kg/cm ³

Element From Node	80	
Detail Type	Insulation	
Detail ID	PP	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm ³

Element From Node	80	
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Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³

Element From Node	80	
Detail Type	Ring	
Detail ID	R1	
Dist. from "FROM" Node / Offset dist	800	mm
Stiffening Ring Moment of Inertia	189384992	mm ⁴
Distance from Shell Surface to Centr	134.62	mm
Stiffening Ring Cross Sectional Area	14581	mm ²
Material Name	SA-516 70	
Stiffening Ring Section Name	W10X77	
Height of Section Ring	269.2	mm
Using Custom Stiffener Section	No	

Element From Node	80	
Detail Type	Nozzle	
Detail ID	K1a 2"	
Dist. from "FROM" Node / Offset dist	200	mm
Nozzle Diameter	2	in.
Nozzle Schedule	40	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B +IT	

Element From Node	80	
Detail Type	Nozzle	
Detail ID	K2a 2"	
Dist. from "FROM" Node / Offset dist	200	mm
Nozzle Diameter	2	in.
Nozzle Schedule	40	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	80	
Detail Type	Nozzle	
Detail ID	K3a 2"	
Dist. from "FROM" Node / Offset dist	200	mm
Nozzle Diameter	2	in.
Nozzle Schedule	40	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	80	
Detail Type	Nozzle	
Detail ID	K4a 2"	
Dist. from "FROM" Node / Offset dist	200	mm
Nozzle Diameter	2	in.
Nozzle Schedule	40	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	



Element From Node 80
 Detail Type Nozzle
 Detail ID K1b 2"
 Dist. from "FROM" Node / Offset dist 2000 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule 40
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B

Element From Node 80
 Detail Type Nozzle
 Detail ID K2b 2"
 Dist. from "FROM" Node / Offset dist 2000 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule 40
 Nozzle Class 300
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B

Element From Node 80
 Detail Type Nozzle
 Detail ID K3b 2"
 Dist. from "FROM" Node / Offset dist 2000 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule 40
 Nozzle Class 300
 Layout Angle 180.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B

Element From Node 80
 Detail Type Nozzle
 Detail ID K4b 2"
 Dist. from "FROM" Node / Offset dist 2000 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule 40
 Nozzle Class 300
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B

Element From Node 80
 Detail Type Nozzle
 Detail ID N10 2"
 Dist. from "FROM" Node / Offset dist 200 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule 40
 Nozzle Class 300
 Layout Angle 300.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B

Element From Node 80
 Detail Type Nozzle
 Detail ID N14 4"
 Dist. from "FROM" Node / Offset dist 700 mm



Nozzle Diameter	4	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	300.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	80	
Detail Type	Nozzle	
Detail ID	N13A 8"	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	8	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	240.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	80	
Detail Type	Nozzle	
Detail ID	MH1 30"	
Dist. from "FROM" Node / Offset dist	800	mm
Nozzle Diameter	30	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	45.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 60 +IT	

Element From Node	80	
Detail Type	Nozzle	
Detail ID	N12 2"	
Dist. from "FROM" Node / Offset dist	200	mm
Nozzle Diameter	2	in.
Nozzle Schedule	40	
Nozzle Class	300	
Layout Angle	330.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	80	
Detail Type	Nozzle	
Detail ID	N13B 8"	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	8	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	80	
Detail Type	Nozzle	
Detail ID	N13C 8"	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	8	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	300.0	
Blind Flange (Y/N)	N	



Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	80	
Detail Type	Weight	
Detail ID	N13A PIPE WT	
Dist. from "FROM" Node / Offset dist	2500	mm
Miscellaneous Weight	16798	N
Offset from Element Centerline	4560	mm
Element From Node	80	
Detail Type	Weight	
Detail ID	N13B PIPE WT	
Dist. from "FROM" Node / Offset dist	2500	mm
Miscellaneous Weight	16798	N
Offset from Element Centerline	4560	mm
Element From Node	80	
Detail Type	Weight	
Detail ID	N13C PIPE WT	
Dist. from "FROM" Node / Offset dist	2500	mm
Miscellaneous Weight	16798	N
Offset from Element Centerline	4560	mm

Element#	9/26
Element From Node	90
Element To Node	100
Element Type	Cylinder
Description	Shell-2
Distance "FROM" to "TO"	3000 mm
Inside Diameter	6706 mm
Element Thickness	26 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	26 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.3472 MPa
Design Temperature Internal Pressure	160 °C
Design External Pressure	0.04903 MPa
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.38
Material Name	SA-516 60
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	90
Detail Type	Platform
Detail ID	PLTF-4
Dist. from "FROM" Node / Offset dist	1000 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	180.0
Platform Wind Area	1957980 mm ²
Platform Weight	34573 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	85 mm
Platform Force Coefficient	1.2
Ladder Layout Angle	180.0
Ladder Start Elevation	0 mm
Ladder End Elevation	4000 mm
Unit Weight of Ladder	0.363 N/mm
Platform Length (top head platform)	0 mm

Element From Node	90
Detail Type	Insulation



Detail ID	PP	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm ³
Element From Node	90	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³
Element From Node	90	
Detail Type	Ring	
Detail ID	R2	
Dist. from "FROM" Node / Offset dist	2900	mm
Stiffening Ring Moment of Inertia	222266992	mm ⁴
Distance from Shell Surface to Centr	137.67	mm
Stiffening Ring Cross Sectional Area	16710	mm ²
Material Name	SA-516 70	
Stiffening Ring Section Name	W10X88	
Height of Section Ring	275.3	mm
Using Custom Stiffener Section	No	
Element From Node	90	
Detail Type	Nozzle	
Detail ID	N2A 84"	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	84	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	200.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	30440	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	
Element From Node	90	
Detail Type	Nozzle	
Detail ID	MH2 30"	
Dist. from "FROM" Node / Offset dist	2800	mm
Nozzle Diameter	30	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	45.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 70 +IT	
Element From Node	90	
Detail Type	Nozzle	
Detail ID	K9a 2"	
Dist. from "FROM" Node / Offset dist	2800	mm
Nozzle Diameter	2	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-105 +IT	
Element From Node	90	
Detail Type	Nozzle	
Detail ID	N2B 84"	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	84	in.



Nozzle Schedule	None
Nozzle Class	150
Layout Angle	110.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	30440 N
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2

Element From Node	90
Detail Type	Weight
Detail ID	SCHOEPENTOETER
Dist. from "FROM" Node / Offset dist	1000 mm
Miscellaneous Weight	40361 N
Offset from Element Centerline	0 mm

Element From Node	90
Detail Type	Weight
Detail ID	N2A INSERT PLT
Dist. from "FROM" Node / Offset dist	1000 mm
Miscellaneous Weight	15690 N
Offset from Element Centerline	0 mm

Element From Node	90
Detail Type	Weight
Detail ID	N2B INSERT PLT
Dist. from "FROM" Node / Offset dist	1000 mm
Miscellaneous Weight	15690 N
Offset from Element Centerline	0 mm

Element#	10/26
Element From Node	100
Element To Node	110
Element Type	Cylinder
Description	Shell-3
Distance "FROM" to "TO"	3000 mm
Element Outside Diameter	6706 mm
Element Thickness	26 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	26 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.3472 MPa
Design Temperature Internal Pressure	160 °C
Design External Pressure	0.04903 MPa
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.38
Material Name	SA-516 60
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	100
Detail Type	Platform
Detail ID	PLTF-5
Dist. from "FROM" Node / Offset dist	2000 mm
Platform Start Angle (degrees)	180.0
Platform End Angle (degrees)	0.0
Platform Wind Area	1957981 mm ²
Platform Weight	34573 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	85 mm
Platform Force Coefficient	1.2
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	4000 mm
Unit Weight of Ladder	0.363 N/mm



Platform Length (top head platform)	0	mm
Element From Node	100	
Detail Type	Packing	
Detail ID	Pack Bed#1-(1)	
Dist. from "FROM" Node / Offset dist	2100	mm
Height of Packed Section	900	mm
Density	0.00025	kg/cm ³
Percent Volume Holdup	15.0	
Specific Gravity of Packing Liquid	1.0	
Element From Node	100	
Detail Type	Insulation	
Detail ID	PP	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm ³
Element From Node	100	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³
Element From Node	100	
Detail Type	Ring	
Detail ID	R3	
Dist. from "FROM" Node / Offset dist	2350	mm
Stiffening Ring Moment of Inertia	189384992	mm ⁴
Distance from Shell Surface to Centr	134.62	mm
Stiffening Ring Cross Sectional Area	14581	mm ²
Material Name	SA-516 70	
Stiffening Ring Section Name	W10X77	
Height of Section Ring	269.2	mm
Using Custom Stiffener Section	No	
Element From Node	100	
Detail Type	Weight	
Detail ID	SR+BEAM GRID1	
Dist. from "FROM" Node / Offset dist	200	mm
Miscellaneous Weight	5050.1	N
Offset from Element Centerline	0	mm
Element From Node	100	
Detail Type	Weight	
Detail ID	SP GRID-BED1 WT	
Dist. from "FROM" Node / Offset dist	900	mm
Miscellaneous Weight	14709	N
Offset from Element Centerline	0	mm

Element#	11/26	
Element From Node	110	
Element To Node	120	
Element Type	Cylinder	
Description	Shell-4	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	6706	mm
Element Thickness	24	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	24	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3472	MPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	0.04903	MPa
Design Temperature External Pressure	150	°C



Effective Diameter Multiplier	1.38	
Material Name	SA-516 60	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	110	
Detail Type	Packing	
Detail ID	Pack Bed#1-(2)	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	3000	mm
Density	0.00025	kg/cm ³
Percent Volume Holdup	15.0	
Specific Gravity of Packing Liquid	1.0	
Element From Node	110	
Detail Type	Insulation	
Detail ID	PP	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm ³
Element From Node	110	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³
Element From Node	110	
Detail Type	Ring	
Detail ID	R4	
Dist. from "FROM" Node / Offset dist	1350	mm
Stiffening Ring Moment of Inertia	113215000	mm ⁴
Distance from Shell Surface to Centr	126.75	mm
Stiffening Ring Cross Sectional Area	9290.3	mm ²
Material Name	SA-516 70	
Stiffening Ring Section Name	W10X49	
Height of Section Ring	253.5	mm
Using Custom Stiffener Section	No	
Element From Node	110	
Detail Type	Weight	
Detail ID	CLIP WT-1	
Dist. from "FROM" Node / Offset dist	1000	mm
Miscellaneous Weight	30153	N
Offset from Element Centerline	0	mm

Element#	12/26	
Element From Node	120	
Element To Node	130	
Element Type	Cylinder	
Description	Shell-5	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	6706	mm
Element Thickness	22	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	22	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3472	MPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	0.04903	MPa
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.38	
Material Name	SA-516 60	
Efficiency, Longitudinal Seam	1.0	



Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	120	
Detail Type	Platform	
Detail ID	PLTF-6	
Dist. from "FROM" Node / Offset dist	0	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	1957980	mm ²
Platform Weight	34541	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.363	N/mm
Platform Length (top head platform)	0	mm
Element From Node	120	
Detail Type	Packing	
Detail ID	Pack Bed#1-(3)	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	3000	mm
Density	0.00025	kg/cm ³
Percent Volume Holdup	15.0	
Specific Gravity of Packing Liquid	1.0	
Element From Node	120	
Detail Type	Insulation	
Detail ID	PP	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm ³
Element From Node	120	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³
Element From Node	120	
Detail Type	Ring	
Detail ID	R5	
Dist. from "FROM" Node / Offset dist	350	mm
Stiffening Ring Moment of Inertia	103225000	mm ⁴
Distance from Shell Surface to Centr	128.27	mm
Stiffening Ring Cross Sectional Area	8580.6	mm ²
Material Name	SA-516 70	
Stiffening Ring Section Name	W10X45	
Height of Section Ring	256.5	mm
Using Custom Stiffener Section	No	
Element From Node	120	
Detail Type	Ring	
Detail ID	R6	
Dist. from "FROM" Node / Offset dist	2350	mm
Stiffening Ring Moment of Inertia	222266992	mm ⁴
Distance from Shell Surface to Centr	137.67	mm
Stiffening Ring Cross Sectional Area	16710	mm ²
Material Name	SA-516 70	
Stiffening Ring Section Name	W10X88	
Height of Section Ring	275.3	mm



Using Custom Stiffener Section

No

Element#	13/26
Element From Node	130
Element To Node	140
Element Type	Cylinder
Description	Shell-6
Distance "FROM" to "TO"	3000 mm
Inside Diameter	6706 mm
Element Thickness	20 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	20 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.3472 MPa
Design Temperature Internal Pressure	160 °C
Design External Pressure	0.04903 MPa
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.38
Material Name	SA-516 60
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	130
Detail Type	Platform
Detail ID	PLTF-7
Dist. from "FROM" Node / Offset dist	1000 mm
Platform Start Angle (degrees)	180.0
Platform End Angle (degrees)	0.0
Platform Wind Area	1957980 mm ²
Platform Weight	34526 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	85 mm
Platform Force Coefficient	1.2
Ladder Layout Angle	0.0
Ladder Start Elevation	0 mm
Ladder End Elevation	4000 mm
Unit Weight of Ladder	0.363 N/mm
Platform Length (top head platform)	0 mm
Element From Node	130
Detail Type	Packing
Detail ID	Pack Bed#1-(4)
Dist. from "FROM" Node / Offset dist	0 mm
Height of Packed Section	2100 mm
Density	0.00025 kg/cm ³
Percent Volume Holdup	15.0
Specific Gravity of Packing Liquid	1.0
Element From Node	130
Detail Type	Insulation
Detail ID	PP
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	40 mm
Density	0.000192 kg/cm ³
Element From Node	130
Detail Type	Lining
Detail ID	Clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	3000 mm
Thickness of Lining	3 mm
Density	0.008027 kg/cm ³



Element From Node	130	
Detail Type	Ring	
Detail ID	R7	
Dist. from "FROM" Node / Offset dist	1350	mm
Stiffening Ring Moment of Inertia	103225000	mm ⁴
Distance from Shell Surface to Centr	128.27	mm
Stiffening Ring Cross Sectional Area	8580.6	mm ²
Material Name	SA-516 70	
Stiffening Ring Section Name	W10X45	
Height of Section Ring	256.5	mm
Using Custom Stiffener Section	No	

Element From Node	130	
Detail Type	Nozzle	
Detail ID	K9b 2"	
Dist. from "FROM" Node / Offset dist	2600	mm
Nozzle Diameter	2	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	130	
Detail Type	Weight	
Detail ID	SR+BEAM BL1	
Dist. from "FROM" Node / Offset dist	2100	mm
Miscellaneous Weight	11179	N
Offset from Element Centerline	0	mm

Element From Node	130	
Detail Type	Weight	
Detail ID	BL-BED1 WT	
Dist. from "FROM" Node / Offset dist	2100	mm
Miscellaneous Weight	75996	N
Offset from Element Centerline	0	mm

Element From Node	130	
Detail Type	Weight	
Detail ID	SR+BEAM DIST-1	
Dist. from "FROM" Node / Offset dist	2300	mm
Miscellaneous Weight	3667.4	N
Offset from Element Centerline	0	mm

Element From Node	130	
Detail Type	Weight	
Detail ID	DIST-1 DEAD WT	
Dist. from "FROM" Node / Offset dist	2300	mm
Miscellaneous Weight	28535	N
Offset from Element Centerline	0	mm

Element#	14/26	
Element From Node	140	
Element To Node	150	
Element Type	Cylinder	
Description	Shell-7	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	6706	mm
Element Thickness	20	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	20	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3472	MPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	0.04903	MPa
Design Temperature External Pressure	150	°C



Effective Diameter Multiplier	1.38	
Material Name	SA-516 60	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	140	
Detail Type	Platform	
Detail ID	PLTF-8	
Dist. from "FROM" Node / Offset dist	2000	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	1957980	mm ²
Platform Weight	34526	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.363	N/mm
Platform Length (top head platform)	0	mm
Element From Node	140	
Detail Type	Insulation	
Detail ID	PP	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm ³
Element From Node	140	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³
Element From Node	140	
Detail Type	Ring	
Detail ID	R8	
Dist. from "FROM" Node / Offset dist	350	mm
Stiffening Ring Moment of Inertia	141935008	mm ⁴
Distance from Shell Surface to Centr	129.79	mm
Stiffening Ring Cross Sectional Area	11355	mm ²
Material Name	SA-516 70	
Stiffening Ring Section Name	W10X60	
Height of Section Ring	259.6	mm
Using Custom Stiffener Section	No	
Element From Node	140	
Detail Type	Nozzle	
Detail ID	MH3 30"	
Dist. from "FROM" Node / Offset dist	100	mm
Nozzle Diameter	30	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	45.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 60 +IT	
Element From Node	140	
Detail Type	Nozzle	
Detail ID	N4 20"	



Dist. from "FROM" Node / Offset dist	100	mm
Nozzle Diameter	20	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	200.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 60	+IT

Element From Node	140	
Detail Type	Nozzle	
Detail ID	N3a 24"	
Dist. from "FROM" Node / Offset dist	1820	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	6396.6	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 60	+IT

Element From Node	140	
Detail Type	Nozzle	
Detail ID	N3B 24"	
Dist. from "FROM" Node / Offset dist	1820	mm
Nozzle Diameter	24	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	5228.4	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 60	+IT

Element From Node	140	
Detail Type	Nozzle	
Detail ID	K5a 2"	
Dist. from "FROM" Node / Offset dist	2020	mm
Nozzle Diameter	2	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	200.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	140	
Detail Type	Nozzle	
Detail ID	K6a 2"	
Dist. from "FROM" Node / Offset dist	2020	mm
Nozzle Diameter	2	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	+IT

Element From Node	140	
Detail Type	Nozzle	
Detail ID	K14a 2"	
Dist. from "FROM" Node / Offset dist	2020	mm
Nozzle Diameter	2	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	220.0	



Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	140	
Detail Type	Nozzle	
Detail ID	K14b 2"	
Dist. from "FROM" Node / Offset dist	2850	mm
Nozzle Diameter	2	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	220.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	140	
Detail Type	Nozzle	
Detail ID	K6b	
Dist. from "FROM" Node / Offset dist	2850	mm
Nozzle Diameter	2	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	140	
Detail Type	Nozzle	
Detail ID	K5b 2"	
Dist. from "FROM" Node / Offset dist	2850	mm
Nozzle Diameter	2	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	200.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	140	
Detail Type	Nozzle	
Detail ID	MH4 30"	
Dist. from "FROM" Node / Offset dist	2500	mm
Nozzle Diameter	30	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	200.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 60 +IT	
Element From Node	140	
Detail Type	Nozzle	
Detail ID	N15 38"	
Dist. from "FROM" Node / Offset dist	1500	mm
Nozzle Diameter	38	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	245.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	6293.9	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 60 +IT	



Element From Node	140	
Detail Type	Weight	
Detail ID	N4 PIPE WT	
Dist. from "FROM" Node / Offset dist	100	mm
Miscellaneous Weight	75526	N
Offset from Element Centerline	4704	mm

Element From Node	140	
Detail Type	Weight	
Detail ID	N3A PIPE WT	
Dist. from "FROM" Node / Offset dist	1820	mm
Miscellaneous Weight	102296	N
Offset from Element Centerline	4755	mm

Element From Node	140	
Detail Type	Weight	
Detail ID	N3B PIPE WT	
Dist. from "FROM" Node / Offset dist	1820	mm
Miscellaneous Weight	102296	N
Offset from Element Centerline	4755	mm

Element From Node	140	
Detail Type	Weight	
Detail ID	N4 INT PIPE WT	
Dist. from "FROM" Node / Offset dist	100	mm
Miscellaneous Weight	36880	N
Offset from Element Centerline	0	mm

Element From Node	140	
Detail Type	Weight	
Detail ID	SR+BEAM CT-1	
Dist. from "FROM" Node / Offset dist	2450	mm
Miscellaneous Weight	22063	N
Offset from Element Centerline	0	mm

Element From Node	140	
Detail Type	Weight	
Detail ID	CT-1 DEAD WT	
Dist. from "FROM" Node / Offset dist	2450	mm
Miscellaneous Weight	49030	N
Offset from Element Centerline	0	mm

Element From Node	140	
Detail Type	Weight	
Detail ID	CLIP WT-2	
Dist. from "FROM" Node / Offset dist	2000	mm
Miscellaneous Weight	30153	N
Offset from Element Centerline	0	mm

Element From Node	140	
Detail Type	Weight	
Detail ID	SP GRID-BED2 WT	
Dist. from "FROM" Node / Offset dist	1500	mm
Miscellaneous Weight	29418	N
Offset from Element Centerline	0	mm

Element From Node	140	
Detail Type	Weight	
Detail ID	SR+BEAM GRID2	
Dist. from "FROM" Node / Offset dist	650	mm
Miscellaneous Weight	5050.1	N
Offset from Element Centerline	0	mm

Element From Node	140	
Detail Type	Weight	
Detail ID	N15 PIPE WT	
Dist. from "FROM" Node / Offset dist	1500	mm
Miscellaneous Weight	271793	N
Offset from Element Centerline	4933	mm



Element# 15/26
 Element From Node 150
 Element To Node 160
 Element Type Cylinder
 Description Shell-9
 Distance "FROM" to "TO" 3000 mm
 Inside Diameter 6706 mm
 Element Thickness 18 mm
 Internal Corrosion Allowance 1 mm
 Nominal Thickness 18 mm
 External Corrosion Allowance 0 mm
 Design Internal Pressure 0.3472 MPa
 Design Temperature Internal Pressure 160 °C
 Design External Pressure 0.04903 MPa
 Design Temperature External Pressure 150 °C
 Effective Diameter Multiplier 1.38
 Material Name SA-516 60
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Weld is pre-Heated No

Element From Node 150
 Detail Type Packing
 Detail ID Pack Bed#2-(2)
 Dist. from "FROM" Node / Offset dist 0 mm
 Height of Packed Section 3000 mm
 Density 0.00025 kg/cm³
 Percent Volume Holdup 15.0
 Specific Gravity of Packing Liquid 1.0

Element From Node 150
 Detail Type Insulation
 Detail ID PP
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Insulation 3000 mm
 Thickness of Insulation 40 mm
 Density 0.000192 kg/cm³

Element From Node 150
 Detail Type Lining
 Detail ID Clad
 Dist. from "FROM" Node / Offset dist 0 mm
 Height/Length of Lining 3000 mm
 Thickness of Lining 3 mm
 Density 0.008027 kg/cm³

Element From Node 150
 Detail Type Ring
 Detail ID R9
 Dist. from "FROM" Node / Offset dist 350 mm
 Stiffening Ring Moment of Inertia 141935008 mm⁴
 Distance from Shell Surface to Centr 129.79 mm
 Stiffening Ring Cross Sectional Area 11355 mm²
 Material Name SA-516 70
 Stiffening Ring Section Name W10X60
 Height of Section Ring 259.6 mm
 Using Custom Stiffener Section No

Element From Node 150
 Detail Type Ring
 Detail ID R10
 Dist. from "FROM" Node / Offset dist 2350 mm
 Stiffening Ring Moment of Inertia 113215000 mm⁴
 Distance from Shell Surface to Centr 126.75 mm
 Stiffening Ring Cross Sectional Area 9290.3 mm²
 Material Name SA-516 70
 Stiffening Ring Section Name W10X49
 Height of Section Ring 253.5 mm



Using Custom Stiffener Section

No

Element#	16/26
Element From Node	160
Element To Node	170
Element Type	Cylinder
Description	Shell-10
Distance "FROM" to "TO"	3000 mm
Inside Diameter	6706 mm
Element Thickness	18 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.3472 MPa
Design Temperature Internal Pressure	160 °C
Design External Pressure	0.04903 MPa
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.38
Material Name	SA-516 60
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	160
Detail Type	Platform
Detail ID	PLTF-10
Dist. from "FROM" Node / Offset dist	1000 mm
Platform Start Angle (degrees)	0.0
Platform End Angle (degrees)	180.0
Platform Wind Area	1957981 mm ²
Platform Weight	34510 N
Platform Railing Weight	0 N/mm
Platform Grating Weight	0.017 kg/cm ²
Platform Width	1500 mm
Platform Height	1000 mm
Platform Clearance or End Offset	85 mm
Platform Force Coefficient	1.2
Ladder Layout Angle	180.0
Ladder Start Elevation	0 mm
Ladder End Elevation	4000 mm
Unit Weight of Ladder	0.363 N/mm
Platform Length (top head platform)	0 mm
Element From Node	160
Detail Type	Packing
Detail ID	Pack Bed#2-(3)
Dist. from "FROM" Node / Offset dist	0 mm
Height of Packed Section	3000 mm
Density	0.00025 kg/cm ³
Percent Volume Holdup	15.0
Specific Gravity of Packing Liquid	1.0
Element From Node	160
Detail Type	Insulation
Detail ID	PP
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	3000 mm
Thickness of Insulation	40 mm
Density	0.000192 kg/cm ³
Element From Node	160
Detail Type	Lining
Detail ID	Clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	3000 mm
Thickness of Lining	3 mm
Density	0.008027 kg/cm ³



Element From Node	160	
Detail Type	Ring	
Detail ID	R11	
Dist. from "FROM" Node / Offset dist	1350	mm
Stiffening Ring Moment of Inertia	103225000	mm ⁴
Distance from Shell Surface to Centr	128.27	mm
Stiffening Ring Cross Sectional Area	8580.6	mm ²
Material Name	SA-516 70	
Stiffening Ring Section Name	W10X45	
Height of Section Ring	256.5	mm
Using Custom Stiffener Section	No	

Element#	17/26	
Element From Node	170	
Element To Node	180	
Element Type	Cylinder	
Description	Shell-11	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	6706	mm
Element Thickness	16	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3472	MPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	0.04903	MPa
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.38	
Material Name	SA-516 60	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	170	
Detail Type	Platform	
Detail ID	PLTF-11	
Dist. from "FROM" Node / Offset dist	2000	mm
Platform Start Angle (degrees)	180.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	1957980	mm ²
Platform Weight	34494	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.363	N/mm
Platform Length (top head platform)	0	mm

Element From Node	170	
Detail Type	Packing	
Detail ID	Pack Bed#2-(4)	
Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	3000	mm
Density	0.00025	kg/cm ³
Percent Volume Holdup	15.0	
Specific Gravity of Packing Liquid	1.0	

Element From Node	170	
Detail Type	Insulation	
Detail ID	PP	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	40	mm



Density	0.000192	kg/cm ³
Element From Node	170	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³
Element From Node	170	
Detail Type	Ring	
Detail ID	R12	
Dist. from "FROM" Node / Offset dist	350	mm
Stiffening Ring Moment of Inertia	103225000	mm ⁴
Distance from Shell Surface to Centr	128.27	mm
Stiffening Ring Cross Sectional Area	8580.6	mm ²
Material Name	SA-516 70	
Stiffening Ring Section Name	W10X45	
Height of Section Ring	256.5	mm
Using Custom Stiffener Section	No	
Element From Node	170	
Detail Type	Ring	
Detail ID	R13	
Dist. from "FROM" Node / Offset dist	2350	mm
Stiffening Ring Moment of Inertia	103225000	mm ⁴
Distance from Shell Surface to Centr	128.27	mm
Stiffening Ring Cross Sectional Area	8580.6	mm ²
Material Name	SA-516 70	
Stiffening Ring Section Name	W10X45	
Height of Section Ring	256.5	mm
Using Custom Stiffener Section	No	
Element From Node	170	
Detail Type	Weight	
Detail ID	SR+BEAM BL2	
Dist. from "FROM" Node / Offset dist	2550	mm
Miscellaneous Weight	11179	N
Offset from Element Centerline	0	mm
Element From Node	170	
Detail Type	Weight	
Detail ID	BL-BED2 WT	
Dist. from "FROM" Node / Offset dist	2550	mm
Miscellaneous Weight	75996	N
Offset from Element Centerline	0	mm
Element From Node	170	
Detail Type	Weight	
Detail ID	SR+BEAM DIST-2	
Dist. from "FROM" Node / Offset dist	2750	mm
Miscellaneous Weight	3667.4	N
Offset from Element Centerline	0	mm
Element From Node	170	
Detail Type	Weight	
Detail ID	DIST-2 DEAD WT	
Dist. from "FROM" Node / Offset dist	2750	mm
Miscellaneous Weight	28535	N
Offset from Element Centerline	0	mm
Element From Node	170	
Detail Type	Weight	
Detail ID	CLIP WT-3	
Dist. from "FROM" Node / Offset dist	0	mm
Miscellaneous Weight	30153	N
Offset from Element Centerline	0	mm



Element#	18/26	
Element From Node	180	
Element To Node	190	
Element Type	Cylinder	
Description	Shell-12	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	6706	mm
Element Thickness	16	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3472	MPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	0.04903	MPa
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.38	
Material Name	SA-516 60	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	
Element From Node	180	
Detail Type	Insulation	
Detail ID	PP	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm ³
Element From Node	180	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³
Element From Node	180	
Detail Type	Ring	
Detail ID	R14	
Dist. from "FROM" Node / Offset dist	1350	mm
Stiffening Ring Moment of Inertia	103225000	mm ⁴
Distance from Shell Surface to Centr	128.27	mm
Stiffening Ring Cross Sectional Area	8580.6	mm ²
Material Name	SA-516 70	
Stiffening Ring Section Name	W10X45	
Height of Section Ring	256.5	mm
Using Custom Stiffener Section	No	
Element From Node	180	
Detail Type	Nozzle	
Detail ID	K10B	
Dist. from "FROM" Node / Offset dist	250	mm
Nozzle Diameter	2	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	200.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	180	
Detail Type	Nozzle	
Detail ID	N5 14"	
Dist. from "FROM" Node / Offset dist	1950	mm
Nozzle Diameter	14	in.
Nozzle Schedule	None	
Nozzle Class	150	



Layout Angle	250.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	180	
Detail Type	Nozzle	
Detail ID	MH5 30"	
Dist. from "FROM" Node / Offset dist	1950	mm
Nozzle Diameter	30	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	45.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	180	
Detail Type	Nozzle	
Detail ID	N6a 18"	
Dist. from "FROM" Node / Offset dist	2020	mm
Nozzle Diameter	18	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 60 +IT	
Element From Node	180	
Detail Type	Nozzle	
Detail ID	N6b 18"	
Dist. from "FROM" Node / Offset dist	2020	mm
Nozzle Diameter	18	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	210.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 60 +IT	
Element From Node	180	
Detail Type	Nozzle	
Detail ID	K7a	
Dist. from "FROM" Node / Offset dist	0.002326	mm
Nozzle Diameter	2	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	180	
Detail Type	Nozzle	
Detail ID	K8a	
Dist. from "FROM" Node / Offset dist	2820	mm
Nozzle Diameter	2	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	30.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	



Element From Node	180	
Detail Type	Nozzle	
Detail ID	K15a	
Dist. from "FROM" Node / Offset dist	2820	mm
Nozzle Diameter	2	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	70.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element From Node	180	
Detail Type	Weight	
Detail ID	N5 PIPE WT	
Dist. from "FROM" Node / Offset dist	1950	mm
Miscellaneous Weight	85224	N
Offset from Element Centerline	4624	mm

Element From Node	180	
Detail Type	Weight	
Detail ID	N6A PIPE WT	
Dist. from "FROM" Node / Offset dist	2020	mm
Miscellaneous Weight	127174	N
Offset from Element Centerline	4675	mm

Element From Node	180	
Detail Type	Weight	
Detail ID	N6B PIPE WT	
Dist. from "FROM" Node / Offset dist	2020	mm
Miscellaneous Weight	127174	N
Offset from Element Centerline	4675	mm

Element From Node	180	
Detail Type	Weight	
Detail ID	N5 INT PIPE WT	
Dist. from "FROM" Node / Offset dist	1950	mm
Miscellaneous Weight	29820	N
Offset from Element Centerline	0	mm

Element From Node	180	
Detail Type	Weight	
Detail ID	MANWAY DAVIT	
Dist. from "FROM" Node / Offset dist	1500	mm
Miscellaneous Weight	3922.4	N
Offset from Element Centerline	0	mm

Element#	19/26	
Element From Node	190	
Element To Node	200	
Element Type	Cylinder	
Description	Shell-13	
Distance "FROM" to "TO"	3000	mm
Inside Diameter	6706	mm
Element Thickness	16	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3472	MPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	0.04903	MPa
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.38	
Material Name	SA-516 60	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	



Weld is pre-Heated

No

Element From Node	190	
Detail Type	Platform	
Detail ID	PLTF-12	
Dist. from "FROM" Node / Offset dist	0.004651	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	1957980	mm ²
Platform Weight	34494	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.363	N/mm
Platform Length (top head platform)	0	mm

Element From Node	190	
Detail Type	Packing	
Detail ID	Pack Bed#3-(1)	
Dist. from "FROM" Node / Offset dist	2470	mm
Height of Packed Section	530	mm
Density	0.00025	kg/cm ³
Percent Volume Holdup	15.0	
Specific Gravity of Packing Liquid	1.0	

Element From Node	190	
Detail Type	Insulation	
Detail ID	PP	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm ³

Element From Node	190	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	3000	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³

Element From Node	190	
Detail Type	Ring	
Detail ID	R15	
Dist. from "FROM" Node / Offset dist	350	mm
Stiffening Ring Moment of Inertia	103225000	mm ⁴
Distance from Shell Surface to Centr	128.27	mm
Stiffening Ring Cross Sectional Area	8580.6	mm ²
Material Name	SA-516 70	
Stiffening Ring Section Name	W10X45	
Height of Section Ring	256.5	mm
Using Custom Stiffener Section	No	

Element From Node	190	
Detail Type	Ring	
Detail ID	R16	
Dist. from "FROM" Node / Offset dist	2350	mm
Stiffening Ring Moment of Inertia	103225000	mm ⁴
Distance from Shell Surface to Centr	128.27	mm
Stiffening Ring Cross Sectional Area	8580.6	mm ²
Material Name	SA-516 70	
Stiffening Ring Section Name	W10X45	
Height of Section Ring	256.5	mm
Using Custom Stiffener Section	No	



Element From Node 190
 Detail Type Nozzle
 Detail ID K7b
 Dist. from "FROM" Node / Offset dist 750.01 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule STD
 Nozzle Class 300
 Layout Angle 0.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B

Element From Node 190
 Detail Type Nozzle
 Detail ID K8b
 Dist. from "FROM" Node / Offset dist 750.01 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule STD
 Nozzle Class 300
 Layout Angle 30.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B

Element From Node 190
 Detail Type Nozzle
 Detail ID K15b
 Dist. from "FROM" Node / Offset dist 750.01 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule STD
 Nozzle Class 300
 Layout Angle 70.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B

Element From Node 190
 Detail Type Nozzle
 Detail ID MH6 30"
 Dist. from "FROM" Node / Offset dist 1170 mm
 Nozzle Diameter 30 in.
 Nozzle Schedule None
 Nozzle Class 150
 Layout Angle 45.0
 Blind Flange (Y/N) Y
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B

Element From Node 190
 Detail Type Nozzle
 Detail ID K11a 2"
 Dist. from "FROM" Node / Offset dist 1170 mm
 Nozzle Diameter 2 in.
 Nozzle Schedule STD
 Nozzle Class 300
 Layout Angle 120.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 0 N
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-106 B

Element From Node 190
 Detail Type Weight
 Detail ID SR+BEAM CT-2
 Dist. from "FROM" Node / Offset dist 285.01 mm



Miscellaneous Weight	22063	N
Offset from Element Centerline	0	mm
Element From Node	190	
Detail Type	Weight	
Detail ID	CT-2 DEAD WT	
Dist. from "FROM" Node / Offset dist	285.01	mm
Miscellaneous Weight	49030	N
Offset from Element Centerline	0	mm
Element From Node	190	
Detail Type	Weight	
Detail ID	SR+BEAM GRID3	
Dist. from "FROM" Node / Offset dist	1570	mm
Miscellaneous Weight	5050.1	N
Offset from Element Centerline	0	mm
Element From Node	190	
Detail Type	Weight	
Detail ID	SP GRID-BED3 WT	
Dist. from "FROM" Node / Offset dist	2270	mm
Miscellaneous Weight	11767	N
Offset from Element Centerline	0	mm

Element#	20/26	
Element From Node	200	
Element To Node	210	
Element Type	Cylinder	
Description	Shell-14	
Distance "FROM" to "TO"	2500	mm
Inside Diameter	6706	mm
Element Thickness	16	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3432	MPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	0.04903	MPa
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.38	
Material Name	SA-516 60	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	200	
Detail Type	Platform	
Detail ID	PLTF-13	
Dist. from "FROM" Node / Offset dist	1000	mm
Platform Start Angle (degrees)	180.0	
Platform End Angle (degrees)	0.0	
Platform Wind Area	1957980	mm ²
Platform Weight	34494	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.363	N/mm
Platform Length (top head platform)	0	mm

Element From Node	200	
Detail Type	Packing	
Detail ID	Pack Bed#3-(2)	



Dist. from "FROM" Node / Offset dist	0	mm
Height of Packed Section	2470	mm
Density	0.00025	kg/cm ³
Percent Volume Holdup	15.0	
Specific Gravity of Packing Liquid	1.0	
Element From Node	200	
Detail Type	Insulation	
Detail ID	PP	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	2500	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm ³
Element From Node	200	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	2500	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm ³
Element From Node	200	
Detail Type	Ring	
Detail ID	R17	
Dist. from "FROM" Node / Offset dist	1350	mm
Stiffening Ring Moment of Inertia	103225000	mm ⁴
Distance from Shell Surface to Centr	128.27	mm
Stiffening Ring Cross Sectional Area	8580.6	mm ²
Material Name	SA-516 70	
Stiffening Ring Section Name	W10X45	
Height of Section Ring	256.5	mm
Using Custom Stiffener Section	No	
Element From Node	200	
Detail Type	Weight	
Detail ID	SR+BEAM BL3	
Dist. from "FROM" Node / Offset dist	2470	mm
Miscellaneous Weight	11277	N
Offset from Element Centerline	0	mm
Element From Node	200	
Detail Type	Weight	
Detail ID	BL-BED3 WT	
Dist. from "FROM" Node / Offset dist	2470	mm
Miscellaneous Weight	75996	N
Offset from Element Centerline	0	mm
Element From Node	200	
Detail Type	Weight	
Detail ID	CLIP WT-4	
Dist. from "FROM" Node / Offset dist	1000	mm
Miscellaneous Weight	30153	N
Offset from Element Centerline	0	mm

Element#	21/26
Element From Node	210
Element To Node	220
Element Type	Cylinder
Description	Shell-15
Distance "FROM" to "TO"	1650 mm
Element Outside Diameter	6738 mm
Element Thickness	20 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	20 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.3472 MPa
Design Temperature Internal Pressure	160 °C



Design External Pressure	0.04903	MPa
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.38	
Material Name	SA-516 60	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	

Element From Node	210	
Detail Type	Insulation	
Detail ID	PP	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	1650	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm³

Element From Node	210	
Detail Type	Lining	
Detail ID	Clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	1650	mm
Thickness of Lining	3	mm
Density	0.008027	kg/cm³

Element From Node	210	
Detail Type	Ring	
Detail ID	R18	
Dist. from "FROM" Node / Offset dist	350	mm
Stiffening Ring Moment of Inertia	86992304	mm⁴
Distance from Shell Surface to Centr	125.98	mm
Stiffening Ring Cross Sectional Area	7419.3	mm²
Material Name	SA-516 70	
Stiffening Ring Section Name	W10X39	
Height of Section Ring	252	mm
Using Custom Stiffener Section	No	

Element From Node	210	
Detail Type	Ring	
Detail ID	R19	
Dist. from "FROM" Node / Offset dist	1586.8	mm
Inside Diameter of Ring	6738	mm
Thickness of Ring	20	mm
Outside Diameter of Ring	7038	mm
Material Name	SA-516 70	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	

Element From Node	210	
Detail Type	Nozzle	
Detail ID	N7 16"	
Dist. from "FROM" Node / Offset dist	1000	mm
Nozzle Diameter	16	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	240.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 60 +IT	

Element From Node	210	
Detail Type	Nozzle	
Detail ID	MH7	
Dist. from "FROM" Node / Offset dist	1320	mm
Nozzle Diameter	30	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	45.0	
Blind Flange (Y/N)	Y	
Weight of Nozzle (Used if > 0)	0	N



Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	210	
Detail Type	Nozzle	
Detail ID	K16a	
Dist. from "FROM" Node / Offset dist	1320	mm
Nozzle Diameter	2	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	210	
Detail Type	Nozzle	
Detail ID	K16b	
Dist. from "FROM" Node / Offset dist	1320	mm
Nozzle Diameter	2	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	210	
Detail Type	Nozzle	
Detail ID	K11b	
Dist. from "FROM" Node / Offset dist	1320	mm
Nozzle Diameter	2	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	120.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	210	
Detail Type	Nozzle	
Detail ID	K12	
Dist. from "FROM" Node / Offset dist	1320	mm
Nozzle Diameter	2	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	150.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	210	
Detail Type	Nozzle	
Detail ID	K13b	
Dist. from "FROM" Node / Offset dist	1320	mm
Nozzle Diameter	2	in.
Nozzle Schedule	STD	
Nozzle Class	300	
Layout Angle	200.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	210	
Detail Type	Weight	



Detail ID	DEMISTER	
Dist. from "FROM" Node / Offset dist	1000	mm
Miscellaneous Weight	14709	N
Offset from Element Centerline	0	mm
Element From Node	210	
Detail Type	Weight	
Detail ID	SR+BEAM DIST-3	
Dist. from "FROM" Node / Offset dist	170	mm
Miscellaneous Weight	3667.4	N
Offset from Element Centerline	0	mm
Element From Node	210	
Detail Type	Weight	
Detail ID	DIST-3 DEAD WT	
Dist. from "FROM" Node / Offset dist	170	mm
Miscellaneous Weight	28535	N
Offset from Element Centerline	0	mm
Element From Node	210	
Detail Type	Weight	
Detail ID	SR+BEAM DIST-3	
Dist. from "FROM" Node / Offset dist	170	mm
Miscellaneous Weight	22063	N
Offset from Element Centerline	0	mm
Element From Node	210	
Detail Type	Weight	
Detail ID	N7 PIPE WT	
Dist. from "FROM" Node / Offset dist	1000	mm
Miscellaneous Weight	105228	N
Offset from Element Centerline	4653	mm
Element From Node	210	
Detail Type	Weight	
Detail ID	N7 INT PIPE WT	
Dist. from "FROM" Node / Offset dist	1000	mm
Miscellaneous Weight	29820	N
Offset from Element Centerline	0	mm

Element#	22/26	
Element From Node	220	
Element To Node	230	
Element Type	Conical	
Description	Cone-1	
Distance "FROM" to "TO"	961	mm
Element Outside Diameter	6738	mm
Element Thickness	18	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	22	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3472	MPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	0.04903	MPa
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.38	
Material Name	SA-516 60	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Cone Diameter at "To" End	5672.2	mm
Design Length of Cone	961	mm
Half Apex Angle of Cone	29.008617	degrees
Toriconical (Y/N)	N	
Weld is pre-Heated	No	
Element From Node	220	
Detail Type	Platform	
Detail ID	PLTF-14	



Dist. from "FROM" Node / Offset dist	300	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	1957980	mm ²
Platform Weight	34636	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	180.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4000	mm
Unit Weight of Ladder	0.363	N/mm
Platform Length (top head platform)	0	mm

Element From Node	220	
Detail Type	Insulation	
Detail ID	Ins	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	961	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm ³

Element From Node	220	
Detail Type	Weight	
Detail ID	SR+BEAM MIST	
Dist. from "FROM" Node / Offset dist	170	mm
Miscellaneous Weight	11277	N
Offset from Element Centerline	0	mm

Element From Node	220	
Detail Type	Weight	
Detail ID	TRUNION WT	
Dist. from "FROM" Node / Offset dist	170	mm
Miscellaneous Weight	49030	N
Offset from Element Centerline	3400	mm

Element#	23/26	
Element From Node	230	
Element To Node	240	
Element Type	Conical	
Description	Cone-2	
Distance "FROM" to "TO"	1000	mm
Element Outside Diameter	5672.2	mm
Element Thickness	14	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	18	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3472	MPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	0.04903	MPa
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.38	
Material Name	SA-516 60	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Cone Diameter at "To" End	4563.6	mm
Design Length of Cone	1000	mm
Half Apex Angle of Cone	29.00004	degrees
Toriconical (Y/N)	N	
Weld is pre-Heated	No	

Element From Node	230	
Detail Type	Insulation	
Detail ID	Ins	
Dist. from "FROM" Node / Offset dist	0	mm



Height/Length of Insulation	1000	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm ³

Element#	24/26
Element From Node	240
Element To Node	250
Element Type	Conical
Description	Cone-3
Distance "FROM" to "TO"	1000 mm
Element Outside Diameter	4563.2 mm
Element Thickness	14 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	18 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.3472 MPa
Design Temperature Internal Pressure	160 °C
Design External Pressure	0.04903 MPa
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.38
Material Name	SA-516 60
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Cone Diameter at "To" End	3455 mm
Design Length of Cone	1000 mm
Half Apex Angle of Cone	28.991724 degrees
Toriconical (Y/N)	N
Weld is pre-Heated	No

Element From Node	240
Detail Type	Insulation
Detail ID	Ins
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1000 mm
Thickness of Insulation	40 mm
Density	0.000192 kg/cm ³

Element#	25/26
Element From Node	250
Element To Node	260
Element Type	Cylinder
Description	Shell-16
Distance "FROM" to "TO"	3000 mm
Inside Diameter	3422.4 mm
Element Thickness	16 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	0.3472 MPa
Design Temperature Internal Pressure	160 °C
Design External Pressure	0.04903 MPa
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.38
Material Name	SA-516 60
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	250
Detail Type	Platform
Detail ID	PLTF-15
Dist. from "FROM" Node / Offset dist	107.99 mm
Platform Start Angle (degrees)	180.0
Platform End Angle (degrees)	0.0
Platform Wind Area	1752730 mm ²
Platform Weight	21772 N



Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4500	mm
Unit Weight of Ladder	0.363	N/mm
Platform Length (top head platform)	0	mm

Element From Node	250	
Detail Type	Platform	
Detail ID	PLTF-16	
Dist. from "FROM" Node / Offset dist	2258	mm
Platform Start Angle (degrees)	0.0	
Platform End Angle (degrees)	180.0	
Platform Wind Area	1752730	mm ²
Platform Weight	21772	N
Platform Railing Weight	0	N/mm
Platform Grating Weight	0.017	kg/cm ²
Platform Width	1500	mm
Platform Height	1000	mm
Platform Clearance or End Offset	85	mm
Platform Force Coefficient	1.2	
Ladder Layout Angle	0.0	
Ladder Start Elevation	0	mm
Ladder End Elevation	4500	mm
Unit Weight of Ladder	0.363	N/mm
Platform Length (top head platform)	0	mm

Element From Node	250	
Detail Type	Insulation	
Detail ID	Ins	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	3000	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm ³

Element From Node	250	
Detail Type	Ring	
Detail ID	R20	
Dist. from "FROM" Node / Offset dist	36	mm
Inside Diameter of Ring	3454.4	mm
Thickness of Ring	14	mm
Outside Diameter of Ring	3954.4	mm
Material Name	SA-516 70	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	

Element From Node	250	
Detail Type	Nozzle	
Detail ID	P1(frm V-1403)	
Dist. from "FROM" Node / Offset dist	199.99	mm
Nozzle Diameter	4	in.
Nozzle Schedule	STD	
Nozzle Class	150	
Layout Angle	285.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B +IT	

Element From Node	250	
Detail Type	Nozzle	
Detail ID	P2(frm ME-1702)	
Dist. from "FROM" Node / Offset dist	199.99	mm
Nozzle Diameter	4	in.
Nozzle Schedule	STD	



Nozzle Class	150	
Layout Angle	245.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	
Element From Node	250	
Detail Type	Nozzle	
Detail ID	P3 (from C-1402)	
Dist. from "FROM" Node / Offset dist	1200	mm
Nozzle Diameter	32	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	45.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-516 60 +IT	
Element From Node	250	
Detail Type	Weight	
Detail ID	P3 Fy weight	
Dist. from "FROM" Node / Offset dist	1200	mm
Miscellaneous Weight	395711	N
Offset from Element Centerline	3214	mm
Element From Node	250	
Detail Type	Weight	
Detail ID	P4 FY weight	
Dist. from "FROM" Node / Offset dist	1200	mm
Miscellaneous Weight	395711	N
Offset from Element Centerline	3214	mm

Element#	26/26	
Element From Node	260	
Element To Node	270	
Element Type	Elliptical	
Description	Top head	
Distance "FROM" to "TO"	50	mm
Inside Diameter	3422.4	mm
Element Thickness	14	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	18	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	0.3472	MPa
Design Temperature Internal Pressure	160	°C
Design External Pressure	0.04903	MPa
Design Temperature External Pressure	150	°C
Effective Diameter Multiplier	1.38	
Material Name	SA-516 60	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	

Element From Node	260	
Detail Type	Insulation	
Detail ID	Ins	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	905.6	mm
Thickness of Insulation	40	mm
Density	0.000192	kg/cm³



XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
	Skirt-1	...	1300	...	1300
	Skirt-2	...	4300	...	3000
	Skirt-3	...	7300	...	3000
	Skirt-4	...	10300	...	3000
	Skirt-5	...	13300	...	3000
	Skirt-6	...	14300	...	1000
	Btm Dish End	...	14350	...	50
	Shell-1	...	17350	...	3000
	Shell-2	...	20350	...	3000
	Shell-3	...	23350	...	3000
	Shell-4	...	26350	...	3000
	Shell-5	...	29350	...	3000
	Shell-6	...	32350	...	3000
	Shell-7	...	35350	...	3000
	Shell-9	...	38350	...	3000
	Shell-10	...	41350	...	3000
	Shell-11	...	44350	...	3000
	Shell-12	...	47350	...	3000
	Shell-13	...	50350	...	3000
	Shell-14	...	52850	...	2500
	Shell-15	...	54500	...	1650
	Cone-1	...	55461	...	961
	Cone-2	...	56461	...	1000
	Cone-3	...	57461	...	1000
	Shell-16	...	60461	...	3000
	Top head	...	60511	...	50

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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**Internal Pressure Results Summary:****Element Thickness, Pressure, Diameter and Allowable Stress :**

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa	Element#
	Skirt-1	...	56	1	6791	...	1
	Skirt-2	...	34	1	6769	...	2
	Skirt-3	...	34	1	6769	...	3
	Skirt-4	...	32	1	6767	...	4
	Skirt-5	...	32	1	6767	...	5
	Skirt-6	...	36	1	6771	...	6
Btm	Dish End	0.3766	46	1	6706	118	7
	Shell-1	0.3661	28	1	6762	118	8
	Shell-2	0.3472	26	1	6706	118	9
	Shell-3	0.3472	26	1	6706	118	10
	Shell-4	0.3472	24	1	6706	118	11
	Shell-5	0.3472	22	1	6706	118	12
	Shell-6	0.3472	20	1	6706	118	13
	Shell-7	0.3472	20	1	6706	118	14
	Shell-9	0.3472	18	1	6706	118	15
	Shell-10	0.3472	18	1	6706	118	16
	Shell-11	0.3472	16	1	6706	118	17
	Shell-12	0.3472	16	1	6706	118	18
	Shell-13	0.3472	16	1	6706	118	19
	Shell-14	0.3432	16	1	6706	118	20
	Shell-15	0.3472	20	1	6738	118	21
	Cone-1	0.3472	22	1	6738	118	22
	Cone-2	0.3472	18	1	5672.2	118	23
	Cone-3	0.3472	18	1	4563.2	118	24
	Shell-16	0.3472	16	1	3422.4	118	25
	Top head	0.3472	18	1	3422.4	118	26

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm	Element#
Skirt-1	...		No Calc	No Calc	56	No Calc	1
Skirt-2	...		No Calc	No Calc	34	No Calc	2
Skirt-3	...		No Calc	No Calc	34	No Calc	3
Skirt-4	...		No Calc	No Calc	32	No Calc	4
Skirt-5	...		No Calc	No Calc	32	No Calc	5
Skirt-6	...		No Calc	No Calc	36	No Calc	6
Btm Dish End	0.34724	0.69779	0.74644	38	20.1584		7
Shell-1	0.34724	0.92644	0.98046	28	11.4777		8
Shell-2	0.34724	0.87562	0.91075	26	10.8876		9
Shell-3	0.34724	0.88243	0.91783	26	10.8556		10
Shell-4	0.34724	0.80585	0.84099	24	10.8876		11
Shell-5	0.34724	0.73604	0.77118	22	10.8876		12
Shell-6	0.34724	0.66618	0.70132	20	10.8876		13
Shell-7	0.34724	0.66618	0.70132	20	10.8876		14
Shell-9	0.34724	0.59627	0.63142	18	10.8876		15
Shell-10	0.34724	0.59627	0.63142	18	10.8876		16
Shell-11	0.34724	0.52631	0.56146	16	10.8876		17
Shell-12	0.34724	0.52631	0.56146	16	10.8876		18
Shell-13	0.34724	0.52631	0.56146	16	10.8876		19
Shell-14	0.34324	0.52631	0.56146	16	10.7735		20
Shell-15	0.34724	0.66697	0.70216	20	10.9027		21
Cone-1	0.34724	0.52164	0.55238	18	12.3232		22
Cone-2	0.34724	0.47382	0.51033	14	10.5314		23
Cone-3	0.34724	0.58924	0.63467	14	8.66726		24
Shell-16	0.34724	1.02833	1.09714	16	6.04758		25
Top head	0.34724	0.89592	0.9646	14	6.03623		26



Minimum	0	1
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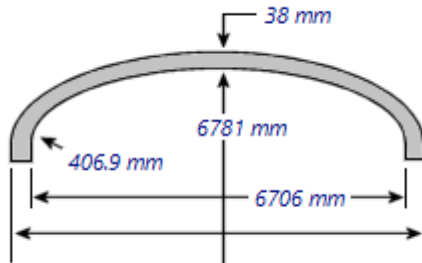
MAWP: 0 MPa, limited by: Cone-2.

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Btm Dish End

Torispherical Head Design Information:							
Design Pressure	P	0.377 MPa		Int. Design Temperature		160 °C	
Material		SA-516 60		Ext. Design Temperature		150 °C	
Length		50.000 mm		External Pressure Chart		CS-2	
				UNS Number		K02100	
Int. Corr. All.	c	1.0000 mm		Allowable Stress (ope)	S	118.0 MPa	
Ext. Corr. All.	ce	0.0000 mm		Allowable Stress (amb)	Sa	118.0 MPa	
Inside Diameter	Di	6706.000 mm		Actual Stress		61.1 MPa	
Spec. Min. Thk t,e, or sa		38.0000 mm		Spec. Nominal Thk.		46.000 mm	
Surface Area		44508196.0 mm ²		SG of Contents		1.054	
Element Volume		26129.7 ltr		Weight of Contents		210531.4 N	
Empty Weight		152783.0 N		Operating Weight		363314.4 N	
				Comp. All. Ope. Stress		110.6 MPa	
Ins. Crn. Rad. L or R		6781.0000 mm		Ins. Knuckle Radius	r	406.860 mm	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 83.825 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 41.912 mm

Inside Corroded Head Depth [h]:

$$= L - \sqrt{\left(\left(\frac{L - D_i}{2}\right) \cdot \left(\frac{L + D_i}{2} - 2 \cdot r\right)\right)}$$

$$= 6782.0 - \sqrt{\left(\left(\frac{6782.0 - 6708.0}{2}\right) \cdot \left(\frac{6782.0 + 6708.0}{2} - 2 \cdot 407.86\right)\right)}$$

$$= 1129.574 \text{ mm}$$

M factor for Torispherical Heads (Corroded):

$$= \frac{3 + \sqrt{\left(\frac{L + C}{r + C}\right)}}{4} \text{ per Appendix 1-4 (b \& d)}$$

$$= \frac{3 + \sqrt{\left(\frac{6781.0 + 1.0}{406.86 + 1.0}\right)}}{4}$$

$$= 1.7694$$

Required Thickness due to Internal Pressure [tr]:

$$= \frac{P \cdot L \cdot M}{2 \cdot S \cdot E - 0.2 \cdot P} \text{ per Appendix 1-4 (d)}$$

$$= \frac{0.377 \cdot 6782.0 \cdot 1.7694}{2 \cdot 118.0 \cdot 1.0 - 0.2 \cdot 0.377}$$

$$= 19.1584 + 1.0000 = 20.1584 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.029 MPa

$$= \frac{2 \cdot S \cdot E \cdot t}{M \cdot L + 0.2 \cdot t} \text{ per Appendix 1-4 (d)}$$

$$= \frac{2 \cdot 118.0 \cdot 1.0 \cdot 37.0}{1.7694 \cdot 6782.0 + 0.2 \cdot 37.0}$$

$$= 1 - 0 = 1 \text{ MPa}$$



M factor for Torispherical Heads (New & Cold):
 $= (3 + \sqrt{L/r})/4$ per Appendix 1-4 (b & d)
 $= (3 + \sqrt{(6781.0/406.86)})/4$
 $= 1.7706$

Maximum Allowable Pressure, New and Cold [MAPNC]:
 $= (2 \cdot S \cdot E \cdot t) / (M \cdot L + 0.2 \cdot t)$ per Appendix 1-4 (d)
 $= (2 \cdot 118.0 \cdot 1.0 \cdot 38.0) / (1.7706 \cdot 6781.0 + 0.2 \cdot 38.0)$
 $= 1 \text{ MPa}$

Actual stress at given pressure and thickness, corroded [Sact]:
 $= (P \cdot (M \cdot L + 0.2 \cdot t)) / (2 \cdot E \cdot t)$
 $= (0.377 \cdot (1.7694 \cdot 6782.0 + 0.2 \cdot 37.0)) / (2 \cdot 1.0 \cdot 37.0)$
 $= 61.117 \text{ MPa}$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (0.377 \cdot 3354.0) / (118.0 \cdot 1.0 - 0.6 \cdot 0.377) + 1.0$
 $= 11.726 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:
 Less Operating Hydrostatic Head Pressure of 0.029 MPa
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (118.0 \cdot 1.0 \cdot 45.0) / (3354.0 + 0.6 \cdot 45.0)$
 $= 2 - 0 = 2 \text{ MPa}$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 8.026 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 38.0$, $t_r = 25.599$, $c = 1.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.692$, Temp. Reduction = 17 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 46.0$, $t_r = 14.34$, $c = 1.0 \text{ mm}$, $E^* = 1.0$
 Thickness Ratio $= t_r \cdot E^* / (t_g - c) = 0.319$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	0.503 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	46.000 mm	Basic MDMT	-6 °C
Stress/Thk. Ratio	0.319	Temperature Difference	98 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 83.825 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 41.912 mm

EXTERNAL PRESSURE RESULTS for: Btm Dish End



Torispherical Head From 70 to 80 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Crown Rad	Ro/t	Factor A	Factor B
37.000	6819.00	184.30	0.0006783	67.76

$$MAEP = B/(Ro/t) = 67.7637/184.2973 = 0.3677 \text{ MPa}$$

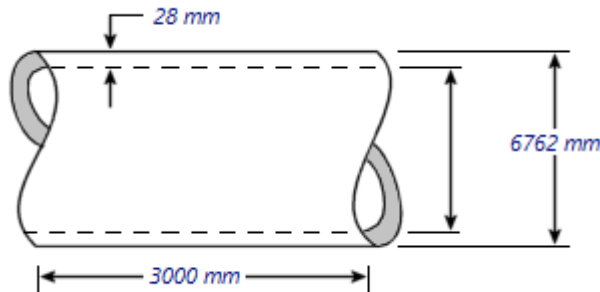
Results for Required Thickness				Tca
Tca	Crown Rad	Ro/t	Factor A	Factor B
13.512	6819.00	504.66	0.0002477	24.75

$$MAEP = B/(Ro/t) = 24.7466/504.6625 = 0.049 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-1

Cylindrical Shell Design Information:					
Design Pressure	P	0.366 MPa	Int. Design Temperature	160 °C	
Material		SA-516 60	Ext. Design Temperature	150 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02100	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Outside Diameter	Do	6762.000 mm	Actual Stress	45.7 MPa	
Spec. Min. Thk t _e , or s _a		28.0000 mm	Spec. Nominal Thk.	28.000 mm	
Surface Area		63730344.0 mm ²	SG of Contents	1.054	
Element Volume		105978.1 ltr	Weight of Contents	665775.8 N	
Empty Weight		135059.2 N	Operating Weight	800835.1 N	
			Comp. All. Ope. Stress	100.3 MPa	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot Ro) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (0.366 \cdot 3381.0) / (118.0 \cdot 1.0 + 0.4 \cdot 0.366)$$

$$= 10.4777 + 1.0000 = 11.4777 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.019 MPa

$$= (S \cdot E \cdot t) / (Ro - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 27.0) / (3381.0 - 0.4 \cdot 27.0)$$

$$= 1 - 0 = 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (Ro - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 28.0) / (3381.0 - 0.4 \cdot 28.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (Ro - 0.4 \cdot t)) / (E \cdot t)$$

$$= (0.366 \cdot (3381.0 - 0.4 \cdot 27.0)) / (1.0 \cdot 27.0)$$

$$= 45.701 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / Ro)) 0.416 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 28.0, tr = 14.094, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr * E^* / (tg - c) = 0.522$, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	0.493 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	28.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.522	Temperature Difference	30 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-1

Cylindrical Shell From 80 to R1 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
27.000	6762.00	1226.19	250.44	0.1813	0.0021428	105.03

$$MAEP = (4*B)/(3*(Do/t)) = (4*105.0309)/(3*250.4444) = 0.5592 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.892	6762.00	1226.19	856.78	0.1813	0.0003154	31.51

$$MAEP = (4*B)/(3*(Do/t)) = (4*31.5092)/(3*856.7781) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
27.000	6762.00	23247.25	250.44	3.4379	0.0000922	9.21

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.2149)/(3*250.4444) = 0.0491 \text{ MPa}$$

Cylindrical Shell From R1 to 90 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
27.000	6762.00	5100.00	250.44	0.7542	0.0004480	44.76

$$MAEP = (4*B)/(3*(Do/t)) = (4*44.7644)/(3*250.4444) = 0.2383 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
14.343	6762.00	5100.00	471.45	0.7542	0.0001735	17.34

$$MAEP = (4*B)/(3*(Do/t)) = (4*17.3382)/(3*471.447) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
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PV Elite 28 U2 Licensee: SL User
FileName : C-1401_Ope_Rev 0
Element: Shell-1

Step: 5 4:01pm Jul 21,2026

Page 69 of 564



HEXAGON

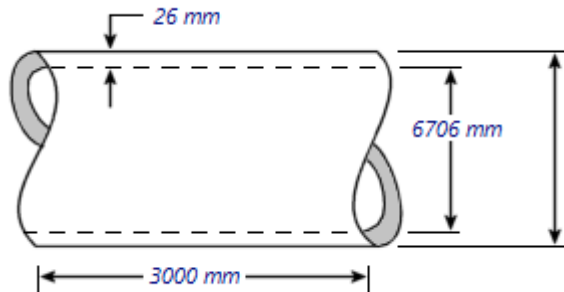
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
27.000	6762.00	23248.65	250.44	3.4381	0.0000922	9.21

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.2143)/(3*250.4444) = 0.0491 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-2

Cylindrical Shell Design Information:					
Design Pressure	P	0.347 MPa	Int. Design Temperature		160 °C
Material		SA-516 60	Ext. Design Temperature		150 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02100
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Inside Diameter	Di	6706.000 mm	Actual Stress		46.8 MPa
Spec. Min. Thk t _e , or s _a		26.0000 mm	Spec. Nominal Thk.		26.000 mm
Surface Area		63692648.0 mm ²	SG of Contents		0.000
Element Volume		105978.1 ltr	Weight of Contents		0.0 N
Empty Weight		125375.3 N	Operating Weight		125375.3 N
			Comp. All. Ope. Stress		97.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.347 \cdot 3354.0) / (118.0 \cdot 1.0 - 0.6 \cdot 0.347)$$

$$= 9.8876 + 1.0000 = 10.8876 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 25.0) / (3354.0 + 0.6 \cdot 25.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 26.0) / (3353.0 + 0.6 \cdot 26.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.347 \cdot (3354.0 + 0.6 \cdot 25.0)) / (1.0 \cdot 25.0)$$

$$= 46.795 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.386 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 26.0, tr = 13.5, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.54, Temp. Reduction = 28 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -20 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	0.474 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	26.000 mm	Basic MDMT	-20 °C
Stress/Thk. Ratio	0.540	Temperature Difference	28 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-2

Cylindrical Shell From 90 to R2 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	6758.00	5100.00	270.32	0.7547	0.0004020	40.17

$$MAEP = (4*B)/(3*(Do/t)) = (4*40.1673)/(3*270.32) = 0.1981 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
14.338	6758.00	5100.00	471.34	0.7547	0.0001735	17.33

$$MAEP = (4*B)/(3*(Do/t)) = (4*17.3341)/(3*471.337) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	6758.00	19498.39	270.32	2.8852	0.0000996	9.95

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.9496)/(3*270.32) = 0.0491 \text{ MPa}$$

Cylindrical Shell From R2 to 100 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	6758.00	2450.00	270.32	0.3625	0.0008785	81.07

$$MAEP = (4*B)/(3*(Do/t)) = (4*81.0731)/(3*270.32) = 0.3999 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.560	6758.00	2450.00	639.95	0.3625	0.0002356	23.54

$$MAEP = (4*B)/(3*(Do/t)) = (4*23.5358)/(3*639.947) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
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PV Elite 28 U2 Licensee: SL User
FileName : C-1401_Ope_Rev 0
Element: Shell-2

Page 72 of 564

Step: 6 4:01pm Jul 21, 2026



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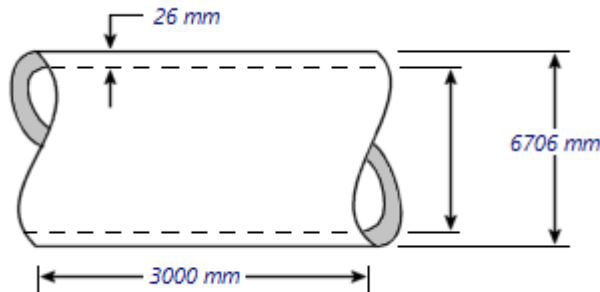
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	6758.00	19502.23	270.32	2.8858	0.0000996	9.95

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.9476)/(3*270.32) = 0.0491 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-3

Cylindrical Shell Design Information:					
Design Pressure	P	0.347 MPa	Int. Design Temperature		160 °C
Material		SA-516 60	Ext. Design Temperature		150 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02100
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Outside Diameter	Do	6706.000 mm	Actual Stress		46.4 MPa
Spec. Min. Thk t _e , or s _a		26.0000 mm	Spec. Nominal Thk.		26.000 mm
Surface Area		63202556.0 mm ²	SG of Contents		0.000
Element Volume		104340.9 ltr	Weight of Contents		0.0 N
Empty Weight		124405.8 N	Operating Weight		124405.8 N
			Comp. All. Ope. Stress		97.9 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (0.347 \cdot 3353.0) / (118.0 \cdot 1.0 + 0.4 \cdot 0.347)$$

$$= 9.8556 + 1.0000 = 10.8556 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 25.0) / (3353.0 - 0.4 \cdot 25.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 26.0) / (3353.0 - 0.4 \cdot 26.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

$$= (0.347 \cdot (3353.0 - 0.4 \cdot 25.0)) / (1.0 \cdot 25.0)$$

$$= 46.434 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.389 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 26.0, t_r = 13.442, c = 1.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.538, Temp. Reduction = 28 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -20 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	0.474 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	26.000 mm	Basic MDMT	-20 °C
Stress/Thk. Ratio	0.538	Temperature Difference	28 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-3

Cylindrical Shell From 100 to R3 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	6706.00	2450.00	268.24	0.3653	0.0008811	81.17

$$MAEP = (4*B)/(3*(Do/t)) = (4*81.1693)/(3*268.24) = 0.4035 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
10.516	6706.00	2450.00	637.72	0.3653	0.0002348	23.45

$$MAEP = (4*B)/(3*(Do/t)) = (4*23.4538)/(3*637.7158) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	6706.00	19705.86	268.24	2.9385	0.0000988	9.87

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.8711)/(3*268.24) = 0.0491 \text{ MPa}$$

Cylindrical Shell From R3 to 110 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	6706.00	2000.00	268.24	0.2982	0.0011041	88.11

$$MAEP = (4*B)/(3*(Do/t)) = (4*88.1065)/(3*268.24) = 0.4379 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.619	6706.00	2000.00	697.18	0.2982	0.0002566	25.64

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.6402)/(3*697.1848) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
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PV Elite 28 U2 Licensee: SL User
FileName : C-1401_Ope_Rev 0
Element: Shell-3

Page 75 of 564

Step: 7 4:01pm Jul 21, 2026



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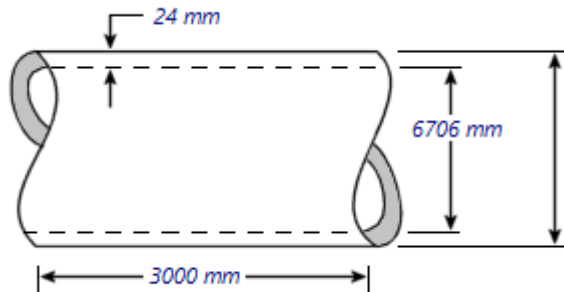
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
25.000	6706.00	19703.87	268.24	2.9382	0.0000988	9.87

$$\text{MAEP} = (4*B)/(3*(Do/t)) = (4*9.8721)/(3*268.24) = 0.0491 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-4

Cylindrical Shell Design Information:					
Design Pressure	P	0.347 MPa	Int. Design Temperature		160 °C
Material		SA-516 60	Ext. Design Temperature		150 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02100
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Inside Diameter	Di	6706.000 mm	Actual Stress		50.8 MPa
Spec. Min. Thk t _e or s _a		24.0000 mm	Spec. Nominal Thk.		24.000 mm
Surface Area		63654952.0 mm ²	SG of Contents		0.000
Element Volume		105978.1 ltr	Weight of Contents		0.0 N
Empty Weight		115696.4 N	Operating Weight		115696.4 N
			Comp. All. Ope. Stress		94.6 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.347 \cdot 3354.0) / (118.0 \cdot 1.0 - 0.6 \cdot 0.347)$$

$$= 9.8876 + 1.0000 = 10.8876 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 23.0) / (3354.0 + 0.6 \cdot 23.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 24.0) / (3353.0 + 0.6 \cdot 24.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.347 \cdot (3354.0 + 0.6 \cdot 23.0)) / (1.0 \cdot 23.0)$$

$$= 50.846 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.357 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 24.0, tr = 13.5, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.587, Temp. Reduction = 24 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -22 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -45 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	0.474 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	24.000 mm	Basic MDMT	-22 °C
Stress/Thk. Ratio	0.587	Temperature Difference	24 °C
UCS-66(c) red.	No	Computed Min. Temp.	-45 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-4

Cylindrical Shell From 110 to R4 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
23.000	6754.00	2000.00	293.65	0.2961	0.0009628	83.97

$$MAEP = (4*B)/(3*(Do/t)) = (4*83.9684)/(3*293.6522) = 0.3813 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.658	6754.00	2000.00	699.30	0.2961	0.0002574	25.72

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.7179)/(3*699.3026) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
23.000	6754.00	15933.48	293.65	2.3591	0.0001082	10.81

$$MAEP = (4*B)/(3*(Do/t)) = (4*10.8066)/(3*293.6522) = 0.0491 \text{ MPa}$$

Cylindrical Shell From R4 to 120 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
23.000	6754.00	2000.00	293.65	0.2961	0.0009628	83.97

$$MAEP = (4*B)/(3*(Do/t)) = (4*83.9684)/(3*293.6522) = 0.3813 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.658	6754.00	2000.00	699.30	0.2961	0.0002574	25.72

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.7179)/(3*699.3023) = 0.049 \text{ MPa}$$

PV Elite 28 U2 Licensee: SL User
FileName : C-1401_Ope_Rev 0
Element: Shell-4

Step: 8 4:01pm Jul 21,2026

Page 78 of 564



HEXAGON

Results for Maximum Stiffened Length

Slen

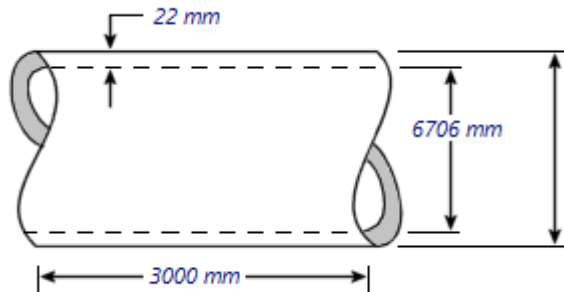
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
23.000	6754.00	15933.48	293.65	2.3591	0.0001082	10.81

$$MAEP = (4*B)/(3*(Do/t)) = (4*10.8066)/(3*293.6522) = 0.0491 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-5

Cylindrical Shell Design Information:					
Design Pressure	P	0.347 MPa	Int. Design Temperature	160 °C	
Material		SA-516 60	Ext. Design Temperature	150 °C	
Length		3000.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02100	
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Inside Diameter	Di	6706.000 mm	Actual Stress	55.7 MPa	
Spec. Min. Thk t _e or s _a		22.0000 mm	Spec. Nominal Thk.	22.000 mm	
Surface Area		63617244.0 mm ²	SG of Contents	0.000	
Element Volume		105978.1 ltr	Weight of Contents	0.0 N	
Empty Weight		106023.1 N	Operating Weight	106023.1 N	
			Comp. All. Ope. Stress	91.2 MPa	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.347 \cdot 3354.0) / (118.0 \cdot 1.0 - 0.6 \cdot 0.347)$$

$$= 9.8876 + 1.0000 = 10.8876 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 21.0) / (3354.0 + 0.6 \cdot 21.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 22.0) / (3353.0 + 0.6 \cdot 22.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.347 \cdot (3354.0 + 0.6 \cdot 21.0)) / (1.0 \cdot 21.0)$$

$$= 55.668 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.327 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 22.0, tr = 13.5, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.643, Temp. Reduction = 20 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -24 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -44 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	0.474 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	22.000 mm	Basic MDMT	-24 °C
Stress/Thk. Ratio	0.643	Temperature Difference	20 °C
UCS-66(c) red.	No	Computed Min. Temp.	-44 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-5

Cylindrical Shell From 120 to R5 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure						MAEP
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.000	6750.00	2000.00	321.43	0.2963	0.0008449	79.81

$$MAEP = (4*B)/(3*(Do/t)) = (4*79.8114)/(3*321.4286) = 0.3311 \text{ MPa}$$

Results for Required Thickness						Tca
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.655	6750.00	2000.00	699.13	0.2963	0.0002573	25.71

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.7113)/(3*699.1285) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length						Slen
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.000	6750.00	12940.24	321.43	1.9171	0.0001184	11.83

$$MAEP = (4*B)/(3*(Do/t)) = (4*11.8282)/(3*321.4286) = 0.0491 \text{ MPa}$$

Cylindrical Shell From R5 to R6 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure						MAEP
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.000	6750.00	2000.00	321.43	0.2963	0.0008449	79.81

$$MAEP = (4*B)/(3*(Do/t)) = (4*79.8114)/(3*321.4286) = 0.3311 \text{ MPa}$$

Results for Required Thickness						Tca
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.655	6750.00	2000.00	699.13	0.2963	0.0002573	25.71

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.7113)/(3*699.1285) = 0.049 \text{ MPa}$$



Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.000	6750.00	12940.24	321.43	1.9171	0.0001184	11.83

$$MAEP = (4*B)/(3*(Do/t)) = (4*11.8282)/(3*321.4286) = 0.0491 \text{ MPa}$$

Cylindrical Shell From R6 to 130 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.000	6750.00	2000.00	321.43	0.2963	0.0008449	79.81

$$MAEP = (4*B)/(3*(Do/t)) = (4*79.8114)/(3*321.4286) = 0.3311 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.655	6750.00	2000.00	699.13	0.2963	0.0002573	25.71

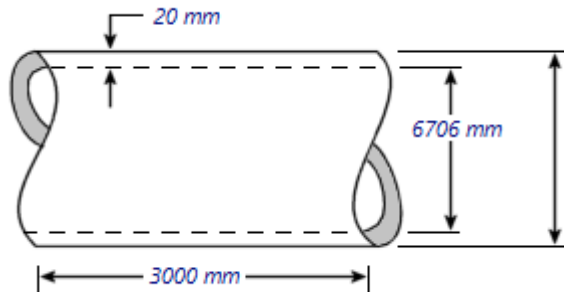
$$MAEP = (4*B)/(3*(Do/t)) = (4*25.7113)/(3*699.1286) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
21.000	6750.00	12940.24	321.43	1.9171	0.0001184	11.83

$$MAEP = (4*B)/(3*(Do/t)) = (4*11.8282)/(3*321.4286) = 0.0491 \text{ MPa}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-6

Cylindrical Shell Design Information:					
Design Pressure	P	0.347 MPa	Int. Design Temperature		160 °C
Material		SA-516 60	Ext. Design Temperature		150 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02100
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Inside Diameter	Di	6706.000 mm	Actual Stress		61.5 MPa
Spec. Min. Thk t _e , or s _a		20.0000 mm	Spec. Nominal Thk.		20.000 mm
Surface Area		63579548.0 mm ²	SG of Contents		0.000
Element Volume		105978.1 ltr	Weight of Contents		0.0 N
Empty Weight		96356.0 N	Operating Weight		96356.0 N
			Comp. All. Ope. Stress		83.9 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.347 \cdot 3354.0) / (118.0 \cdot 1.0 - 0.6 \cdot 0.347)$$

$$= 9.8876 + 1.0000 = 10.8876 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 19.0) / (3354.0 + 0.6 \cdot 19.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 20.0) / (3353.0 + 0.6 \cdot 20.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.347 \cdot (3354.0 + 0.6 \cdot 19.0)) / (1.0 \cdot 19.0)$$

$$= 61.506 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.297 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 20.0, tr = 13.5, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.711, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -27 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -43 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	0.474 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	20.000 mm	Basic MDMT	-27 °C
Stress/Thk. Ratio	0.711	Temperature Difference	16 °C
UCS-66(c) red.	No	Computed Min. Temp.	-43 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-6

Cylindrical Shell From 130 to R7 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure						MAEP
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.000	6746.00	2000.00	355.05	0.2965	0.0007293	72.87

$$MAEP = (4*B)/(3*(Do/t)) = (4*72.8679)/(3*355.0526) = 0.2736 \text{ MPa}$$

Results for Required Thickness						Tca
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.652	6746.00	2000.00	698.95	0.2965	0.0002573	25.70

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.7047)/(3*698.9548) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length						Slen
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.000	6746.00	10238.70	355.05	1.5177	0.0001308	13.06

$$MAEP = (4*B)/(3*(Do/t)) = (4*13.0643)/(3*355.0526) = 0.0491 \text{ MPa}$$

Cylindrical Shell From R7 to 140 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure						MAEP
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.000	6746.00	2000.00	355.05	0.2965	0.0007293	72.87

$$MAEP = (4*B)/(3*(Do/t)) = (4*72.8677)/(3*355.0526) = 0.2736 \text{ MPa}$$

Results for Required Thickness						Tca
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.652	6746.00	2000.00	698.95	0.2965	0.0002573	25.70

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.7047)/(3*698.9541) = 0.049 \text{ MPa}$$

PV Elite 28 U2 Licensee: SL User
FileName : C-1401_Ope_Rev 0
Element: Shell-6

Step: 10 4:01pm Jul 21, 2026

Page 84 of 564



HEXAGON

Results for Maximum Stiffened Length

Slen

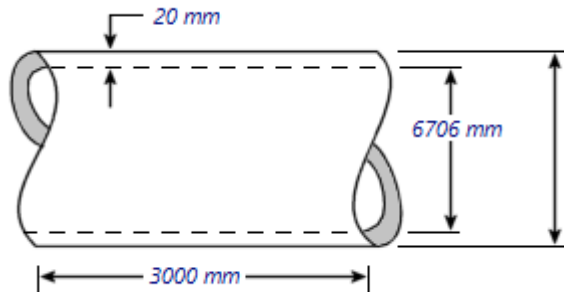
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.000	6746.00	10238.70	355.05	1.5177	0.0001308	13.06

$$MAEP = (4*B)/(3*(Do/t)) = (4*13.0644)/(3*355.0526) = 0.0491 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-7

Cylindrical Shell Design Information:					
Design Pressure	P	0.347 MPa	Int. Design Temperature		160 °C
Material		SA-516 60	Ext. Design Temperature		150 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02100
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Inside Diameter	Di	6706.000 mm	Actual Stress		61.5 MPa
Spec. Min. Thk t _e or s _a		20.0000 mm	Spec. Nominal Thk.		20.000 mm
Surface Area		63579548.0 mm ²	SG of Contents		0.000
Element Volume		105978.1 ltr	Weight of Contents		0.0 N
Empty Weight		96356.0 N	Operating Weight		96356.0 N
			Comp. All. Ope. Stress		83.9 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.347 \cdot 3354.0) / (118.0 \cdot 1.0 - 0.6 \cdot 0.347)$$

$$= 9.8876 + 1.0000 = 10.8876 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 19.0) / (3354.0 + 0.6 \cdot 19.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 20.0) / (3353.0 + 0.6 \cdot 20.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.347 \cdot (3354.0 + 0.6 \cdot 19.0)) / (1.0 \cdot 19.0)$$

$$= 61.506 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.297 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 20.0, tr = 13.5, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.711, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -27 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -43 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	0.474 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	20.000 mm	Basic MDMT	-27 °C
Stress/Thk. Ratio	0.711	Temperature Difference	16 °C
UCS-66(c) red.	No	Computed Min. Temp.	-43 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-7

Cylindrical Shell From 140 to R8 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.000	6746.00	2000.00	355.05	0.2965	0.0007293	72.87

$$MAEP = (4*B)/(3*(Do/t)) = (4*72.8677)/(3*355.0526) = 0.2736 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.652	6746.00	2000.00	698.95	0.2965	0.0002573	25.70

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.7047)/(3*698.9541) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.000	6746.00	10238.70	355.05	1.5177	0.0001308	13.06

$$MAEP = (4*B)/(3*(Do/t)) = (4*13.0644)/(3*355.0526) = 0.0491 \text{ MPa}$$

Cylindrical Shell From R8 to 150 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.000	6746.00	3000.00	355.05	0.4447	0.0004695	46.91

$$MAEP = (4*B)/(3*(Do/t)) = (4*46.908)/(3*355.0526) = 0.1762 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.524	6746.00	3000.00	585.40	0.4447	0.0002155	21.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*21.5287)/(3*585.4022) = 0.049 \text{ MPa}$$

PV Elite 28 U2 Licensee: SL User
FileName : C-1401_Ope_Rev 0
Element: Shell-7

Step: 11 4:01pm Jul 21,2026

Page 87 of 564



HEXAGON

Results for Maximum Stiffened Length

Slen

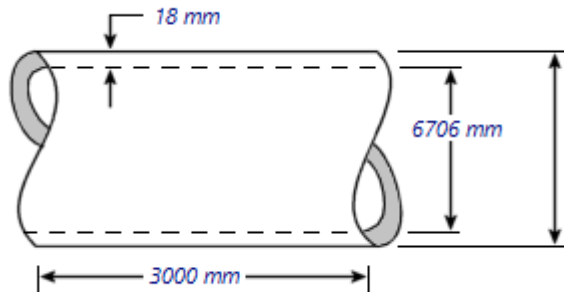
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.000	6746.00	10235.50	355.05	1.5173	0.0001308	13.07

$$MAEP = (4*B)/(3*(Do/t)) = (4*13.0686)/(3*355.0526) = 0.0491 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-9

Cylindrical Shell Design Information:					
Design Pressure	P	0.347 MPa	Int. Design Temperature		160 °C
Material		SA-516 60	Ext. Design Temperature		150 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02100
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Inside Diameter	Di	6706.000 mm	Actual Stress		68.7 MPa
Spec. Min. Thk t _e , or s _a		18.0000 mm	Spec. Nominal Thk.		18.000 mm
Surface Area		63541856.0 mm ²	SG of Contents		0.000
Element Volume		105978.1 ltr	Weight of Contents		0.0 N
Empty Weight		86695.7 N	Operating Weight		86695.7 N
			Comp. All. Ope. Stress		75.1 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.347 \cdot 3354.0) / (118.0 \cdot 1.0 - 0.6 \cdot 0.347)$$

$$= 9.8876 + 1.0000 = 10.8876 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 17.0) / (3354.0 + 0.6 \cdot 17.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 18.0) / (3353.0 + 0.6 \cdot 18.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.347 \cdot (3354.0 + 0.6 \cdot 17.0)) / (1.0 \cdot 17.0)$$

$$= 68.718 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.268 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 18.0, tr = 13.5, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.794, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -41 °C

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	0.474 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	18.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.794	Temperature Difference	11 °C
UCS-66(c) red.	No	Computed Min. Temp.	-41 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-9

Cylindrical Shell From 150 to R9 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
17.000	6742.00	3000.00	396.59	0.4450	0.0003896	38.93

$$MAEP = (4*B)/(3*(Do/t)) = (4*38.9273)/(3*396.5882) = 0.1309 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.520	6742.00	3000.00	585.26	0.4450	0.0002154	21.52

$$MAEP = (4*B)/(3*(Do/t)) = (4*21.5241)/(3*585.2574) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
17.000	6742.00	7703.47	396.59	1.1426	0.0001461	14.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*14.5947)/(3*396.5882) = 0.0491 \text{ MPa}$$

Cylindrical Shell From R9 to R10 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
17.000	6742.00	2000.00	396.59	0.2966	0.0006041	60.35

$$MAEP = (4*B)/(3*(Do/t)) = (4*60.3509)/(3*396.5882) = 0.2029 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.648	6742.00	2000.00	698.77	0.2966	0.0002572	25.70

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.6988)/(3*698.7688) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
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Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
17.000	6742.00	7705.12	396.59	1.1429	0.0001460	14.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*14.5914)/(3*396.5882) = 0.0491 \text{ MPa}$$

Cylindrical Shell From R10 to 160 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
17.000	6742.00	2000.00	396.59	0.2966	0.0006041	60.35

$$MAEP = (4*B)/(3*(Do/t)) = (4*60.351)/(3*396.5882) = 0.2029 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.648	6742.00	2000.00	698.77	0.2966	0.0002572	25.70

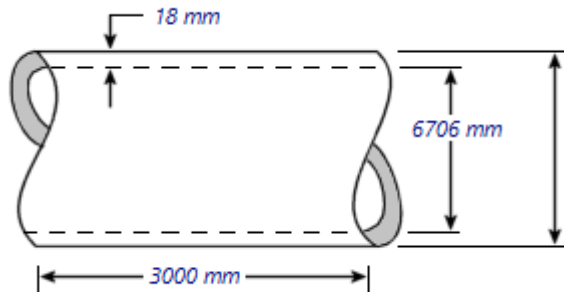
$$MAEP = (4*B)/(3*(Do/t)) = (4*25.6988)/(3*698.7696) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
17.000	6742.00	7705.12	396.59	1.1429	0.0001460	14.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*14.5914)/(3*396.5882) = 0.0491 \text{ MPa}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-10

Cylindrical Shell Design Information:					
Design Pressure	P	0.347 MPa	Int. Design Temperature		160 °C
Material		SA-516 60	Ext. Design Temperature		150 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02100
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Inside Diameter	Di	6706.000 mm	Actual Stress		68.7 MPa
Spec. Min. Thk t _e , or s _a		18.0000 mm	Spec. Nominal Thk.		18.000 mm
Surface Area		63541856.0 mm ²	SG of Contents		0.000
Element Volume		105978.1 ltr	Weight of Contents		0.0 N
Empty Weight		86695.7 N	Operating Weight		86695.7 N
			Comp. All. Ope. Stress		75.1 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.347 \cdot 3354.0) / (118.0 \cdot 1.0 - 0.6 \cdot 0.347)$$

$$= 9.8876 + 1.0000 = 10.8876 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 17.0) / (3354.0 + 0.6 \cdot 17.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 18.0) / (3353.0 + 0.6 \cdot 18.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.347 \cdot (3354.0 + 0.6 \cdot 17.0)) / (1.0 \cdot 17.0)$$

$$= 68.718 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.268 %

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 18.0, t_r = 13.5, c = 1.0 mm, E* = 1.0
 Thickness Ratio = t_r * E* / (t_g - c) = 0.794, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -41 °C

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	0.474 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	18.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.794	Temperature Difference	11 °C
UCS-66(c) red.	No	Computed Min. Temp.	-41 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-10

Cylindrical Shell From 160 to R11 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
17.000	6742.00	2000.00	396.59	0.2966	0.0006041	60.35

$$MAEP = (4*B)/(3*(Do/t)) = (4*60.351)/(3*396.5882) = 0.2029 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.648	6742.00	2000.00	698.77	0.2966	0.0002572	25.70

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.6988)/(3*698.7696) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
17.000	6742.00	7705.12	396.59	1.1429	0.0001460	14.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*14.5914)/(3*396.5882) = 0.0491 \text{ MPa}$$

Cylindrical Shell From R11 to 170 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
17.000	6742.00	2000.00	396.59	0.2966	0.0006041	60.35

$$MAEP = (4*B)/(3*(Do/t)) = (4*60.3508)/(3*396.5882) = 0.2029 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.648	6742.00	2000.00	698.77	0.2966	0.0002572	25.70

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.6988)/(3*698.7685) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
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PV Elite 28 U2 Licensee: SL User
FileName : C-1401_Ope_Rev 0
Element: Shell-10

Step: 13 4:01pm Jul 21,2026

Page 93 of 564



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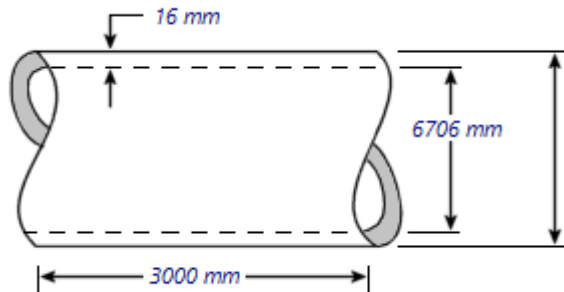
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
17.000	6742.00	7705.12	396.59	1.1429	0.0001460	14.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*14.5914)/(3*396.5882) = 0.0491 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-11

Cylindrical Shell Design Information:					
Design Pressure	P	0.347 MPa	Int. Design Temperature		160 °C
Material		SA-516 60	Ext. Design Temperature		150 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02100
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Inside Diameter	Di	6706.000 mm	Actual Stress		77.9 MPa
Spec. Min. Thk t _e , or s _a		16.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		63504160.0 mm ²	SG of Contents		0.000
Element Volume		105978.1 ltr	Weight of Contents		0.0 N
Empty Weight		77040.4 N	Operating Weight		77040.4 N
			Comp. All. Ope. Stress		66.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.347 \cdot 3354.0) / (118.0 \cdot 1.0 - 0.6 \cdot 0.347)$$

$$= 9.8876 + 1.0000 = 10.8876 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 15.0) / (3354.0 + 0.6 \cdot 15.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 16.0) / (3353.0 + 0.6 \cdot 16.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.347 \cdot (3354.0 + 0.6 \cdot 15.0)) / (1.0 \cdot 15.0)$$

$$= 77.852 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.238 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 16.0, tr = 13.5, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.9, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -32 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -38 °C

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	0.474 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	16.000 mm	Basic MDMT	-32 °C
Stress/Thk. Ratio	0.900	Temperature Difference	6 °C
UCS-66(c) red.	No	Computed Min. Temp.	-38 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-11

Cylindrical Shell From 170 to R12 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	6738.00	2000.00	449.20	0.2968	0.0005028	50.24

$$MAEP = (4*B)/(3*(Do/t)) = (4*50.2354)/(3*449.2) = 0.1491 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.645	6738.00	2000.00	698.60	0.2968	0.0002572	25.69

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.6921)/(3*698.5962) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	6738.00	5725.19	449.20	0.8497	0.0001655	16.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.5326)/(3*449.2) = 0.0491 \text{ MPa}$$

Cylindrical Shell From R12 to R13 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	6738.00	2000.00	449.20	0.2968	0.0005028	50.24

$$MAEP = (4*B)/(3*(Do/t)) = (4*50.2355)/(3*449.2) = 0.1491 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.645	6738.00	2000.00	698.60	0.2968	0.0002572	25.69

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.6921)/(3*698.5968) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
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Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	6738.00	5725.19	449.20	0.8497	0.0001655	16.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.5326)/(3*449.2) = 0.0491 \text{ MPa}$$

Cylindrical Shell From R13 to 180 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	6738.00	2000.00	449.20	0.2968	0.0005028	50.24

$$MAEP = (4*B)/(3*(Do/t)) = (4*50.2354)/(3*449.2) = 0.1491 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.645	6738.00	2000.00	698.60	0.2968	0.0002572	25.69

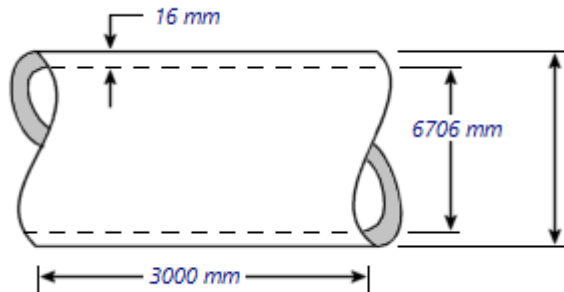
$$MAEP = (4*B)/(3*(Do/t)) = (4*25.6921)/(3*698.5962) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	6738.00	5725.19	449.20	0.8497	0.0001655	16.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.5326)/(3*449.2) = 0.0491 \text{ MPa}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-12

Cylindrical Shell Design Information:					
Design Pressure	P	0.347 MPa	Int. Design Temperature		160 °C
Material		SA-516 60	Ext. Design Temperature		150 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02100
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Inside Diameter	Di	6706.000 mm	Actual Stress		77.9 MPa
Spec. Min. Thk t _e , or s _a		16.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		63504160.0 mm ²	SG of Contents		0.000
Element Volume		105978.1 ltr	Weight of Contents		0.0 N
Empty Weight		77040.4 N	Operating Weight		77040.4 N
			Comp. All. Ope. Stress		66.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.347 \cdot 3354.0) / (118.0 \cdot 1.0 - 0.6 \cdot 0.347)$$

$$= 9.8876 + 1.0000 = 10.8876 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 15.0) / (3354.0 + 0.6 \cdot 15.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 16.0) / (3353.0 + 0.6 \cdot 16.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.347 \cdot (3354.0 + 0.6 \cdot 15.0)) / (1.0 \cdot 15.0)$$

$$= 77.852 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.238 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 16.0, tr = 13.5, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.9, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -32 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -38 °C

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	0.474 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	16.000 mm	Basic MDMT	-32 °C
Stress/Thk. Ratio	0.900	Temperature Difference	6 °C
UCS-66(c) red.	No	Computed Min. Temp.	-38 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-12

Cylindrical Shell From 180 to R14 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	6738.00	2000.00	449.20	0.2968	0.0005028	50.24

$$MAEP = (4*B)/(3*(Do/t)) = (4*50.2354)/(3*449.2) = 0.1491 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.645	6738.00	2000.00	698.60	0.2968	0.0002572	25.69

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.6921)/(3*698.5962) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	6738.00	5725.19	449.20	0.8497	0.0001655	16.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.5326)/(3*449.2) = 0.0491 \text{ MPa}$$

Cylindrical Shell From R14 to 190 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	6738.00	2000.00	449.20	0.2968	0.0005028	50.24

$$MAEP = (4*B)/(3*(Do/t)) = (4*50.2354)/(3*449.2) = 0.1491 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.645	6738.00	2000.00	698.60	0.2968	0.0002572	25.69

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.6921)/(3*698.5966) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
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PV Elite 28 U2 Licensee: SL User
FileName : C-1401_Ope_Rev 0
Element: Shell-12

Step: 15 4:01pm Jul 21,2026

Page 99 of 564



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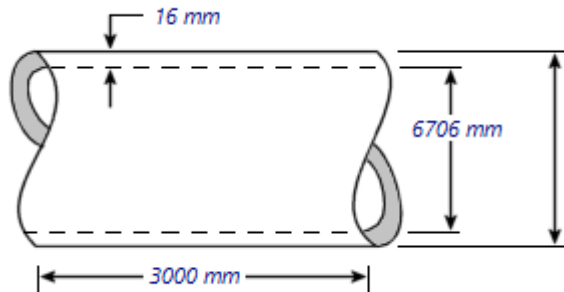
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	6738.00	5725.19	449.20	0.8497	0.0001655	16.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.5326)/(3*449.2) = 0.0491 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-13

Cylindrical Shell Design Information:					
Design Pressure	P	0.347 MPa	Int. Design Temperature		160 °C
Material		SA-516 60	Ext. Design Temperature		150 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02100
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Inside Diameter	Di	6706.000 mm	Actual Stress		77.9 MPa
Spec. Min. Thk t,e, or sa		16.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		63504160.0 mm²	SG of Contents		0.000
Element Volume		105978.1 ltr	Weight of Contents		0.0 N
Empty Weight		77040.4 N	Operating Weight		77040.4 N
			Comp. All. Ope. Stress		66.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.347 \cdot 3354.0) / (118.0 \cdot 1.0 - 0.6 \cdot 0.347)$$

$$= 9.8876 + 1.0000 = 10.8876 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 15.0) / (3354.0 + 0.6 \cdot 15.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 16.0) / (3353.0 + 0.6 \cdot 16.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.347 \cdot (3354.0 + 0.6 \cdot 15.0)) / (1.0 \cdot 15.0)$$

$$= 77.852 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * tnom / Rf * (1 - Rf / Ro)) 0.238 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 16.0, tr = 13.5, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.9, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -32 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -38 °C

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	0.474 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	16.000 mm	Basic MDMT	-32 °C
Stress/Thk. Ratio	0.900	Temperature Difference	6 °C
UCS-66(c) red.	No	Computed Min. Temp.	-38 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-13

Cylindrical Shell From 190 to R15 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	6738.00	2000.00	449.20	0.2968	0.0005028	50.24

$$MAEP = (4*B)/(3*(Do/t)) = (4*50.2354)/(3*449.2) = 0.1491 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.645	6738.00	2000.00	698.60	0.2968	0.0002572	25.69

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.6921)/(3*698.5966) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	6738.00	5725.19	449.20	0.8497	0.0001655	16.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.5326)/(3*449.2) = 0.0491 \text{ MPa}$$

Cylindrical Shell From R15 to R16 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	6738.00	2000.00	449.20	0.2968	0.0005028	50.24

$$MAEP = (4*B)/(3*(Do/t)) = (4*50.2354)/(3*449.2) = 0.1491 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.645	6738.00	2000.00	698.60	0.2968	0.0002572	25.69

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.6921)/(3*698.5964) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
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Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	6738.00	5725.19	449.20	0.8497	0.0001655	16.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.5326)/(3*449.2) = 0.0491 \text{ MPa}$$

Cylindrical Shell From R16 to 200 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	6738.00	1999.99	449.20	0.2968	0.0005028	50.24

$$MAEP = (4*B)/(3*(Do/t)) = (4*50.2356)/(3*449.2) = 0.1491 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.645	6738.00	1999.99	698.60	0.2968	0.0002572	25.69

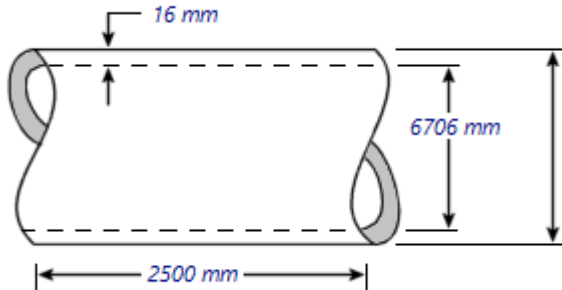
$$MAEP = (4*B)/(3*(Do/t)) = (4*25.6922)/(3*698.5977) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	6738.00	5725.19	449.20	0.8497	0.0001655	16.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.5326)/(3*449.2) = 0.0491 \text{ MPa}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-14

Cylindrical Shell Design Information:					
Design Pressure	P	0.343 MPa	Int. Design Temperature		160 °C
Material		SA-516 60	Ext. Design Temperature		150 °C
Length		2500.000 mm	External Pressure Chart		CS-2
			UNS Number		K02100
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Inside Diameter	Di	6706.000 mm	Actual Stress		77.0 MPa
Spec. Min. Thk t _e , or s _a		16.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		52920132.0 mm ²	SG of Contents		0.000
Element Volume		88315.1 ltr	Weight of Contents		0.0 N
Empty Weight		64200.6 N	Operating Weight		64200.6 N
			Comp. All. Ope. Stress		66.3 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.343 \cdot 3354.0) / (118.0 \cdot 1.0 - 0.6 \cdot 0.343)$$

$$= 9.7735 + 1.0000 = 10.7735 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 15.0) / (3354.0 + 0.6 \cdot 15.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 16.0) / (3353.0 + 0.6 \cdot 16.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.343 \cdot (3354.0 + 0.6 \cdot 15.0)) / (1.0 \cdot 15.0)$$

$$= 76.955 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.238 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 16.0, tr = 13.5, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.9, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -32 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -38 °C

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	0.474 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	16.000 mm	Basic MDMT	-32 °C
Stress/Thk. Ratio	0.900	Temperature Difference	6 °C
UCS-66(c) red.	No	Computed Min. Temp.	-38 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-14

Cylindrical Shell From 200 to R17 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	6738.00	1999.99	449.20	0.2968	0.0005028	50.24

$$MAEP = (4*B)/(3*(Do/t)) = (4*50.2356)/(3*449.2) = 0.1491 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
9.645	6738.00	1999.99	698.60	0.2968	0.0002572	25.69

$$MAEP = (4*B)/(3*(Do/t)) = (4*25.6922)/(3*698.5975) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	6738.00	5725.19	449.20	0.8497	0.0001655	16.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.5326)/(3*449.2) = 0.0491 \text{ MPa}$$

Cylindrical Shell From R17 to 210 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	6738.00	1500.00	449.20	0.2226	0.0006873	68.66

$$MAEP = (4*B)/(3*(Do/t)) = (4*68.6635)/(3*449.2) = 0.2038 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.591	6738.00	1500.00	784.31	0.2226	0.0002887	28.84

$$MAEP = (4*B)/(3*(Do/t)) = (4*28.8439)/(3*784.3085) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
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PV Elite 28 U2 Licensee: SL User
FileName : C-1401_Ope_Rev 0
Element: Shell-14

Step: 17 4:01pm Jul 21,2026

Page 105 of 564



HEXAGON

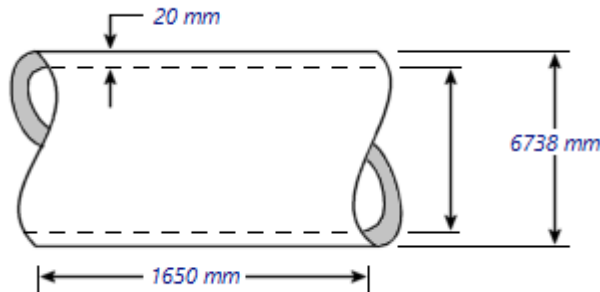
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	6738.00	5726.93	449.20	0.8499	0.0001654	16.53

$$MAEP = (4*B)/(3*(Do/t)) = (4*16.5275)/(3*449.2) = 0.0491 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-15

Cylindrical Shell Design Information:					
Design Pressure	P	0.347 MPa	Int. Design Temperature		160 °C
Material		SA-516 60	Ext. Design Temperature		150 °C
Length		1650.000 mm	External Pressure Chart		CS-2
			UNS Number		K02100
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Outside Diameter	Do	6738.000 mm	Actual Stress		61.4 MPa
Spec. Min. Thk t _e , or s _a		20.0000 mm	Spec. Nominal Thk.		20.000 mm
Surface Area		34927288.0 mm ²	SG of Contents		0.000
Element Volume		58149.0 ltr	Weight of Contents		0.0 N
Empty Weight		52932.8 N	Operating Weight		52932.8 N
			Comp. All. Ope. Stress		84.0 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R_o) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (0.347 \cdot 3369.0002) / (118.0 \cdot 1.0 + 0.4 \cdot 0.347)$$

$$= 9.9027 + 1.0000 = 10.9027 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 19.0) / (3369.0002 - 0.4 \cdot 19.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R_o - 0.4 \cdot t) \text{ per Appendix 1-1 (a)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 20.0) / (3369.0002 - 0.4 \cdot 20.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R_o - 0.4 \cdot t)) / (E \cdot t)$$

$$= (0.347 \cdot (3369.0002 - 0.4 \cdot 19.0)) / (1.0 \cdot 19.0)$$

$$= 61.433 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.298 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 20.0, tr = 13.506, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.711, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -27 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -43 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	0.474 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	20.000 mm	Basic MDMT	-27 °C
Stress/Thk. Ratio	0.711	Temperature Difference	16 °C
UCS-66(c) red.	No	Computed Min. Temp.	-43 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-15

Cylindrical Shell From 210 to R18 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure						MAEP
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.000	6738.00	1500.00	354.63	0.2226	0.0010004	85.15

$$MAEP = (4*B)/(3*(Do/t)) = (4*85.1496)/(3*354.6316) = 0.3201 \text{ MPa}$$

Results for Required Thickness						Tca
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.591	6738.00	1500.00	784.31	0.2226	0.0002887	28.84

$$MAEP = (4*B)/(3*(Do/t)) = (4*28.8436)/(3*784.3132) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length						Slen
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.000	6738.00	10254.65	354.63	1.5219	0.0001306	13.05

$$MAEP = (4*B)/(3*(Do/t)) = (4*13.0508)/(3*354.6316) = 0.0491 \text{ MPa}$$

Cylindrical Shell From R18 to R19 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure						MAEP
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.000	6738.00	1236.75	354.63	0.1835	0.0012443	91.54

$$MAEP = (4*B)/(3*(Do/t)) = (4*91.5406)/(3*354.6316) = 0.3442 \text{ MPa}$$

Results for Required Thickness						Tca
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.907	6738.00	1236.75	852.15	0.1835	0.0003137	31.34

$$MAEP = (4*B)/(3*(Do/t)) = (4*31.3395)/(3*852.1512) = 0.049 \text{ MPa}$$

PV Elite 28 U2 Licensee: SL User
FileName : C-1401_Ope_Rev 0
Element: Shell-15

Step: 18 4:01pm Jul 21,2026

Page 108 of 564



HEXAGON

Results for Maximum Stiffened Length

Slen

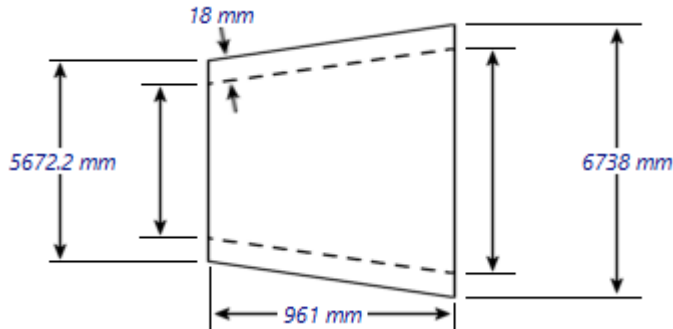
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.000	6738.00	10253.54	354.63	1.5217	0.0001306	13.05

$$MAEP = (4*B)/(3*(Do/t)) = (4*13.0523)/(3*354.6316) = 0.0491 \text{ MPa}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Cone-1

Conical Head/Section Design Information:					
Design Pressure	P	0.347 MPa	Int. Design Temperature		160 °C
Material		SA-516 60	Ext. Design Temperature		150 °C
Length		961.000 mm	External Pressure Chart		CS-2
			UNS Number		K02100
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Outside Diameter	Do	6738.000 mm	Actual Stress		78.5 MPa
Spec. Min. Thk t _e , or s _a		18.0000 mm	Spec. Nominal Thk.		22.000 mm
Surface Area		21452614.0 mm ²	SG of Contents		0.000
Element Volume		28753.5 ltr	Weight of Contents		0.0 N
Empty Weight		35724.2 N	Operating Weight		35724.2 N
			Comp. All. Ope. Stress		85.9 MPa
			To End Diameter		5672.240 mm
Line of Support	Both Ends				

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D_o) / (2 \cdot \cos(a) \cdot (S \cdot E + 0.4 \cdot P))$$
 per Appendix 1-4 (e)

$$= (0.347 \cdot 6738.0005) / (2 \cdot 0.8745 \cdot (118.0 \cdot 1.0 + 0.4 \cdot 0.35))$$

$$= 11.3232 + 1.0000 = 12.3232 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D - 0.8 \cdot t \cdot \cos(a))$$
 per Appendix 1-4 (e)(5)

$$= (2 \cdot 118.0 \cdot 1.0 \cdot (17.0) \cdot 0.8745) / (6738.0005 - 0.8 \cdot (17.0) \cdot 0.8745)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D - 0.8 \cdot t \cdot \cos(a))$$
 per Appendix 1-4(e)

$$= (2 \cdot 118.0 \cdot 1.0 \cdot 18.0 \cdot 0.8745) / (6738.0005 - 0.8 \cdot 18.0 \cdot 0.8745)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (D - 0.8 \cdot t \cdot \cos(a))) / (2 \cdot E \cdot t \cdot \cos(a))$$

$$= (0.347 \cdot (6738.0005 - 0.8 \cdot (17.0) \cdot 0.8745)) / (2 \cdot 1.0 \cdot (17.0) \cdot 0.8745)$$

$$= 78.548 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.390 %

Minimum Design Metal Temperature Results:



Govrn. thk, tg = 18.0, tr = 15.444, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.908$, Temp. Reduction = 5 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -34 °C

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	0.474 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	18.000 mm	Basic MDMT	-29 °C
Stress/Thk. Ratio	0.908	Temperature Difference	5 °C
UCS-66(c) red.	No	Computed Min. Temp.	-34 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Cone-1

Cone From 220 to 230 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
14.867	6738.00	884.94	453.21	0.1313	0.0012376	91.39

$$MAEP = (4*B) / (3*(Do/t)) = (4*91.3904) / (3*453.2095) = 0.2689 \text{ MPa}$$

Note:

The cone thickness used in the calculation has been modified per UG-33(f), $te = t * \cos(\alpha)$.

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.853	6738.00	884.94	983.19	0.1313	0.0003619	36.16

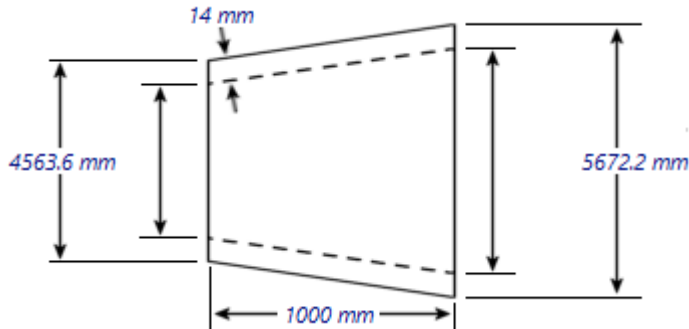
$$MAEP = (4*B) / (3*(Do/t)) = (4*36.1576) / (3*983.1897) = 0.049 \text{ MPa}$$

Note:

The cone thickness used in the calculation has been modified per UG-33(f), $te = t * \cos(\alpha)$.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Cone-2

Conical Head/Section Design Information:					
Design Pressure	P	0.347 MPa	Int. Design Temperature		160 °C
Material		SA-516 60	Ext. Design Temperature		150 °C
Length		1000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02100
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Outside Diameter	Do	5672.240 mm	Actual Stress		86.5 MPa
Spec. Min. Thk t _e , or s _a		14.0000 mm	Spec. Nominal Thk.		18.000 mm
Surface Area		18416226.0 mm ²	SG of Contents		0.000
Element Volume		20399.6 ltr	Weight of Contents		0.0 N
Empty Weight		25092.7 N	Operating Weight		25092.7 N
			Comp. All. Ope. Stress		78.1 MPa
			To End Diameter		4563.620 mm
Line of Support	Both Ends				

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D_o) / (2 \cdot \cos(a) \cdot (S \cdot E + 0.4 \cdot P))$$
 per Appendix 1-4 (e)

$$= (0.347 \cdot 5672.2402) / (2 \cdot 0.8746 \cdot (118.0 \cdot 1.0 + 0.4 \cdot 0.35))$$

$$= 9.5314 + 1.0000 = 10.5314 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D - 0.8 \cdot t \cdot \cos(a))$$
 per Appendix 1-4 (e)(5)

$$= (2 \cdot 118.0 \cdot 1.0 \cdot (13.0) \cdot 0.8746) / (5672.2402 - 0.8 \cdot (13.0) \cdot 0.8746)$$

$$= 0 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D - 0.8 \cdot t \cdot \cos(a))$$
 per Appendix 1-4(e)

$$= (2 \cdot 118.0 \cdot 1.0 \cdot 14.0 \cdot 0.8746) / (5672.2402 - 0.8 \cdot 14.0 \cdot 0.8746)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (D - 0.8 \cdot t \cdot \cos(a))) / (2 \cdot E \cdot t \cdot \cos(a))$$

$$= (0.347 \cdot (5672.2402 - 0.8 \cdot (13.0) \cdot 0.8746)) / (2 \cdot 1.0 \cdot (13.0) \cdot 0.8746)$$

$$= 86.477 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.396 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 14.0, tr = 13.0, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 1., Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -36 °C

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	0.474 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	14.000 mm	Basic MDMT	-36 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-36 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Cone-2

Cone From 230 to 240 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.370	5672.24	902.21	498.88	0.1591	0.0008392	79.59

$$MAEP = (4*B)/(3*(Do/t)) = (4*79.5924)/(3*498.8755) = 0.2127 \text{ MPa}$$

Note:

The cone thickness used in the calculation has been modified
 per UG-33(f), $t_e = t * \cos(\alpha)$.

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
6.252	5672.24	902.21	907.26	0.1591	0.0003340	33.37

$$MAEP = (4*B)/(3*(Do/t)) = (4*33.3661)/(3*907.2581) = 0.049 \text{ MPa}$$

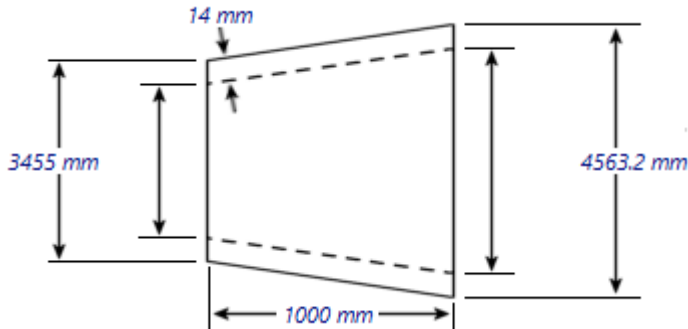
Note:

The cone thickness used in the calculation has been modified
 per UG-33(f), $t_e = t * \cos(\alpha)$.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Cone-3

Conical Head/Section Design Information:					
Design Pressure	P	0.347 MPa	Int. Design Temperature		160 °C
Material		SA-516 60	Ext. Design Temperature		150 °C
Length		1000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02100
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Outside Diameter	Do	4563.240 mm	Actual Stress		69.5 MPa
Spec. Min. Thk t,e, or sa		14.0000 mm	Spec. Nominal Thk.		18.000 mm
Surface Area		14432268.0 mm²	SG of Contents		0.000
Element Volume		12505.6 ltr	Weight of Contents		0.0 N
Empty Weight		19642.7 N	Operating Weight		19642.7 N
			Comp. All. Ope. Stress		92.9 MPa
			To End Diameter		3455.000 mm
Line of Support	Both Ends				

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot Do) / (2 \cdot \cos(a) \cdot (S \cdot E + 0.4 \cdot P))$ per Appendix 1-4 (e)
 $= (0.347 \cdot 4563.2402) / (2 \cdot 0.8747 \cdot (118.0 \cdot 1.0 + 0.4 \cdot 0.35))$
 $= 7.6673 + 1.0000 = 8.6673 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
 $= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D - 0.8 \cdot t \cdot \cos(a))$ per Appendix 1-4 (e)(5)
 $= (2 \cdot 118.0 \cdot 1.0 \cdot (13.0) \cdot 0.8747) / (4563.2402 - 0.8 \cdot (13.0) \cdot 0.8747)$
 $= 1 \text{ MPa}$

Maximum Allowable Pressure, New and Cold [MAPNC]:
 $= (2 \cdot S \cdot E \cdot t \cdot \cos(a)) / (D - 0.8 \cdot t \cdot \cos(a))$ per Appendix 1-4(e)
 $= (2 \cdot 118.0 \cdot 1.0 \cdot 14.0 \cdot 0.8747) / (4563.2402 - 0.8 \cdot 14.0 \cdot 0.8747)$
 $= 1 \text{ MPa}$

Actual stress at given pressure and thickness, corroded [Sact]:
 $= (P \cdot (D - 0.8 \cdot t \cdot \cos(a))) / (2 \cdot E \cdot t \cdot \cos(a))$
 $= (0.347 \cdot (4563.2402 - 0.8 \cdot (13.0) \cdot 0.8747)) / (2 \cdot 1.0 \cdot (13.0) \cdot 0.8747)$
 $= 69.537 \text{ MPa}$

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 0.524 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 14.0, tr = 10.457, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.804$, Temp. Reduction = 11 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -36 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -46 °C

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	0.474 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	14.000 mm	Basic MDMT	-36 °C
Stress/Thk. Ratio	0.804	Temperature Difference	11 °C
UCS-66(c) red.	No	Computed Min. Temp.	-46 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Cone-3

Cone From 240 to 250 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.371	4563.24	878.46	401.31	0.1925	0.0009506	83.57

$$MAEP = (4*B)/(3*(Do/t)) = (4*83.5698)/(3*401.3063) = 0.2777 \text{ MPa}$$

Note:

The cone thickness used in the calculation has been modified per UG-33(f), $te = t * \cos(\alpha)$.

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.471	4563.24	878.46	834.09	0.1925	0.0003070	30.68

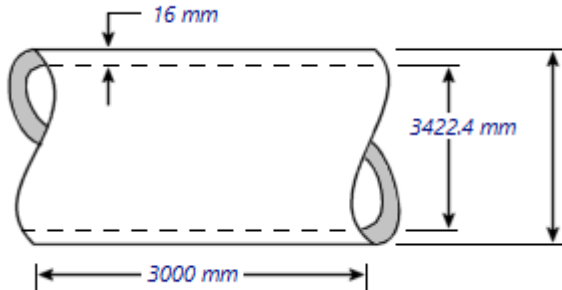
$$MAEP = (4*B)/(3*(Do/t)) = (4*30.6751)/(3*834.0914) = 0.049 \text{ MPa}$$

Note:

The cone thickness used in the calculation has been modified per UG-33(f), $te = t * \cos(\alpha)$.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Shell-16

Cylindrical Shell Design Information:					
Design Pressure	P	0.347 MPa	Int. Design Temperature		160 °C
Material		SA-516 60	Ext. Design Temperature		150 °C
Length		3000.000 mm	External Pressure Chart		CS-2
			UNS Number		K02100
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Inside Diameter	Di	3422.400 mm	Actual Stress		39.8 MPa
Spec. Min. Thk t _e , or s _a		16.0000 mm	Spec. Nominal Thk.		16.000 mm
Surface Area		32556954.0 mm ²	SG of Contents		0.000
Element Volume		27602.6 ltr	Weight of Contents		0.0 N
Empty Weight		39406.7 N	Operating Weight		39406.7 N
			Comp. All. Ope. Stress		103.2 MPa

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (0.347 \cdot 1712.2) / (118.0 \cdot 1.0 - 0.6 \cdot 0.347)$$

$$= 5.0476 + 1.0000 = 6.0476 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 15.0) / (1712.2 + 0.6 \cdot 15.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 16.0) / (1711.2 + 0.6 \cdot 16.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (0.347 \cdot (1712.2 + 0.6 \cdot 15.0)) / (1.0 \cdot 15.0)$$

$$= 39.845 \text{ MPa}$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 0.465 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 16.0, tr = 6.892, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E* / (tg - c) = 0.459, Temp. Reduction = 38 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -32 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	0.474 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	16.000 mm	Basic MDMT	-32 °C
Stress/Thk. Ratio	0.459	Temperature Difference	16 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Shell-16

Cylindrical Shell From R20 to 260 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	3454.40	3299.20	230.29	0.9551	0.0004022	40.19

$$MAEP = (4*B)/(3*(Do/t)) = (4*40.1869)/(3*230.2933) = 0.2327 \text{ MPa}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.069	3454.40	3299.20	428.10	0.9551	0.0001576	15.74

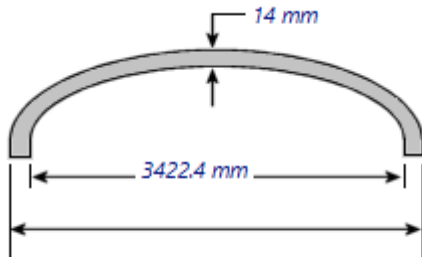
$$MAEP = (4*B)/(3*(Do/t)) = (4*15.7442)/(3*428.1043) = 0.049 \text{ MPa}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	3454.40	14627.68	230.29	4.2345	0.0000848	8.48

$$MAEP = (4*B)/(3*(Do/t)) = (4*8.4761)/(3*230.2933) = 0.0491 \text{ MPa}$$

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Top head

Elliptical Head Design Information:					
Design Pressure	P	0.347 MPa	Int. Design Temperature		160 °C
Material		SA-516 60	Ext. Design Temperature		150 °C
Length		50.000 mm	External Pressure Chart		CS-2
			UNS Number		K02100
Int. Corr. All.	c	1.0000 mm	Allowable Stress (ope)	S	118.0 MPa
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 MPa
Inside Diameter	Di	3422.400 mm	Actual Stress		45.7 MPa
Spec. Min. Thk t _e , or s _a		14.0000 mm	Spec. Nominal Thk.		18.000 mm
Surface Area		13508284.0 mm ²	SG of Contents		0.000
Element Volume		5708.2 ltr	Weight of Contents		0.0 N
Empty Weight		20119.3 N	Operating Weight		20119.3 N
			Comp. All. Ope. Stress		98.3 MPa
			Aspect Ratio		2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 42.780 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 21.390 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (0.347 \cdot 3424.3999 \cdot 0.999) / (2 \cdot 118.0 \cdot 1.0 - 0.2 \cdot 0.347)$$

$$= 5.0362 + 1.0000 = 6.0362 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 118.0 \cdot 1.0 \cdot 13.0) / (0.999 \cdot 3424.3999 + 0.2 \cdot 13.0)$$

$$= 1 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 118.0 \cdot 1.0 \cdot 14.0) / (1.0 \cdot 3422.3999 + 0.2 \cdot 14.0)$$

$$= 1 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (0.347 \cdot (0.999 \cdot 3424.3999 + 0.2 \cdot 13.0)) / (2 \cdot 1.0 \cdot 13.0)$$

$$= 45.734 \text{ MPa}$$

Straight Flange Required Thickness:



$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (0.347 \cdot 1712.2) / (118.0 \cdot 1.0 - 0.6 \cdot 0.347) + 1.0$$

$$= 6.048 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 17.0) / (1712.2 + 0.6 \cdot 17.0)$$

$$= 1 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (3424.4 / (2 \cdot 856.6))^2) / 6$$

$$= 0.999222$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 2.285 %

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 14.0$, $t_r = 6.873$, $c = 1.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.529$, Temp. Reduction = 29 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -36 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, $t_g = 18.0$, $t_r = 6.892$, $c = 1.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.405$, Temp. Reduction = 49 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	0.474 MPa
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-1 °C
Governing Thk.	14.000 mm	Basic MDMT	-36 °C
Stress/Thk. Ratio	0.529	Temperature Difference	13 °C
UCS-66(c) red.	No	Computed Min. Temp.	-48 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 42.780 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 21.390 mm

EXTERNAL PRESSURE RESULTS for: Top head

Elliptical Head From 260 to 270 Ext. Chart: CS-2 at 150 °C

Elastic Modulus from Chart: CS-2 at 150 °C: 199818 MPa

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
13.000	3450.40	265.42	0.0005233	52.28

$$MAEP = B / (K_0 \cdot Do / t) = 52.2815 / (0.9 \cdot 265.4153) = 0.2189 \text{ MPa}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
6.153	3450.40	560.73	0.0002477	24.75



$$MAEP = B/(K0*Do/t) = 24.7469/(0.9 * 560.7295) = 0.049 \text{ MPa}$$

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	0.616 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.451 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	0.598 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	0.521 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.497 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.497 MPa
App 27-4	= M.A.W.P.	0.474 MPa

UG-99(b), Test Pressure Calculation:
 = Test Factor • MAWP • Stress Ratio
 = 1.3 • 0.474 • 1.0
 = 0.616 MPa

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Btm Dish End	107.8	198.0	0.544	0.68
Shell-1	82.1	198.0	0.415	0.68
Shell-2	88.3	198.0	0.446	0.68
Shell-3	87.6	198.0	0.443	0.68
Shell-4	95.6	198.0	0.483	0.68
Shell-5	104.3	198.0	0.527	0.68
Shell-6	114.7	198.0	0.579	0.68
Shell-7	114.7	198.0	0.579	0.68
Shell-9	127.4	198.0	0.643	0.68
Shell-10	127.4	198.0	0.643	0.68
Shell-11	143.3	198.0	0.724	0.68
Shell-12	143.3	198.0	0.724	0.68
Shell-13	143.3	198.0	0.724	0.68
Shell-14	143.3	198.0	0.724	0.68
Shell-15	114.6	198.0	0.579	0.68
Cone-1	145.7	198.0	0.736	0.68
Cone-2	155.3	198.0	0.784	0.67
Cone-3	122.8	198.0	0.620	0.66
Shell-16	69.9	198.0	0.353	0.65
Top head	79.5	198.0	0.401	0.65

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1 22"	Nozzle	118.00	118.00	1.000
N1 22"	Pad	138.00	138.00	1.000
K1a 2"	Nozzle	118.00	118.00	1.000
K2a 2"	Nozzle	118.00	118.00	1.000
K3a 2"	Nozzle	118.00	118.00	1.000
K4a 2"	Nozzle	118.00	118.00	1.000
K1b 2"	Nozzle	118.00	118.00	1.000
K2b 2"	Nozzle	118.00	118.00	1.000
K3b 2"	Nozzle	118.00	118.00	1.000
K4b 2"	Nozzle	118.00	118.00	1.000
N10 2"	Nozzle	118.00	118.00	1.000
N14 4"	Nozzle	118.00	118.00	1.000
N14 4"	Pad	138.00	138.00	1.000
N13A 8"	Nozzle	118.00	118.00	1.000
N13A 8"	Pad	138.00	138.00	1.000
MH1 30"	Nozzle	118.00	118.00	1.000
MH1 30"	Pad	118.00	118.00	1.000
N12 2"	Nozzle	118.00	118.00	1.000



N13B 8"	Nozzle	118.00	118.00	1.000
N13B 8"	Pad	138.00	138.00	1.000
N13C 8"	Nozzle	118.00	118.00	1.000
N13C 8"	Pad	138.00	138.00	1.000
N2A 84"	Nozzle	138.00	138.00	1.000
MH2 30"	Nozzle	138.00	138.00	1.000
MH2 30"	Pad	138.00	138.00	1.000
K9a 2"	Nozzle	138.00	138.00	1.000
N2B 84"	Nozzle	138.00	138.00	1.000
K9b 2"	Nozzle	118.00	118.00	1.000
MH3 30"	Nozzle	118.00	118.00	1.000
MH3 30"	Pad	118.00	118.00	1.000
N4 20"	Nozzle	118.00	118.00	1.000
N4 20"	Pad	118.00	118.00	1.000
N3a 24"	Nozzle	118.00	118.00	1.000
N3a 24"	Pad	118.00	118.00	1.000
N3B 24"	Nozzle	118.00	118.00	1.000
N3B 24"	Pad	118.00	118.00	1.000
K5a 2"	Nozzle	118.00	118.00	1.000
K6a 2"	Nozzle	118.00	118.00	1.000
K14a 2"	Nozzle	118.00	118.00	1.000
K14b 2"	Nozzle	118.00	118.00	1.000
K6b	Nozzle	118.00	118.00	1.000
K5b 2"	Nozzle	118.00	118.00	1.000
MH4 30"	Nozzle	118.00	118.00	1.000
MH4 30"	Pad	118.00	118.00	1.000
N15 38"	Nozzle	118.00	118.00	1.000
N15 38"	Pad	118.00	118.00	1.000
K10B	Nozzle	118.00	118.00	1.000
N5 14"	Nozzle	118.00	118.00	1.000
N5 14"	Pad	138.00	138.00	1.000
MH5 30"	Nozzle	118.00	118.00	1.000
MH5 30"	Pad	138.00	138.00	1.000
N6a 18"	Nozzle	118.00	118.00	1.000
N6a 18"	Pad	138.00	138.00	1.000
N6b 18"	Nozzle	118.00	118.00	1.000
N6b 18"	Pad	138.00	138.00	1.000
K7a	Nozzle	118.00	118.00	1.000
K8a	Nozzle	118.00	118.00	1.000
K15a	Nozzle	118.00	118.00	1.000
K7b	Nozzle	118.00	118.00	1.000
K8b	Nozzle	118.00	118.00	1.000
K15b	Nozzle	118.00	118.00	1.000
MH6 30"	Nozzle	118.00	118.00	1.000
MH6 30"	Pad	138.00	138.00	1.000
K11a 2"	Nozzle	118.00	118.00	1.000
N7 16"	Nozzle	118.00	118.00	1.000
N7 16"	Pad	118.00	118.00	1.000
MH7	Nozzle	118.00	118.00	1.000
MH7	Pad	138.00	138.00	1.000
K16a	Nozzle	118.00	118.00	1.000
K16b	Nozzle	118.00	118.00	1.000
K11b	Nozzle	118.00	118.00	1.000
K12	Nozzle	118.00	118.00	1.000
K13b	Nozzle	118.00	118.00	1.000
P1(frm V-1403)	Nozzle	118.00	118.00	1.000
P1(frm V-1403)	Pad	137.90	137.90	1.000
P2(frm ME-1702)	Nozzle	117.90	117.90	1.000
P2(frm ME-1702)	Pad	137.90	137.90	1.000
P3 (from C-1402)	Nozzle	118.00	118.00	1.000
P3 (from C-1402)	Pad	118.00	118.00	1.000

Minimum

1.000

Stress ratios for Stiffening Ring Materials (MPa):

Description	Ambient	Operating	Ratio
R1	138.00	138.00	1.000
R2	138.00	138.00	1.000



R3	138.00	138.00	1.000
R4	138.00	138.00	1.000
R5	138.00	138.00	1.000
R6	138.00	138.00	1.000
R7	138.00	138.00	1.000
R8	138.00	138.00	1.000
R9	138.00	138.00	1.000
R10	138.00	138.00	1.000
R11	138.00	138.00	1.000
R12	138.00	138.00	1.000
R13	138.00	138.00	1.000
R14	138.00	138.00	1.000
R15	138.00	138.00	1.000
R16	138.00	138.00	1.000
R17	138.00	138.00	1.000
R18	138.00	138.00	1.000
R19	138.00	138.00	1.000
R20	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Btm Dish End	118.00	118.00	1.000
Shell-1	118.00	118.00	1.000
Shell-2	118.00	118.00	1.000
Shell-3	118.00	118.00	1.000
Shell-4	118.00	118.00	1.000
Shell-5	118.00	118.00	1.000
Shell-6	118.00	118.00	1.000
Shell-7	118.00	118.00	1.000
Shell-9	118.00	118.00	1.000
Shell-10	118.00	118.00	1.000
Shell-11	118.00	118.00	1.000
Shell-12	118.00	118.00	1.000
Shell-13	118.00	118.00	1.000
Shell-14	118.00	118.00	1.000
Shell-15	118.00	118.00	1.000
Cone-1	118.00	118.00	1.000
Cone-2	118.00	118.00	1.000
Cone-3	118.00	118.00	1.000
Shell-16	118.00	118.00	1.000
Top head	118.00	118.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N1 22"	18.77	216.00	0.087
K1a 2"	4.99	216.00	0.023
K2a 2"	4.99	216.00	0.023
K3a 2"	4.99	216.00	0.023
K4a 2"	4.99	216.00	0.023
K1b 2"	4.99	216.00	0.023
K2b 2"	4.99	216.00	0.023
K3b 2"	4.99	216.00	0.023
K4b 2"	4.99	216.00	0.023
N10 2"	4.99	216.00	0.023
N14 4"	6.20	216.00	0.029
N13A 8"	8.86	216.00	0.041
MH1 30"	18.29	198.00	0.092
N12 2"	4.99	216.00	0.023
N13B 8"	8.86	216.00	0.041
N13C 8"	8.86	216.00	0.041
N2A 84"	36.09	225.00	0.160
MH2 30"	14.16	234.00	0.060



K9a 2"	4.98	225.00	0.022
N2B 84"	36.09	225.00	0.160
K9b 2"	4.98	216.00	0.023
MH3 30"	12.71	198.00	0.064
N4 20"	17.04	198.00	0.086
N3a 24"	17.04	198.00	0.086
N3B 24"	20.51	198.00	0.104
K5a 2"	4.98	216.00	0.023
K6a 2"	4.98	216.00	0.023
K14a 2"	4.98	216.00	0.023
K14b 2"	4.98	216.00	0.023
K6b	4.98	216.00	0.023
K5b 2"	4.98	216.00	0.023
MH4 30"	12.71	198.00	0.064
N15 38"	16.18	198.00	0.082
K10B	4.98	216.00	0.023
N5 14"	11.85	216.00	0.055
MH5 30"	12.71	216.00	0.059
N6a 18"	15.31	198.00	0.077
N6b 18"	15.31	198.00	0.077
K7a	4.98	216.00	0.023
K8a	4.98	216.00	0.023
K15a	4.98	216.00	0.023
K7b	4.98	216.00	0.023
K8b	4.98	216.00	0.023
K15b	4.98	216.00	0.023
MH6 30"	12.71	216.00	0.059
K11a 2"	4.98	216.00	0.023
N7 16"	13.59	198.00	0.069
MH7	12.72	216.00	0.059
K16a	4.99	216.00	0.023
K16b	4.99	216.00	0.023
K11b	4.99	216.00	0.023
K12	4.99	216.00	0.023
K13b	4.99	216.00	0.023
P1(frm V-1403)	5.91	216.00	0.027
P2(frm ME-1702)	5.91	217.20	0.027
P3 (from C-1402)	14.40	198.00	0.073

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	0.616 MPa
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	0.451 MPa
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	0.606 MPa
UG100	= 1.10 * M.A.W.P. * Sa/S	0.521 MPa
PED	= max(1.43*DP, 1.25*DP*ratio)	0.497 MPa
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	0.497 MPa
App 27-4	= M.A.W.P.	0.474 MPa

UG-99(b) Note 35, Test Pressure Calculation:
 = Test Factor • Design Pressure • Stress Ratio
 = 1.3 • 0.347 • 1.0
 = 0.451 MPa

Vertical Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
Btm Dish End	149.9	198.0	0.757	0.92
Shell-1	113.9	198.0	0.575	0.91
Shell-2	119.0	198.0	0.601	0.88
Shell-3	114.1	198.0	0.576	0.85
Shell-4	120.7	198.0	0.609	0.82
Shell-5	127.4	198.0	0.643	0.79
Shell-6	135.5	198.0	0.685	0.77
Shell-7	130.3	198.0	0.658	0.74
Shell-9	139.8	198.0	0.706	0.71
Shell-10	134.0	198.0	0.677	0.68
Shell-11	145.2	198.0	0.733	0.65
Shell-12	138.6	198.0	0.700	0.62
Shell-13	132.0	198.0	0.667	0.59
Shell-14	125.4	198.0	0.633	0.56
Shell-15	94.6	198.0	0.478	0.53
Cone-1	117.3	198.0	0.593	0.52
Cone-2	126.8	198.0	0.641	0.51
Cone-3	100.0	198.0	0.505	0.50
Shell-16	56.2	198.0	0.284	0.49
Top head	60.6	198.0	0.306	0.46

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1 22"	Nozzle	118.00	118.00	1.000
N1 22"	Pad	138.00	138.00	1.000
K1a 2"	Nozzle	118.00	118.00	1.000
K2a 2"	Nozzle	118.00	118.00	1.000
K3a 2"	Nozzle	118.00	118.00	1.000
K4a 2"	Nozzle	118.00	118.00	1.000
K1b 2"	Nozzle	118.00	118.00	1.000
K2b 2"	Nozzle	118.00	118.00	1.000
K3b 2"	Nozzle	118.00	118.00	1.000
K4b 2"	Nozzle	118.00	118.00	1.000
N10 2"	Nozzle	118.00	118.00	1.000
N14 4"	Nozzle	118.00	118.00	1.000
N14 4"	Pad	138.00	138.00	1.000
N13A 8"	Nozzle	118.00	118.00	1.000
N13A 8"	Pad	138.00	138.00	1.000
MH1 30"	Nozzle	118.00	118.00	1.000
MH1 30"	Pad	118.00	118.00	1.000
N12 2"	Nozzle	118.00	118.00	1.000



N13B 8"	Nozzle	118.00	118.00	1.000
N13B 8"	Pad	138.00	138.00	1.000
N13C 8"	Nozzle	118.00	118.00	1.000
N13C 8"	Pad	138.00	138.00	1.000
N2A 84"	Nozzle	138.00	138.00	1.000
MH2 30"	Nozzle	138.00	138.00	1.000
MH2 30"	Pad	138.00	138.00	1.000
K9a 2"	Nozzle	138.00	138.00	1.000
N2B 84"	Nozzle	138.00	138.00	1.000
K9b 2"	Nozzle	118.00	118.00	1.000
MH3 30"	Nozzle	118.00	118.00	1.000
MH3 30"	Pad	118.00	118.00	1.000
N4 20"	Nozzle	118.00	118.00	1.000
N4 20"	Pad	118.00	118.00	1.000
N3a 24"	Nozzle	118.00	118.00	1.000
N3a 24"	Pad	118.00	118.00	1.000
N3B 24"	Nozzle	118.00	118.00	1.000
N3B 24"	Pad	118.00	118.00	1.000
K5a 2"	Nozzle	118.00	118.00	1.000
K6a 2"	Nozzle	118.00	118.00	1.000
K14a 2"	Nozzle	118.00	118.00	1.000
K14b 2"	Nozzle	118.00	118.00	1.000
K6b	Nozzle	118.00	118.00	1.000
K5b 2"	Nozzle	118.00	118.00	1.000
MH4 30"	Nozzle	118.00	118.00	1.000
MH4 30"	Pad	118.00	118.00	1.000
N15 38"	Nozzle	118.00	118.00	1.000
N15 38"	Pad	118.00	118.00	1.000
K10B	Nozzle	118.00	118.00	1.000
N5 14"	Nozzle	118.00	118.00	1.000
N5 14"	Pad	138.00	138.00	1.000
MH5 30"	Nozzle	118.00	118.00	1.000
MH5 30"	Pad	138.00	138.00	1.000
N6a 18"	Nozzle	118.00	118.00	1.000
N6a 18"	Pad	138.00	138.00	1.000
N6b 18"	Nozzle	118.00	118.00	1.000
N6b 18"	Pad	138.00	138.00	1.000
K7a	Nozzle	118.00	118.00	1.000
K8a	Nozzle	118.00	118.00	1.000
K15a	Nozzle	118.00	118.00	1.000
K7b	Nozzle	118.00	118.00	1.000
K8b	Nozzle	118.00	118.00	1.000
K15b	Nozzle	118.00	118.00	1.000
MH6 30"	Nozzle	118.00	118.00	1.000
MH6 30"	Pad	138.00	138.00	1.000
K11a 2"	Nozzle	118.00	118.00	1.000
N7 16"	Nozzle	118.00	118.00	1.000
N7 16"	Pad	118.00	118.00	1.000
MH7	Nozzle	118.00	118.00	1.000
MH7	Pad	138.00	138.00	1.000
K16a	Nozzle	118.00	118.00	1.000
K16b	Nozzle	118.00	118.00	1.000
K11b	Nozzle	118.00	118.00	1.000
K12	Nozzle	118.00	118.00	1.000
K13b	Nozzle	118.00	118.00	1.000
P1(frm V-1403)	Nozzle	118.00	118.00	1.000
P1(frm V-1403)	Pad	137.90	137.90	1.000
P2(frm ME-1702)	Nozzle	117.90	117.90	1.000
P2(frm ME-1702)	Pad	137.90	137.90	1.000
P3 (from C-1402)	Nozzle	118.00	118.00	1.000
P3 (from C-1402)	Pad	118.00	118.00	1.000

Minimum

1.000

Stress ratios for Stiffening Ring Materials (MPa):

Description	Ambient	Operating	Ratio
R1	138.00	138.00	1.000
R2	138.00	138.00	1.000



R3	138.00	138.00	1.000
R4	138.00	138.00	1.000
R5	138.00	138.00	1.000
R6	138.00	138.00	1.000
R7	138.00	138.00	1.000
R8	138.00	138.00	1.000
R9	138.00	138.00	1.000
R10	138.00	138.00	1.000
R11	138.00	138.00	1.000
R12	138.00	138.00	1.000
R13	138.00	138.00	1.000
R14	138.00	138.00	1.000
R15	138.00	138.00	1.000
R16	138.00	138.00	1.000
R17	138.00	138.00	1.000
R18	138.00	138.00	1.000
R19	138.00	138.00	1.000
R20	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (MPa):

Description	Ambient	Operating	Ratio
Btm Dish End	118.00	118.00	1.000
Shell-1	118.00	118.00	1.000
Shell-2	118.00	118.00	1.000
Shell-3	118.00	118.00	1.000
Shell-4	118.00	118.00	1.000
Shell-5	118.00	118.00	1.000
Shell-6	118.00	118.00	1.000
Shell-7	118.00	118.00	1.000
Shell-9	118.00	118.00	1.000
Shell-10	118.00	118.00	1.000
Shell-11	118.00	118.00	1.000
Shell-12	118.00	118.00	1.000
Shell-13	118.00	118.00	1.000
Shell-14	118.00	118.00	1.000
Shell-15	118.00	118.00	1.000
Cone-1	118.00	118.00	1.000
Cone-2	118.00	118.00	1.000
Cone-3	118.00	118.00	1.000
Shell-16	118.00	118.00	1.000
Top head	118.00	118.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
N1 22"	25.44	216.00	0.118
K1a 2"	6.67	216.00	0.031
K2a 2"	6.67	216.00	0.031
K3a 2"	6.67	216.00	0.031
K4a 2"	6.67	216.00	0.031
K1b 2"	6.67	216.00	0.031
K2b 2"	6.67	216.00	0.031
K3b 2"	6.67	216.00	0.031
K4b 2"	6.67	216.00	0.031
N10 2"	6.67	216.00	0.031
N14 4"	8.30	216.00	0.038
N13A 8"	11.85	216.00	0.055
MH1 30"	24.46	198.00	0.124
N12 2"	6.67	216.00	0.031
N13B 8"	11.85	216.00	0.055
N13C 8"	11.85	216.00	0.055
N2A 84"	46.74	225.00	0.208
MH2 30"	18.33	234.00	0.078



K9a 2"	6.45	225.00	0.029
N2B 84"	46.74	225.00	0.208
K9b 2"	5.59	216.00	0.026
MH3 30"	13.72	198.00	0.069
N4 20"	18.40	198.00	0.093
N3a 24"	18.40	198.00	0.093
N3B 24"	22.13	198.00	0.112
K5a 2"	5.38	216.00	0.025
K6a 2"	5.38	216.00	0.025
K14a 2"	5.38	216.00	0.025
K14b 2"	5.38	216.00	0.025
K6b	5.38	216.00	0.025
K5b 2"	5.38	216.00	0.025
MH4 30"	13.72	198.00	0.069
N15 38"	17.46	198.00	0.088
K10B	4.52	216.00	0.021
N5 14"	10.74	216.00	0.050
MH5 30"	11.53	216.00	0.053
N6a 18"	13.89	198.00	0.070
N6b 18"	13.89	198.00	0.070
K7a	4.52	216.00	0.021
K8a	4.52	216.00	0.021
K15a	4.52	216.00	0.021
K7b	4.30	216.00	0.020
K8b	4.30	216.00	0.020
K15b	4.30	216.00	0.020
MH6 30"	10.98	216.00	0.051
K11a 2"	4.30	216.00	0.020
N7 16"	10.66	198.00	0.054
MH7	9.98	216.00	0.046
K16a	3.91	216.00	0.018
K16b	3.91	216.00	0.018
K11b	3.91	216.00	0.018
K12	3.91	216.00	0.018
K13b	3.91	216.00	0.018
P1(frm V-1403)	4.45	216.00	0.021
P2(frm ME-1702)	4.45	217.20	0.021
P3 (from C-1402)	10.86	198.00	0.055

Elements Suitable for Test Pressure.

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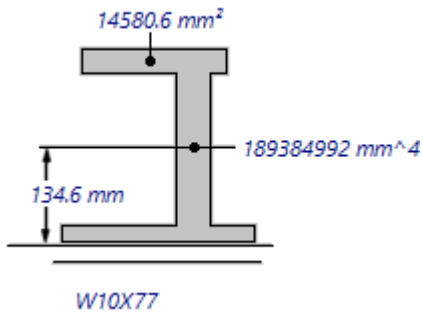


External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

Stiffening Ring Calculations for: R1, W10X77, SA-516 70

Material treatment options:



Effective Length of Shell: 470 mm

	Area (mm²)	Distance (mm)	Area*Dist
Shell:	12690.419	13.5000	171320.672
Ring :	14580.600	161.6200	2356516.250
Total:	27271.018		2527837.000

Centroid of Ring plus Shell: 93 mm

	Inertia	Distance	A*Dist²
Shell:	770942.250	79.1932	79588712.000
Ring :	189384992.000	-68.9268	69271064.000
Total:	190155936.000		148859776.000

Available Moment of Inertia, Ring + Shell: 0.339E+09 mm⁴

Required Stress in Ring plus Shell Breq 7.87 MPa
 Required Strain in Ring plus Shell Areq 0.0000790

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6762.0^2 \cdot 3163.0959(27.0 + 14580.5996/3163.0959)0.000079) / 10.9$$

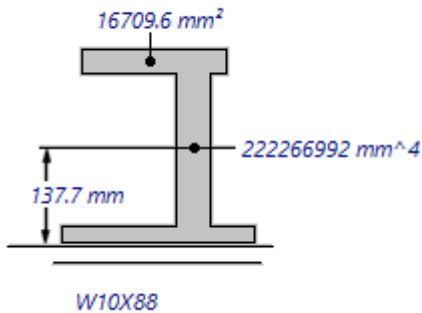
$$= 33134616 \text{ mm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	17.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	155.08	N/mm
The Radial Shear Load	V	10486.75	N
The First Moment of the Area (Ring + Shell)	Q100	4994.44	mm³
Weld Shear Flow due to Rad. Shear Load	VQ/I	31.09	N/mm
The Weld Allowable Stress	0.55*S	64.90	MPa
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	0.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		216.00	mm
The Weld Allowable Load	WLeg*0.55*S	1103.19	N/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	158.17	N/mm
MAEP of Stiffening Ring Weld	Weld MAEP	0.34	MPa

Stiffening Ring Calculations for: R2, W10X88, SA-516 70

Material treatment options:



Effective Length of Shell: 452 mm

	Area (mm²)	Distance (mm)	Area*Dist
Shell:	11303.470	12.5000	141293.375
Ring :	16709.600	162.6680	2718117.000
Total:	28013.070		2859410.500

Centroid of Ring plus Shell: 102 mm

	Inertia	Distance	A*Dist²
Shell:	588721.875	89.5742	90693728.000
Ring :	222266992.000	-60.5938	61351196.000
Total:	222855728.000		152044928.000

Available Moment of Inertia, Ring + Shell: 0.375E+09 mm⁴

Required Stress in Ring plus Shell Breq 8.45 MPa
 Required Strain in Ring plus Shell Areq 0.0000845

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6758.0^2 \cdot 3775.0(25.0 + 16709.5996/3775.0)0.000085) / 10.9$$

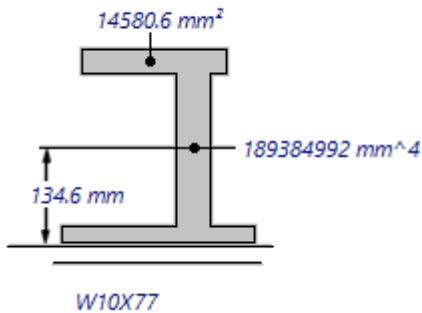
$$= 39345020 \text{ mm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	17.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	185.08	N/mm
The Radial Shear Load	V	12508.01	N
The First Moment of the Area (Ring + Shell)	Q10	12498.81	mm³
Weld Shear Flow due to Rad. Shear Load	VQ/I	33.78	N/mm
The Weld Allowable Stress	0.55*S	64.90	MPa
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	0.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		200.00	mm
The Weld Allowable Load	WLeg*0.55*S	1103.19	N/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	188.14	N/mm
MAEP of Stiffening Ring Weld	Weld MAEP	0.29	MPa

Stiffening Ring Calculations for: R3, W10X77, SA-516 70

Material treatment options:



Effective Length of Shell: 450 mm

	Area (mm²)	Distance (mm)	Area*Dist
Shell:	11259.898	12.5000	140748.734
Ring :	14580.600	159.6200	2327355.250
Total:	25840.500		2468104.000

Centroid of Ring plus Shell: 96 mm

	Inertia	Distance	A*Dist²
Shell:	586452.500	83.0130	77593776.000
Ring :	189384992.000	-64.1070	59921952.000
Total:	189971440.000		137515728.000

Available Moment of Inertia, Ring + Shell: 0.327E+09 mm⁴

Required Stress in Ring plus Shell Breq 7.82 MPa
 Required Strain in Ring plus Shell Areq 0.0000780

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6706.0^2 \cdot 2225.0(25.0 + 14580.5996/2225.0)0.000078) / 10.9$$

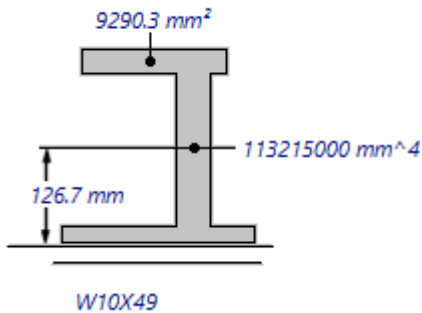
$$= 22592618 \text{ mm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	17.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	109.09	N/mm
The Radial Shear Load	V	7315.55	N
The First Moment of the Area (Ring + Shell)	Q	934718.12	mm³
Weld Shear Flow due to Rad. Shear Load	VQ/I	20.88	N/mm
The Weld Allowable Stress	0.55*S	64.90	MPa
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	0.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		200.00	mm
The Weld Allowable Load	WLeg*0.55*S	1103.19	N/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	111.07	N/mm
MAEP of Stiffening Ring Weld	Weld MAEP	0.49	MPa

Stiffening Ring Calculations for: R4, W10X49, SA-516 70

Material treatment options:



Effective Length of Shell: 434 mm

	Area (mm²)	Distance (mm)	Area*Dist
Shell:	9971.604	11.5000	114673.445
Ring :	9290.300	149.7460	1391185.250
Total:	19261.902		1505858.750

Centroid of Ring plus Shell: 78 mm

	Inertia	Distance	A*Dist²
Shell:	439581.125	66.6781	44333424.000
Ring :	113215000.000	-71.5679	47584608.000
Total:	113654576.000		91918032.000

Available Moment of Inertia, Ring + Shell: 0.206E+09 mm⁴

Required Stress in Ring plus Shell Breq 8.98 MPa
 Required Strain in Ring plus Shell Areq 0.0000900

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6754.0^2 \cdot 2000.0(23.0 + 9290.2998/2000.0)0.00009) / 10.9$$

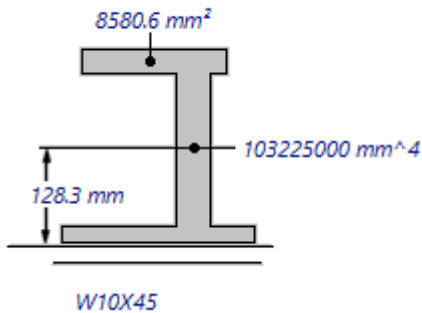
$$= 20825078 \text{ mm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	17.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	98.06	N/mm
The Radial Shear Load	V	6622.84	N
The First Moment of the Area (Ring + Shell)	Q	664887.38	mm³
Weld Shear Flow due to Rad. Shear Load	VQ/I	21.42	N/mm
The Weld Allowable Stress	0.55*S	64.90	MPa
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	0.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		184.00	mm
The Weld Allowable Load	WLeg*0.55*S	1103.19	N/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	100.37	N/mm
MAEP of Stiffening Ring Weld	Weld MAEP	0.54	MPa

Stiffening Ring Calculations for: R5, W10X45, SA-516 70

Material treatment options:



Effective Length of Shell: 414 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	8697.081	10.5000	91319.352
Ring :	8580.630	149.2700	1280830.750
Total:	17277.711		1372150.125

Centroid of Ring plus Shell: 79 mm

	Inertia	Distance	A*Dist ²
Shell:	319617.438	68.9173	41307664.000
Ring :	103225000.000	-69.8527	41868272.000
Total:	103544616.000		83175936.000

Available Moment of Inertia, Ring + Shell: 0.187E+09 mm⁴

Required Stress in Ring plus Shell Breq 9.82 MPa
 Required Strain in Ring plus Shell Areq 0.0000980

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6750.0^2 \cdot 2000.0(21.0 + 8580.6299/2000.0)0.000098) / 10.9$$

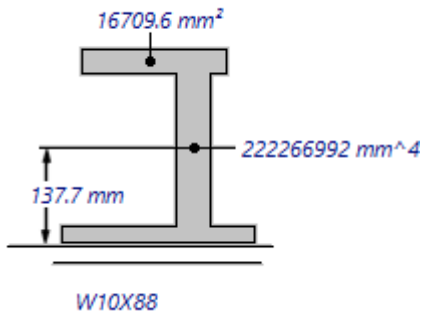
$$= 20720058 \text{ mm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	17.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	98.06	N/mm
The Radial Shear Load	V	6618.92	N
The First Moment of the Area (Ring + Shell)	Q	599379.69	mm ³
Weld Shear Flow due to Rad. Shear Load	VQ/I	21.25	N/mm
The Weld Allowable Stress	0.55*S	64.90	MPa
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	0.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		168.00	mm
The Weld Allowable Load	WLeg*0.55*S	1103.19	N/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	100.33	N/mm
MAEP of Stiffening Ring Weld	Weld MAEP	0.54	MPa

Stiffening Ring Calculations for: R6, W10X88, SA-516 70

Material treatment options:



Effective Length of Shell: 414 mm

	Area (mm²)	Distance (mm)	Area*Dist
Shell:	8697.081	10.5000	91319.352
Ring :	16709.600	158.6680	2651278.750
Total:	25406.682		2742598.000

Centroid of Ring plus Shell: 108 mm

	Inertia	Distance	A*Dist²
Shell:	319617.438	97.4479	82588320.000
Ring :	222266992.000	-50.7201	42985904.000
Total:	222586608.000		125574224.000

Available Moment of Inertia, Ring + Shell: 0.348E+09 mm⁴

Required Stress in Ring plus Shell Breq 8.46 MPa
 Required Strain in Ring plus Shell Areq 0.0000850

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6750.0^2 \cdot 2000.0(21.0 + 16709.5996/2000.0)0.000085) / 10.9$$

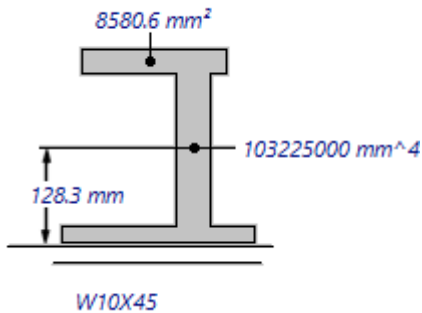
$$= 20859732 \text{ mm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	17.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	98.06	N/mm
The Radial Shear Load	V	6618.92	N
The First Moment of the Area (Ring + Shell)	Q	847512.38	mm³
Weld Shear Flow due to Rad. Shear Load	VQ/I	16.11	N/mm
The Weld Allowable Stress	0.55*S	64.90	MPa
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	0.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		168.00	mm
The Weld Allowable Load	WLeg*0.55*S	1103.19	N/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	99.37	N/mm
MAEP of Stiffening Ring Weld	Weld MAEP	0.54	MPa

Stiffening Ring Calculations for: R7, W10X45, SA-516 70

Material treatment options:



Effective Length of Shell: 394 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	7482.491	9.5000	71083.664
Ring :	8580.630	147.2700	1263669.375
Total:	16063.121		1334753.000

Centroid of Ring plus Shell: 83 mm

	Inertia	Distance	A*Dist ²
Shell:	225098.062	73.5942	40526020.000
Ring :	103225000.000	-64.1758	35339564.000
Total:	103450104.000		75865584.000

Available Moment of Inertia, Ring + Shell: 0.179E+09 mm⁴

Required Stress in Ring plus Shell Breq 10.65 MPa
 Required Strain in Ring plus Shell Areq 0.0001070

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6746.0^2 \cdot 2000.0017(19.0 + 8580.6299/2000.0017)0.000107) / 10.9$$

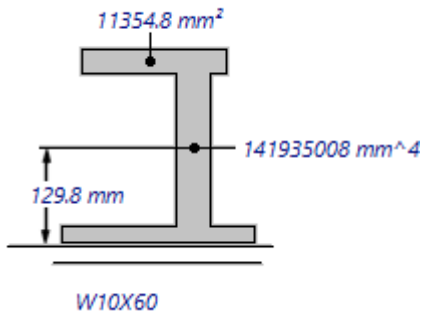
$$= 20809194 \text{ mm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	17.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	98.06	N/mm
The Radial Shear Load	V	6615.00	N
The First Moment of the Area (Ring + Shell)	Q	550668.31	mm ³
Weld Shear Flow due to Rad. Shear Load	VQ/I	20.31	N/mm
The Weld Allowable Stress	0.55*S	64.90	MPa
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	0.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		152.00	mm
The Weld Allowable Load	WLeg*0.55*S	1103.19	N/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	100.14	N/mm
MAEP of Stiffening Ring Weld	Weld MAEP	0.54	MPa

Stiffening Ring Calculations for: R8, W10X60, SA-516 70

Material treatment options:



Effective Length of Shell: 394 mm

	Area (mm²)	Distance (mm)	Area*Dist
Shell:	7482.491	9.5000	71083.664
Ring :	11354.800	148.7940	1689526.250
Total:	18837.291		1760609.875

Centroid of Ring plus Shell: 93 mm

	Inertia	Distance	A*Dist²
Shell:	225098.062	83.9641	52751300.000
Ring :	141935008.000	-55.3299	34761604.000
Total:	142160112.000		87512904.000

Available Moment of Inertia, Ring + Shell: 0.230E+09 mm⁴

Required Stress in Ring plus Shell Breq 10.54 MPa
 Required Strain in Ring plus Shell Areq 0.0001050

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6746.0^2 \cdot 2500.0015(19.0 + 11354.7998/2500.0015)0.000105) / 10.9$$

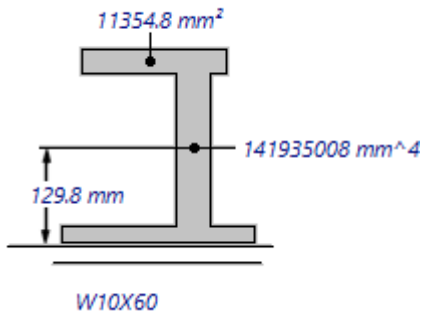
$$= 25801040 \text{ mm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	17.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	122.57	N/mm
The Radial Shear Load	V	8268.75	N
The First Moment of the Area (Ring + Shell)	Q	628260.38	mm³
Weld Shear Flow due to Rad. Shear Load	VQ/I	22.62	N/mm
The Weld Allowable Stress	0.55*S	64.90	MPa
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	0.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		152.00	mm
The Weld Allowable Load	WLeg*0.55*S	1103.19	N/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	124.64	N/mm
MAEP of Stiffening Ring Weld	Weld MAEP	0.43	MPa

Stiffening Ring Calculations for: R9, W10X60, SA-516 70

Material treatment options:



Effective Length of Shell: 372 mm

	Area (mm²)	Distance (mm)	Area*Dist
Shell:	6330.827	8.5000	53812.031
Ring :	11354.800	146.7940	1666816.625
Total:	17685.627		1720628.625

Centroid of Ring plus Shell: 97 mm

	Inertia	Distance	A*Dist²
Shell:	152467.281	88.7897	49909740.000
Ring :	141935008.000	-49.5043	27826980.000
Total:	142087472.000		77736720.000

Available Moment of Inertia, Ring + Shell: 0.220E+09 mm⁴

Required Stress in Ring plus Shell Breq 11.51 MPa
 Required Strain in Ring plus Shell Areq 0.0001150

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6742.0^2 \cdot 2500.0(17.0 + 11354.7998/2500.0)0.000115) / 10.9$$

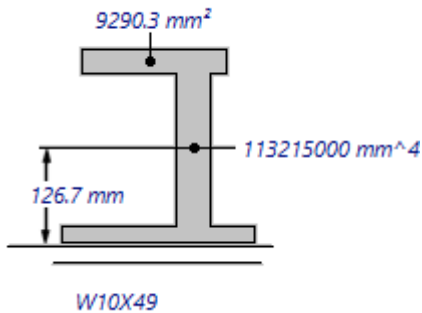
$$= 25826942 \text{ mm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	17.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	122.57	N/mm
The Radial Shear Load	V	8263.84	N
The First Moment of the Area (Ring + Shell)	Q	562112.00	mm³
Weld Shear Flow due to Rad. Shear Load	VQ/I	21.13	N/mm
The Weld Allowable Stress	0.55*S	64.90	MPa
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	0.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		136.00	mm
The Weld Allowable Load	WLeg*0.55*S	1103.19	N/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	124.38	N/mm
MAEP of Stiffening Ring Weld	Weld MAEP	0.43	MPa

Stiffening Ring Calculations for: R10, W10X49, SA-516 70

Material treatment options:



Effective Length of Shell: 372 mm

	Area (mm²)	Distance (mm)	Area*Dist
Shell:	6330.827	8.5000	53812.031
Ring :	9290.300	143.7460	1335443.500
Total:	15621.126		1389255.500

Centroid of Ring plus Shell: 89 mm

	Inertia	Distance	A*Dist²
Shell:	152467.281	80.4344	40958504.000
Ring :	113215000.000	-54.8116	27910962.000
Total:	113367464.000		68869464.000

Available Moment of Inertia, Ring + Shell: 0.182E+09 mm⁴

Required Stress in Ring plus Shell Breq 11.45 MPa
 Required Strain in Ring plus Shell Areq 0.0001150

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6742.0^2 \cdot 1999.9984(17.0 + 9290.2998/1999.9984)0.000115) / 10.9$$

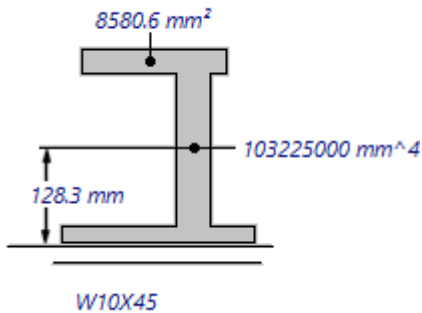
$$= 20760554 \text{ mm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	17.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	98.06	N/mm
The Radial Shear Load	V	6611.07	N
The First Moment of the Area (Ring + Shell)	Q	509216.28	mm³
Weld Shear Flow due to Rad. Shear Load	VQ/I	18.47	N/mm
The Weld Allowable Stress	0.55*S	64.90	MPa
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	0.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		136.00	mm
The Weld Allowable Load	WLeg*0.55*S	1103.19	N/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	99.78	N/mm
MAEP of Stiffening Ring Weld	Weld MAEP	0.54	MPa

Stiffening Ring Calculations for: R11, W10X45, SA-516 70

Material treatment options:



Effective Length of Shell: 372 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	6330.827	8.5000	53812.031
Ring :	8580.630	145.2700	1246508.125
Total:	14911.456		1300320.125

Centroid of Ring plus Shell: 87 mm

	Inertia	Distance	A*Dist ²
Shell:	152467.281	78.7028	39213928.000
Ring :	103225000.000	-58.0672	28932210.000
Total:	103377472.000		68146136.000

Available Moment of Inertia, Ring + Shell: 0.172E+09 mm⁴

Required Stress in Ring plus Shell Breq 11.65 MPa
 Required Strain in Ring plus Shell Areq 0.0001160

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6742.0^2 \cdot 2000.0(17.0 + 8580.6299/2000.0)0.000116) / 10.9$$

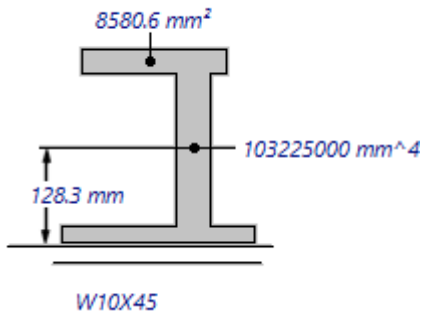
$$= 20597792 \text{ mm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	17.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	98.06	N/mm
The Radial Shear Load	V	6611.07	N
The First Moment of the Area (Ring + Shell)	Q	498253.56	mm ³
Weld Shear Flow due to Rad. Shear Load	VQ/I	19.20	N/mm
The Weld Allowable Stress	0.55*S	64.90	MPa
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	0.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		136.00	mm
The Weld Allowable Load	WLeg*0.55*S	1103.19	N/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	99.92	N/mm
MAEP of Stiffening Ring Weld	Weld MAEP	0.54	MPa

Stiffening Ring Calculations for: R12, W10X45, SA-516 70

Material treatment options:



Effective Length of Shell: 350 mm

	Area (mm²)	Distance (mm)	Area*Dist
Shell:	5245.599	7.5000	39341.996
Ring :	8580.630	143.2700	1229346.875
Total:	13826.229		1268688.875

Centroid of Ring plus Shell: 92 mm

	Inertia	Distance	A*Dist²
Shell:	98354.883	84.2596	37242052.000
Ring :	103225000.000	-51.5104	22767200.000
Total:	103323360.000		60009252.000

Available Moment of Inertia, Ring + Shell: 0.163E+09 mm⁴

Required Stress in Ring plus Shell Breq 12.85 MPa
 Required Strain in Ring plus Shell Areq 0.0001280

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6738.0005^2 \cdot 2000.0007(15.0 + 8580.6299/2000.0007)0.000128) / 10.9$$

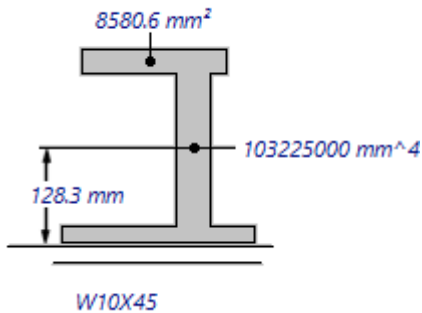
$$= 20569066 \text{ mm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	17.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	98.06	N/mm
The Radial Shear Load	V	6607.15	N
The First Moment of the Area (Ring + Shell)	Q	441991.91	mm³
Weld Shear Flow due to Rad. Shear Load	VQ/I	17.88	N/mm
The Weld Allowable Stress	0.55*S	64.90	MPa
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	0.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		120.00	mm
The Weld Allowable Load	WLeg*0.55*S	1103.19	N/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	99.67	N/mm
MAEP of Stiffening Ring Weld	Weld MAEP	0.54	MPa

Stiffening Ring Calculations for: R13, W10X45, SA-516 70

Material treatment options:



Effective Length of Shell: 350 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	5245.599	7.5000	39341.996
Ring :	8580.630	143.2700	1229346.875
Total:	13826.229		1268688.875

Centroid of Ring plus Shell: 92 mm

	Inertia	Distance	A*Dist ²
Shell:	98354.883	84.2596	37242052.000
Ring :	103225000.000	-51.5104	22767200.000
Total:	103323360.000		60009252.000

Available Moment of Inertia, Ring + Shell: 0.163E+09 mm⁴

Required Stress in Ring plus Shell Breq 12.85 MPa
 Required Strain in Ring plus Shell Areq 0.0001280

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6738.0005^2 \cdot 2000.0007(15.0 + 8580.6299/2000.0007)0.000128) / 10.9$$

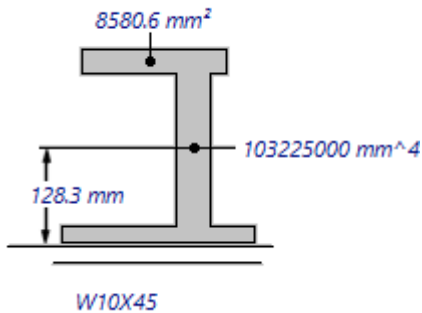
$$= 20569066 \text{ mm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	17.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	98.06	N/mm
The Radial Shear Load	V	6607.15	N
The First Moment of the Area (Ring + Shell)	Q	441991.91	mm ³
Weld Shear Flow due to Rad. Shear Load	VQ/I	17.88	N/mm
The Weld Allowable Stress	0.55*S	64.90	MPa
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	0.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		120.00	mm
The Weld Allowable Load	WLeg*0.55*S	1103.19	N/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	99.67	N/mm
MAEP of Stiffening Ring Weld	Weld MAEP	0.54	MPa

Stiffening Ring Calculations for: R14, W10X45, SA-516 70

Material treatment options:



Effective Length of Shell: 350 mm

	Area (mm²)	Distance (mm)	Area*Dist
Shell:	5245.599	7.5000	39341.996
Ring :	8580.630	143.2700	1229346.875
Total:	13826.229		1268688.875

Centroid of Ring plus Shell: 92 mm

	Inertia	Distance	A*Dist²
Shell:	98354.883	84.2596	37242052.000
Ring :	103225000.000	-51.5104	22767200.000
Total:	103323360.000		60009252.000

Available Moment of Inertia, Ring + Shell: 0.163E+09 mm⁴

Required Stress in Ring plus Shell Breq 12.85 MPa
 Required Strain in Ring plus Shell Areq 0.0001280

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6738.0005^2 \cdot 2000.0017(15.0 + 8580.6299/2000.0017)0.000128) / 10.9$$

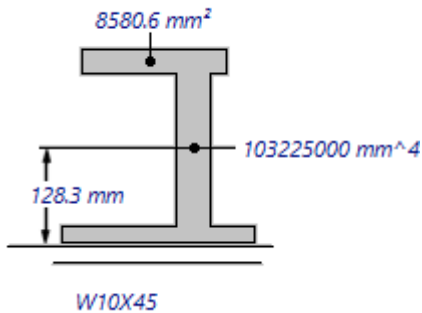
$$= 20569074 \text{ mm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	17.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	98.06	N/mm
The Radial Shear Load	V	6607.16	N
The First Moment of the Area (Ring + Shell)	Q	441991.91	mm³
Weld Shear Flow due to Rad. Shear Load	VQ/I	17.88	N/mm
The Weld Allowable Stress	0.55*S	64.90	MPa
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	0.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		120.00	mm
The Weld Allowable Load	WLeg*0.55*S	1103.19	N/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	99.67	N/mm
MAEP of Stiffening Ring Weld	Weld MAEP	0.54	MPa

Stiffening Ring Calculations for: R15, W10X45, SA-516 70

Material treatment options:



Effective Length of Shell: 350 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	5245.599	7.5000	39341.996
Ring :	8580.630	143.2700	1229346.875
Total:	13826.229		1268688.875

Centroid of Ring plus Shell: 92 mm

	Inertia	Distance	A*Dist ²
Shell:	98354.883	84.2596	37242052.000
Ring :	103225000.000	-51.5104	22767200.000
Total:	103323360.000		60009252.000

Available Moment of Inertia, Ring + Shell: 0.163E+09 mm⁴

Required Stress in Ring plus Shell Breq 12.85 MPa
 Required Strain in Ring plus Shell Areq 0.0001280

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6738.0005^2 \cdot 2000.0007(15.0 + 8580.6299/2000.0007)0.000128) / 10.9$$

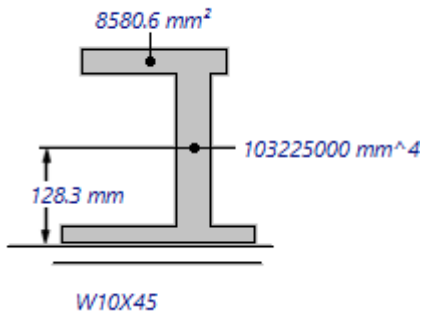
$$= 20569066 \text{ mm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	17.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	98.06	N/mm
The Radial Shear Load	V	6607.15	N
The First Moment of the Area (Ring + Shell)	Q	441991.91	mm ³
Weld Shear Flow due to Rad. Shear Load	VQ/I	17.88	N/mm
The Weld Allowable Stress	0.55*S	64.90	MPa
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	0.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		120.00	mm
The Weld Allowable Load	WLeg*0.55*S	1103.19	N/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	99.67	N/mm
MAEP of Stiffening Ring Weld	Weld MAEP	0.54	MPa

Stiffening Ring Calculations for: R16, W10X45, SA-516 70

Material treatment options:



Effective Length of Shell: 350 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	5245.599	7.5000	39341.996
Ring :	8580.630	143.2700	1229346.875
Total:	13826.229		1268688.875

Centroid of Ring plus Shell: 92 mm

	Inertia	Distance	A*Dist ²
Shell:	98354.883	84.2596	37242052.000
Ring :	103225000.000	-51.5104	22767200.000
Total:	103323360.000		60009252.000

Available Moment of Inertia, Ring + Shell: 0.163E+09 mm⁴

Required Stress in Ring plus Shell Breq 12.85 MPa
 Required Strain in Ring plus Shell Areq 0.0001280

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6738.0005^2 \cdot 1999.9973(15.0 + 8580.6299/1999.9973)0.000128) / 10.9$$

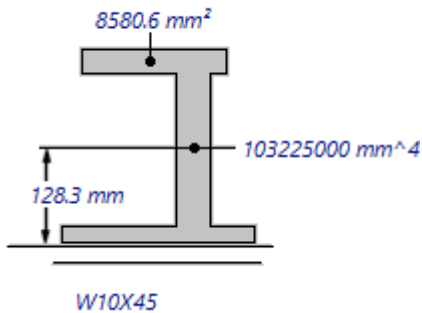
$$= 20569038 \text{ mm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	17.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	98.06	N/mm
The Radial Shear Load	V	6607.14	N
The First Moment of the Area (Ring + Shell)	Q	441991.91	mm ³
Weld Shear Flow due to Rad. Shear Load	VQ/I	17.88	N/mm
The Weld Allowable Stress	0.55*S	64.90	MPa
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	0.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		120.00	mm
The Weld Allowable Load	WLeg*0.55*S	1103.19	N/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	99.67	N/mm
MAEP of Stiffening Ring Weld	Weld MAEP	0.54	MPa

Stiffening Ring Calculations for: R17, W10X45, SA-516 70

Material treatment options:



Effective Length of Shell: 350 mm

	Area (mm ²)	Distance (mm)	Area*Dist
Shell:	5245.599	7.5000	39341.996
Ring :	8580.630	143.2700	1229346.875
Total:	13826.229		1268688.875

Centroid of Ring plus Shell: 92 mm

	Inertia	Distance	A*Dist ²
Shell:	98354.883	84.2596	37242052.000
Ring :	103225000.000	-51.5104	22767200.000
Total:	103323360.000		60009252.000

Available Moment of Inertia, Ring + Shell: 0.163E+09 mm⁴

Required Stress in Ring plus Shell Breq 12.45 MPa
 Required Strain in Ring plus Shell Areq 0.0001240

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6738.0005^2 \cdot 1749.9961(15.0 + 8580.6299/1749.9961)0.000124) / 10.9$$

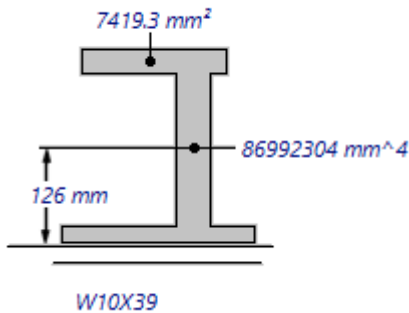
$$= 17989436 \text{ mm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	17.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	85.80	N/mm
The Radial Shear Load	V	5781.25	N
The First Moment of the Area (Ring + Shell)	Q	441991.91	mm ³
Weld Shear Flow due to Rad. Shear Load	VQ/I	15.64	N/mm
The Weld Allowable Stress	0.55*S	64.90	MPa
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	0.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		120.00	mm
The Weld Allowable Load	WLeg*0.55*S	1103.19	N/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	87.22	N/mm
MAEP of Stiffening Ring Weld	Weld MAEP	0.62	MPa

Stiffening Ring Calculations for: R18, W10X39, SA-516 70

Material treatment options:



Effective Length of Shell: 394 mm

	Area (mm²)	Distance (mm)	Area*Dist
Shell:	7478.054	9.5000	71041.508
Ring :	7419.340	144.9840	1075685.625
Total:	14897.394		1146727.125

Centroid of Ring plus Shell: 77 mm

	Inertia	Distance	A*Dist²
Shell:	224964.594	67.4750	34046660.000
Ring :	86992304.000	-68.0090	34316100.000
Total:	87217272.000		68362760.000

Available Moment of Inertia, Ring + Shell: 0.156E+09 mm⁴

Required Stress in Ring plus Shell Breq 10.15 MPa
 Required Strain in Ring plus Shell Areq 0.0001020

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (6738.0005^2 \cdot 1368.3756(19.0 + 7419.3398/1368.3756)0.000102) / 10.9$$

$$= 14197833 \text{ mm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	17.000	mm
Stiffening Ring Attachment Style	INTERMITTENT		
Location of Stiffening Ring	EXTERNAL		
Radial Pressure Load	Pext*Slen	67.09	N/mm
The Radial Shear Load	V	4520.53	N
The First Moment of the Area (Ring + Shell)	Q	504581.75	mm³
Weld Shear Flow due to Rad. Shear Load	VQ/I	14.66	N/mm
The Weld Allowable Stress	0.55*S	64.90	MPa
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	0.00	mm
Max Clear Spacing bet Toes of Adj Weld Segments, 8t		152.00	mm
The Weld Allowable Load	WLeg*0.55*S	1103.19	N/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	68.67	N/mm
MAEP of Stiffening Ring Weld	Weld MAEP	0.79	MPa

MDMT Calculations for: R19, Shell to Ring Junction, min(Curve:B, Curve:C)

Govrn. thk, tg = 20.0, tr = 13.506, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.711, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -23 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Calculations for: R20, Shell to Ring Junction, min(Curve:B, Curve:C)

Govrn. thk, tg = 14.0, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.459, Temp. Reduction = 38 °C
 Attachment governing, Attachment stress = Shell stress per Code



Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B MPa
10	20	14300	...	...	No Calc	No Calc
20	30	14300	...	...	No Calc	No Calc
30	40	14300	...	...	No Calc	No Calc
40	50	14300	...	...	No Calc	No Calc
50	60	14300	...	...	No Calc	No Calc
60	70	14300	...	...	No Calc	No Calc
70	80	No Calc	6819	37	0.00067825	67.7637
80	Ring	1226.19	6762	27	0.0021428	105.031
Ring	90	5100	6762	27	0.00044805	44.7644
90	Ring	5100	6758	25	0.00040204	40.1673
Ring	100	2450	6758	25	0.00087846	81.0731
100	Ring	2450	6706	25	0.0008811	81.1693
Ring	110	2000	6706	25	0.0011041	88.1065
110	Ring	2000	6754	23	0.00096282	83.9684
Ring	120	2000	6754	23	0.00096282	83.9684
120	Ring	2000	6750	21	0.00084485	79.8114
Ring	Ring	2000	6750	21	0.00084485	79.8114
Ring	130	2000	6750	21	0.00084485	79.8114
130	Ring	2000	6746	19	0.00072934	72.8679
Ring	140	2000	6746	19	0.00072934	72.8677
140	Ring	2000	6746	19	0.00072934	72.8677
Ring	150	3000	6746	19	0.00046951	46.908
150	Ring	3000	6742	17	0.00038963	38.9273
Ring	Ring	2000	6742	17	0.00060406	60.3509
Ring	160	2000	6742	17	0.00060406	60.351
160	Ring	2000	6742	17	0.00060406	60.351
Ring	170	2000	6742	17	0.00060406	60.3508
170	Ring	2000	6738	15	0.00050281	50.2354
Ring	Ring	2000	6738	15	0.00050281	50.2355
Ring	180	2000	6738	15	0.00050281	50.2354
180	Ring	2000	6738	15	0.00050281	50.2354
Ring	190	2000	6738	15	0.00050281	50.2354
190	Ring	2000	6738	15	0.00050281	50.2354
Ring	Ring	2000	6738	15	0.00050281	50.2354
Ring	200	1999.99	6738	15	0.00050281	50.2356
200	Ring	1999.99	6738	15	0.00050281	50.2356
Ring	210	1500	6738	15	0.00068726	68.6635
210	Ring	1500	6738	19	0.0010004	85.1496
Ring	Ring	1236.75	6738	19	0.0012443	91.5406
Ring	220	No Calc	6738	19	0.007005	122.384
220	230	884.94	6738	17	0.0012376	91.3904
230	240	902.207	5672.24	13	0.0008392	79.5924
240	250	878.462	4563.24	13	0.00095056	83.5698
250	Ring	No Calc	3454.4	15	0.015294	122.677
Ring	260	3299.2	3454.4	15	0.00040223	40.1869
260	270	No Calc	3450.4	13	0.00052329	52.2815

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure MPa	External M.A.W.P. MPa
10	20	...	No Calc	...	No Calc
20	30	...	No Calc	...	No Calc
30	40	...	No Calc	...	No Calc
40	50	...	No Calc	...	No Calc
50	60	...	No Calc	...	No Calc
60	70	...	No Calc	...	No Calc



70	80	38	14.512	0.049033	0.36769
80	Ring	28	8.89236	0.049033	0.55917
Ring	90	28	15.3431	0.049033	0.23832
90	Ring	26	15.3379	0.049033	0.19812
Ring	100	26	11.5602	0.049033	0.39989
100	Ring	26	11.5157	0.049033	0.40347
Ring	110	26	10.6187	0.049033	0.43795
110	Ring	24	10.6582	0.049033	0.38126
Ring	120	24	10.6582	0.049033	0.38126
120	Ring	22	10.6549	0.049033	0.33107
Ring	Ring	22	10.6549	0.049033	0.33107
Ring	130	22	10.6549	0.049033	0.33107
130	Ring	20	10.6516	0.049033	0.27364
Ring	140	20	10.6516	0.049033	0.27364
140	Ring	20	10.6516	0.049033	0.27364
Ring	150	20	12.5237	0.049033	0.17615
150	Ring	18	12.5197	0.049033	0.13087
Ring	Ring	18	10.6484	0.049033	0.2029
Ring	160	18	10.6484	0.049033	0.2029
160	Ring	18	10.6484	0.049033	0.2029
Ring	170	18	10.6484	0.049033	0.2029
170	Ring	16	10.6451	0.049033	0.14911
Ring	Ring	16	10.645	0.049033	0.14911
Ring	180	16	10.6451	0.049033	0.14911
180	Ring	16	10.6451	0.049033	0.14911
Ring	190	16	10.6451	0.049033	0.14911
190	Ring	16	10.6451	0.049033	0.14911
Ring	Ring	16	10.6451	0.049033	0.14911
Ring	200	16	10.645	0.049033	0.14911
200	Ring	16	10.645	0.049033	0.14911
Ring	210	16	9.59101	0.049033	0.20381
210	Ring	20	9.59096	0.049033	0.32014
Ring	Ring	20	8.90705	0.049033	0.34417
Ring	220	20	No Calc	0.049033	No Calc
220	230	18	8.83629	0.049033	0.26887
230	240	14	8.14833	0.049033	0.21272
240	250	14	7.25469	0.049033	0.27766
250	Ring	16	No Calc	0.049033	No Calc
Ring	260	16	9.06906	0.049033	0.23267
260	270	14	7.15341	0.049033	0.21887
R1					0
R2					0
R3					0
R4					1
R5					1
R6					1
R7					1
R8					0
R9					0
R10					1
R11					1
R12					1
R13					1
R14					1
R15					1
R16					1
R17					1
R18					1

Minimum

0

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required mm ⁴	Ring Inertia Available mm ⁴
10	20	14300	No Calc	No Calc	No Calc
20	30	14300	No Calc	No Calc	No Calc



30	40	14300	No Calc	No Calc	No Calc
40	50	14300	No Calc	No Calc	No Calc
50	60	14300	No Calc	No Calc	No Calc
60	70	14300	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc
80	Ring	1226.19	23247.3	No Calc	No Calc
Ring	90	5100	23248.6	33134616	339015552
90	Ring	5100	19498.4	No Calc	No Calc
Ring	100	2450	19502.2	39345020	374900512
100	Ring	2450	19705.9	No Calc	No Calc
Ring	110	2000	19703.9	22592618	327487040
110	Ring	2000	15933.5	No Calc	No Calc
Ring	120	2000	15933.5	20825078	205572512
120	Ring	2000	12940.2	No Calc	No Calc
Ring	Ring	2000	12940.2	20720058	186720480
Ring	130	2000	12940.2	20859732	348160704
130	Ring	2000	10238.7	No Calc	No Calc
Ring	140	2000	10238.7	20809194	179315616
140	Ring	2000	10238.7	No Calc	No Calc
Ring	150	3000	10235.5	25801040	229672928
150	Ring	3000	7703.47	No Calc	No Calc
Ring	Ring	2000	7705.12	25826942	219824128
Ring	160	2000	7705.12	20760554	182236864
160	Ring	2000	7705.12	No Calc	No Calc
Ring	170	2000	7705.12	20597792	171523552
170	Ring	2000	5725.19	No Calc	No Calc
Ring	Ring	2000	5725.19	20569066	163332544
Ring	180	2000	5725.19	20569066	163332544
180	Ring	2000	5725.19	No Calc	No Calc
Ring	190	2000	5725.19	20569074	163332544
190	Ring	2000	5725.19	No Calc	No Calc
Ring	Ring	2000	5725.19	20569066	163332544
Ring	200	1999.99	5725.19	20569038	163332544
200	Ring	1999.99	5725.19	No Calc	No Calc
Ring	210	1500	5726.93	17989436	163332544
210	Ring	1500	10254.6	No Calc	No Calc
Ring	Ring	1236.75	10253.5	14197833	155579952
Ring	220	No Calc	No Calc	No Calc	No Calc
220	230	884.94	884.94	No Calc	No Calc
230	240	902.207	902.207	No Calc	No Calc
240	250	878.462	878.462	No Calc	No Calc
250	Ring	No Calc	No Calc	No Calc	No Calc
Ring	260	3299.2	14627.7	No Calc	No Calc
260	270	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kgm	Element ID Volume ltr	Corroded Metal Wgt. kgm	Corroded ID Volume ltr	Extra due Misc % kgm
10	20	20901	...	20689.5	...	836.039
20	30	16726.9	...	16237.3	...	669.076
30	40	16726.9	...	16237.3	...	669.076
40	50	15743	...	15253.3	...	629.721
50	60	15743	...	15253.3	...	629.721
60	70	5903.65	...	5740.54	...	236.146
70	80	15580.6	26129.7	15248	26172.6	623.223
80	90	13773.1	105978	13283.2	106041	550.925
90	100	12785.6	105978	12295.6	106041	511.423
100	110	12686.7	104341	12200.6	104404	507.468
110	120	11798.5	105978	11308.6	106041	471.941
120	130	10812.1	105978	10322.1	106041	432.483
130	140	9826.23	105978	9336.29	106041	393.049
140	150	9826.23	105978	9336.29	106041	393.049
150	160	8841.09	105978	8351.16	106041	353.644
160	170	8841.09	105978	8351.16	106041	353.644
170	180	7856.46	105978	7366.53	106041	314.258
180	190	7856.46	105978	7366.53	106041	314.258
190	200	7856.46	105978	7366.53	106041	314.258
200	210	6547.07	88315.1	6138.74	88367.8	261.883
210	220	5398	58149	5128.84	58183.7	215.92
220	230	3643.1	28753.5	3478.13	28774.8	145.724
230	240	2558.91	20399.6	2417.3	20417.9	102.356
240	250	2003.13	12505.6	1892.39	12519.9	80.1252
250	260	4018.63	27602.6	3768.57	27634.9	160.745
260	270	2051.73	5708.22	1937.74	5717.97	82.0692
Total		256305	1537663.12	246305	1538647.62	10252

Weight of Details:

From	Type	Weight of Detail kgm	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Wght	2500	3400	300	...	TAILING LUG WT
10	Wght	5000	...	300	...	BRACING WT
20	Plat	1852.63	2693.09	2700	2693.09	SK PLT1
40	Plat	1851.43	2693.72	700	2693.72	PLAT:[1 OF 1]
50	Plat	1814.48	2693.72	1000	2693.72	SKIRT PLTF-2
70	Liqd	21469.7	...	-458.486	...	Liquid: 50
70	Insl	411.465	...	-458.488	...	PP
70	Lini	1033.92	...	-540.205	...	Clad
70	Nozl	1266.85	...	-1459.57	...	N1 22"
70	Wght	4173	...	...	-4755	N1 PIPE WT
80	Plat	6914.34	0.37480E-04	0.0011308	...	BTL PLTF
80	Liqd	67894.8	...	913.5	...	Liquid: 20
80	Insl	492.92	...	1500	...	PP
80	Lini	1523.19	...	1500	...	Clad
80	Ring	2496.24	...	800.001	...	R1
80	Nozl	5.51209	3453	200	...	K1a 2"
80	Nozl	5.51209	...	200	-3453	K2a 2"
80	Nozl	5.51209	-3453	200	...	K3a 2"
80	Nozl	5.51209	...	200	3453	K4a 2"
80	Nozl	5.51209	3453	2000	...	K1b 2"
80	Nozl	5.51209	...	2000	-3453	K2b 2"
80	Nozl	5.51209	-3453	2000	...	K3b 2"
80	Nozl	5.51209	...	2000	3453	K4b 2"
80	Nozl	5.51209	1726.5	200	2990.39	N10 2"
80	Nozl	35.486	1726.5	700	2990.39	N14 4"
80	Nozl	57.3461	-1726.5	2500	2990.39	N13A 8"
80	Nozl	426.104	2512.35	800	-2512.35	MH1 30"
80	Nozl	5.51209	2990.39	200	1726.5	N12 2"



80	Nozl	57.3461	...	2500	3453
80	Nozl	57.3461	1726.5	2500	2990.39
80	Wght	1713	-2280	2500	3949.08
80	Wght	1713	...	2500	4560
80	Wght	1713	2280	2500	3949.08
90	Plat	3525.67	...	1000	-2694.67
90	Insl	492.63	...	1500	...
90	Lini	1523.19	...	1500	...
90	Ring	2861.59	...	2900	...
90	Nozl	3104.26	-3526.67	1000	1283.6
90	Nozl	442.139	2512.35	2800	-2512.35
90	Nozl	5.50134	...	2800	-3453
90	Nozl	3104.26	-1283.6	1000	-3526.67
90	Wght	4116	...	1000	...
90	Wght	1600	...	1000	...
90	Wght	1600	...	1000	...
100	Plat	3525.67	...	2000	2678.29
100	Pack	7819.21	...	2550	...
100	Insl	488.861	...	1500	...
100	Lini	1511.32	...	1500	...
100	Ring	2476.36	...	2350	...
100	Wght	515	...	200.001	...
100	Wght	1500	...	900.001	...
110	Pack	26473.4	...	1500	...
110	Insl	492.34	...	1500	...
110	Lini	1523.19	...	1500	...
110	Ring	1585.16	...	1350	...
110	Wght	3075	...	1000	...
120	Plat	3522.47	...	...	-2694.67
120	Pack	26473.4	...	1500	...
120	Insl	492.05	...	1500	...
120	Lini	1523.19	...	1500	...
120	Ring	1463.87	...	350	...
120	Ring	2858.33	...	2350	...
130	Plat	3520.87	...	999.999	2694.67
130	Pack	18531.3	...	1050	...
130	Insl	491.76	...	1500	...
130	Lini	1523.19	...	1500	...
130	Ring	1463.03	...	1350	...
130	Nozl	5.46911	...	2600	-3453
130	Wght	1140	...	2100	...
130	Wght	7750	...	2100	...
130	Wght	374	...	2300	...
130	Wght	2910	...	2300	...
140	Plat	3520.87	...	2000	-2694.67
140	Insl	491.76	...	1500	...
140	Lini	1523.19	...	1500	...
140	Ring	1936.89	...	350.003	...
140	Nozl	1043.89	2512.35	100.002	-2512.35
140	Nozl	270.556	-3338.73	100.002	1215.2
140	Nozl	652.313	...	1820	-3553
140	Nozl	533.187	...	1820	3553
140	Nozl	5.46911	-3244.76	2020	1181
140	Nozl	5.46911	-3453	2020	...
140	Nozl	5.46911	-2645.15	2020	2219.55
140	Nozl	5.46911	-2645.15	2850	2219.55
140	Nozl	5.46911	-3453	2850	...
140	Nozl	5.46911	-3244.76	2850	1181
140	Nozl	1026.35	-3338.73	2500	1215.2
140	Nozl	641.84	-1501.56	1500	3220.11
140	Wght	7702	-4420.31	100.002	1608.86
140	Wght	10432	...	1820	-4755
140	Wght	10432	...	1820	4755
140	Wght	3761	...	100.002	...
140	Wght	2250	...	2450	...
140	Wght	5000	...	2450	...
140	Wght	3075	...	2000	...
140	Wght	3000	...	1500	...
140	Wght	515	...	650.002	...
140	Wght	27717	-2084.78	1500	4470.82

N13B 8"
 N13C 8"
 N13A PIPE WT
 N13B PIPE WT
 N13C PIPE WT
 PLTF-4
 PP
 Clad
 R2
 N2A 84"
 MH2 30"
 K9a 2"
 N2B 84"
 SCHOEPENTOETER
 N2A INSERT PLT
 N2B INSERT PLT
 PLTF-5
 Pack Bed#1-(1)
 PP
 Clad
 R3
 SR+BEAM GRID1
 SP GRID-BED1 WT
 Pack Bed#1-(2)
 PP
 Clad
 R4
 CLIP WT-1
 PLTF-6
 Pack Bed#1-(3)
 PP
 Clad
 R5
 R6
 PLTF-7
 Pack Bed#1-(4)
 PP
 Clad
 R7
 K9b 2"
 SR+BEAM BL1
 BL-BED1 WT
 SR+BEAM DIST-1
 DIST-1 DEAD WT
 PLTF-8
 PP
 Clad
 R8
 MH3 30"
 N4 20"
 N3a 24"
 N3B 24"
 K5a 2"
 K6a 2"
 K14a 2"
 K14b 2"
 K6b
 K5b 2"
 MH4 30"
 N15 38"
 N4 PIPE WT
 N3A PIPE WT
 N3B PIPE WT
 N4 INT PIPE WT
 SR+BEAM CT-1
 CT-1 DEAD WT
 CLIP WT-2
 SP GRID-BED2 WT
 SR+BEAM GRID2
 N15 PIPE WT



150	Pack	26473.4	...	1500	...
150	Insl	491.471	...	1500	...
150	Lini	1523.19	...	1500	...
150	Ring	1935.78	...	350.002	...
150	Ring	1582.44	...	2350	...
160	Plat	3519.26	...	999.999	-2694.67
160	Pack	26473.4	...	1500	...
160	Insl	491.471	...	1500	...
160	Lini	1523.19	...	1500	...
160	Ring	1462.2	...	1350	...
170	Plat	3517.65	...	2000	2694.67
170	Pack	26473.3	...	1500	...
170	Insl	491.179	...	1500	...
170	Lini	1523.19	...	1500	...
170	Ring	1461.36	...	350.002	...
170	Ring	1461.36	...	2350	...
170	Wght	1140	...	2550	...
170	Wght	7750	...	2550	...
170	Wght	374	...	2750	...
170	Wght	2910	...	2750	...
170	Wght	3075	...	...	...
180	Insl	491.179	...	1500	...
180	Lini	1523.19	...	1500	...
180	Ring	1461.36	...	1350	...
180	Nozl	5.44761	-3244.76	250.003	1181
180	Nozl	145.302	-1215.2	1950	3338.73
180	Nozl	1042.45	2512.35	1950	-2512.35
180	Nozl	252.039	...	2020	-3553
180	Nozl	252.039	-3076.99	2020	1776.5
180	Nozl	5.44761	3453	0.0023256	...
180	Nozl	5.44761	2990.39	2820	-1726.5
180	Nozl	5.44761	1181	2820	-3244.76
180	Wght	8691	-1581.5	1950	4345.14
180	Wght	12969	...	2020	-4675
180	Wght	12969	...	2020	4675
180	Wght	3041	...	1950	...
180	Wght	400	...	1500	...
190	Plat	3517.65	...	0.0046512	-2694.67
190	Pack	4685.34	...	2735	...
190	Insl	491.179	...	1500	...
190	Lini	1523.19	...	1500	...
190	Ring	1461.36	...	350.004	...
190	Ring	1461.36	...	2350.01	...
190	Nozl	5.44761	3453	750.005	...
190	Nozl	5.44761	2990.39	750.005	-1726.5
190	Nozl	5.44761	1181	750.005	-3244.76
190	Nozl	831.744	2512.35	1170.01	-2512.35
190	Nozl	5.44761	-1726.5	1170.01	-2990.39
190	Wght	2250	...	285.006	...
190	Wght	5000	...	285.006	...
190	Wght	515	...	1570	...
190	Wght	1200	...	2270.01	...
200	Plat	3517.65	...	999.999	2694.67
200	Pack	21796.4	...	1235	...
200	Insl	409.316	...	1250	...
200	Lini	1269.32	...	1250	...
200	Ring	1461.36	...	1350	...
200	Wght	1150	...	2470	...
200	Wght	7750	...	2470	...
200	Wght	3075	...	999.999	...
210	Insl	270.149	...	825	...
210	Lini	836.749	...	825	...
210	Ring	1262.76	...	349.997	...
210	Ring	503.145	...	1586.75	...
210	Nozl	199.361	-1774.5	999.999	3073.52
210	Nozl	397.424	2509.52	1320	-2509.52
210	Nozl	5.46911	...	1320	-3449
210	Nozl	5.46911	...	1320	-3449
210	Nozl	5.46911	-1724.5	1320	-2986.92
210	Nozl	5.46911	-2986.92	1320	-1724.5

Pack Bed#2-(2)
 PP
 Clad
 R9
 R10
 PLTF-10
 Pack Bed#2-(3)
 PP
 Clad
 R11
 PLTF-11
 Pack Bed#2-(4)
 PP
 Clad
 R12
 R13
 SR+BEAM BL2
 BL-BED2 WT
 SR+BEAM DIST-2
 DIST-2 DEAD WT
 CLIP WT-3
 PP
 Clad
 R14
 K10B
 N5 14"
 MH5 30"
 N6a 18"
 N6b 18"
 K7a
 K8a
 K15a
 N5 PIPE WT
 N6A PIPE WT
 N6B PIPE WT
 N5 INT PIPE WT
 MANWAY DAVIT
 PLTF-12
 Pack Bed#3-(1)
 PP
 Clad
 R15
 R16
 K7b
 K8b
 K15b
 MH6 30"
 K11a 2"
 SR+BEAM CT-2
 CT-2 DEAD WT
 SR+BEAM GRID3
 SP GRID-BED3 WT
 PLTF-13
 Pack Bed#3-(2)
 PP
 Clad
 R17
 SR+BEAM BL3
 BL-BED3 WT
 CLIP WT-4
 PP
 Clad
 R18
 R19
 N7 16"
 MH7
 K16a
 K16b
 K11b
 K12



210	Nozl	5.46911	-3241	1320	1179.63
210	Wght	1500	...	999.999	...
210	Wght	374	...	170.001	...
210	Wght	2910	...	170.001	...
210	Wght	2250	...	170.001	...
210	Wght	10731	-2326.5	999.999	4029.62
210	Wght	3041	...	999.999	...
220	Plat	3532.08	...	300	-2691.78
220	Insl	202.522	...	480.5	...
220	Wght	1150	...	169.999	...
220	Wght	5000	3400	169.999	...
230	Insl	183.06	...	500	...
240	Insl	143.632	...	500	...
250	Plat	2220.23	...	107.993	1667.84
250	Plat	2220.23	...	2257.99	-1667.84
250	Insl	253.229	...	1500	...
250	Ring	315.694	...	36	...
250	Nozl	20.0669	496.091	199.992	1851.44
250	Nozl	19.0648	-812.737	199.992	1742.92
250	Nozl	441.233	1298.39	1200	-1298.39
250	Wght	40354	2272.64	1200	-2272.64
250	Wght	40354	-2272.64	1200	-2272.64
260	Insl	147.674	...	452.8	...
100	Pliq	4692.48	...	2550.000	...
110	Pliq	15887.01	...	1500.000	...
120	Pliq	15887.01	...	1500.000	...
130	Pliq	11120.91	...	1050.000	...
150	Pliq	15887.01	...	1500.000	...
160	Pliq	15887.01	...	1500.000	...
170	Pliq	15887.00	...	1499.999	...
190	Pliq	2806.71	...	2735.003	...
200	Pliq	13080.31	...	1235.000	...

K13b
 DEMISTER
 SR+BEAM DIST-3
 DIST-3 DEAD WT
 SR+BEAM DIST-3
 N7 PIPE WT
 N7 INT PIPE WT
 PLTF-14
 Ins
 SR+BEAM MIST
 TRUNION WT
 Ins
 Ins
 PLTF-15
 PLTF-16
 Ins
 R20
 P1(frm V-1403)
 P2(frm ME-1702)
 P3 (from C-1402)
 P3 Fy weight
 P4 FY weight
 Ins

Total Weight of Each Detail Type:

Platforms	52093.2
Packing	185199.1
Packing Liquid	111135.5
Liquid	89364.4
Insulation	7919.8
Lining	20569.6
Cladding & Overlay	836.7
Stiffeners	32971.7
Nozzles	16489.8
Weights	297199.0

Sum of the Detail Weights 813778.9 kgm

Weight Summation Results: (kgm)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	266557.8	266557.8	266557.8	266557.8	266557.8	266557.8
Cladding	836.7	836.7	836.7	836.7	836.7	836.7
Stif. Rings	32971.7	32971.7	32971.7	32971.7	32971.7	32971.7
Nozzles	16489.8	16489.8	16489.8	16489.8	16489.8	16489.8
Wld Weights	52240.0	52240.0	52240.0	52240.0	52240.0	52240.0
Platforms	...	...	...	52093.2	52093.2	52093.2
Packing	...	...	...	...	185199.1	185199.1
Insulation	...	...	7919.8	7919.8	7919.8	7919.8
Lining	20569.6	20569.6	20569.6	20569.6	20569.6	20569.6
Empty Weights	...	...	...	1500.0	1500.0	...
Test Weights	...	1500.0	...	...	...	...
Ope Weights	...	...	...	...	...	244959.0
Ope. Liquid	...	...	...	...	...	89364.4
Packing Liq	...	...	...	...	...	111135.5
Test Liquid	...	1534062.0	...	...	...	...
Totals	389665.7	1925227.9	397585.6	451178.7	636377.8	1080336.6



Field Installation Options:

Platforms installed before lifting.

* Packing installed after lifting.

Miscellaneous Weight Percent input values:

Saddles:	0.0%	Platforms:	0.0%	Rings:	0.0%	Nozzles:	0.0%
Trays:	0.0%	Legs:	0.0%	Clips:	0.0%	Sup Lugs:	0.0%
1/2 Pipes:	0.0%	Tubesheets:	0.0%	Tubes:	0.0%	Floating Heads:	0.0%
Jackets:	0.0%	Shells etc:	4.0%				

Note that the above values for the miscellaneous weight percents have been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	389665.7 kgm
Shop Test Wt.	- Fabricated weight + water (full)	1925227.9 kgm
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	397585.6 kgm
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	451178.7 kgm
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	636377.8 kgm
Operating Wt.	- Empty weight + operating liq. uncorroded	1080336.6 kgm
Field Test Wt.	- Empty weight + test medium (full)	1985240.6 kgm

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	376147.8 kgm
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	780228.0 kgm
Mass of the Upper 1/3 of the Vertical Vessel (empty)	188657.9 kgm

Surface Areas of Elements:

From	To	Outside Surface Area mm ²	Inside Surface Area mm ²
10	20	27734924	27277504
20	30	63796320	63155440
30	40	63796320	63155440
40	50	63777472	63174288
50	60	63777472	63174288
60	70	21271724	21045530
70	80	44508196	42899652
80	90	63730344	63202556
90	100	63692648	63202556
100	110	63202556	62712472
110	120	63654952	63202556
120	130	63617244	63202556
130	140	63579548	63202556
140	150	63579548	63202556
150	160	63541856	63202556
160	170	63541856	63202556
170	180	63504160	63202556
180	190	63504160	63202556
190	200	63504160	63202556
200	210	52920132	52668800
210	220	34927288	34719944
220	230	21452614	21278930
230	240	18416226	18268378
240	250	14432268	14284445
250	260	32556954	32255360
260	270	13508284	13234116

Total	1299529344.0	1288532352.0 mm ²
Total	1299.5	1288.5 m ²

Element and Detail Weights:



From	To	Total Ele. Empty Wgt. kgm	Total. Ele. Oper. Wgt. kgm	Total. Ele. Hydro. Wgt. kgm	Total Dtl. Offset Mom. N-mm	Oper. Wgt. No Liquid kgm
10	20	29237	29237	29237	83384768	29237
20	30	17396	19248.6	19248.6	48944936	19248.6
30	40	17396	17396	17396	...	17396
40	50	16372.7	18224.2	18224.2	48924616	18224.2
50	60	16372.7	18187.2	18187.2	47948248	18187.2
60	70	6139.79	6139.79	6139.79	...	6139.79
70	80	18916	44558.7	45029.8	...	23089
80	90	19525.2	99473.3	132353	11357876	31578.5
90	100	28030.6	35672.2	137470	135588656	35672.2
100	110	18185.7	35723.1	125989	...	31030.6
110	120	18946.2	61306.5	124860	...	45419.5
120	130	17582	63464.8	127018	...	47577.8
130	140	15216.7	59049.9	124651	...	47928.9
140	150	24212.1	95776.9	133646	928046272	95776.9
150	160	14727.6	57088	120641	...	41201
160	170	12671.6	58551.2	122104	...	42664.2
170	180	17696.8	74234.8	127128	...	58347.8
180	190	16801.1	51430.1	122714	118250016	51430.1
190	200	16726.3	33936	126157	20814418	31129.3
200	210	14174	60318.3	105953	...	47238
210	220	16275.8	29916.9	74389.3	239025712	29916.9
220	230	10141.3	13673.4	42409.4	166769536	13673.4
230	240	2844.33	2844.33	23231.5	...	2844.33
240	250	2226.89	2226.89	14724.8	...	2226.89
250	260	5228.66	90377.1	37254.9	5565747	90377.1
260	270	2281.47	2281.47	7986.21	...	2281.47

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kgm	Cumulative Oper. Wgt. kgm	Cumulative Hydro. Wgt. kgm
10	20	879837	1080336	1984142
20	30	850600	1051099	1954905
30	40	831351	1031851	1935656
40	50	813955	1014455	1918260
50	60	795731	996231	1900036
60	70	777544	978044	1881849
70	80	771404	971904	1875709
80	90	748315	927345	1830679
90	100	716736	827872	1698326
100	110	681064	792200	1560857
110	120	650034	756477	1434868
120	130	604614	695170	1310009
130	140	557036	631705	1182991
140	150	509107	572656	1058340
150	160	413331	476878	924694
160	170	372130	419791	804053
170	180	329465	361239	681948
180	190	271118	287004	554821
190	200	219687	235574	432106
200	210	188558	201638	305949
210	220	141320	141320	199996
220	230	111403	111403	125607
230	240	97729.8	97729.8	83197.5
240	250	94885.5	94885.5	59966
250	260	92658.6	92658.6	45241.1
260	270	2281.47	2281.47	7986.21

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. N-mm	Cumulative Oper. Mom. N-mm	Cumulative Hydro. Mom. N-mm
10	20	1854620800	1854620800	1854620800
20	30	1771235968	1771235968	1771235968
30	40	1722290944	1722290944	1722290944
40	50	1722290944	1722290944	1722290944
50	60	1673366400	1673366400	1673366400
60	70	1625418240	1625418240	1625418240
70	80	1625418240	1625418240	1625418240
80	90	1625418240	1625418240	1625418240
90	100	1614060288	1614060288	1614060288
100	110	1478471680	1478471680	1478471680
110	120	1478471680	1478471680	1478471680
120	130	1478471680	1478471680	1478471680
130	140	1478471680	1478471680	1478471680
140	150	1478471680	1478471680	1478471680
150	160	550425472	550425472	550425472
160	170	550425472	550425472	550425472
170	180	550425472	550425472	550425472
180	190	550425472	550425472	550425472
190	200	432175424	432175424	432175424
200	210	411361024	411361024	411361024
210	220	411361024	411361024	411361024
220	230	172335296	172335296	172335296
230	240	5565747	5565747	5565747
240	250	5565747	5565747	5565747
250	260	5565747	5565747	5565747
260	270	...	...	...

Nozzle Flange MAWP Results: (MPa & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N1 22"	1.54	1.96	160	150	GR 1.1	...	...	...	...
K1a 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K2a 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K3a 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K4a 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K1b 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K2b 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K3b 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K4b 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
N10 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
N14 4"	4.48	5.10	160	300	GR 1.1	4.321	2.405	4.484	2.405
N13A 8"	4.48	5.10	160	300	GR 1.1	2.936	3.791	4.484	3.791
MH1 30"	1.54	1.96	160	150	GR 1.1	...	...	...	...
N12 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
N13B 8"	4.48	5.10	160	300	GR 1.1	2.936	3.791	4.484	3.791
N13C 8"	4.48	5.10	160	300	GR 1.1	2.936	3.791	4.484	3.791
N2A 84"	1.54	1.96	160	150	GR 1.1	...	...	...	...
MH2 30"	1.54	1.96	160	150	GR 1.1	...	...	...	...
K9a 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
N2B 84"	1.54	1.96	160	150	GR 1.1	...	...	...	...
K9b 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
MH3 30"	1.54	1.96	160	150	GR 1.1	...	...	...	...
N4 20"	1.54	1.96	160	150	GR 1.1	1.658	1.540	1.540	0.652
N3a 24"	1.54	1.96	160	150	GR 1.1	1.411	1.540	1.540	0.899
N3B 24"	1.54	1.96	160	150	GR 1.1	1.411	1.540	1.540	0.899
K5a 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K6a 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K14a 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K14b 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
K6b	4.48	5.10	160	300	GR 1.1	...	...	...	...
K5b 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
MH4 30"	1.54	1.96	160	150	GR 1.1	...	...	...	...
N15 38"	1.54	1.96	160	150	GR 1.1	0.754	1.215	1.540	1.540
K10B	4.48	5.10	160	300	GR 1.1	...	...	...	...
N5 14"	1.54	1.96	160	150	GR 1.1	2.319	1.070	1.540	-0.008
MH5 30"	1.54	1.96	160	150	GR 1.1	...	...	...	...
N6a 18"	1.54	1.96	160	150	GR 1.1	1.796	1.540	1.540	0.515
N6b 18"	1.54	1.96	160	150	GR 1.1	1.796	1.540	1.540	0.515
K7a	4.48	5.10	160	300	GR 1.1	...	...	...	...
K8a	4.48	5.10	160	300	GR 1.1	...	...	...	...
K15a	4.48	5.10	160	300	GR 1.1	...	...	...	...
K7b	4.48	5.10	160	300	GR 1.1	...	...	...	...
K8b	4.48	5.10	160	300	GR 1.1	...	...	...	...
K15b	4.48	5.10	160	300	GR 1.1	...	...	...	...
MH6 30"	1.54	1.96	160	150	GR 1.1	...	...	...	...
K11a 2"	4.48	5.10	160	300	GR 1.1	...	...	...	...
N7 16"	1.54	1.96	160	150	GR 1.1	2.050	1.338	1.540	0.260
MH7	1.54	1.96	160	150	GR 1.1	...	...	...	...
K16a	4.48	5.10	160	300	GR 1.1	...	...	...	...
K16b	4.48	5.10	160	300	GR 1.1	...	...	...	...
K11b	4.48	5.10	160	300	GR 1.1	...	...	...	...
K12	4.48	5.10	160	300	GR 1.1	...	...	...	...
K13b	4.48	5.10	160	300	GR 1.1	...	...	...	...
P1(frm V-1403)	1.54	1.96	160	150	GR 1.1	...	...	...	...
P2(frm ME-1702	1.54	1.96	160	150	GR 1.1	...	...	...	...
P3 (from C-140	1.54	1.96	160	150	GR 1.1	1.184	1.280	1.540	1.126
Min Rating	1.070	1.960, [for Core Elements]					...	...	-0.008

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum



allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME N-mm	Axial Force FE N	MAWP PD MPa	Maximum Pressure Pmax MPa	Pass Fail
N14 4"	138.176	0.50	1979899	7500	0.474	2.405	Pass
N13A 8"	248.539	0.50	7919597	15000	0.474	3.791	Pass
N13B 8"	248.539	0.50	7919597	15000	0.474	3.791	Pass
N13C 8"	248.539	0.50	7919597	15000	0.474	3.791	Pass
N4 20"	551.688	1.20	49497480	37500	0.474	1.540	Pass
N3a 24"	657.225	1.20	71276336	45000	0.474	1.540	Pass
N3B 24"	657.225	1.20	71276336	45000	0.474	1.540	Pass
N15 38"	992.124	0.28	126980240	71250	0.474	1.215	Pass
N5 14"	389.001	1.20	24253776	26250	0.474	1.070	Pass
N6a 18"	500.888	1.20	40093044	33750	0.474	1.540	Pass
N6b 18"	500.888	1.20	40093044	33750	0.474	1.540	Pass
N7 16"	442.976	1.20	31678346	30000	0.474	1.338	Pass
P3 (from C-1402)	843.026	0.60	126713584	60000	0.474	1.280	Pass

Evaluating Flange on Nozzle [N14 4"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(1979899) + 4 \cdot \text{abs}(7500) \cdot 138.176 \leq \pi \cdot 138.176^3 \cdot ((4.484 - 0.474) + .500 \cdot 4.484)$$

$$= 0.35825E+08 \leq 0.51838E+08 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N14 4" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.484, 0.500 \cdot 4.484 + 4.484 - 35825308.0 / (\pi \cdot 138.176^3))$$

$$= \min(4.484, 2.405)$$

$$= 2.405 \text{ MPa}$$

Evaluating Flange on Nozzle [N13A 8"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(7919597) + 4 \cdot \text{abs}(15000) \cdot 248.539 \leq \pi \cdot 248.539^3 \cdot ((4.484 - 0.474) + .500 \cdot 4.484)$$

$$= 0.14163E+09 \leq 0.30167E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N13A 8" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.484, 0.500 \cdot 4.484 + 4.484 - 141631872.0 / (\pi \cdot 248.539^3))$$

$$= \min(4.484, 3.791)$$

$$= 3.791 \text{ MPa}$$

Evaluating Flange on Nozzle [N13B 8"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(7919597) + 4 \cdot \text{abs}(15000) \cdot 248.539 \leq \pi \cdot 248.539^3 \cdot ((4.484 - 0.474) + .500 \cdot 4.484)$$

$$= 0.14163E+09 \leq 0.30167E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N13B 8" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.484, 0.500 \cdot 4.484 + 4.484 - 141631872.0 / (\pi \cdot 248.539^3))$$

$$= \min(4.484, 3.791)$$

$$= 3.791 \text{ MPa}$$

Evaluating Flange on Nozzle [N13C 8"]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 \cdot ((PR - PD) + FM \cdot PR)$$

$$= 16(7919597) + 4 \cdot \text{abs}(15000) \cdot 248.539 \leq \pi \cdot 248.539^3 \cdot ((4.484 - 0.474) + .500 \cdot 4.484)$$

$$= 0.14163E+09 \leq 0.30167E+09 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N13C 8" [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(4.484, 0.500 \cdot 4.484 + 4.484 - 141631872.0 / (\pi \cdot 248.539^3))$$

$$= \min(4.484, 3.791)$$

$$= 3.791 \text{ MPa}$$



Evaluating Flange on Nozzle [N4 20"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(49497480) + 4 \cdot \text{abs}(37500) 551.688 \leq \pi \cdot 551.688^3 [(1.540 - 0.474) + 1.200 \cdot 1.540] \\ &= 0.87475E+09 \leq 0.15379E+10 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N4 20" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(1.540, 1.200 \cdot 1.540 + 1.540 - 874746368.0 / (\pi \cdot 551.688^3)) \\ &= \min(1.540, 1.731) \\ &= 1.540 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N3a 24"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(71276336) + 4 \cdot \text{abs}(45000) 657.225 \leq \pi \cdot 657.225^3 [(1.540 - 0.474) + 1.200 \cdot 1.540] \\ &= 0.12588E+10 \leq 0.26001E+10 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N3a 24" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(1.540, 1.200 \cdot 1.540 + 1.540 - 1258769664.0 / (\pi \cdot 657.225^3)) \\ &= \min(1.540, 1.977) \\ &= 1.540 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N3B 24"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(71276336) + 4 \cdot \text{abs}(45000) 657.225 \leq \pi \cdot 657.225^3 [(1.540 - 0.474) + 1.200 \cdot 1.540] \\ &= 0.12588E+10 \leq 0.26001E+10 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N3B 24" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(1.540, 1.200 \cdot 1.540 + 1.540 - 1258769664.0 / (\pi \cdot 657.225^3)) \\ &= \min(1.540, 1.977) \\ &= 1.540 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N15 38"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(126980240) + 4 \cdot \text{abs}(71250) 992.124 \leq \pi \cdot 992.124^3 [(1.540 - 0.474) + 1.278 \cdot 1.540] \\ &= 0.23146E+10 \leq 0.45888E+10 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N15 38" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(1.540, 0.278 \cdot 1.540 + 1.540 - 2314553088.0 / (\pi \cdot 992.124^3)) \\ &= \min(1.540, 1.215) \\ &= 1.215 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N5 14"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(24253776) + 4 \cdot \text{abs}(26250) 389.001 \leq \pi \cdot 389.001^3 [(1.540 - 0.474) + 1.200 \cdot 1.540] \\ &= 0.42892E+09 \leq 0.53914E+09 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N5 14" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(1.540, 1.200 \cdot 1.540 + 1.540 - 428922016.0 / (\pi \cdot 389.001^3)) \\ &= \min(1.540, 1.070) \\ &= 1.070 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N6a 18"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(40093044) + 4 \cdot \text{abs}(33750) 500.888 \leq \pi \cdot 500.888^3 [(1.540 - 0.474) + 1.200 \cdot 1.540] \\ &= 0.70914E+09 \leq 0.11510E+10 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N6a 18" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(1.540, 1.200 \cdot 1.540 + 1.540 - 709136000.0 / (\pi \cdot 500.888^3)) \\ &= \min(1.540, 1.593) \\ &= 1.540 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N6b 18"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(40093044) + 4 \cdot \text{abs}(33750) 500.888 \leq \pi \cdot 500.888^3 [(1.540 - 0.474) + 1.200 \cdot 1.540] \end{aligned}$$



= 0.70914E+09 ≤ 0.11510E+10 Check Passes

Maximum Pressure for Flange on Nozzle N6b 18" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(1.540, 1.200 \cdot 1.540 + 1.540 - 709136000.0 / (\pi \cdot 500.888^3)) \\ &= \min(1.540, 1.593) \\ &= 1.540 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [N7 16"]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(31678346) + 4 \cdot \text{abs}(30000)442.976 \leq \pi \cdot 442.976^3 [(1.540 - 0.474) + 1.200 \cdot 1.540] \\ &= 0.56003E+09 \leq 0.79614E+09 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle N7 16" [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(1.540, 1.200 \cdot 1.540 + 1.540 - 560032128.0 / (\pi \cdot 442.976^3)) \\ &= \min(1.540, 1.338) \\ &= 1.338 \text{ MPa} \end{aligned}$$

Evaluating Flange on Nozzle [P3 (from C-1402)]:

$$\begin{aligned} &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\ &= 16(126713584) + 4 \cdot \text{abs}(60000)843.026 \leq \pi \cdot 843.026^3 [(1.540 - 0.474) + 1.200 \cdot 1.540] \\ &= 0.22298E+10 \leq 0.37476E+10 \text{ Check Passes} \end{aligned}$$

Maximum Pressure for Flange on Nozzle P3 (from C-1402) [Pmax]:

$$\begin{aligned} &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\ &= \min(1.540, 0.600 \cdot 1.540 + 1.540 - 2229825536.0 / (\pi \cdot 843.026^3)) \\ &= \min(1.540, 1.280) \\ &= 1.280 \text{ MPa} \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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The Natural Frequencies for the vessel have been computed iteratively by solving a system of matrices. These matrices describe the mass and the stiffness of the vessel. This is the generalized eigenvalue/eigenvector problem and is referenced in some mathematical texts.

The Natural Frequency for the Vessel (Empty.) is 0.92778 Hz.

The Natural Frequency for the Vessel (Ope...) is 0.88361 Hz.

The Natural Frequency for the Vessel (Filled) is 0.67734 Hz.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	300 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Dynamic Gust-Effect Factor, Operating Case [Gf]:

$$= 0.925[(1+1.7 \cdot I_{zbar} \cdot \sqrt{(gQ^2 + gR^2 \cdot R^2)}) / (1+1.7 \cdot gV \cdot I_{zbar})]$$

$$= 0.925[(1+1.7 \cdot 0.161 \cdot \sqrt{(3.4^2 \cdot 0.874^2 + 4.16^2 \cdot 0.378^2)}) / (1+1.7 \cdot 3.4 \cdot 0.161)]$$

$$= 0.920$$

Natural Frequency of Vessel (Operating)	0.884 Hz
Natural Frequency of Vessel (Empty)	0.928 Hz
Natural Frequency of Vessel (Test)	0.677 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.614
Structure Height to Diameter ratio	9.446
Height to top of Structure	61420.598 mm

This is classified as a flexible structure. Dynamic analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Zg = 274320.0$ mm

Effective Height $[z]$:
 $=$ Centroid Height + Vessel Base Elevation
 $= 650.0 + 300.0 = 950.0$ mm
 $= 3.117$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (3.117 ft.) $<$ 15 ft.
 $= 2.01(15 / Zg)^{2/\alpha}$
 $= 2.01(15/900.0)^{2/9.5}$
 $= 0.849$

Type of Hill: No Hill

Wind Directionality Factor $[Kd]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[Kzt]$:



K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0.0 \cdot 0.0 \cdot 0.0)^2$$

$$= 1.0$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 0.849 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$$

$$= 16.0 \text{ psf [0.766] kPa}$$

Force on the first element [F]:

$$= qz \cdot Gf \cdot Cf \cdot \text{WindArea}$$

$$= 16.0 \cdot 0.92 \cdot 0.614 \cdot 131.137$$

$$= 1184.4 \text{ lbs. [5268.1] N}$$

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz kPa
Skirt-1	950.0	0.000	0.000	0.000	0.849	1.000	0.766
Skirt-2	3100.0	0.000	0.000	0.000	0.849	1.000	0.766
Skirt-3	6100.0	0.000	0.000	0.000	0.902	1.000	0.766
Skirt-4	9100.0	0.000	0.000	0.000	0.981	1.000	0.766
Skirt-5	12100.0	0.000	0.000	0.000	1.042	1.000	0.766
Skirt-6	14100.0	0.000	0.000	0.000	1.076	1.000	0.766
Btm Dish End	14625.0	0.000	0.000	0.000	1.084	1.000	0.766
Shell-1	16150.0	0.000	0.000	0.000	1.107	1.000	0.766
Shell-2	19150.0	0.000	0.000	0.000	1.148	1.000	0.766
Shell-3	22150.0	0.000	0.000	0.000	1.183	1.000	0.766
Shell-4	25150.0	0.000	0.000	0.000	1.215	1.000	0.766
Shell-5	28150.0	0.000	0.000	0.000	1.245	1.000	0.766
Shell-6	31150.0	0.000	0.000	0.000	1.271	1.000	0.766
Shell-7	34150.0	0.000	0.000	0.000	1.296	1.000	0.766
Shell-9	37150.0	0.000	0.000	0.000	1.319	1.000	0.766
Shell-10	40150.0	0.000	0.000	0.000	1.341	1.000	0.766
Shell-11	43150.0	0.000	0.000	0.000	1.362	1.000	0.766
Shell-12	46150.0	0.000	0.000	0.000	1.381	1.000	0.766
Shell-13	49150.0	0.000	0.000	0.000	1.400	1.000	0.766
Shell-14	51900.0	0.000	0.000	0.000	1.416	1.000	0.766
Shell-15	53975.0	0.000	0.000	0.000	1.427	1.000	0.766
Cone-1	55266.7	0.000	0.000	0.000	1.435	1.000	0.766
Cone-2	56242.8	0.000	0.000	0.000	1.440	1.000	0.766
Cone-3	57237.8	0.000	0.000	0.000	1.445	1.000	0.766
Shell-16	59261.0	0.000	0.000	0.000	1.456	1.000	0.766
Top head	61170.2	0.000	0.000	0.000	1.466	1.000	0.766

Wind Vibration Calculations

This evaluation is based on work by Kanti Mahajan and Ed Zorilla

Nomenclature

- Cf - Correction factor for natural frequency
- D - Average outer insulated diameter of vessel, mm
- Df - Damping Factor < 0.75 Unstable, > 0.95 Stable
- Dr - Average outer insulated of top half of vessel, mm
- f - Natural frequency of vibration (Hertz)
- f1 - Natural frequency of bare vessel based on a unit value of $(D/L^2)(104)$
- L - Total height of structure mm
- Lc - Total length of conical section(s) of vessel, mm
- tb - Uncorroded plate thickness at bottom of vessel, mm
- V30 - Design Wind Speed provided by user, km/hr
- Vc - Critical wind velocity, km/hr
- Vw - Maximum wind speed at top of structure, km/hr
- W - Total corroded weight of structure, N
- Ws - Cor. vessel weight excl. weight of parts which do not effect stiff. N
- Z - Maximum amplitude of vibration at top of vessel, mm
- Dl - Logarithmic decrement (taken as 0.03 for Welded Structures)



Vp - Vib. Chance, ≤ 0.000003 (High); $0.000003 < 0.000004$ (Probable)
 P30 - wind pressure 30 feet above the base

Check other Conditions and Basic Assumptions:

:: Total Cone Length / Total Length < 0.5

:: $2961.0/60510.996 = 0.049$

:: $[D / L^2] \cdot 10^4 < 8$ (English Units, ft.)

:: $[21.54/198.53^2] \cdot 10^4 = 5.464$

Compute the vibration possibility. If Vp > 0.000004 no chance. [Vp]:

= $W / (L \cdot Dr^2)$

= $10491797 / (60511.0 \cdot 6302.141^2)$

= 0.43656E-05

Since Vp is > 0.000004 no further vibration analysis is required !

Platform Load Calculations

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N	Cf
SK PLT1	157981.50	4300.00	0.77	121.02	1.20
PLAT:[1 OF 1]	157981.50	8300.00	0.77	121.02	1.20
SKIRT PLTF-2	568486.06	11600.00	0.77	435.49	1.20
BTL PLTF	2368490.00	14650.00	0.77	1814.37	1.20
PLTF-4	1957980.00	18650.00	0.77	1499.90	1.20
PLTF-5	1957981.50	22650.00	0.77	1499.90	1.20
PLTF-6	1957980.00	26650.00	0.77	1499.90	1.20
PLTF-7	1957980.00	30650.00	0.77	1499.90	1.20
PLTF-8	1957980.00	34650.00	0.77	1499.90	1.20
PLTF-10	1957981.50	39650.00	0.77	1499.90	1.20
PLTF-11	1957980.00	43650.00	0.77	1499.90	1.20
PLTF-12	1957980.00	47650.00	0.77	1499.90	1.20
PLTF-13	1957980.00	51649.99	0.77	1499.90	1.20
PLTF-14	1957980.00	55100.00	0.77	1499.90	1.20
PLTF-15	1752730.00	57868.99	0.77	1342.67	1.20
PLTF-16	1752730.00	60018.98	0.77	1342.67	1.20

Wind Loads on Masses/Equipment/Piping

ID	Wind Area mm ²	Elevation mm	Pressure kPa	Force N
TAILING LUG W	0.00	600.00	0.77	0.00
BRACING WT	0.00	600.00	0.77	0.00
N1 PIPE WT	0.00	14600.00	0.77	0.00
N13A PIPE WT	0.00	17150.00	0.77	0.00
N13B PIPE WT	0.00	17150.00	0.77	0.00
N13C PIPE WT	0.00	17150.00	0.77	0.00
SCHOEPENTOETE	0.00	18650.00	0.77	0.00
N2A INSERT PL	0.00	18650.00	0.77	0.00
N2B INSERT PL	0.00	18650.00	0.77	0.00
SR+BEAM GRID1	0.00	20850.00	0.77	0.00
SP GRID-BED1	0.00	21550.00	0.77	0.00
CLIP WT-1	0.00	24650.00	0.77	0.00
SR+BEAM BL1	0.00	31750.00	0.77	0.00
BL-BED1 WT	0.00	31750.00	0.77	0.00
SR+BEAM DIST-	0.00	31950.00	0.77	0.00
DIST-1 DEAD W	0.00	31950.00	0.77	0.00
N4 PIPE WT	0.00	32750.00	0.77	0.00
N3A PIPE WT	0.00	34470.00	0.77	0.00
N3B PIPE WT	0.00	34470.00	0.77	0.00
N4 INT PIPE W	0.00	32750.00	0.77	0.00
SR+BEAM CT-1	0.00	35100.00	0.77	0.00
CT-1 DEAD WT	0.00	35100.00	0.77	0.00
CLIP WT-2	0.00	34650.00	0.77	0.00
SP GRID-BED2	0.00	34150.00	0.77	0.00



SR+BEAM GRID2	0.00	33300.00	0.77	0.00
N15 PIPE WT	0.00	34150.00	0.77	0.00
SR+BEAM BL2	0.00	44200.00	0.77	0.00
BL-BED2 WT	0.00	44200.00	0.77	0.00
SR+BEAM DIST-	0.00	44400.00	0.77	0.00
DIST-2 DEAD W	0.00	44400.00	0.77	0.00
CLIP WT-3	0.00	41650.00	0.77	0.00
N5 PIPE WT	0.00	46600.00	0.77	0.00
N6A PIPE WT	0.00	46670.00	0.77	0.00
N6B PIPE WT	0.00	46670.00	0.77	0.00
N5 INT PIPE W	0.00	46600.00	0.77	0.00
MANWAY DAVIT	0.00	46150.00	0.77	0.00
SR+BEAM CT-2	0.00	47935.00	0.77	0.00
CT-2 DEAD WT	0.00	47935.00	0.77	0.00
SR+BEAM GRID3	0.00	49220.00	0.77	0.00
SP GRID-BED3	0.00	49920.00	0.77	0.00
SR+BEAM BL3	0.00	53119.99	0.77	0.00
BL-BED3 WT	0.00	53119.99	0.77	0.00
CLIP WT-4	0.00	51649.99	0.77	0.00
DEMISTER	0.00	54150.00	0.77	0.00
SR+BEAM DIST-	0.00	53320.00	0.77	0.00
DIST-3 DEAD W	0.00	53320.00	0.77	0.00
SR+BEAM DIST-	0.00	53320.00	0.77	0.00
N7 PIPE WT	0.00	54150.00	0.77	0.00
N7 INT PIPE W	0.00	54150.00	0.77	0.00
SR+BEAM MIST	0.00	54970.00	0.77	0.00
TRUNION WT	0.00	54970.00	0.77	0.00
P3 Fy weight	0.00	58960.99	0.77	0.00
P4 FY weight	0.00	58960.99	0.77	0.00

The Natural Frequency for the Vessel (Ope...) is 0.88361 Hz.

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area mm ²	Wind Pressure kPa	Element Wind Load N
10	20	950	9371.58	12183054	0.76606	5794.88
20	30	3100	9341.22	28023658	0.76606	13514.1
30	40	6100	9341.22	28023658	0.76606	13329.5
40	50	9100	9338.46	28015382	0.76606	13447
50	60	12100	9338.46	28015382	0.76606	13784.8
60	70	14100	9343.98	9343980	0.76606	4444.47
70	80	14625	9469.56	473478	0.76606	225.21
80	90	16150	9441.96	28325880	0.76606	15283.7
90	100	19150	9436.44	28309318	0.76606	15072.2
100	110	22150	9364.68	28094036	0.76606	15050.1
110	120	25150	9430.92	28292760	0.76606	13457.5
120	130	28150	9425.4	28276200	0.76606	15011.6
130	140	31150	9419.88	28259634	0.76606	15065.1
140	150	34150	9419.88	28259634	0.76606	15115.8
150	160	37150	9414.36	28243078	0.76606	13433.9
160	170	40150	9414.36	28243078	0.76606	15063.2
170	180	43150	9408.84	28226518	0.76606	15095
180	190	46150	9408.84	28226518	0.76606	13426
190	200	49150	9408.84	28226518	0.76606	15025.5
200	210	51900	9408.84	23522100	0.76606	12830.3
210	220	53975	9408.84	15524585	0.76606	7384.29
220	230	55266.7	8666.34	8328352	0.76606	5606.31
230	240	56242.8	7167.6	7167604	0.76606	3409.28
240	250	57237.8	5637.45	5637450	0.76606	2681.46
250	260	59261	4877.47	14632415	0.76606	9898
260	270	61170.2	4871.95	3724111	0.76606	1771.38

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.





Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	1.250
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	2.500
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\ SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_n^x$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(201.334^{0.75})$
 = 1.069 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/0.884)
 = 1.132

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 1.069$, 1.132) per 12.8.2
 = 1.132

Seismic Response Coefficient per equation 12.8-2 [Cs]:
 = $Sds / (R / I)$
 = $0.157 / (1.25 / 1.25)$
 = 0.157

Maximum value of Cs per equation 12.8-3 [Cs]:
 = $Sd1 / (T(R / I_e))$
 = $0.104 / (1.132(1.25 / 1.25))$
 = 0.092

Minimum value of Cs per equation 15.4-1 [Cs]:
 = $\max((0.044 \cdot SDS \cdot I_e), 0.030)$
 = $\max((0.044 \cdot 0.1568 \cdot 1.25), 0.030)$
 = 0.030

Total Base Shear [V]:
 = $C_s \cdot W$ (Equation 12.8-1):
 = $0.0919 \cdot 10593777$
 = 973525.188 N

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.157 \cdot 10593777$



= 332220.8 N

Distribute the Base shear force to each element according to the equations

$F_x = C_{vx} * V$ (eqn. 12.8-11) and the vertical distribution factor

$C_{vx} = W_x * h_x / k / (\text{Sum of } W_i * h_i^k)$ and k is an exponent which is related to the period of Vibration.

In this case, the value of k was 1.3159

The Natural Frequency for the Vessel (Ope...) is 0.88361 Hz.

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight N	Element Ope Load N	Element Emp Load N	Element Vertical Load N
10	20	650	286698	143.193	210.249	332221
20	30	2800	188752	644.117	821.751	323230
30	40	5800	170585	1517.68	2100.81	317311
40	50	8800	178706	2751.83	3383.97	311961
50	60	11800	178344	4039.98	4938.94	306357
60	70	13800	60206.8	1675.87	2266.26	300764
70	80	14325	436942	12774.9	7326.32	298876
80	90	15850	975435	32579.4	8615.47	285174
90	100	18850	349802	14676.7	15465	254584
100	110	21850	350300	17850.3	12137.2	243614
110	120	24850	601172	36285	14925.5	232629
120	130	27850	622336	43640.2	16042.6	213776
130	140	30850	579043	46455.3	15841.5	194260
140	150	33850	939188	85135.6	28408.9	176101
150	160	36850	559805	56744.3	19279.2	146648
160	170	39850	574153	64512	18348.4	129092
170	180	42850	727946	89989.4	28137.9	111087
180	190	45850	504323	68151	29148.3	88258.5
190	200	48850	332777	48880.5	31488.8	72442.9
200	210	51600	591481	93373	28635.4	62007
210	220	53675	293365	48777.4	34595.9	43458.2
220	230	54980.5	134082	23009.8	22234.6	34258.3
230	240	55961	27891.5	4899.1	6379.81	30053.5
240	250	56961	21836.8	3926.06	5110.23	29178.8
250	260	58961	886238	166740	12544.4	28494
260	270	60486	22372.1	4353	5656.78	701.59



The following table is for the Operating Case.

Wind/Earthquake Shear, Bending:

From	To	Distance to Support mm	Cumulative Wind Shear N	Earthquake Shear N	Wind Bending N-mm	Earthquake Bending N-mm
10	20	650	284220	973525	8.545E+09	41.25E+09
20	30	2800	278425	973382	8.179E+09	39.98E+09
30	40	5800	264911	972738	7.364E+09	37.06E+09
40	50	8800	251582	971220	6.589E+09	34.14E+09
50	60	11800	238135	968468	5.854E+09	31.23E+09
60	70	13800	224350	964428	5.160E+09	28.33E+09
70	80	14325	219906	962752	4.938E+09	27.37E+09
80	90	15850	219680	949978	4.927E+09	27.32E+09
90	100	18850	204397	917398	4.291E+09	24.52E+09
100	110	21850	189325	902721	3.700E+09	21.79E+09
110	120	24850	174274	884871	3.154E+09	19.10E+09
120	130	27850	160817	848586	2.651E+09	16.50E+09
130	140	30850	145805	804946	2.191E+09	14.02E+09
140	150	33850	130740	758491	1776219648	11.67E+09
150	160	36850	115624	673355	1406522752	9.526E+09
160	170	39850	102191	616611	1079667712	7.591E+09
170	180	42850	87127.4	552099	795575424	5.837E+09
180	190	45850	72032.4	462109	556738944	4.315E+09
190	200	48850	58606.5	393958	360701312	3.030E+09
200	210	51600	43581	345078	207358128	1921259520
210	220	53675	30750.7	251705	114405792	1174978176
220	230	54980.5	23366.4	202928	69741024	799754048
230	240	55961	17760.1	179918	49971684	615721920
240	250	56961	14350.8	175019	33909712	438181920
250	260	58961	11669.4	171093	20894344	265056272
260	270	60486	1771.38	4353	725072	1781799

Note:

The Wind Shears/Moments and the Earthquake Shears/Moments calculated and printed in the Wind/Earthquake Shear and Bending report have been factored by the input Scalar/Load reductions factors of;
 Wind: 1.100; Earthquake: 1.000.

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Wind Deflection Calculations:

The following table is for Wind Deflection; Operating Case.

Wind Load Deflection:

From	To	Cumulative Shear N	Centroid Deflection mm	Elem. End Deflection mm	Elem. Ang. Rotation
10	20	284220	0.0013825	0.0054902	0.83853E-05
20	30	278425	0.029629	0.076105	0.38170E-04
30	40	264911	0.14376	0.23146	0.64907E-04
40	50	251582	0.33871	0.46497	0.90262E-04
50	60	238135	0.60913	0.77012	0.0001127
60	70	224350	0.82725	0.88589	0.00011879
70	80	219906	0.88886	0.89184	0.00011908
80	90	219680	1.07905	1.28269	0.00014099
90	100	204397	1.50226	1.73721	0.00016152
100	110	189325	1.98661	2.24952	0.00017954
110	120	174274	2.52527	2.81318	0.00019576
120	130	160817	3.11274	3.42342	0.00021059
130	140	145805	3.7447	4.07606	0.00022403
140	150	130740	4.41646	4.76493	0.0002348
150	160	115624	5.12098	5.48408	0.00024421
160	170	102191	5.85333	6.22789	0.0002513
170	180	87127.4	6.60728	6.991	0.00025709
180	190	72032.4	7.37832	7.76858	0.00026101
190	200	58606.5	8.16117	8.55556	0.00026343
200	210	43581	8.88527	9.21569	0.00026456
210	220	30750.7	9.43403	9.65252	0.0002649
220	230	23366.4	9.77983	9.90718	0.00026509
230	240	17760.1	10.0398	10.1724	0.0002654
240	250	14350.8	10.3052	10.4381	0.00026583
250	260	11669.4	10.8372	11.2367	0.00026637
260	270	1771.38	11.2434	11.25	0.00026637

Critical Wind Velocity for Tower Vibration:

From	To	1st Crit. Wind Speed km/hr	2nd Crit. Wind Speed km/hr
10	20	148.654	929.088
20	30	148.173	926.078
30	40	148.173	926.078
40	50	148.129	925.805
50	60	148.129	925.805
60	70	148.216	926.352
70	80	150.208	938.802
80	90	149.77	936.065
90	100	149.683	935.518
100	110	148.545	928.404
110	120	149.595	934.971
120	130	149.508	934.424
130	140	149.42	933.876
140	150	149.42	933.876
150	160	149.333	933.329
160	170	149.333	933.329
170	180	149.245	932.782
180	190	149.245	932.782
190	200	149.245	932.782
200	210	149.245	932.782
210	220	149.245	932.782
220	230	137.467	859.171
230	240	113.694	710.588
240	250	89.4225	558.891
250	260	77.3675	483.547



260	270	77.28	483
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Allowable deflection at the Tower Top (Ope)(6.000"/100ft. Criteria)
Allowable deflection : 302.555 Actual deflection : 11.250 mm

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Longitudinal Stress Constants:

From	To	Metal Area New mm ²	Metal Area Corroded mm ²	Section Modulus New mm ³	Section Modulus Corroded mm ³
10	20	1184883	1163899	1978733184	1944264704
20	30	719390	698334	1205221376	1170294656
30	40	719390	698334	1205221376	1170294656
40	50	677076	656016	1134661376	1099693440
50	60	677076	656016	1134661376	1099693440
60	70	761716	740668	1275752064	1240866560
70	80	805103	784031	1349837696	1314900736
80	90	592353	571281	993108480	958068224
90	100	549881	528809	921899328	886838272
100	110	545629	524723	907686016	873167360
110	120	507432	486360	850729728	815647936
120	130	465003	443931	779593664	744491072
130	140	422606	401534	708509568	673386176
140	150	422606	401534	708509568	673386176
150	160	380236	359165	637471488	602327232
160	170	380236	359165	637471488	602327232
170	180	337891	316819	566476416	531311296
180	190	337891	316819	566476416	531311296
190	200	337891	316819	566476416	531311296
200	210	337891	316819	566476416	531311296
210	220	422103	401059	706826304	671785408
220	230	434351	410294	727208128	687161792
230	240	284437	264173	401078528	372654912
240	250	228650	212370	259021920	240700048
250	260	172833	162078	147882048	138760752
260	270	151140	140386	129319432	120187544



Longitudinal Allowable Stresses:

From	To	Tensile MPa	Hydrotest Tensile MPa	Compressive MPa	Hydrotest Compressive MPa
10	20	98.2782	181.437	-121.864	-124.838
20	30	98.2782	181.437	-107.045	-109.381
30	40	98.2782	181.437	-107.045	-109.381
40	50	98.2782	181.437	-105	-107.233
50	60	98.2782	181.437	-105	-107.233
60	70	70.7987	118.798	-108.924	-111.351
70	80	141.597	237.596	-110.616	-113.123
80	90	141.597	237.596	-100.307	-102.294
90	100	141.597	237.596	-97.5938	-99.4333
100	110	141.597	237.596	-97.8717	-99.7265
110	120	141.597	237.596	-94.5675	-96.2377
120	130	141.597	237.596	-91.1628	-92.6376
130	140	141.597	237.596	-83.8937	-84.4755
140	150	141.597	237.596	-83.8937	-84.4755
150	160	141.597	237.596	-75.1073	-75.6282
160	170	141.597	237.596	-75.1073	-75.6282
170	180	141.597	237.596	-66.3105	-66.7704
180	190	141.597	237.596	-66.3105	-66.7704
190	200	141.597	237.596	-66.3105	-66.7704
200	210	141.597	237.596	-66.3105	-66.7704
210	220	141.597	237.596	-83.9933	-84.5758
220	230	141.597	237.596	-85.9324	-86.5284
230	240	141.597	237.596	-78.0534	-78.5947
240	250	141.597	237.596	-92.9022	-94.4775
250	260	141.597	237.596	-103.201	-105.341
260	270	141.597	237.596	-98.2512	-100.127



Longitudinal Stress Report



Longitudinal Operating and Empty Stresses are computed in the corroded condition. Stresses due to loads in the hydrostatic test cases have also been computed in the corroded condition.

Longitudinal Pressure Stresses due to:

From	To	Longitudinal Stress Internal Pressure MPa	Longitudinal Stress External Pressure MPa	Longitudinal Stress Hydrotest Pressure MPa
10	20	...	...	...
20	30	...	...	...
30	40	...	...	...
40	50	...	...	...
50	60	...	...	...
60	70	...	...	...
70	80	15.6692	-2.25924	20.3699
80	90	21.4983	-3.08234	27.9478
90	100	23.2237	-3.32597	30.1908
100	110	23.0432	-3.30048	29.9561
110	120	25.2492	-3.61197	32.824
120	130	27.6605	-3.9525	35.9587
130	140	30.5795	-4.36466	39.7533
140	150	30.5795	-4.36466	39.7533
150	160	34.1852	-4.87376	44.4408
160	170	34.1852	-4.87376	44.4408
170	180	38.7525	-5.51863	50.3783
180	190	38.7525	-5.51863	50.3783
190	200	38.7525	-5.51863	50.3783
200	210	38.3061	-5.51863	50.3783
210	220	30.5429	-4.35947	39.7058
220	230	34.1299	-4.87263	44.3689
230	240	37.6001	-5.36268	48.8801
240	250	30.1944	-4.31698	39.2528
250	260	19.749	-2.83531	25.6737
260	270	22.7979	-3.26585	29.6373

Longitudinal Stresses due to Weight Loads for these Conditions:

From	To	Wght. Str. Empty MPa	Wght. Str. Operating MPa	Wght. Str. Hydrotest MPa	Wght. Str. Emp. Mom. MPa	Wght. Str. Opr. Mom. MPa
10	20	-7.41337	-9.10275	-16.7181	0.95359	0.95359
20	30	-11.9451	-14.7608	-27.4531	1.51301	1.51301
30	40	-11.6748	-14.4905	-27.1828	1.4712	1.4712
40	50	-12.1679	-15.1652	-28.6762	1.56565	1.56565
50	60	-11.8954	-14.8927	-28.4038	1.52118	1.52118
60	70	-10.2951	-12.9498	-24.9167	1.30949	1.30949
70	80	-9.64889	-11.039	-9.64889	1.23576	1.23576
80	90	-12.8459	-14.7537	-12.8459	1.69601	1.69601
90	100	-13.292	-15.353	-13.292	1.81943	1.81943
100	110	-12.7288	-14.8058	-12.7288	1.69269	1.69269
110	120	-13.1071	-15.2534	-13.1071	1.81205	1.81205
120	130	-13.3565	-15.3569	-13.3565	1.98525	1.98525
130	140	-13.6047	-15.4284	-13.6047	2.19487	2.19487
140	150	-12.4341	-13.9862	-12.4341	2.19487	2.19487
150	160	-11.2858	-13.021	-11.2858	0.91354	0.91354
160	170	-10.1608	-11.4622	-10.1608	0.91354	0.91354
170	180	-10.1983	-11.1818	-10.1983	1.03564	1.03564
180	190	-8.39219	-8.88396	-8.39219	1.03564	1.03564
190	200	-6.80022	-7.29199	-6.80022	0.81315	0.81315
200	210	-5.83664	-6.24153	-5.83664	0.77399	0.77399
210	220	-3.45561	-3.45561	-3.45561	0.61214	0.61214



220	230	-2.66275	-2.66275	-2.66275	0.25071	0.25071
230	240	-3.628	-3.628	-3.628	0.014931	0.014931
240	250	-4.38163	-4.38163	-4.38163	0.023116	0.023116
250	260	-5.60648	-5.60648	-5.60648	0.040098	0.040098
260	270	-0.15938	-0.15938	-0.15938	...	...

Longitudinal Stresses due to Weight Loads and Bending for these Conditions:

From	To	Wght. Str. Hyd. Mom. MPa	Bend. Str. Oper. Wind MPa	Bend. Str. Oper. Equ. MPa	Bend. Str. Hyd. Wind MPa	Bend. Str. Hyd. Equ. MPa
10	20	0.95359	4.39374	21.2075	2.19687	...
20	30	1.51301	6.987	34.1515	3.4935	...
30	40	1.4712	6.29053	31.6569	3.14526	...
40	50	1.56565	5.98982	31.0375	2.99491	...
50	60	1.52118	5.32178	28.3915	2.66089	...
60	70	1.30949	4.15721	22.8246	2.0786	...
70	80	1.23576	3.7542	20.8066	1.8771	...
80	90	1.69601	5.14097	28.5061	2.57049	...
90	100	1.81943	4.83655	27.637	2.41827	...
100	110	1.69269	4.23585	24.9427	2.11792	...
110	120	1.81205	3.86583	23.4139	1.93292	...
120	130	1.98525	3.56012	22.1589	1.78006	...
130	140	2.19487	3.25297	20.8151	1.62648	...
140	150	2.19487	2.6369	17.3321	1.31845	...
150	160	0.91354	2.3344	15.8108	1.1672	...
160	170	0.91354	1.79192	12.5981	0.89596	...
170	180	1.03564	1.4969	10.9822	0.74845	...
180	190	1.03564	1.04752	8.1186	0.52376	...
190	200	0.81315	0.67867	5.70154	0.33934	...
200	210	0.77399	0.39015	3.61491	0.19508	...
210	220	0.61214	0.17025	1.74848	0.085123	...
220	230	0.25071	0.10146	1.16348	0.050729	...
230	240	0.014931	0.13405	1.65173	0.067027	...
240	250	0.023116	0.14083	1.81986	0.070417	...
250	260	0.040098	0.15053	1.90956	0.075265	...
260	270	...	0.0060309	0.01482	0.0030155	...

Longitudinal Stresses due to these Conditions:

From	To	Vortex Shedding Operating Case MPa	Vortex Shedding Empty Case MPa	Vortex Shedding Test Case MPa	Earthquake Empty Case MPa
10	20	...	...	...	7.72145
20	30	...	...	...	12.4126
30	40	...	...	...	11.4553
40	50	...	...	...	11.1759
50	60	...	...	...	10.1686
60	70	...	...	...	8.12901
70	80	...	...	...	7.3964
80	90	...	...	...	10.1326
90	100	...	...	...	9.76677
100	110	...	...	...	8.76295
110	120	...	...	...	8.19336
120	130	...	...	...	7.72994
130	140	...	...	...	7.23701
140	150	...	...	...	5.99888
150	160	...	...	...	5.4326
160	170	...	...	...	4.27739
170	180	...	...	...	3.64572
180	190	...	...	...	2.57358
190	200	...	...	...	1.66319
200	210	...	...	...	0.92401
210	220	...	...	...	0.35549
220	230	...	...	...	0.18131
230	240	...	...	...	0.22908



240	250	...	...	...	0.24455
250	260	...	...	...	0.27461
260	270	...	...	...	0.019259

Longitudinal Stresses due to Applied Axial Forces:

From	To	Longitudinal Stress Y Forces Wind MPa	Longitudinal Stress Y Forces Seismic MPa
10	20	...	0.28546
20	30	...	0.4629
30	40	...	0.45442
40	50	...	0.47558
50	60	...	0.46704
60	70	...	0.40611
70	80	...	0.38124
80	90	...	0.49923
90	100	...	0.48147
100	110	...	0.46431
110	120	...	0.47835
120	130	...	0.48159
130	140	...	0.48383
140	150	...	0.43861
150	160	...	0.40834
160	170	...	0.35945
170	180	...	0.35066
180	190	...	0.2786
190	200	...	0.22868
200	210	...	0.19573
210	220	...	0.10837
220	230	...	0.083504
230	240	...	0.11377
240	250	...	0.13741
250	260	...	0.17582
260	270	...	0.004998

Longitudinal Stresses due to User Forces and Moments:

From	To	Wind For/Mom Corroded MPa	Earthquake For/Mom Corroded MPa	Wind For/Mom No Corrosion MPa	Earthquake For/Mom No Corrosion MPa
10	20	...	...	...	...
20	30	...	...	...	...
30	40	...	...	...	...
40	50	...	...	...	...
50	60	...	...	...	...
60	70	...	...	...	...
70	80	...	...	...	...
80	90	...	...	...	...
90	100	...	...	...	...
100	110	...	...	...	...
110	120	...	...	...	...
120	130	...	...	...	...
130	140	...	...	...	...
140	150	...	...	...	...
150	160	...	...	...	...
160	170	...	...	...	...
170	180	...	...	...	...
180	190	...	...	...	...
190	200	...	...	...	...
200	210	...	...	...	...
210	220	...	...	...	...
220	230	...	...	...	...
230	240	...	...	...	...
240	250	...	...	...	...
250	260	...	...	...	...



260 270

Bending Stresses due to Blast Load:

From	To	Bending Stress MPa
10	20	...
20	30	...
30	40	...
40	50	...
50	60	...
60	70	...
70	80	...
80	90	...
90	100	...
100	110	...
110	120	...
120	130	...
130	140	...
140	150	...
150	160	...
160	170	...
170	180	...
180	190	...
190	200	...
200	210	...
210	220	...
220	230	...
230	240	...
240	250	...
250	260	...
260	270	...



Stress Combination Load Cases for Vertical Vessels:

Load Case Definition Key

IP = Longitudinal Stress due to Internal Pressure
 EP = Longitudinal Stress due to External Pressure
 HP = Longitudinal Stress due to Hydrotest Pressure
 NP = No Pressure
 EW = Longitudinal Stress due to Weight (No Liquid)
 OW = Longitudinal Stress due to Weight (Operating)
 HW = Longitudinal Stress due to Weight (Hydrotest)
 WI = Bending Stress due to Wind Moment (Operating)
 EQ = Bending Stress due to Earthquake Moment (Operating)
 EE = Bending Stress due to Earthquake Moment (Empty)
 HI = Bending Stress due to Wind Moment (Hydrotest)
 HE = Bending Stress due to Earthquake Moment (Hydrotest)
 WE = Bending Stress due to Wind Moment (Empty) (no CA)
 WF = Bending Stress due to Wind Moment (Filled) (no CA)
 CW = Longitudinal Stress due to Weight (Empty) (no CA)
 VO = Bending Stress due to Vortex Shedding Loads (Ope)
 VE = Bending Stress due to Vortex Shedding Loads (Emp)
 VF = Bending Stress due to Vortex Shedding Loads (Test No CA.)
 FW = Axial Stress due to Vertical Forces for the Wind Case
 FS = Axial Stress due to Vertical Forces for the Seismic Case
 BW = Bending Stress due to Lat. Forces for the Wind Case, Corroded
 BS = Bending Stress due to Lat. Forces for the Seismic Case, Corroded
 BN = Bending Stress due to Lat. Forces for the Wind Case, UnCorroded
 BU = Bending Stress due to Lat. Forces for the Seismic Case, UnCorroded
 BL = Bending Stress due to Blast Loads, Corroded

General Notes:

Case types HI and HE are in the Corroded condition.

Case types WE, WF, and CW are in the Un-Corroded condition.

A blank stress and stress ratio indicates that the corresponding stress comprising those components that did not contribute to that type of stress.

An asterisk (*) in the final column denotes overstress.

Analysis of Load Case 1 : NP+EW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		98.28	-12.76	121.86		0.1047
20		98.28	-20.45	107.05		0.1910
30		98.28	-19.44	107.05		0.1816
40		98.28	-19.72	105.00		0.1878
50		98.28	-18.74	105.00		0.1785
60		70.80	-15.76	108.92		0.1447
70		141.60	-14.64	110.62		0.1323
80		141.60	-19.68	100.31		0.1962
90		141.60	-19.95	97.59		0.2044
100		141.60	-18.66	97.87		0.1906
110		141.60	-18.78	94.57		0.1986
120		141.60	-18.90	91.16		0.2073
130		141.60	-19.05	83.89		0.2271
140		141.60	-17.27	83.89		0.2058
150		141.60	-14.53	75.11		0.1935
160		141.60	-12.87	75.11		0.1713
170		141.60	-12.73	66.31		0.1920
180		141.60	-10.48	66.31		0.1580
190		141.60	-8.29	66.31		0.1250
200		141.60	-7.00	66.31		0.1056
210		141.60	-4.24	83.99		0.0505
220		141.60	-3.01	85.93		0.0351
230		141.60	-3.78	78.05		0.0484
240		141.60	-4.55	92.90		0.0489



250	141.60	-5.80	103.20	0.0562
260	141.60	-0.17	98.25	0.0017

Analysis of Load Case 2 : NP+EW+EE+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	1.55	98.28	-15.80	121.86	0.0157	0.1297
20	2.44	98.28	-25.41	107.05	0.0249	0.2374
30	1.71	98.28	-24.15	107.05	0.0174	0.2256
40	1.05	98.28	-24.43	105.00	0.0107	0.2327
50	0.26	98.28	-23.12	105.00	0.0027	0.2202
60		70.80	-19.33	108.92		0.1774
70		141.60	-17.90	110.62		0.1618
80		141.60	-24.18	100.31		0.2410
90		141.60	-24.40	97.59		0.2500
100		141.60	-22.72	97.87		0.2321
110		141.60	-22.63	94.57		0.2393
120		141.60	-22.59	91.16		0.2478
130		141.60	-22.55	83.89		0.2688
140		141.60	-20.19	83.89		0.2407
150		141.60	-17.22	75.11		0.2293
160		141.60	-14.99	75.11		0.1996
170		141.60	-14.53	66.31		0.2191
180		141.60	-11.72	66.31		0.1768
190		141.60	-9.05	66.31		0.1364
200		141.60	-7.34	66.31		0.1107
210		141.60	-4.31	83.99		0.0514
220		141.60	-3.01	85.93		0.0350
230		141.60	-3.76	78.05		0.0481
240		141.60	-4.51	92.90		0.0486
250		141.60	-5.75	103.20		0.0557
260		141.60	-0.17	98.25		0.0018

Analysis of Load Case 3 : NP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		98.28	-14.45	121.86		0.1186
20		98.28	-23.26	107.05		0.2173
30		98.28	-22.25	107.05		0.2079
40		98.28	-22.72	105.00		0.2164
50		98.28	-21.74	105.00		0.2070
60		70.80	-18.42	108.92		0.1691
70		141.60	-16.03	110.62		0.1449
80		141.60	-21.59	100.31		0.2152
90		141.60	-22.01	97.59		0.2255
100		141.60	-20.73	97.87		0.2119
110		141.60	-20.93	94.57		0.2213
120		141.60	-20.90	91.16		0.2293
130		141.60	-20.88	83.89		0.2488
140		141.60	-18.82	83.89		0.2243
150		141.60	-16.27	75.11		0.2166
160		141.60	-14.17	75.11		0.1886
170		141.60	-13.71	66.31		0.2068
180		141.60	-10.97	66.31		0.1654
190		141.60	-8.78	66.31		0.1325
200		141.60	-7.41	66.31		0.1117
210		141.60	-4.24	83.99		0.0505
220		141.60	-3.01	85.93		0.0351
230		141.60	-3.78	78.05		0.0484
240		141.60	-4.55	92.90		0.0489
250		141.60	-5.80	103.20		0.0562
260		141.60	-0.17	98.25		0.0017

Analysis of Load Case 4 : NP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	13.34	98.28	-30.98	121.86	0.1358	0.2542



20	21.37	98.28	-49.96	107.05	0.2174	0.4667
30	19.09	98.28	-47.16	107.05	0.1943	0.4406
40	17.91	98.28	-47.29	105.00	0.1823	0.4504
50	15.49	98.28	-44.34	105.00	0.1576	0.4223
60	11.59	70.80	-36.68	108.92	0.1637	0.3367
70	11.38	141.60	-32.70	110.62	0.0804	0.2956
80	15.95	141.60	-44.46	100.31	0.1126	0.4432
90	14.58	141.60	-44.33	97.59	0.1030	0.4542
100	12.29	141.60	-40.98	97.87	0.0868	0.4187
110	10.45	141.60	-40.00	94.57	0.0738	0.4230
120	9.27	141.60	-39.02	91.16	0.0655	0.4280
130	8.07	141.60	-37.95	83.89	0.0570	0.4524
140	5.98	141.60	-33.07	83.89	0.0422	0.3942
150	4.11	141.60	-29.34	75.11	0.0290	0.3906
160	2.41	141.60	-24.61	75.11	0.0170	0.3277
170	1.19	141.60	-22.85	66.31	0.0084	0.3446
180	0.55	141.60	-17.76	66.31	0.0039	0.2678
190		141.60	-13.58	66.31		0.2048
200		141.60	-10.43	66.31		0.1574
210		141.60	-5.71	83.99		0.0680
220		141.60	-3.99	85.93		0.0465
230		141.60	-5.18	78.05		0.0664
240		141.60	-6.09	92.90		0.0655
250		141.60	-7.38	103.20		0.0715
260		141.60	-0.17	98.25		0.0017

Analysis of Load Case 5 : NP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		181.44	-19.87	124.84		0.1592
20		181.44	-32.46	109.38		0.2968
30		181.44	-31.80	109.38		0.2907
40		181.44	-33.24	107.23		0.3099
50		181.44	-32.59	107.23		0.3039
60		118.80	-28.30	111.35		0.2542
70		237.60	-12.76	113.12		0.1128
80		237.60	-17.11	102.29		0.1673
90		237.60	-17.53	99.43		0.1763
100		237.60	-16.54	99.73		0.1658
110		237.60	-16.85	96.24		0.1751
120		237.60	-17.12	92.64		0.1848
130		237.60	-17.43	84.48		0.2063
140		237.60	-15.95	84.48		0.1888
150		237.60	-13.37	75.63		0.1767
160		237.60	-11.97	75.63		0.1583
170		237.60	-11.98	66.77		0.1795
180		237.60	-9.95	66.77		0.1490
190		237.60	-7.95	66.77		0.1191
200		237.60	-6.81	66.77		0.1019
210		237.60	-4.15	84.58		0.0491
220		237.60	-2.96	86.53		0.0343
230		237.60	-3.71	78.59		0.0472
240		237.60	-4.48	94.48		0.0474
250		237.60	-5.72	105.34		0.0543
260		237.60	-0.16	100.13		0.0016

Analysis of Load Case 6 : NP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		181.44	-17.67	124.84		0.1416
20		181.44	-28.97	109.38		0.2648
30		181.44	-28.65	109.38		0.2620
40		181.44	-30.24	107.23		0.2820
50		181.44	-29.92	107.23		0.2791
60		118.80	-26.23	111.35		0.2355
70		237.60	-10.88	113.12		0.0962
80		237.60	-14.54	102.29		0.1422
90		237.60	-15.11	99.43		0.1520



100	237.60	-14.42	99.73	0.1446
110	237.60	-14.92	96.24	0.1550
120	237.60	-15.34	92.64	0.1656
130	237.60	-15.80	84.48	0.1870
140	237.60	-14.63	84.48	0.1732
150	237.60	-12.20	75.63	0.1613
160	237.60	-11.07	75.63	0.1464
170	237.60	-11.23	66.77	0.1682
180	237.60	-9.43	66.77	0.1412
190	237.60	-7.61	66.77	0.1140
200	237.60	-6.61	66.77	0.0990
210	237.60	-4.07	84.58	0.0481
220	237.60	-2.91	86.53	0.0337
230	237.60	-3.64	78.59	0.0464
240	237.60	-4.40	94.48	0.0466
250	237.60	-5.65	105.34	0.0536
260	237.60	-0.16	100.13	0.0016

Analysis of Load Case 7 : IP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		98.28	-14.45	121.86		0.1186
20		98.28	-23.26	107.05		0.2173
30		98.28	-22.25	107.05		0.2079
40		98.28	-22.72	105.00		0.2164
50		98.28	-21.74	105.00		0.2070
60		70.80	-18.42	108.92		0.1691
70	9.62	141.60	-0.36	110.62	0.0679	0.0033
80	13.58	141.60	-0.09	100.31	0.0959	0.0009
90	14.53	141.60		97.59	0.1026	
100	14.17	141.60		97.87	0.1000	
110	15.67	141.60		94.57	0.1107	
120	17.85	141.60		91.16	0.1261	
130	20.60	141.60		83.89	0.1455	
140	21.43	141.60		83.89	0.1513	
150	24.41	141.60		75.11	0.1724	
160	25.43	141.60		75.11	0.1796	
170	30.10	141.60		66.31	0.2126	
180	31.95	141.60		66.31	0.2257	
190	32.95	141.60		66.31	0.2327	
200	33.23	141.60		66.31	0.2347	
210	27.87	141.60		83.99	0.1968	
220	31.82	141.60		75.21	0.2247	
230	34.12	141.60		68.32	0.2410	
240	25.98	141.60		84.93	0.1835	
250	14.33	141.60		103.20	0.1012	
260	22.64	141.60		98.25	0.1599	

Analysis of Load Case 8 : IP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	13.34	98.28	-30.98	121.86	0.1358	0.2542
20	21.37	98.28	-49.96	107.05	0.2174	0.4667
30	19.09	98.28	-47.16	107.05	0.1943	0.4406
40	17.91	98.28	-47.29	105.00	0.1823	0.4504
50	15.49	98.28	-44.34	105.00	0.1576	0.4223
60	11.59	70.80	-36.68	108.92	0.1637	0.3367
70	27.05	141.60	-17.03	110.62	0.1911	0.1540
80	37.45	141.60	-22.96	100.31	0.2645	0.2289
90	37.81	141.60	-21.10	97.59	0.2670	0.2162
100	35.34	141.60	-17.93	97.87	0.2496	0.1832
110	35.70	141.60	-14.75	94.57	0.2521	0.1560
120	36.93	141.60	-11.36	91.16	0.2608	0.1246
130	38.64	141.60	-7.38	83.89	0.2729	0.0879
140	36.56	141.60	-2.50	83.89	0.2582	0.0297
150	38.30	141.60		75.11	0.2705	
160	36.59	141.60		75.11	0.2584	
170	39.94	141.60		66.31	0.2821	



180	39.30	141.60	66.31	0.2776
190	38.20	141.60	66.31	0.2698
200	36.65	141.60	66.31	0.2588
210	29.56	141.60	83.99	0.2087
220	32.96	141.60	75.21	0.2328
230	35.75	141.60	68.32	0.2525
240	27.79	141.60	84.93	0.1963
250	16.27	141.60	103.20	0.1149
260	22.66	141.60	98.25	0.1600

Analysis of Load Case 9 : EP+OW+WI+FW+BW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		98.28	-14.45	124.54		0.1160
20		98.28	-23.26	109.15		0.2131
30		98.28	-22.25	109.15		0.2039
40		98.28	-22.72	107.01		0.2123
50		98.28	-21.74	107.01		0.2031
60		70.80	-18.42	111.11		0.1657
70		141.60	-18.29	112.88		0.1620
80		141.60	-24.67	102.10		0.2417
90		141.60	-25.33	99.25		0.2553
100		141.60	-24.03	99.54		0.2415
110		141.60	-24.54	96.07		0.2555
120		141.60	-24.85	92.49		0.2687
130		141.60	-25.24	84.42		0.2990
140		141.60	-23.18	84.42		0.2746
150		141.60	-21.14	75.58		0.2798
160		141.60	-19.04	75.58		0.2519
170		141.60	-19.23	66.72		0.2882
180		141.60	-16.49	66.72		0.2471
190		141.60	-14.30	66.72		0.2143
200		141.60	-12.92	66.72		0.1937
210		141.60	-8.60	84.52		0.1017
220		141.60	-7.89	75.68		0.1042
230		141.60	-9.14	68.74		0.1330
240		141.60	-8.86	85.46		0.1037
250		141.60	-8.63	105.13		0.0821
260		141.60	-3.43	99.94		0.0343

Analysis of Load Case 10 : EP+OW+EQ+FS+BS

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10	13.34	98.28	-30.98	124.54	0.1358	0.2487
20	21.37	98.28	-49.96	109.15	0.2174	0.4577
30	19.09	98.28	-47.16	109.15	0.1943	0.4321
40	17.91	98.28	-47.29	107.01	0.1823	0.4419
50	15.49	98.28	-44.34	107.01	0.1576	0.4143
60	11.59	70.80	-36.68	111.11	0.1637	0.3301
70	9.13	141.60	-34.96	112.88	0.0644	0.3097
80	12.87	141.60	-47.54	102.10	0.0909	0.4656
90	11.26	141.60	-47.65	99.25	0.0795	0.4801
100	8.99	141.60	-44.28	99.54	0.0635	0.4448
110	6.84	141.60	-43.61	96.07	0.0483	0.4540
120	5.32	141.60	-42.97	92.49	0.0375	0.4646
130	3.70	141.60	-42.32	84.42	0.0261	0.5013
140	1.61	141.60	-37.44	84.42	0.0114	0.4435
150		141.60	-34.21	75.58		0.4527
160		141.60	-29.49	75.58		0.3902
170		141.60	-28.37	66.72		0.4251
180		141.60	-23.28	66.72		0.3489
190		141.60	-19.10	66.72		0.2862
200		141.60	-15.95	66.72		0.2391
210		141.60	-10.07	84.52		0.1191
220		141.60	-8.87	75.68		0.1172
230		141.60	-10.54	68.74		0.1534
240		141.60	-10.40	85.46		0.1217
250		141.60	-10.22	105.13		0.0972



260 141.60 -3.44 99.94 0.0344

Analysis of Load Case 11 : HP+HW+HI

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		181.44	-19.87	124.84		0.1592
20		181.44	-32.46	109.38		0.2968
30		181.44	-31.80	109.38		0.2907
40		181.44	-33.24	107.23		0.3099
50		181.44	-32.59	107.23		0.3039
60		118.80	-28.30	111.35		0.2542
70	13.83	237.60		113.12	0.0582	
80	19.37	237.60		102.29	0.0815	
90	21.14	237.60		99.43	0.0890	
100	21.04	237.60		99.73	0.0885	
110	23.46	237.60		96.24	0.0987	
120	26.37	237.60		92.64	0.1110	
130	29.97	237.60		84.48	0.1261	
140	30.83	237.60		84.48	0.1298	
150	35.24	237.60		75.63	0.1483	
160	36.09	237.60		75.63	0.1519	
170	41.96	237.60		66.77	0.1766	
180	43.55	237.60		66.77	0.1833	
190	44.73	237.60		66.77	0.1883	
200	45.51	237.60		66.77	0.1915	
210	36.95	237.60		84.58	0.1555	
220	42.01	237.60		86.53	0.1768	
230	45.33	237.60		78.59	0.1908	
240	34.96	237.60		94.48	0.1472	
250	20.18	237.60		105.34	0.0849	
260	29.48	237.60		100.13	0.1241	

Analysis of Load Case 12 : HP+HW+HE

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		181.44	-17.67	124.84		0.1416
20		181.44	-28.97	109.38		0.2648
30		181.44	-28.65	109.38		0.2620
40		181.44	-30.24	107.23		0.2820
50		181.44	-29.92	107.23		0.2791
60		118.80	-26.23	111.35		0.2355
70	11.96	237.60		113.12	0.0503	
80	16.80	237.60		102.29	0.0707	
90	18.72	237.60		99.43	0.0788	
100	18.92	237.60		99.73	0.0796	
110	21.53	237.60		96.24	0.0906	
120	24.59	237.60		92.64	0.1035	
130	28.34	237.60		84.48	0.1193	
140	29.51	237.60		84.48	0.1242	
150	34.07	237.60		75.63	0.1434	
160	35.19	237.60		75.63	0.1481	
170	41.22	237.60		66.77	0.1735	
180	43.02	237.60		66.77	0.1811	
190	44.39	237.60		66.77	0.1868	
200	45.32	237.60		66.77	0.1907	
210	36.86	237.60		84.58	0.1551	
220	41.96	237.60		86.53	0.1766	
230	45.27	237.60		78.59	0.1905	
240	34.89	237.60		94.48	0.1469	
250	20.11	237.60		105.34	0.0846	
260	29.48	237.60		100.13	0.1241	

Analysis of Load Case 13 : IP+WE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		98.28	-8.37	121.86		0.0687
20		98.28	-13.46	107.05		0.1257



30		98.28	-13.15	107.05	0.1228
40		98.28	-13.73	105.00	0.1308
50		98.28	-13.42	105.00	0.1278
60		70.80	-11.60	108.92	0.1065
70	7.26	141.60		110.62	0.0512
80	10.35	141.60		100.31	0.0731
90	11.75	141.60		97.59	0.0830
100	12.01	141.60		97.87	0.0848
110	13.95	141.60		94.57	0.0985
120	16.29	141.60		91.16	0.1150
130	19.17	141.60		83.89	0.1354
140	20.34	141.60		83.89	0.1436
150	23.81	141.60		75.11	0.1682
160	24.94	141.60		75.11	0.1761
170	29.59	141.60		66.31	0.2090
180	31.40	141.60		66.31	0.2217
190	32.77	141.60		66.31	0.2314
200	33.24	141.60		66.31	0.2348
210	27.70	141.60		83.99	0.1956
220	31.72	141.60		75.21	0.2240
230	33.99	141.60		68.32	0.2400
240	25.84	141.60		84.93	0.1825
250	14.18	141.60		103.20	0.1002
260	22.64	141.60		98.25	0.1599

Analysis of Load Case 14 : IP+WF+CW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		98.28	-8.94	121.86		0.0734
20		98.28	-14.33	107.05		0.1339
30		98.28	-14.07	107.05		0.1314
40		98.28	-14.69	105.00		0.1399
50		98.28	-14.43	105.00		0.1374
60		70.80	-12.59	108.92		0.1156
70	4.92	141.60		110.62	0.0347	
80	7.27	141.60		100.31	0.0513	
90	8.46	141.60		97.59	0.0597	
100	8.80	141.60		97.87	0.0622	
110	10.63	141.60		94.57	0.0751	
120	13.00	141.60		91.16	0.0918	
130	15.92	141.60		83.89	0.1124	
140	17.29	141.60		83.89	0.1221	
150	21.89	141.60		75.11	0.1546	
160	23.36	141.60		75.11	0.1650	
170	28.27	141.60		66.31	0.1996	
180	30.42	141.60		66.31	0.2149	
190	31.92	141.60		66.31	0.2254	
200	32.45	141.60		66.31	0.2292	
210	27.26	141.60		83.99	0.1925	
220	31.61	141.60		75.21	0.2233	
230	34.23	141.60		68.32	0.2417	
240	26.12	141.60		84.93	0.1845	
250	14.49	141.60		103.20	0.1023	
260	22.65	141.60		98.25	0.1600	

Analysis of Load Case 15 : IP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		98.28	-10.06	121.86		0.0825
20		98.28	-16.27	107.05		0.1520
30		98.28	-15.96	107.05		0.1491
40		98.28	-16.73	105.00		0.1593
50		98.28	-16.41	105.00		0.1563
60		70.80	-14.26	108.92		0.1309
70	5.87	141.60		110.62	0.0414	
80	8.44	141.60		100.31	0.0596	
90	9.69	141.60		97.59	0.0684	
100	9.93	141.60		97.87	0.0701	



110	11.81	141.60	94.57	0.0834
120	14.29	141.60	91.16	0.1009
130	17.35	141.60	83.89	0.1225
140	18.79	141.60	83.89	0.1327
150	22.08	141.60	75.11	0.1559
160	23.64	141.60	75.11	0.1669
170	28.61	141.60	66.31	0.2020
180	30.90	141.60	66.31	0.2183
190	32.27	141.60	66.31	0.2279
200	32.84	141.60	66.31	0.2319
210	27.70	141.60	83.99	0.1956
220	31.72	141.60	75.21	0.2240
230	33.99	141.60	68.32	0.2400
240	25.84	141.60	84.93	0.1825
250	14.18	141.60	103.20	0.1002
260	22.64	141.60	98.25	0.1599

Analysis of Load Case 16 : IP+VE+EW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		98.28	-8.37	121.86		0.0687
20		98.28	-13.46	107.05		0.1257
30		98.28	-13.15	107.05		0.1228
40		98.28	-13.73	105.00		0.1308
50		98.28	-13.42	105.00		0.1278
60		70.80	-11.60	108.92		0.1065
70	7.26	141.60		110.62	0.0512	
80	10.35	141.60		100.31	0.0731	
90	11.75	141.60		97.59	0.0830	
100	12.01	141.60		97.87	0.0848	
110	13.95	141.60		94.57	0.0985	
120	16.29	141.60		91.16	0.1150	
130	19.17	141.60		83.89	0.1354	
140	20.34	141.60		83.89	0.1436	
150	23.81	141.60		75.11	0.1682	
160	24.94	141.60		75.11	0.1761	
170	29.59	141.60		66.31	0.2090	
180	31.40	141.60		66.31	0.2217	
190	32.77	141.60		66.31	0.2314	
200	33.24	141.60		66.31	0.2348	
210	27.70	141.60		83.99	0.1956	
220	31.72	141.60		75.21	0.2240	
230	33.99	141.60		68.32	0.2400	
240	25.84	141.60		84.93	0.1825	
250	14.18	141.60		103.20	0.1002	
260	22.64	141.60		98.25	0.1599	

Analysis of Load Case 17 : NP+VO+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		98.28	-10.06	121.86		0.0825
20		98.28	-16.27	107.05		0.1520
30		98.28	-15.96	107.05		0.1491
40		98.28	-16.73	105.00		0.1593
50		98.28	-16.41	105.00		0.1563
60		70.80	-14.26	108.92		0.1309
70		141.60	-12.27	110.62		0.1110
80		141.60	-16.45	100.31		0.1640
90		141.60	-17.17	97.59		0.1760
100		141.60	-16.50	97.87		0.1686
110		141.60	-17.07	94.57		0.1805
120		141.60	-17.34	91.16		0.1902
130		141.60	-17.62	83.89		0.2101
140		141.60	-16.18	83.89		0.1929
150		141.60	-13.93	75.11		0.1855
160		141.60	-12.38	75.11		0.1648
170		141.60	-12.22	66.31		0.1842
180		141.60	-9.92	66.31		0.1496



190	141.60	-8.11	66.31	0.1222
200	141.60	-7.02	66.31	0.1058
210	141.60	-4.07	83.99	0.0484
220	141.60	-2.91	85.93	0.0339
230	141.60	-3.64	78.05	0.0467
240	141.60	-4.40	92.90	0.0474
250	141.60	-5.65	103.20	0.0547
260	141.60	-0.16	98.25	0.0016

Analysis of Load Case 18 : FS+BS+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		81.90	-9.77	101.55		0.0962
20		81.90	-15.81	89.20		0.1772
30		81.90	-15.51	89.20		0.1738
40		81.90	-16.26	87.50		0.1858
50		81.90	-15.95	87.50		0.1822
60		59.00	-13.85	90.77		0.1526
70	6.25	118.00		92.18	0.0529	
80	8.94	118.00		83.59	0.0758	
90	10.17	118.00		81.33	0.0862	
100	10.39	118.00		81.56	0.0881	
110	12.29	118.00		78.81	0.1041	
120	14.77	118.00		75.97	0.1252	
130	17.83	118.00		69.91	0.1511	
140	19.23	118.00		69.91	0.1629	
150	22.49	118.00		62.59	0.1906	
160	24.00	118.00		62.59	0.2034	
170	28.96	118.00		55.26	0.2454	
180	31.18	118.00		55.26	0.2643	
190	32.50	118.00		55.26	0.2754	
200	33.03	118.00		55.26	0.2800	
210	27.81	118.00		69.99	0.2357	
220	31.80	118.00		62.67	0.2695	
230	34.10	118.00		56.93	0.2890	
240	25.97	118.00		70.78	0.2201	
250	14.36	118.00		86.00	0.1217	
260	22.64	118.00		81.88	0.1919	

Analysis of Load Case 19 : FS+BS+EP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		81.90	-9.77	103.79		0.0941
20		81.90	-15.81	90.96		0.1738
30		81.90	-15.51	90.96		0.1705
40		81.90	-16.26	89.18		0.1823
50		81.90	-15.95	89.18		0.1788
60		59.00	-13.85	92.59		0.1496
70		118.00	-14.15	94.06		0.1505
80		118.00	-19.03	85.08		0.2237
90		118.00	-20.02	82.71		0.2420
100		118.00	-19.33	82.95		0.2331
110		118.00	-20.20	80.06		0.2523
120		118.00	-20.81	77.08		0.2700
130		118.00	-21.50	70.35		0.3057
140		118.00	-20.11	70.35		0.2858
150		118.00	-18.40	62.98		0.2922
160		118.00	-16.89	62.98		0.2682
170		118.00	-17.39	55.60		0.3127
180		118.00	-15.16	55.60		0.2726
190		118.00	-13.40	55.60		0.2409
200		118.00	-12.34	55.60		0.2219
210		118.00	-8.32	70.43		0.1181
220		118.00	-7.70	63.07		0.1221
230		118.00	-8.89	57.29		0.1552
240		118.00	-8.58	71.22		0.1205
250		118.00	-8.31	87.61		0.0948
260		118.00	-3.42	83.28		0.0411

Analysis of Load Case 20 : BL+IP+OW

From Node	Tensile Stress	All. Tens. Stress	Comp. Stress	All. Comp. Stress	Tens. Ratio	Comp. Ratio
10		98.28	-10.06	121.86		0.0825
20		98.28	-16.27	107.05		0.1520
30		98.28	-15.96	107.05		0.1491
40		98.28	-16.73	105.00		0.1593
50		98.28	-16.41	105.00		0.1563
60		70.80	-14.26	108.92		0.1309
70	5.87	141.60		110.62	0.0414	
80	8.44	141.60		100.31	0.0596	
90	9.69	141.60		97.59	0.0684	
100	9.93	141.60		97.87	0.0701	
110	11.81	141.60		94.57	0.0834	
120	14.29	141.60		91.16	0.1009	
130	17.35	141.60		83.89	0.1225	
140	18.79	141.60		83.89	0.1327	
150	22.08	141.60		75.11	0.1559	
160	23.64	141.60		75.11	0.1669	
170	28.61	141.60		66.31	0.2020	
180	30.90	141.60		66.31	0.2183	
190	32.27	141.60		66.31	0.2279	
200	32.84	141.60		66.31	0.2319	
210	27.70	141.60		83.99	0.1956	
220	31.72	141.60		75.21	0.2240	
230	33.99	141.60		68.32	0.2400	
240	25.84	141.60		84.93	0.1825	
250	14.18	141.60		103.20	0.1002	
260	22.64	141.60		98.25	0.1599	

Absolute Maximum of the all of the Stress Ratio's 0.5013

Governing Element: Shell-6

Governing Load Case 10 : EP+OW+EQ+FS+BS

Element Description	Long Pressure Stress	Weight + Axial Load Stress	Eccentric Moment Stress	Wind or Earthquake Stress	Combined Stress	Which Side
Skirt-1	...	-8.8	1.0	21.2	13.3	+ Tens
Skirt-1	...	-8.8	-1.0	-21.2	-31.0	- Comp
Skirt-2	...	-14.3	1.5	34.2	21.4	+ Tens
Skirt-2	...	-14.3	-1.5	-34.2	-50.0	- Comp
Skirt-3	...	-14.0	1.5	31.7	19.1	+ Tens
Skirt-3	...	-14.0	-1.5	-31.7	-47.2	- Comp
Skirt-4	...	-14.7	1.6	31.0	17.9	+ Tens
Skirt-4	...	-14.7	-1.6	-31.0	-47.3	- Comp
Skirt-5	...	-14.4	1.5	28.4	15.5	+ Tens
Skirt-5	...	-14.4	-1.5	-28.4	-44.3	- Comp
Skirt-6	...	-12.5	1.3	22.8	11.6	+ Tens
Skirt-6	...	-12.5	-1.3	-22.8	-36.7	- Comp
Btm Dish End	-2.3	-10.7	1.2	20.8	9.1	+ Tens
Btm Dish End	-2.3	-10.7	-1.2	-20.8	-35.0	- Comp
Shell-1	-3.1	-14.3	1.7	28.5	12.9	+ Tens
Shell-1	-3.1	-14.3	-1.7	-28.5	-47.5	- Comp
Shell-2	-3.3	-14.9	1.8	27.6	11.3	+ Tens
Shell-2	-3.3	-14.9	-1.8	-27.6	-47.7	- Comp
Shell-3	-3.3	-14.3	1.7	24.9	9.0	+ Tens
Shell-3	-3.3	-14.3	-1.7	-24.9	-44.3	- Comp



Shell-4	-3.6	-14.8	1.8	23.4	6.8	+ Tens
Shell-4	-3.6	-14.8	-1.8	-23.4	-43.6	- Comp
Shell-5	-4.0	-14.9	2.0	22.2	5.3	+ Tens
Shell-5	-4.0	-14.9	-2.0	-22.2	-43.0	- Comp
Shell-6	-4.4	-14.9	2.2	20.8	3.7	+ Tens
Shell-6	-4.4	-14.9	-2.2	-20.8	-42.3	- Comp
Shell-7	-4.4	-13.5	2.2	17.3	1.6	+ Tens
Shell-7	-4.4	-13.5	-2.2	-17.3	-37.4	- Comp
Shell-9	-4.9	-12.6	0.9	15.8	...	+ Tens
Shell-9	-4.9	-12.6	-0.9	-15.8	-34.2	- Comp
Shell-10	-4.9	-11.1	0.9	12.6	...	+ Tens
Shell-10	-4.9	-11.1	-0.9	-12.6	-29.5	- Comp
Shell-11	-5.5	-10.8	1.0	11.0	...	+ Tens
Shell-11	-5.5	-10.8	-1.0	-11.0	-28.4	- Comp
Shell-12	-5.5	-8.6	1.0	8.1	...	+ Tens
Shell-12	-5.5	-8.6	-1.0	-8.1	-23.3	- Comp
Shell-13	-5.5	-7.1	0.8	5.7	...	+ Tens
Shell-13	-5.5	-7.1	-0.8	-5.7	-19.1	- Comp
Shell-14	-5.5	-6.0	0.8	3.6	...	+ Tens
Shell-14	-5.5	-6.0	-0.8	-3.6	-16.0	- Comp
Shell-15	-4.4	-3.3	0.6	1.7	...	+ Tens
Shell-15	-4.4	-3.3	-0.6	-1.7	-10.1	- Comp
Cone-1	-4.9	-2.6	0.3	1.2	...	+ Tens
Cone-1	-4.9	-2.6	-0.3	-1.2	-8.9	- Comp
Cone-2	-5.4	-3.5	...	1.7	...	+ Tens
Cone-2	-5.4	-3.5	...	-1.7	-10.5	- Comp
Cone-3	-4.3	-4.2	...	1.8	...	+ Tens
Cone-3	-4.3	-4.2	...	-1.8	-10.4	- Comp
Shell-16	-2.8	-5.4	...	1.9	...	+ Tens
Shell-16	-2.8	-5.4	...	-1.9	-10.2	- Comp
Top head	-3.3	-0.2	...	...	...	+ Tens
Top head	-3.3	-0.2	...	...	-3.4	- Comp



Skirt Support Analysis Per EN 13445-3 Section 16.12:

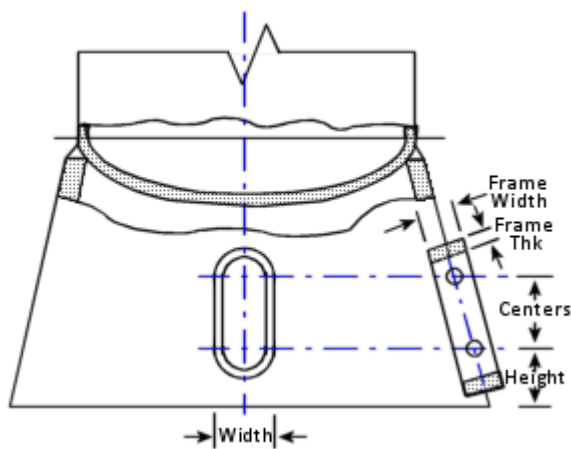
The analysis of this skirt has been conducted referencing the EN 13445-3 design code. It is important to note that the selected design code does not explicitly cover the analysis of this component. Therefore, the results provided herein should be regarded as informational only.

Item: Skirt-6

Total vessel (empty) weight	Fg	8817824.000 N
Head Weight below section 2-2	DeltaFg	226410.906 N
Weight of vessel contents (liquid)	Ff	876307.188 N
Vessel Gross Weight	Fg + Ff	9694131.000 N
Skirt Mean Diameter	Dz	6736.000 mm
Skirt Analysis Thickness (Corroded)	ez	35.000 mm
Skirt Height	H	1000.000 mm
Additional Axial Force at section 1-1	F1	0.000 N
Moment at section 1-1	M1	27367415808.000 N-mm
Applied Moment - Bottom of Skirt	Mb	28331393024.000 N-mm
Weight at Bottom of Skirt	W	9590693.000 N

Structural Type B Skirt per EN 13445 Figure 16.12-2:

Type of Head:		Torispherical
Head Mean Diameter at Skirt Top	Db	6745.000 mm
Head Corroded Thickness	eb	37.000 mm



There are no Access Openings in the Skirt - Plain Skirt Considered

Moment of Inertia	Ixx =	4200816443392.000 mm ⁴
Moment of Inertia	Iyy =	4200816443392.000 mm ⁴
Section Modulus	W =	1247274368.000 mm ³
Section Metal Area	A =	740661.812 mm ²

Analysis of the Section at weakest section Modulus [Z min]:

Radius of Gyration of Skirt at 4-4 [rg]:
 $= \sqrt{\frac{\min(Ixx, Iyy)}{Area}}$
 $= \sqrt{\frac{\min(0.42008E+13, 0.42008E+13)}{740662}}$
 $= 2381.536 \text{ mm}$

Skirt Slenderness Ratio [rg/L]:
 $= rg / L$
 $= 2381.536 / 1000.0$
 $= 2.382$

Correct the Moment [M4]:
 top of skirt is the critical end
 Controlling Moment [M4] = 0.27367E+11 N-mm



Axial Stress at Residual (Weakest) Surface Area [Sigma]:

Downwind Side:

$$= -M / Z - W / A$$

$$= -0.27367E+11 / 4.9891E+10 - 9590693 / 740662$$

$$= -18.434 \text{ MPa}$$

Upwind Side:

$$= +(M) / Z - W / A$$

$$= +.27367E+11 / 4.9891E+10 - 9590693 / 740662$$

$$= -7.466 \text{ MPa}$$

16.12.3.4 Checking at connection areas:

Force at the joint, 16.12-1 [Fzp]:

$$= F1 - Fg - Ff + 4 \cdot M1 / Dz$$

$$= 0.0 - 8817824 - 876307 + 4 \cdot 0.27367E+11 / 6736.0$$

$$= 6550718 \text{ N}$$

Force at the joint, 16.12-2 [Fzq]:

$$= F1 - Fg - Ff - 4 \cdot M1 / Dz$$

$$= 0.0 - 8817824 - 876307 - 4 \cdot 0.27367E+11 / 6736.0$$

$$= -25938978.0 \text{ N}$$

16.12.3.4.1 Membrane stresses:

Membrane stress at 1-1, 16.12-3 & 4:

Stress [σ_{1pm}]:

$$= (Fzp + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (6550718 + 226410.906 + 876307) / (\pi \cdot 6745.0 \cdot 37.0) + 0.347 \cdot 6745.0 / (4 \cdot 37.0)$$

$$= 25.59 \text{ MPa}$$

Stress [σ_{1qm}]:

$$= (Fzq + \Delta Fg + Ff) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (-0.3E+08 + 226410.906 + 876307) / (\pi \cdot 6745.0 \cdot 37.0) + 0.347 \cdot 6745.0 / (4 \cdot 37.0)$$

$$= -15.85 \text{ MPa}$$

Membrane stress at 2-2, 16.12-9:

Stress [σ_{2pm} & σ_{2qm}]:

$$= (Ff + \Delta Fg) / (\pi \cdot Db \cdot eb) + P \cdot Db / (4 \cdot eb)$$

$$= (876307 + 226410.906) / (\pi \cdot 6745.0 \cdot 37.0) + 0.347 \cdot 6745.0 / (4 \cdot 37.0)$$

$$= 17.23 \text{ MPa}$$

Membrane stress at 3-3, 16.12 12 & 13:

Stress [σ_{3pm}]:

$$= Fzp / (\pi \cdot Dz \cdot ez)$$

$$= 6550718 / (\pi \cdot 6736.0 \cdot 35.0)$$

$$= 8.85 \text{ MPa}$$

Stress [σ_{3qm}]:

$$= Fzq / (\pi \cdot Dz \cdot ez)$$

$$= -0.3E+08 / (\pi \cdot 6736.0 \cdot 35.0)$$

$$= -35.02 \text{ MPa}$$

16.12.3.4.2 Bending stresses:

Cosine of angle, 16.12-25 [cos(γ)]:

$$= 1 - 0.5 (Db + eb - Dz + ez) / (r + eb)$$

$$= 1 - 0.5 (6745.0 + 37.0 - 6736.0 + 35.0) / (407.86 + 37.0)$$

$$= 0.909$$

$$\gamma = \arccos(\cos(\gamma)) = \arccos(0.909) = 24.638 \text{ deg}$$

Dimension [a]:

$$= 0.5 \cdot \sqrt{(eb^2 + ez^2 + 2 \cdot eb \cdot ez \cdot \cos(\gamma))}$$

$$= 0.5 \cdot \sqrt{(37.0^2 + 35.0^2 + 2 \cdot 37.0 \cdot 35.0 \cdot 0.909)}$$

$$= 35.172 \text{ mm}$$



Bending moment at verification points p and q [Mp & Mq]:

$$\begin{aligned} M_p &= a \cdot F_{zp} = 35.17 \cdot 6550717.5 = 230.4936 \times 10^6 \text{ N-mm} \\ M_q &= a \cdot F_{zq} = 35.17 \cdot -25938978.0 = -0.9 \times 10^6 \text{ N-mm} \end{aligned}$$

Correction factor [C]:

$$\begin{aligned} &= 0.63 - 0.057 \left(\frac{e_b}{e_z} \right)^2 \\ &= 0.63 - 0.057 \left(\frac{37.0}{35.0} \right)^2 \\ &= 0.566 \end{aligned}$$

Bending stresses in sections 1-1 to 3-3 at the outer surface (o), 16.12-26 to 29:

Stress [$\sigma_{1pb(o)}$ & $\sigma_{2pb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_b \cdot e_b^2) \\ &= 0.566 \cdot 6 \cdot 0.23049 \times 10^9 / (\pi \cdot 6745.0 \cdot 37.0^2) \\ &= 26.99 \text{ MPa} \end{aligned}$$

Stress [$\sigma_{1qb(o)}$ & $\sigma_{2qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_b \cdot e_b^2) \\ &= 0.566 \cdot 6 \cdot -0.9 \times 10^9 / (\pi \cdot 6745.0 \cdot 37.0^2) \\ &= -106.87 \text{ MPa} \end{aligned}$$

Stress [$\sigma_{3pb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_p / (\pi \cdot D_z \cdot e_z^2) \\ &= 0.566 \cdot 6 \cdot 0.23049 \times 10^9 / (\pi \cdot 6736.0 \cdot 35.0^2) \\ &= 30.20 \text{ MPa} \end{aligned}$$

Stress [$\sigma_{3qb(o)}$]:

$$\begin{aligned} &= C \cdot 6 \cdot M_q / (\pi \cdot D_z \cdot e_z^2) \\ &= 0.566 \cdot 6 \cdot -0.9 \times 10^9 / (\pi \cdot 6736.0 \cdot 35.0^2) \\ &= -119.59 \text{ MPa} \end{aligned}$$

Geometry variables, 16.12-32 to 36:

$$\begin{aligned} e_b/D_b &= 1.457/265.551 = 0.0055 \\ y &= 125 \cdot e_b/D_b = 125 \cdot 37.0/6745.0 = 0.6857 \end{aligned}$$

Kloepper (torispherical) type head:

Geometric value [α]:

$$\begin{aligned} &= 6.37181 \cdot 2.71828 - 16.1 \cdot y + 3.6366 \cdot 2.71828 - 1.61536 \cdot y + 6.6736 \\ &= 6.37181 \cdot 2.71828 - 16.1 \cdot 0.686 + 3.6366 \cdot 2.71828 - 1.61536 \cdot 0.686 + 6.6736 \\ &= 7.875 \end{aligned}$$

Bending stress in sections 1-1 and 2-2 Equation 16.12-31 [$\sigma_{1b(p)}$ & $\sigma_{2b(p)}$]:

$$\begin{aligned} &= (P + Ph) \cdot D_b \left(\left(\frac{y}{y_A} \right) \cdot \alpha - 1 \right) / (4 \cdot e_b) \\ &= (0.347 + 0.0) \cdot 6745.0 \left(\left(\frac{24.64}{40.0} \right) \cdot 7.875 - 1 \right) / (4 \cdot 37.0) \\ &= 60.94 \text{ MPa} \end{aligned}$$

16.12.3.4.3 Total stresses and strength conditions:

Structural shape B and C:

Stress [σ_{1pitot}]:

$$\begin{aligned} &= \sigma_{1pm} - \sigma_{1pb(o)} + \sigma_{1b(p)} \\ &= 25.588 - 26.989 + 60.938 \\ &= 59.54 \text{ MPa} \end{aligned}$$

Stress [σ_{1potot}]:

$$\begin{aligned} &= \sigma_{1pm} + \sigma_{1pb(o)} - \sigma_{1b(p)} \\ &= 25.588 + 26.989 - 60.938 \\ &= -8.36 \text{ MPa} \end{aligned}$$

Stress [σ_{1qitot}]:

$$\begin{aligned} &= \sigma_{1qm} - \sigma_{1qb(o)} + \sigma_{1b(p)} \\ &= -15.855 - 106.868 + 60.938 \\ &= 151.95 \text{ MPa} \end{aligned}$$

Stress [σ_{1qotot}]:

$$\begin{aligned} &= \sigma_{1qm} + \sigma_{1qb(o)} - \sigma_{1b(p)} \\ &= -15.855 + 106.868 - 60.938 \end{aligned}$$



= -183.66 MPa

Stress [σ_{2pitot}]:

$$\begin{aligned} &= \sigma_{2pm} + \sigma_{2pb(o)} + \sigma_{2b(p)} \\ &= 17.232 + 26.989 + 60.938 \\ &= 105.16 \text{ MPa} \end{aligned}$$

Stress [σ_{2potot}]:

$$\begin{aligned} &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 17.232 - 26.989 - 60.938 \\ &= -70.69 \text{ MPa} \end{aligned}$$

Stress [σ_{2qitot}]:

$$\begin{aligned} &= \sigma_{2qm} + \sigma_{2qb(o)} + \sigma_{2b(p)} \\ &= 17.232 + -106.868 + 60.938 \\ &= -28.70 \text{ MPa} \end{aligned}$$

Stress [σ_{2qotot}]:

$$\begin{aligned} &= \sigma_{2pm} - \sigma_{2pb(o)} - \sigma_{2b(p)} \\ &= 17.232 - -106.868 - 60.938 \\ &= 63.16 \text{ MPa} \end{aligned}$$

Stress [σ_{3pitot}]:

$$\begin{aligned} &= \sigma_{3pm} - \sigma_{3pb} \\ &= 8.845 - 30.202 \\ &= -21.36 \text{ MPa} \end{aligned}$$

Stress [σ_{3potot}]:

$$\begin{aligned} &= \sigma_{3pm} + \sigma_{3pb} \\ &= 8.845 + 30.202 \\ &= 39.05 \text{ MPa} \end{aligned}$$

Stress [σ_{3qitot}]:

$$\begin{aligned} &= \sigma_{3qm} - \sigma_{3qb} \\ &= -35.024 - -119.59 \\ &= 84.57 \text{ MPa} \end{aligned}$$

Stress [σ_{3qotot}]:

$$\begin{aligned} &= \sigma_{3qm} + \sigma_{3qb} \\ &= -35.024 + -119.59 \\ &= -154.61 \text{ MPa} \end{aligned}$$

Allowable stresses:

For σ_{1pitot} and σ_{1potot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1pm} / f)^2 \\ &= 117.998(3 - (1 / 1.5)) \cdot (25.588 / 117.998)^2 \\ &= 350.29 \text{ MPa} \end{aligned}$$

For σ_{1qitot} and σ_{1qotot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5)) \cdot (\sigma_{1qm} / f)^2 \\ &= 117.998(3 - (1 / 1.5)) \cdot (-15.855 / 117.998)^2 \\ &= 352.57 \text{ MPa} \end{aligned}$$

For σ_{2pitot} and σ_{2potot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2pm} / f)^2 \\ &= 117.998(3 - (1 / 1.5)) \cdot (17.232 / 117.998)^2 \\ &= 352.32 \text{ MPa} \end{aligned}$$

For σ_{2qitot} and σ_{2qotot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5)) \cdot (\sigma_{2qm} / f)^2 \\ &= 117.998(3 - (1 / 1.5)) \cdot (17.232 / 117.998)^2 \\ &= 352.32 \text{ MPa} \end{aligned}$$

For σ_{3pitot} & σ_{3potot} , the stress must not exceed:

$$\begin{aligned} &= fs(3 - (1 / 1.5)) \cdot (\sigma_{3pm} / f)^2 \\ &= 117.998(3 - (1 / 1.5)) \cdot (8.845 / 117.998)^2 \\ &= 353.55 \text{ MPa} \end{aligned}$$

For σ_{3qitot} & σ_{3qotot} , the stress must not exceed:



$$\begin{aligned} &= fs(3 - (1 / 1.5) \cdot (\sigma_{qm} / f)^2) \\ &= 117.998(3 - (1/1.5) \cdot (-35.024/117.998)^2) \\ &= 347.06 \text{ MPa} \end{aligned}$$

Please note: None of the following stresses in the table below are in the region where there is an access opening.

Summary of Stresses: (MPa)

Stress	Actual	Allowed	Result	Equation
Stot1pi	59.54	350.29	Pass	46
Stot1po	-8.36	350.29	Pass	47
Stot1qi	151.95	352.57	Pass	48
Stot1qo	-183.66	352.57	Pass	49
Stot2pi	105.16	352.32	Pass	50
Stot2po	-70.69	352.32	Pass	51
Stot2qi	-28.70	352.32	Pass	52
Stot2qo	63.16	352.32	Pass	53
Stot3pi	-21.36	353.55	Pass	54
Stot3po	39.05	353.55	Pass	55
Stot3qi	84.57	347.06	Pass	56
Stot3qo	-154.61	347.06	Pass	57

The stresses are satisfactory in the region without openings.

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Shop/Field Installation Options :

Platform(s) installed in the Field before being lifted.
 Packing is installed in the Field after being lifted.
 Insulation is installed in the Shop.
 Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Platforms	32599.051 mm
Center of Gravity of the Packing	35969.297 mm
Center of Gravity of the Liquid	14921.867 mm
Center of Gravity of the Insulation	36060.766 mm
Center of Gravity of the Lining	33432.949 mm
Center of Gravity of the Stiffening Rings	34257.023 mm
Center of Gravity of the Nozzles	28779.756 mm
Center of Gravity of the Added Weights (Operating)	43587.883 mm
Center of Gravity of the Added Weights (Empty)	36114.793 mm
Center of Gravity of Bare Shell New and Cold	23205.283 mm
Center of Gravity of Bare Shell Corroded	22962.602 mm
Vessel CG in the Operating Condition	32472.670 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	26890.770 mm
Vessel CG in the Test Condition	33704.914 mm

Rigging Analysis Results:

- Lifting weight based on the user selected calculated erected weight

Total Effective Length of Vessel for this analysis	60511.01 mm
Total vessel weight (No Liquid)	Twt 4424256.50 N
Impact weight multiplication factor	Imp 2.00
Design lifting weight, DWT = Imp * Twt	8848513.00 N
Elevation of the Tailing Lug (bottom)	0.00 mm
Elevation of the Lifting Lug (top)	45000.00 mm
Design Reaction force at the tailing lug	3560883.50 N
Design Reaction force at the lifting lug	5287629.50 N
CG distance during field lifting operation	26890.77 mm
CG distance from tailing lug	26890.77 mm
CG distance from the nearer lifting lug/trunnion	18109.23 mm
Compressive Allowable Increase Factor	1.00

Critical Values: (MPa, mm)

	Max Stress	Elevation	Allowables
Bending	36.37	26350.01	78.75 (UG-23)
Shear	9.96	44950.03	82.60 (0.7*S)

The compressive bending allowable stress is the minimum of the tensile allowable and the compressive allowable times the specified compressive allowable increase factor.

Forces and Moments at selected elevations (not all analysis points shown):

Distance mm	Bending Moment N-mm	Bending Stress MPa	Shear Force N	Shear Stress MPa
0.00	28934.4	0.0	3236892.5	2.7
13300.00	28100456448.0	22.0	1070799.9	1.4
23350.00	30680020992.0	36.1	580450.4	1.1
38350.02	6925266432.0	10.9	2610234.2	6.9
52850.03	2106087424.0	3.0	826941.7	2.0
60461.04	14033600.0	0.1	5106.6	0.0

Unity Check (Actual Stress / Allowable Stress):



Maximum Unity Check is 0.4726 at elevation 29350.0098 mm - Must be ≤ 1

Note: The rigging analysis is performed using a uniformly distributed load.

--- Plot data successfully generated ...----

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Skirt Data :

Skirt Outside Diameter at Base	SOD, D6	6791.0000	mm
Skirt Thickness	STHK	56.0000	mm
Skirt Internal Corrosion Allowance	SCA	1.0000	mm
Skirt External Corrosion Allowance		0.0000	mm
Skirt Material		E250BR	

Basing Input: Type of Geometry: Continuous Top Ring W/Gussets

Thickness of Basing	TBA, ea4	114.0000	mm
Design Temperature of the Basing		38.00	°C
Basing Material		E250BR	
Basing Ope. All. Stress	BASOPE, f4	117.00	MPa
Basing Yield Stress		240.00	MPa
Inside Diameter of Basing	DI, D4	6623.0005	mm
Outside Diameter of Basing	DOU, D5	7223.0000	mm
Nominal Diameter of Bolts	BND, d(b0)	64.0000	mm
Bolt Corrosion Allowance	BCA	1.5000	mm
Bolt Material		SA-36	
Bolt Operating Allowable Stress	SA, fB	284.00	MPa
Number of Bolts	RN, nb	56	
Diameter of Bolt Circle	DC, DBC	7007.0000	mm
Bolt Allowable Shear Stress		170.400	MPa
Ultimate Comp. Strength of Concrete	FPC	41.0	MPa
Allowable Comp. Strength of Concrete	FC	22.5	MPa
Modular ratio Steel/Concrete		9.833	
Thickness of Gusset Plates	TGA, e7	26.0000	mm
Width of Gussets at Top Plate	TWDT	206.0000	mm
Width of Gussets at Base Plate	BWDT	206.0000	mm
Gusset Plate Elastic Modulus	E	201746.0	MPa
Gusset Plate Yield Stress	SY	240.0	MPa
Height of Chair	h1	400.0000	mm
Distance between Gussets	RG, b6	120.0000	mm
Dist. from Bolt Center to Gusset (Rg/2)	CG	60.0000	mm
Number of Gussets per bolt	NG	2	
Thickness of Top Plate or Ring	TTA, e5	58.0000	mm
Radial Width of the Top Plate	TOPWTH, b5	216.0000	mm
Anchor Bolt Hole Dia. in Top Plate	dia,H	90.0000	mm
External Corrosion Allowance	CA	1.0000	mm
Dead Weight of Vessel	DW	6240318.0	N
Operating Weight of Vessel	ROW	10593777.0	N
Earthquake Moment on Basing	EQMOM	42276552704.0	N-mm
Wind Moment on Basing	WIMOM	9575769088.0	N-mm
Percent Bolt Preload	ppl	100.0	
Use AISC A5.2 Increase in Fc and Bolt Stress		No	
Use Allowable Weld Stress per AISC J2.5		No	
Factor for Increase of Allowables	Fact	1.0000	

Results for Brownell and Young Basing Analysis : Analyze Option

Note: This analysis is based on Neutral Axis shift method for Steel on Concrete (or a material with significantly different Young's modulus).

Calculation of Bolt Loads for Multiple Load Conditions:

Bolt Load, Earthquake Operating Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.42277E+11/7007.0) - 10261556) / 56$$

$$= 247545.7 \text{ N}$$

Bolt Load, Windload Empty Case:



$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0.95758E+10/7007.0) - 6240318) / 56$$

$$= -13859.5 \text{ N}$$

Bolt Load, Windload Vortex Shedding Case:

$$= ((4 \cdot M/DC) - W) / nb \quad \text{per Jawad \& Farr, Eq. 12.3}$$

$$= ((4 \cdot 0/7007.0) - 10593777) / 56$$

$$= -189174.6 \text{ N}$$

Governing Bolt Load Condition, Earthquake + Operating Condition:

Area Available in 1 Bolt (corroded) Abss : 2210.1030 mm²
 Area Available in all the Bolts Abss * RN : 123765.7656 mm²

Trial#	k	knew	Cc	Ct	z	j	Ft	Fc	Fs
2	0.711	0.567	2.466	1.479	0.341	0.781	3248533.0	13510089.0	71942.9
4	0.495	0.458	1.988	2.012	0.394	0.785	2531685.0	12793241.0	41205.7
6	0.477	0.468	1.946	2.053	0.398	0.785	2477080.5	12738636.0	39513.3
8	0.472	0.470	1.936	2.063	0.399	0.785	2463526.8	12725083.0	39103.2
10	0.469	0.468	1.928	2.071	0.400	0.785	2453386.0	12714941.0	38798.9
12	0.468	0.468	1.927	2.072	0.400	0.785	2452541.8	12714097.0	38773.6

The Actual Stress in a Single Bolt [Sbolt]:

$$= 2 \cdot Ft / (T1 \cdot Dc \cdot Ct)$$

$$= 2 \cdot 2452542 / (5.622 \cdot 7007.0 \cdot 2.072)$$

$$= 60.104 \text{ MPa, Should be } \leq 284.0 \text{ MPa}$$

Thickness of the Band of Bolting Steel [T1]

$$= RN \cdot \text{Bolt Area} / (\pi \cdot Dc)$$

$$= 56 \cdot 2210.103 / (\pi \cdot 7007.0)$$

$$= 5.622 \text{ mm}$$

Check the Bearing Stress in the Concrete [fc(max)]

$$= fc[(2kd + t3) / (2kd)]$$

$$= 5.386[(2 \cdot 0.4684 \cdot 7007.0 + 300.0) / (2 \cdot 0.4684 \cdot 7007.0)]$$

$$= 5.632 \text{ MPa, Should be } \leq 22.5 \text{ MPa}$$

Values for table 10.3, l = 216.0, b = 196.546, l/b = 1.09898

Maximum Moment per unit width [Mmax]:

$$= \max(Mx, My) = \max(22256.691, 31526.627) = 31526.627 \text{ N}$$

Reqd Thickness of Basing, Brownell & Young Method [T]:

$$= \sqrt{(6 \cdot Mmax / fallow) + Ca}$$

$$= \sqrt{(6 \cdot 31526.627 / 158.4) + 1.0}$$

$$= 35.559 \text{ mm}$$

Nomenclature:

a = (Dc - Ds) / 2	Skirt Distance to Bolt Circle
P = Sa * Abss	Maximum Load on one Bolt
l	Radial Top Plate Width
lavg	Average Gusset Plate Width
g1 = Gamma 1	Constant Term f(b/l)
g2 = Gamma 2	Constant Term f(b/l)
g = Flat distance / 2	Nut 1/2 Dimension (from Tema)
Fb	Allowable Bending Stress

Values for table 10.6, l = 216.0, b = 120.0, b/l = 0.555556

As b/l (0.556) is less than 1, inverting b/l = 1.8.

Moment Term, based on geometry [Mo]:

$$= P / (4 \cdot \pi) [1.3(\ln((2l \sin(\pi \cdot a / l) / (\pi \cdot g))) + 1) - [(0.7 - g2)P / (4\pi)]]$$

$$= 627616 / (4 \cdot \pi) [1.3(\ln((2 \cdot 216.0 \cdot \sin(\pi \cdot 108.0 / 216.0)) / (\pi \cdot 50.0))) + 1] - [(0.7 - 0.04) \cdot 627616 / (4 \cdot \pi)]]$$

$$= 82650.8203 \text{ N}$$

Required Thickness of Continuous Top Ring [Tc]:

$$= \sqrt{(6 \cdot \text{Abs}(Mo) / Fb) + Ca}$$

$$= \sqrt{(6 \cdot \text{Abs}(82650.82) / 160.0) + 1.0}$$

$$= 56.6751 \text{ mm}$$



Required Gusset Plate Thickness [tg]:
= $P / (\text{Stress Term} \cdot l_{avg}) + C_a$
= $627616 / (124.11 \cdot 206.0) + 1.0$
= 25.55 (not less than $9.525 + 1.0$) mm

Bolt spacing [m]:
= $\pi \cdot \text{Bolt Circle Diameter} / \text{number of Bolts}$
= $\pi \cdot 7007.0 / 56 = 393.092$ mm

Req. Skirt Thk. to withstand Local Bending, (Brownell and Young) [t]:
= $1.76(P \cdot a / (m \cdot (h + t_{ba}) \cdot 1.5 \cdot S_{ktope}))^{2/3} \cdot r^{1/3} + C_a$
= $1.76(627616 \cdot 108.0 / (393.09 \cdot 342.0 \cdot 175))^{2/3} \cdot 3395.51 / 3 + C_a$
= $53.464 + 1.0 = 54.463$ mm

Shear Stress in a Single Bolt [tb]:
= $\text{Shear Force} / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $973525 / (2210.103 \cdot 56)$
= 7.9 MPa. Must be less than 170.4 MPa.

Summary of Basing Thickness Calculations

Required Basing Thickness (tension)	35.5588	mm
Actual Basing Thickness as entered by user	114.0000	mm
Required Thickness of Chair Cap	56.6751	mm
Actual Top Ring Thickness as entered by user	58.0000	mm
Required Gusset thickness, + CA	25.5503	mm
Actual Gusset Thickness as entered by user	26.0000	mm
Required Thickness of Skirt for Local Stress	54.4635	mm
Given Thickness of Skirt	56.0000	mm
Required Gusset Height to meet local stress	213.7692	mm

Weld Size Calculations per Steel Plate Engineering Data - Vol. 2

Compute the Weld load at the Skirt/Base Junction [W]:
= $\text{SkirtStress} \cdot (\text{SkirtThickness} - C_A)$
= $30.978 \cdot (56.0 - 1.0)$
= 1703.66 N/mm

Results for Computed Minimum Basing Weld Size [BWeld]:
= $W / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
= $1704 / [(0.4 \cdot 211) \cdot 2 \cdot 0.707]$
= 14.29 mm

Results for Computed Minimum Gusset and Top Plate to Skirt Weld Size

Vertical Plate Load [Wv]:
= $\text{Bolt Load} / (C_{mwth} + 2 \cdot (H_g + T_{ta}))$
= $627616 / (210.0 + 2 \cdot (228.0 + 58.0))$
= 802.578 N/mm

Horizontal Plate Load [Wh]:
= $\text{Bolt Load} \cdot e / (C_{mwth} \cdot (H_g + T_{ta}) + 0.6667 \cdot (H_g + T_{ta})^2)$
= $627616 \cdot 108.0 / (210.0 \cdot (286.0) + 0.6667 \cdot (286.0)^2)$
= 591.517 N/mm

Resultant Weld Load [Wr]:
= $\sqrt{ W_v^2 + W_h^2 }$
= $\sqrt{ 802.58^2 + 591.52^2 }$
= 997.007 N/mm

Results for Computed Min Gusset and Top Plate to Skirt Weld Size [GsWeld]:
= $W_r / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707]$
= $997.01 / [(0.4 \cdot 211) \cdot 2 \cdot 0.707]$
= 8.363 mm

Results for Computed Minimum Gusset to Top Plate Weld Size



Weld Load [Wv]:

$$\begin{aligned} &= \text{Bolt Load} / (2 \cdot \text{TopWth}) \\ &= 627616 / (2 \cdot 206.0) \\ &= 1523.339 \text{ N/mm} \end{aligned}$$

Weld Load [Wh]:

$$\begin{aligned} &= \text{Bolt Load} \cdot e / (2 \cdot \text{Hgt} \cdot \text{TopWth}) \\ &= 627616 \cdot 108.0 / (2 \cdot 286.0 \cdot 206.0) \\ &= 575.247 \text{ N/mm} \end{aligned}$$

Resultant Weld Load [Wr]:

$$\begin{aligned} &= \sqrt{Wv^2 + Wh^2} \\ &= \sqrt{1523.34^2 + 575.25^2} \\ &= 1628.334 \text{ N/mm} \end{aligned}$$

Results for Computed Min Gusset to Top Plate Weld Size [GtpWeld]:

$$\begin{aligned} &= Wr / [(0.4 \cdot \text{Yield}) \cdot 2 \cdot 0.707] \\ &= 1628.33 / [(0.4 \cdot 211) \cdot 2 \cdot 0.707] \\ &= 13.659 \text{ mm} \end{aligned}$$

Note: The calculated weld sizes need not exceed the component thickness framing into the weld. At the same time, the weld must meet a minimum size specification which is 3/16 in. (4.76 mm) or 1/4 in. (6.35 mm), depending on the component thickness.

Summary of Required Weld Sizes:

Required Basing to Skirt Double Fillet Weld Size	14.2904	mm
Required Gusset to Skirt Double Fillet Weld Size	8.3630	mm
Required Top Plate to Skirt Weld Size	13.6586	mm
Required Gusset to Top Plate Double Fillet Weld Size	13.6586	mm

Review notes about nozzle loadings and supports in the Warnings report.

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Conical Reinforcement Calculations, ASME VIII Div. 1, App. 1

Input Values:

Cone-1

Design Internal Pressure, Small End	0.347	MPa
Design Internal Pressure, Large End	0.347	MPa
Temperature for Internal Pressure	160.0	°C
Design External Pressure	0.049	MPa
Temperature for External Pressure	150.0	°C

CONE Properties:

Cone Material		SA-516 60	
Cone Allowable Stress at Temperature	Ss	118.00	MPa
Cone Allowable Stress At Ambient		118.00	MPa
Longitudinal Joint Efficiency of Cone	Ec,E2	1.00	
Circumferential Joint Efficiency of Cone		1.00	
Actual Thickness of Cone	Tc	18.0000	mm
Corrosion Allowance for Cone	Cac	1.0000	mm
Diameter Basis for Cone and Cylinders		ID	
Diameter of Small End of Cone	Ds	3422.989	mm
Diameter of Large End of Cone	Dl	6698.000	mm
Half Apex Angle for Cone		29.01	degrees
Axial Length of Cone	Lc	2961.000	mm

SMALL-END Properties:

Attached Small End Material Specification		SA-516 60	
Operating Allowable Stress	Ss	118.00	MPa
Ambient Allowable Stress		118.00	MPa
Longitudinal Joint Efficiency	Es,E1	1.00	
Actual Thickness	Ts	18.0000	mm
Corrosion Allowance	Cas	1.0000	mm
Axial Length of Attached Element	Ls	902.208	mm

The element type attached to the small end of the cone was not a cylinder. Therefore, the material properties printed above reflect those of the cone.

LARGE-END Properties:

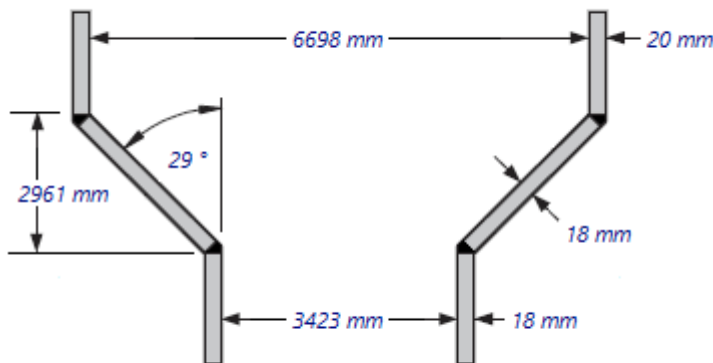
Attached Large End Material Specification		SA-516 60	
Operating Allowable Stress	Sl	118.00	MPa
Ambient Allowable Stress		118.00	MPa
Longitudinal Joint Efficiency	El,E1	1.00	
Actual Thickness	Tl	20.0000	mm
Corrosion Allowance	Cal	1.0000	mm
Axial Length of Attached Element	Ll	1236.753	mm

Type of Reinforcement at Large End of Cone: Bar

Large End Reinforcing/Knuckle Material		SA-516 70	
Large Reinforcing/Knuckle Allowable, Operating		138.00	MPa
Large Reinforcing/Knuckle Allowable, Ambient		138.00	MPa
Location of Reinforcement at Large End of Cone:		Shell	
Radial Width of Reinforcing Bar(Large End)	Rwlb	149.9998	mm
Axial Thickness of Reinforcing Bar	Rtlb	20.0000	mm

Type of Reinforcement at Small End of Cone: None

Force Arl & Ars calculation, cone -> flange	Yes
Force Arl & Ars calculation, cone -> cylinder	Yes



Cone Geometry

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 150 °C

Elastic Modulus of Cone Material	0.195E+06 MPa at 150 °C
Elastic Modulus of Small Cylinder Material	0.195E+06 MPa at 150 °C
Elastic Modulus of Large Cylinder Material	0.195E+06 MPa at 150 °C
Elastic Modulus of Large End Reinforcement	0.195E+06 MPa at 150 °C
Elastic Modulus of Small End Reinforcement	0.195E+06 MPa at 150 °C

Elastic Modulus Data from ASME Section II Part D at 160 °C

Elastic Modulus of Cone Material	0.195E+06 MPa at 160 °C
Elastic Modulus of Small Cylinder Material	0.195E+06 MPa at 160 °C
Elastic Modulus of Large Cylinder Material	0.195E+06 MPa at 160 °C
Elastic Modulus of Large End Reinforcement	0.195E+06 MPa at 160 °C
Elastic Modulus of Small End Reinforcement	0.195E+06 MPa at 160 °C

Forces and Moments on the ends of the Cone:

Axial Force on Small End of Cone	928284.75	N
Axial Force on Large End of Cone	1058161.50	N
Moment on Small End of Cone	615721920.00	N-mm
Moment on Large End of Cone	799754048.00	N-mm

The moments above are computed from the Wind & Seismic loads as well as any user defined moments and include effects due to eccentric masses. The vertical forces include the weight of the elements and details above and below the cone as well as any user defined vertical forces. Vertical forces also include loads cause from vertical accelerations due to earthquake loading.

Note: Both ends of the Cone are Lines of Support

Maximum Centroid Reinforcement Distance Large End	63.2510	mm
Maximum Centroid Reinforcement Distance Small End	42.8671	mm

Note: No ring was found close enough to the small end to be considered.

Reinforcement Calculations for Cone / Large Cylinder:

Required Area of Reinforcement for Large End Under Internal Pressure

Note: Large end is in compression, 1.0 will be used for E1 and E2.

Large end ratio of pressure to allowable stress [P/Ss E1]: 0.00294

Large End Maximum half apex angle w/o reinforcement [δ]:
 $= 326.6 \cdot \sqrt{(P / (Ss \cdot E1))}$ per 1-5 equation 1
 $= 326.6 \cdot \sqrt{(0.347 / (117.998 \cdot 1.0))}$



= 17.717 degrees

Large end actual half apex angle [alpha]: 29.009 degrees

Intermediate Value [k]:

$$\begin{aligned} &= \max(Y / (S_r \cdot E_r), 1) \\ &= \max(22959742 / (138.0 \cdot 194578), 1) \\ &= 1.0000 \end{aligned}$$

where [Y] is:

$$\begin{aligned} &= \text{Large End All. Stress} \cdot \text{Large End Elastic Modulus (Int. temp.)} \\ &= 118.0 \cdot 194578 \\ &= 22959742.0 \text{ (MPa)}^2 \end{aligned}$$

Decay Length, Cone Large End:

$$\begin{aligned} &= 2 \cdot \sqrt{RL(t_s - c_a)} \\ &= 2 \cdot \sqrt{3350.0(20.0 - 1.0)} \\ &= 504.579 \text{ mm} \end{aligned}$$

Required Area of Reinforcement, Large End, Internal [Ar]:

$$\begin{aligned} &= (k \cdot QL \cdot RI / (S_s \cdot E_1)) \cdot (1 - \delta/\alpha) \cdot \tan(\alpha) \\ &= (1.0 \cdot 553.988 \cdot 3350.0 / (118 \cdot 1.0)) \cdot (1 - 17.72/29.01) \cdot 0.5545 \\ &= 3395 \text{ mm}^2 \end{aligned}$$

Force per Length, Cone Large End, tensile side [QL+]:

$$\begin{aligned} &= P(RI/2) + f_1 \\ &= 0.347(3350.0/2) + -27.6 \\ &= 553.988 \text{ N/mm} \end{aligned}$$

Axial Load per Unit Circumference excluding Pressure [f1]:

$$\begin{aligned} &= F_{axial} / (\pi(DI + TI)) + \text{Moment} / (\pi(RI + TI/2)^2) \\ &= -1058162 / (\pi(6700.0 + 20.0)) + 79975E+09 / (\pi(3350.0 + 20.0/2)^2) \\ &= -27.6 \text{ N/mm} \end{aligned}$$

Force per Length, Cone Large End, compressive side [QL-]:

$$\begin{aligned} &= P(RI/2) + f_1 \\ &= 0.347(3350.0/2) + -72.68 \\ &= 508.908 \text{ N/mm} \end{aligned}$$

Axial Load per Unit Circumference excluding Pressure [f1]:

$$\begin{aligned} &= F_{axial} / (\pi(DI + TI)) - \text{Moment} / (\pi(RI + TI/2)^2) \\ &= -1058162 / (\pi(6700.0 + 20.0)) - 0.79975E+09 / (\pi(3350.0 + 20.0/2)^2) \\ &= -72.7 \text{ N/mm} \end{aligned}$$

Area of Reinforcement Available in Large end Shell [Ael]:

$$\begin{aligned} &= (T_s - t) \cdot \sqrt{(RI \cdot T_s) + (T_c - T_r) \cdot (RI \cdot T_c / \cos(\alpha))} \\ &= (19.0 - 9.8758) \cdot \sqrt{(3350.0 \cdot 19.0) + (17.0 - 11.2925) \cdot (3350.0 \cdot 17.0 / 0.8745)} \\ &= 3758.3958 \text{ mm}^2 \end{aligned}$$

Summary of Reinforcement area, Large end, Internal Pressure:

Area of reinforcement required per App. 1-5(1)	3394.9429	mm ²
Area of reinforcement in shell per App. 1-5(2)	3758.3958	mm ²
Area of reinforcement in stiffening ring	2999.9949	mm ²

Required Area of Reinforcement for Large End Under External Pressure

Large end ratio of pressure to allowable stress [P/Ss E1]: 0.00042

Large End Maximum half apex angle w/o reinforcement [δ]:

$$\begin{aligned} &= 104 \cdot \sqrt{P / (S_s \cdot E_1)} \text{ per 1-8 equation 1} \\ &= 104 \cdot \sqrt{0.049 / (117.998 \cdot 1.0)} \\ &= 2.120 \text{ degrees} \end{aligned}$$

Large end actual half apex angle [alpha]: 29.009 degrees

Intermediate Value [k]:

$$\begin{aligned} &= \max(Y / (S_r \cdot E_r), 1) \\ &= \max(23018316 / (138.0 \cdot 195074), 1) \\ &= 1.0000 \end{aligned}$$



where [Y] is:

$$= \text{Large End All. Stress} \cdot \text{Large End Elastic Modulus (Ext. temp.)}$$

$$= 118.0 \cdot 195074$$

$$= 23018316.0 \text{ (MPa)}^2$$

Allowable Stress of Large End Material (Ext. Temp)	118.0 MPa
Allowable Stress of Cone Material (Ext. Temp)	118.0 MPa

Required Area of Reinforcement, Large End, External [Arl]:

$$= (k \cdot QL \cdot RI \cdot \tan(\alpha)) / (Ss \cdot E1) \cdot (1 - 1/4 \cdot ((P \cdot RI - QL) / QL) \cdot (\delta / \alpha))$$

$$= (1.0 \cdot 155.2802 \cdot 3369.0002 \cdot 0.555 / (118 \cdot 1.0)) \cdot$$

$$(1 - 1/4 \cdot ((0.05 \cdot 3369.0 - 155.28) / 155.28) \cdot (2.12 / 29.009))$$

$$= 2456 \text{ mm}^2$$

Force per Length, Cone Large End External Pressure, compressive side [QL-]:

$$= P_{ext}(RI/2) + f1$$

$$= 0.049(3369.0/2) + 72.69$$

$$= 155.280 \text{ N/mm}$$

Axial Load per Unit Circumference excluding Pressure [f1]:

$$= F_{axial} / (\pi(DI + TI)) + \text{Moment} / (\pi(RI - TI/2)^2)$$

$$= +1058162 / (\pi(6738.0 - 20.0)) + 7.9975E+09 / (\pi(3369.0 - 20.0/2)^2)$$

$$= 72.7 \text{ N/mm}$$

Force per Length, Cone Large End External Pressure, tensile side [QL+]:

$$= P_{ext}(RI/2) + f1$$

$$= 0.049(3369.0/2) + 27.58$$

$$= 110.174 \text{ N/mm}$$

Axial Load per Unit Circumference excluding Pressure [f1]:

$$= F_{axial} / (\pi(DI + TI)) - \text{Moment} / (\pi(RI - TI/2)^2)$$

$$= +1058162 / (\pi(6738.0 - 20.0)) - 0.79975E+09 / (\pi(3369.0 - 20.0/2)^2)$$

$$= 27.6 \text{ N/mm}$$

Available Area of Reinforcement, Large End, External [Ael]:

$$= 0.55[\sqrt{(DI \cdot ts)(ts - t)} + \sqrt{(DI \cdot tc / \cos(\alpha))(tc - tr)}] \text{ per 1-8 eqn. 3}$$

$$= 0.55[\sqrt{(6738.0 \cdot 19.0)(19.0 - 7.907)} +$$

$$\sqrt{(6738.0 \cdot 17.0/0.875)(17.0 - 7.836)}]$$

$$= 4007.0237 \text{ mm}^2$$

Summary of Reinforcement Area, Large End, External Pressure:

Area of reinforcement required per App. 1-8(1)	2455.7246	mm ²
Area of reinforcement in shell per App. 1-8(2)	4007.0237	mm ²
Area of reinforcement in stiffening ring	2999.9949	mm ²

Intermediate Results, Large End, External Pressure:

Area Available in Cone, Shell, and Reinforcement [AT]:

$$= LL \cdot ts/2 + Lc \cdot tc/2 + As$$

$$= 1236.753 \cdot 19.0/2 + 3384.598 \cdot 17.0/2 + 2999.995$$

$$= 43518.230 \text{ mm}^2$$

Determine the Meridian Length [M]:

$$= -RL \cdot \tan(\alpha)/2 + LL/2 + (RL^2 - RS^2) / (3 \cdot RL \cdot \tan(\alpha))$$

$$= -3369.0 \cdot \tan(29.01)/2 + 1236.753/2 + (3369.0^2 - 1729.494^2) / (3 \cdot 3369.0 \cdot \tan(29.009))$$

$$= 1175.823 \text{ mm}$$

Force per Unit Length on the Cone/Shell Junction [FL]:

$$= P \cdot M + \tan(\alpha)(\text{Force} / (\pi \cdot D) + \text{Moment} / (\pi \cdot \text{Radius}^2))$$

$$= 0.049 \cdot 1175.8 + \tan(29.009)(1058162 / (\pi \cdot 6738.0) + 0.79975E+09 / (\pi \cdot 3369.0^2))$$

$$= 97.800 \text{ N/mm}$$

Actual Buckling Stress associated with force FL, [B]:

$$= 0.75(FL \cdot DL / ATL)$$

$$= 0.75(97.8 \cdot 6738.0 / 43518.23)$$

$$= 11.358 \text{ MPa}$$

Material Strain associated with this stress [A]: 0.000116



Required Moment of Inertia, Large End, External Pressure [I's]:

$$= A \cdot D^2 \cdot Atl / 10.9$$

$$= 0.000116 \cdot 6738.0005 \cdot 6738.0005 \cdot 43518.23/10.9$$

$$= 21107430.00 \text{ mm}^4$$

Available Moment of Inertia, Large End, External Pressure:

	Area	Centroid	Ar*Ce	Dist	I	Ar*Di ²
Shl	0.374E+04	0.000E+00	0.000E+00	0.424E+01	0.112E+06	0.672E+05
Con	0.383E+04	-0.546E+02	-0.209E+06	0.588E+02	0.392E+07	0.132E+08
Sec	0.300E+04	0.845E+02	0.253E+06	-0.803E+02	0.562E+07	0.193E+08
TOT	0.106E+05		0.448E+05		0.965E+07	0.326E+08

Centroid of Section 0.424E+01 Moment of Inertia 0.423E+08

Summary of Large End Inertia Calculations

Available Moment of Inertia (Large End)	0.4227E+08	mm ⁴
Required Moment of Inertia (Large End)	0.2111E+08	mm ⁴



Because the 1/2 apex angle is less than 30 degrees, the following discontinuity stress evaluation results are for information only.

The following calculations are only required per 1-5(g) and do include external loads due to wind or seismic. These discontinuity stresses are computed at the shell/cone junction and do not include effects of local stiffening from a junction ring.

Results for Discontinuity Stresses per Bednar p. 236 2nd Edition:

Stress Type	Stress	Allowable	Location
Tensile Stress	264.10	387.60	Large Cyl. Long.
Compres. Stress	-205.60	-387.60	Large Cyl. Long.
Membrane Stress	-84.96	177.00	Large End Tang.
Tensile Stress	330.74	387.60	Cone Longitudinal
Compres. Stress	-255.98	-387.60	Cone Longitudinal
Tensile Stress	-67.89	177.00	Cone Tangential

An asterisk (*) denotes that this stress was not applicable for this combination of loads.

Maximum Allowable Pressure Calculations for Cone to Shell Junction:

Pressure Case	Pressure MPa	Reason for Failure at this Pressure
MAWP	0.523	Thickness due to internal pressure, Cone Large End
MAPnc	0.554	Thickness due to internal pressure, Cone Large End

These pressures were determined by iteration.

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Conical Reinforcement Calculations, ASME VIII Div. 1, App. 1

Input Values:

Cone-2

Design Internal Pressure, Small End	0.347	MPa
Design Internal Pressure, Large End	0.347	MPa
Temperature for Internal Pressure	160.0	°C
Design External Pressure	0.049	MPa
Temperature for External Pressure	150.0	°C

CONE Properties:

Cone Material		SA-516 60	
Cone Allowable Stress at Temperature	Ss	118.00	MPa
Cone Allowable Stress At Ambient		118.00	MPa
Longitudinal Joint Efficiency of Cone	Ec,E2	1.00	
Circumferential Joint Efficiency of Cone		1.00	
Actual Thickness of Cone	Tc	18.0000	mm
Corrosion Allowance for Cone	Cac	1.0000	mm
Diameter Basis for Cone and Cylinders		ID	
Diameter of Small End of Cone	Ds	3422.989	mm
Diameter of Large End of Cone	Dl	6696.836	mm
Half Apex Angle for Cone		29.00	degrees
Axial Length of Cone	Lc	2961.000	mm

SMALL-END Properties:

Attached Small End Material Specification		SA-516 60	
Operating Allowable Stress	Ss	118.00	MPa
Ambient Allowable Stress		118.00	MPa
Longitudinal Joint Efficiency	Es,E1	1.00	
Actual Thickness	Ts	14.0000	mm
Corrosion Allowance	Cas	1.0000	mm
Axial Length of Attached Element	Ls	878.462	mm

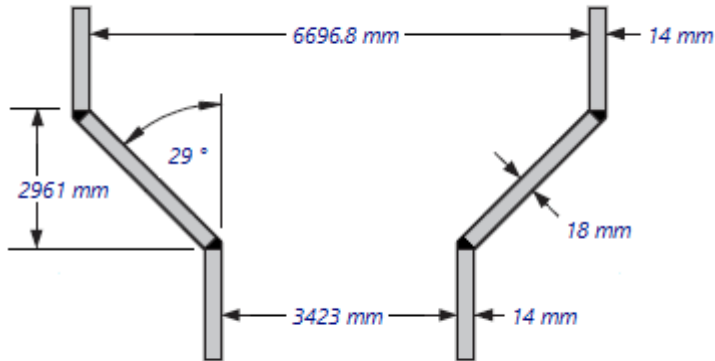
The element type attached to the small end of the cone was not a cylinder. Therefore, the material properties printed above reflect those of the cone.

LARGE-END Properties:

Attached Large End Material Specification		SA-516 60	
Operating Allowable Stress	Sl	118.00	MPa
Ambient Allowable Stress		118.00	MPa
Longitudinal Joint Efficiency	El,E1	1.00	
Actual Thickness	Tl	14.0000	mm
Corrosion Allowance	Cal	1.0000	mm
Axial Length of Attached Element	Ll	884.940	mm

The element type attached to the Large end of the cone was not a cylinder. Therefore, the material properties printed above reflect those of the cone.

Type of Reinforcement at Large End of Cone:	None
Type of Reinforcement at Small End of Cone:	None
Force Arl & Ars calculation, cone -> flange	Yes
Force Arl & Ars calculation, cone -> cylinder	Yes



Cone Geometry

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 150 °C

Elastic Modulus of Cone Material	0.195E+06 MPa at 150 °C
Elastic Modulus of Small Cylinder Material	0.195E+06 MPa at 150 °C
Elastic Modulus of Large Cylinder Material	0.195E+06 MPa at 150 °C
Elastic Modulus of Large End Reinforcement	0.195E+06 MPa at 150 °C
Elastic Modulus of Small End Reinforcement	0.195E+06 MPa at 150 °C

Elastic Modulus Data from ASME Section II Part D at 160 °C

Elastic Modulus of Cone Material	0.195E+06 MPa at 160 °C
Elastic Modulus of Small Cylinder Material	0.195E+06 MPa at 160 °C
Elastic Modulus of Large Cylinder Material	0.195E+06 MPa at 160 °C
Elastic Modulus of Large End Reinforcement	0.195E+06 MPa at 160 °C
Elastic Modulus of Small End Reinforcement	0.195E+06 MPa at 160 °C

Forces and Moments on the ends of the Cone:

Axial Force on Small End of Cone	901267.94	N
Axial Force on Large End of Cone	928284.75	N
Moment on Small End of Cone	438181920.00	N-mm
Moment on Large End of Cone	615721920.00	N-mm

The moments above are computed from the Wind & Seismic loads as well as any user defined moments and include effects due to eccentric masses. The vertical forces include the weight of the elements and details above and below the cone as well as any user defined vertical forces. Vertical forces also include loads cause from vertical accelerations due to earthquake loading.

Note: Both ends of the Cone are Lines of Support

Maximum Centroid Reinforcement Distance Large End	52.2682	mm
Maximum Centroid Reinforcement Distance Small End	37.4428	mm

Note: No ring was found close enough to the small end to be considered.

Maximum Allowable Pressure Calculations for Cone to Shell Junction:

Pressure Case	Pressure MPa	Reason for Failure at this Pressure
MAWP	0.522	Thickness due to internal pressure, Cone Large End
MAPnc	0.553	Thickness due to internal pressure, Cone Large End

These pressures were determined by iteration.



Conical Reinforcement Calculations, ASME VIII Div. 1, App. 1

Input Values:

Cone-3

Design Internal Pressure, Small End	0.347	MPa
Design Internal Pressure, Large End	0.347	MPa
Temperature for Internal Pressure	160.0	°C
Design External Pressure	0.049	MPa
Temperature for External Pressure	150.0	°C

CONE Properties:

Cone Material		SA-516 60	
Cone Allowable Stress at Temperature	Ss	118.00	MPa
Cone Allowable Stress At Ambient		118.00	MPa
Longitudinal Joint Efficiency of Cone	Ec,E2	1.00	
Circumferential Joint Efficiency of Cone		1.00	
Actual Thickness of Cone	Tc	18.0000	mm
Corrosion Allowance for Cone	Cac	1.0000	mm
Diameter Basis for Cone and Cylinders		ID	
Diameter of Small End of Cone	Ds	3422.400	mm
Diameter of Large End of Cone	Dl	6696.836	mm
Half Apex Angle for Cone		28.99	degrees
Axial Length of Cone	Lc	2961.000	mm

SMALL-END Properties:

Attached Small End Material Specification		SA-516 60	
Operating Allowable Stress	Ss	118.00	MPa
Ambient Allowable Stress		118.00	MPa
Longitudinal Joint Efficiency	Es,E1	1.00	
Actual Thickness	Ts	16.0000	mm
Corrosion Allowance	Cas	1.0000	mm
Axial Length of Attached Element	Ls	3299.200	mm

LARGE-END Properties:

Attached Large End Material Specification		SA-516 60	
Operating Allowable Stress	Sl	118.00	MPa
Ambient Allowable Stress		118.00	MPa
Longitudinal Joint Efficiency	El,E1	1.00	
Actual Thickness	Tl	14.0000	mm
Corrosion Allowance	Cal	1.0000	mm
Axial Length of Attached Element	Ll	902.208	mm

The element type attached to the Large end of the cone was not a cylinder. Therefore, the material properties printed above reflect those of the cone.

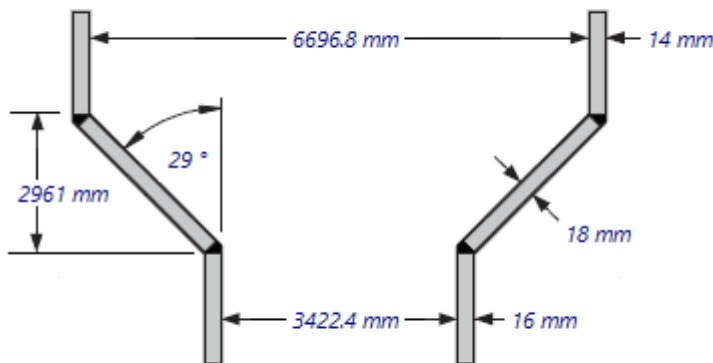
Type of Reinforcement at Large End of Cone: None

Type of Reinforcement at Small End of Cone: Bar

Small End Reinforcing/Knuckle Material		SA-516 70	
Small Reinforcing/Knuckle Allowable, Operating		138.00	MPa
Small Reinforcing/Knuckle Allowable, Ambient		138.00	MPa
Location of Reinforcement at Small End of Cone:		Shell	
Radial Width of Reinforcing Bar(Small End)	Rwsb	250.0000	mm
Axial Thickness of Reinforcing Bar	Rtsb	14.0000	mm

Force Arl & Ars calculation, cone -> flange Yes

Force Arl & Ars calculation, cone -> cylinder Yes



Cone Geometry

Calculated Values:

Elastic Modulus Data from ASME Section II Part D at 150 °C

Elastic Modulus of Cone Material	0.195E+06 MPa at 150 °C
Elastic Modulus of Small Cylinder Material	0.195E+06 MPa at 150 °C
Elastic Modulus of Large Cylinder Material	0.195E+06 MPa at 150 °C
Elastic Modulus of Large End Reinforcement	0.195E+06 MPa at 150 °C
Elastic Modulus of Small End Reinforcement	0.195E+06 MPa at 150 °C

Elastic Modulus Data from ASME Section II Part D at 160 °C

Elastic Modulus of Cone Material	0.195E+06 MPa at 160 °C
Elastic Modulus of Small Cylinder Material	0.195E+06 MPa at 160 °C
Elastic Modulus of Large Cylinder Material	0.195E+06 MPa at 160 °C
Elastic Modulus of Large End Reinforcement	0.195E+06 MPa at 160 °C
Elastic Modulus of Small End Reinforcement	0.195E+06 MPa at 160 °C

Forces and Moments on the ends of the Cone:

Axial Force on Small End of Cone	880115.88	N
Axial Force on Large End of Cone	901267.94	N
Moment on Small End of Cone	270622048.00	N-mm
Moment on Large End of Cone	443747680.00	N-mm

The moments above are computed from the Wind & Seismic loads as well as any user defined moments and include effects due to eccentric masses. The vertical forces include the weight of the elements and details above and below the cone as well as any user defined vertical forces. Vertical forces also include loads cause from vertical accelerations due to earthquake loading.

Note: Both ends of the Cone are Lines of Support

Maximum Centroid Reinforcement Distance Large End	52.2682	mm
Maximum Centroid Reinforcement Distance Small End	40.2399	mm

Reinforcement Calculations for Cone / Small Cylinder:

Required Area of Reinforcement for Small End under Internal Pressure

Small end ratio of pressure to allowable stress [P/Ss E1]: 0.00294

Small End Maximum half apex angle w/o reinforcement [δ]:
 $= 89 \cdot \sqrt{(P / (Ss \cdot E1))}$ per 1-5 equation 6
 $= 89 \cdot \sqrt{(0.347 / (117.998 \cdot 1.0))}$
 $= 4.828$ degrees

Small end actual half apex angle [alpha]: 28.992 degrees



Intermediate Value [k]:

$$= \max(Y / (S_r \cdot E_{rs}), 1)$$

$$= \max(22959742 / (138.0 \cdot 194578), 1)$$

$$= 1.0000$$

where [Y] is:

$$= \text{Small End All. Stress} \cdot \text{Small End Elastic Modulus (Int. temp.)}$$

$$= 118.0 \cdot 194578$$

$$= 22959742.0 \text{ (MPa)}^2$$

Decay Length, Cone Small End:

$$= 1.4 \cdot \sqrt{(R_s(t_s - c_a))}$$

$$= 1.4 \cdot \sqrt{(1712.2(16.0 - 1.0))}$$

$$= 224.363 \text{ mm}$$

Required Area of Reinforcement, Small End, Internal [Ars]:

$$= k \cdot Q_s \cdot R_s / (S_s \cdot E_1) \cdot (1 - \delta / \alpha) \cdot \tan(\alpha) \text{ per 1-5 equation 7}$$

$$= 1.0 \cdot 244.8733 \cdot 1712.2 / (118 \cdot 1.0) \cdot (1 - 4.83 / 28.99) \cdot 0.5541$$

$$= 1641 \text{ mm}^2$$

Force per Length, Cone Small End, tensile side [QS+]:

$$= P(R_s / 2) + f_2$$

$$= 0.347(1712.2 / 2) + -52.38$$

$$= 244.873 \text{ N/mm}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= F_{axial} / (\pi(D_s + T_s)) + \text{Moment} / (\pi(R_s + T_s/2)^2)$$

$$= -880116 / (\pi(3424.4 + 16.0)) + .27062E+09 / (\pi(1712.2 + 16.0/2)^2)$$

$$= -52.4 \text{ N/mm}$$

Force per Length, Cone Small End, compressive side [QS-]:

$$= P(R_s / 2) + f_2$$

$$= 0.347(1712.2 / 2) + -110.58$$

$$= 186.675 \text{ N/mm}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= F_{axial} / (\pi(D_s + T_s)) - \text{Moment} / (\pi(R_s + T_s/2)^2)$$

$$= -880116 / (\pi(3424.4 + 16.0)) - 0.27062E+09 / (\pi(1712.2 + 16.0/2)^2)$$

$$= -110.6 \text{ N/mm}$$

Area of Reinforcement Available in Small End Shell [Aes]:

$$= 0.78[\sqrt{(R_s \cdot t_s)} \cdot (t_s - t) + \sqrt{(R_s \cdot t_c / \cos(\alpha))} \cdot (t_c - t_r)] \text{ per 1-5 eqn. 8}$$

$$= 0.78[\sqrt{(1712.2 \cdot 15.0)} \cdot (15.0 - 5.048) + \sqrt{(1712.2 \cdot 17.0 / 0.875)} \cdot (17.0 - 5.771)]$$

$$= 2841.8701 \text{ mm}^2$$

Summary of Reinforcement Area, Small End, Internal Pressure:

Area of reinforcement required per App. 1-5(7)	1641.1632	mm ²
Area of reinforcement in shell per App. 1-5(8)	2841.8701	mm ²
Area of reinforcement in stiffening ring	3500.0007	mm ²

Required Area of Reinforcement for Small End Under External Pressure

Note: Small end is in compression, 1.0 will be used for E1 and E2.

Allowable Stress of Small End Material (Ext. Temp)	118.0	MPa
Allowable Stress of Cone Material (Ext. Temp)	118.0	MPa

Intermediate Value [k]:

$$= \max(Y / (S_{rs} \cdot E_{rs}), 1)$$

$$= \max(23018316 / (138.0 \cdot 195074), 1)$$

$$= 1.0000$$

where [Y] is:

$$= \text{Small End All. Stress} \cdot \text{Small End Elastic Modulus (Ext. temp.)}$$

$$= 118.0 \cdot 195074$$

$$= 23018316.0 \text{ (MPa)}^2$$

Area of Reinforcement Required in Small End Shell [Ars]:

$$= k \cdot Q_s \cdot R_s \cdot \tan(\alpha) / (S_s \cdot E_1)$$



$$= (1.0 \cdot 152.9511 \cdot 1727.2 \cdot 0.5541 / (118 \cdot 1.0))$$

$$= 1241 \text{ mm}^2$$

Force per Length, Cone Small End External Pressure, compressive side [QS-]:

$$= P_{ext}(R_s/2) + f_2$$

$$= 0.049(1727.2/2) + 110.61$$

$$= 152.951 \text{ N/mm}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= F_{axial} / (\pi(D_s + T_s)) + \text{Moment} / (\pi(R_s - T_s/2)^2)$$

$$= +880116 / (\pi(3454.4 - 16.0)) + .27062E+09 / (\pi(1727.2 - 16.0/2)^2)$$

$$= 110.6 \text{ N/mm}$$

Force per Length, Cone Small End External Pressure, tensile side [QS+]:

$$= P_{ext}(R_s/2) + f_2$$

$$= 0.049(1727.2/2) + 52.34$$

$$= 94.685 \text{ N/mm}$$

Axial Load per Unit Circumference excluding Pressure [f2]:

$$= F_{axial} / (\pi(D_s + T_s)) - \text{Moment} / (\pi(R_s - T_s/2)^2)$$

$$= +880116 / (\pi(3454.4 - 16.0)) - 0.27062E+09 / (\pi(1727.2 - 16.0/2)^2)$$

$$= 52.3 \text{ N/mm}$$

Area of Reinforcement Available in Small End Shell [Aes]:

$$= 0.55[\sqrt{(D_s \cdot t_s)(t_s - t)} + \sqrt{(D_s \cdot t_c / \cos(\alpha))(t_c - t_r)}]$$

$$= 0.55[\sqrt{(3454.4 \cdot 15.0)(15.0 - 8.069)} + \sqrt{(3454.4 \cdot 17.0/0.875)(17.0 - 2.586)}]$$

$$= 2921.8542 \text{ mm}^2$$

Summary of Reinforcement Area, Small End, External Pressure:

Area of reinforcement required per App. 1-8(6)	1240.6869	mm ²
Area of reinforcement in shell per App. 1-8(7)	2921.8542	mm ²
Area of reinforcement in stiffening ring	3500.0007	mm ²

Intermediate Results, Small End, External Pressure:

Area Available in Cone, Shell, and Reinforcement	56995.44	mm ²
Force per Unit Length on Shell / Cone Junction	236.33	N/mm
Actual Buckling Stress associated with this Force	10.74	MPa
Material Strain associated with this stress	0.000110	

Required Moment of Inertia, Small End, External Pressure [I's]:

$$= A \cdot D_s^2 \cdot A_{ts} / 10.9$$

$$= 0.00011 \cdot 3454.3999 \cdot 3454.3999 \cdot 56995.44 / 10.9$$

$$= 6872937.50 \text{ mm}^4$$

Available Moment of Inertia, Small End, External Pressure:

	Area	Centroid	Ar*Ce	Dist	I	Ar*Di ²
Shl	0.188E+04	0.000E+00	0.000E+00	0.702E+02	0.352E+05	0.925E+07
Con	0.243E+04	0.347E+02	0.844E+05	0.355E+02	0.105E+07	0.306E+07
Sec	0.350E+04	0.133E+03	0.464E+06	-0.623E+02	0.182E+08	0.136E+08
TOT	0.781E+04		0.548E+06		0.193E+08	0.259E+08

Centroid of Section 0.702E+02 Moment of Inertia 0.452E+08

Summary of Small End Inertia Calculations

Available Moment of Inertia (Small End)	0.4522E+08	mm ⁴
Required Moment of Inertia (Small End)	0.6873E+07	mm ⁴



Because the 1/2 apex angle is less than 30 degrees, the following discontinuity stress evaluation results are for information only.

The following calculations are only required per 1-5(g) and do include external loads due to wind or seismic. These discontinuity stresses are computed at the shell/cone junction and do not include effects of local stiffening from a junction ring.

Results for Discontinuity Stresses per Bednar p. 236 2nd Edition:

Stress Type	Stress	Allowable	Location
-------------	--------	-----------	----------



Tensile Stress	128.97	387.60	Small Cyl. Long.
Compres. Stress	-96.14	-387.60	Small Cyl. Long.
Membrane Stress	94.89	177.00	Small End Tang.
Tensile Stress	104.19	387.60	Cone Longitudinal
Compres. Stress	-71.07	-387.60	Cone Longitudinal
Tensile Stress	95.24	177.00	Cone Tangential

Maximum Allowable Pressure Calculations for Cone to Shell Junction:

Pressure Case	Pressure MPa	Reason for Failure at this Pressure
MAWP	0.555	Thickness due to internal pressure, Cone Large End
MAPnc	0.600	Thickness due to internal pressure, Cone Large End

These pressures were determined by iteration.

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Input, Nozzle Desc: N1 22"

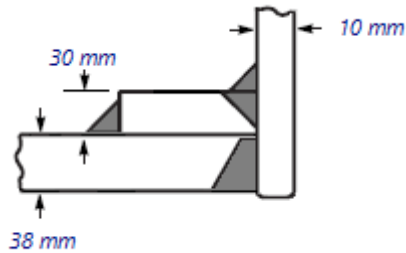
From: 70

Pressure for Reinforcement Calculations	P	0.5032	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C
Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa
Inside Crown Radius of Torispherical Head	L	6781.0000	mm
Inside Knuckle Radius of Torispherical Head	r	406.8600	mm
Head Finished (Minimum) Thickness	t	38.0000	mm
Head Internal Corrosion Allowance	cas	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		-1.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		22.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	600.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	28.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	658.8000	mm
Thickness of Pad	te	30.0000	mm
Weld leg size between Pad and Shell	Wp	20.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	30.0000	mm
Reinforcing Pad Width		50.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	
Piping Exit Angle [North Clockwise reference]		0.0	deg
Bend Radius Multiplier		1.50	
Horizontal Run Length		7000.0	mm
Centerline Distance to Tangent		2500.0	mm

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: N1 22"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	22.000 in.
Actual Thickness Used in Calculation	0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Torispherical Head, tr [Int. Press]
 $= P \cdot L \cdot M / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ Appendix 1-4 (d)
 $= 0.5 \cdot 6782.0 \cdot 1.0 / (2 \cdot 118 \cdot 1.0 - 0.2 \cdot 0.5)$
 $= 14.4674 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.5 \cdot 279.4 / (118 \cdot 1.0 + 0.4 \cdot 0.5)$
 $= 1.1895 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3729 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1077.6000 mm
Parallel to Vessel Wall, opening length	d	538.8000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		55.0000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

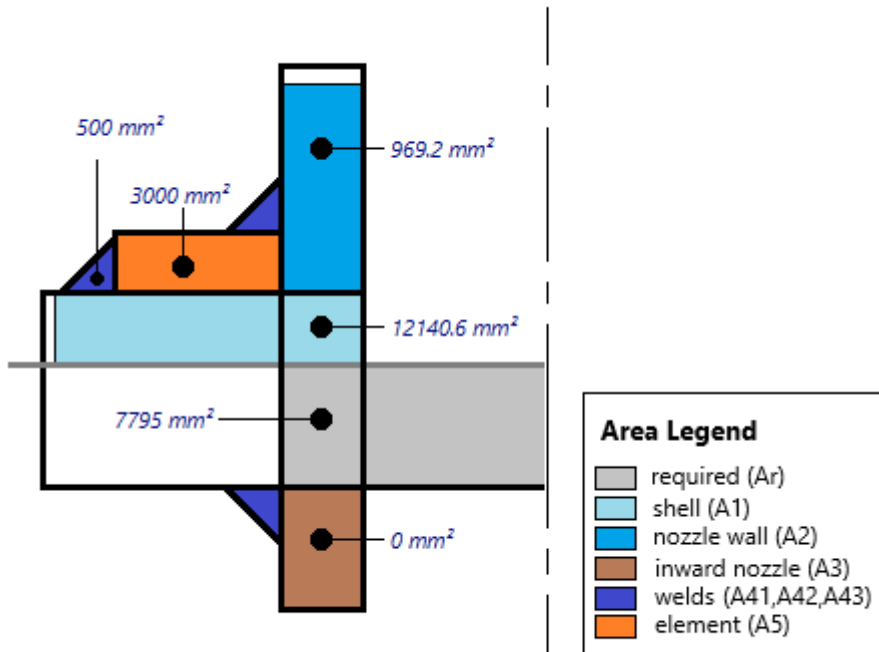


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	7795.037	3640.133	NA
Area in Shell	A1	12140.563	12655.335	NA
Area in Nozzle Wall	A2	969.157	948.984	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		500.000	500.000	NA
Area in Element	A5	3000.000	3000.000	NA
TOTAL AREA AVAILABLE	Atot	16609.721	17104.318	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (538.8 \cdot 14.4674 \cdot 1.0 + 2 \cdot 10.0 \cdot 14.4674 \cdot 1.0 \cdot (1 - 1.0))$
 $= 7795.037 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 538.8 (1.0 \cdot 37.0 - 1.0 \cdot 14.467) - 2 \cdot 10.0$
 $(1.0 \cdot 37.0 - 1.0 \cdot 14.4674) \cdot (1 - 1.0)$
 $= 12140.563 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 55.0) \cdot (10.0 - 1.19) \cdot 1.0$
 $= 969.157 \text{ mm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 20.0^2 \cdot 1.0$



= 500.000 mm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) • fr4
= (658.8 - 558.8) • 30.0 • 1.0
= 3000.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 1.3729 mm
Wall Thickness per UG16(b), tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure trb1 = 26.5993 mm
Wall Thickness tb1 = max(trb1, tr16b) = 26.5993 mm
Wall Thickness, shell/head, external pressure trb2 = 3.4934 mm
Wall Thickness tb2 = max(trb2, tr16b) = 3.4934 mm
Wall Thickness per table UG-45 tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[8.331, max(26.5993, 3.4934)]
= 8.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(1.3729, 8.3312)
= 8.3312 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	30.0,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	265.0 MPa	Passed
Occasional	:	6.7,	Allowable	:	156.9 MPa	Passed
Shear	:	17.9,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 10.0, tr = 1.189, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.119, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-24 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 10.0, tr = 1.189, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.119, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 10.0, tr = 1.189, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.119, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-46 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:C)



Govrn. thk, tg = 30.0, c = 1.0 mm, $E* = 1.0$
 Thickness Ratio = $tr * E*/(tg - c) = 0.391$, Temp. Reduction = 53 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	4 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 10.0, tr = 1.189, c = 0.0 mm, $E* = 1.0$
 Thickness Ratio = $tr * E*/(tg - c) = 0.119$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -48 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.50/1.96 = 0.257$

Weld Size Calculations, Description: N1 22"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 * Wo$ mm
Pad Weld	9.5000 = $0.5 * TminPad$	14.1400 = $0.7 * Wp$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)$
 $= \max(0, (7795.0366 - 12140.5635 + 2 * 10.0 * 1.0 * (1.0 * 37.0 - 14.4674))118)$
 $= \max(0, -459547.78) N$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2*fr2)*Sv$
 $= (969.1575 + 3000.0 + 500.0 - 0.0 * 1.0) * 118$
 $= 527306.25 N$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 * tn * t * fr1)) * Sv$
 $= (969.1575 + 0.0 + 100.0 + (740.0)) * 118$
 $= 213458.56 N$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2*tn*t*fr1))*S$
 $= (969.1575 + 0.0 + 500.0 + 3000.0 + (740.0)) * 118$
 $= 614617.25 N$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) * Dlo * Wo * 0.49 * Snw$
 $= (3.1416/2.0) * 558.8 * 10.0 * 0.49 * 118$
 $= 507478. N$



Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 658.8 \cdot 20.0 \cdot 0.49 \cdot 118$
 $= 1196566. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 274.4) \cdot (10.0 - 0.0) \cdot 0.7 \cdot 118$
 $= 711995. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 558.8 \cdot 30.0 \cdot 0.74 \cdot 138$
 $= 2688880. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 558.8 \cdot (28.0 - 0.0) \cdot 0.74 \cdot 118$
 $= 2145908. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNW}) = (1196566 + 711995) = 1908562 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (507478 + 2688880 + 2145908 + 0) = 5342267 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Spew} + \text{Tngw} + \text{Sinw})$
 $= (1196566 + 2145908 + 0) = 3342475 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 1908561 N	, must exceed W = 0 N	or W1 = 527306 N
Path 2-2 = 5342266 N	, must exceed W = 0 N	or W2 = 213458 N
Path 3-3 = 3342474 N	, must exceed W = 0 N	or W3 = 614617 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 5.7260 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 643.7260 mm

Input Echo, WRC107/537 Item 1, Description: N1 22" :

Diameter Basis for Vessel	Vbasis	ID
Cylindrical or Spherical Vessel	Cylsph	Spherical
Internal Corrosion Allowance	Cas	1.0000 mm
Vessel Diameter	Dv	13562.000 mm
Vessel Thickness	Tv	38.000 mm
Design Temperature	T1	160.0 C
Vessel Material		SA-516 60
Vessel UNS Number		K02100
Vessel Cold S.I. Allowable	Smc	118.00 MPa
Vessel Hot S.I. Allowable	Smh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Attachment Type	Type	Round
WRC107 Attachment Classification	Holsol	Hollow



Diameter Basis for Nozzle	Nbasis	OD	
Corrosion Allowance for Nozzle	Can	0.0000	mm
Nozzle Diameter	Dn	558.800	mm
Nozzle Thickness	Tn	10.000	mm
Nozzle Material		SA-106 B	
Nozzle UNS Number		K03006	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa

Thickness of Reinforcing Pad	Tpad	30.000	mm
Diameter of Reinforcing Pad	Dpad	658.800	mm

Design Internal Pressure	Dp	0.503	MPa
Include Pressure Thrust		No	

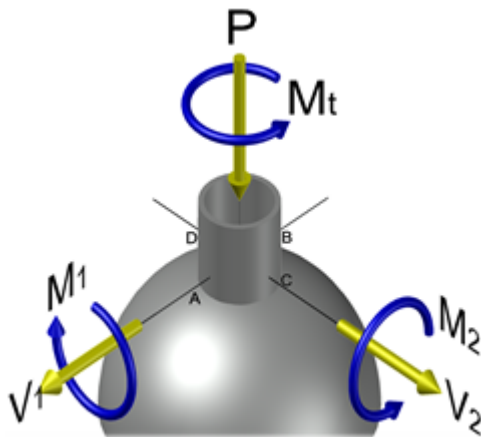
External Forces and Moments in WRC 107/537 Convention:

Radial Load (SUS)	P	-41250.0	N
Longitudinal Shear (SUS)	(V1) V1	27500.0	N
Circumferential Shear (SUS)	(Vc) V2	27500.0	N
Circumferential Moment (SUS)	(Mc) M1	42350060.0	N-mm
Longitudinal Moment (SUS)	(Ml) M2	42350060.0	N-mm
Torsional Moment (SUS)	Mt	60499944.0	N-mm

Use Interactive Control	No
WRC107 Version	Version March 1979

Include Pressure Stress Indices per Div. 2	No
Compute Pressure Stress per WRC-368	No
Local Loads applied at end of Nozzle/Attachment	No

WRC Bulletin 537 provides equations for the dimensionless curves found in bulletin 107. As noted in the foreword to bulletin 537, "537 is equivalent to WRC 107". Where 107 is printed in the results below, "537" can be interchanged with "107".



WRC 107/537 has dimensional limitations that can be found in the bulletin's Foreword and which can also be inferred by review of the non-dimensional curve limits.

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 558.8 + 2 \cdot 1.65 \cdot \sqrt{(6800.5 (38.0 - 1.0))} \\
 &= 2214.133 \text{ mm}
 \end{aligned}$$

WRC 107 Stress Calculation for SUSTained loads:

Radial Load	P	-41250.0	N
Circumferential Shear	(VC) V2	27500.0	N
Longitudinal Shear	(VL) V1	27500.0	N
Circumferential Moment	(MC) M1	42350060.0	N-mm
Longitudinal Moment	(ML) M2	42350060.0	N-mm
Torsional Moment	MT	60499944.0	N-mm

Dimensionless Parameters: U = 0.41 TAU = 27.44 RHO = 6.70

Dimensionless Loads for Spherical Shells at Attachment Junction:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SP 7	0.02918	(A,B,C,D)
M(x) / P	SP 7	0.01600	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SM 7	0.08877	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SM 7	0.04716	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SM 7	0.08877	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SM 7	0.04716	(A,B,C,D)
N(y) * T / P	SP 7	0.35339	(A,B,C,D)
M(y) / P	SP 7	0.10073	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SM 7	0.42385	(A,B,C,D)
M(y) * SQRT(Rm * T) / MC	SM 7	0.36019	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / ML	SM 7	0.42385	(A,B,C,D)
M(y) * SQRT(Rm * T) / ML	SM 7	0.36019	(A,B,C,D)

Stress Concentration Factors: Kn = 1.00, Kb = 1.00

Stresses in the Vessel at the Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Memb. P	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
Rad.	Bend. P	0.9	-0.9	0.9	-0.9	0.9	-0.9	0.9	-0.9
Rad.	Memb. MC	0.0	0.0	0.0	0.0	-1.2	-1.2	1.2	1.2
Rad.	Bend. MC	0.0	0.0	0.0	0.0	-3.9	3.9	3.9	-3.9
Rad.	Memb. ML	-1.2	-1.2	1.2	1.2	0.0	0.0	0.0	0.0
Rad.	Bend. ML	-3.9	3.9	3.9	-3.9	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-4.0	2.1	6.3	-3.3	-4.0	2.1	6.3	-3.3
Tang.	Memb. P	3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2
Tang.	Bend. P	5.6	-5.6	5.6	-5.6	5.6	-5.6	5.6	-5.6
Tang.	Memb. MC	0.0	0.0	0.0	0.0	-5.9	-5.9	5.9	5.9
Tang.	Bend. MC	0.0	0.0	0.0	0.0	-30.2	30.2	30.2	-30.2
Tang.	Memb. ML	-5.9	-5.9	5.9	5.9	0.0	0.0	0.0	0.0
Tang.	Bend. ML	-30.2	30.2	30.2	-30.2	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-27.3	21.9	44.9	-26.6	-27.3	21.9	44.9	-26.6
Shear VC		0.5	0.5	-0.5	-0.5	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.5	-0.5	0.5	0.5
Shear MT		1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8
Tot. Shear		2.3	2.3	1.4	1.4	1.4	1.4	2.3	2.3
Str. Int.		27.5	22.2	44.9	26.6	27.4	22.0	45.0	26.8

Unitless Prm: U = 0.66 TAU = 0.00 (32.44) RHO = 0.00 (3.70)

Dimensionless Loads for Spherical Shells at Pad edge:

Curves read for 1979	Figure	Value	Location
N(x) * T / P	SR 2	0.10587	(A,B,C,D)
M(x) / P	SR 2	0.06018	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / MC	SR 3	0.12028	(A,B,C,D)
M(x) * SQRT(Rm * T) / MC	SR 3	0.16108	(A,B,C,D)
N(x) * T * SQRT(Rm * T) / ML	SR 3	0.12028	(A,B,C,D)
M(x) * SQRT(Rm * T) / ML	SR 3	0.16108	(A,B,C,D)
N(y) * T / P	SR 2	0.03171	(A,B,C,D)
M(y) / P	SR 2	0.01858	(A,B,C,D)
N(y) * T * SQRT(Rm * T) / MC	SR 3	0.03634	(A,B,C,D)

$M(y) * \sqrt{R_m * T} / MC$ SR 3 0.04738 (A,B,C,D)
 $N(y) * T * \sqrt{R_m * T} / ML$ SR 3 0.03634 (A,B,C,D)
 $M(y) * \sqrt{R_m * T} / ML$ SR 3 0.04738 (A,B,C,D)

Stress Concentration Factors: $K_n = 1.00$, $K_b = 1.00$

Stresses in the Vessel at the Edge of Reinforcing Pad (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Memb. P		3.2	3.2	3.2	3.2	3.2	3.2	3.2	3.2
Rad. Bend. P		10.9	-10.9	10.9	-10.9	10.9	-10.9	10.9	-10.9
Rad. Memb. MC		0.0	0.0	0.0	0.0	-7.4	-7.4	7.4	7.4
Rad. Bend. MC		0.0	0.0	0.0	0.0	-59.6	59.6	59.6	-59.6
Rad. Memb. ML		-7.4	-7.4	7.4	7.4	0.0	0.0	0.0	0.0
Rad. Bend. ML		-59.6	59.6	59.6	-59.6	0.0	0.0	0.0	0.0
Tot. Rad. Str.		-52.9	44.5	81.1	-59.9	-52.9	44.5	81.1	-59.9
Tang. Memb. P		1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Tang. Bend. P		3.4	-3.4	3.4	-3.4	3.4	-3.4	3.4	-3.4
Tang. Memb. MC		0.0	0.0	0.0	0.0	-2.2	-2.2	2.2	2.2
Tang. Bend. MC		0.0	0.0	0.0	0.0	-17.5	17.5	17.5	-17.5
Tang. Memb. ML		-2.2	-2.2	2.2	2.2	0.0	0.0	0.0	0.0
Tang. Bend. ML		-17.5	17.5	17.5	-17.5	0.0	0.0	0.0	0.0
Tot. Tang. Str.		-15.5	12.9	24.1	-17.7	-15.5	12.9	24.1	-17.7
Shear VC		0.7	0.7	-0.7	-0.7	0.0	0.0	0.0	0.0
Shear VL		0.0	0.0	0.0	0.0	-0.7	-0.7	0.7	0.7
Shear MT		2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4
Tot. Shear		3.1	3.1	1.7	1.7	1.7	1.7	3.1	3.1
Str. Int.		53.2	44.8	81.1	59.9	53.0	44.6	81.2	60.1

WRC 107/537 Stress Summations:

Vessel Stress Summation at Attachment Junction (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad. Pm (SUS)		25.5	25.5	25.5	25.5	25.5	25.5	25.5	25.5
Rad. Pl (SUS)		-1.0	-1.0	1.5	1.5	-1.0	-1.0	1.5	1.5
Rad. Q (SUS)		-3.1	3.1	4.8	-4.8	-3.1	3.1	4.8	-4.8
Long. Pm (SUS)		25.5	25.5	25.5	25.5	25.5	25.5	25.5	25.5
Long. Pl (SUS)		-2.7	-2.7	9.2	9.2	-2.7	-2.7	9.2	9.2
Long. Q (SUS)		-24.6	24.6	35.7	-35.7	-24.6	24.6	35.7	-35.7
Shear Pm (SUS)		0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear Pl (SUS)		0.5	0.5	-0.5	-0.5	-0.5	-0.5	0.5	0.5
Shear Q (SUS)		1.8	1.8	1.8	1.8	1.8	1.8	1.8	1.8
Pm (SUS)		25.5	25.5	25.5	25.5	25.5	25.5	25.5	25.5
Pm+Pl (SUS)		24.6	24.6	34.7	34.7	24.6	24.6	34.7	34.7
Pm+Pl+Q (Total)		23.7	47.7	70.4	23.4	23.4	47.5	70.5	23.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
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Pm (SUS)	25.47	118.00	Passed
Pm+Pl (SUS)	34.66	177.00	Passed
Pm+Pl+Q (TOTAL)	70.49	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 107/537 Stress Summations:

Vessel Stress Summation at Reinforcing Pad Edge (MPa)

Type of Stress	Load	Stress Intensity Values at							
		Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Rad.	Pm (SUS)	46.1	46.1	46.1	46.1	46.1	46.1	46.1	46.1
Rad.	Pl (SUS)	-4.2	-4.2	10.6	10.6	-4.2	-4.2	10.6	10.6
Rad.	Q (SUS)	-48.7	48.7	70.5	-70.5	-48.7	48.7	70.5	-70.5
Long.	Pm (SUS)	46.1	46.1	46.1	46.1	46.1	46.1	46.1	46.1
Long.	Pl (SUS)	-1.3	-1.3	3.2	3.2	-1.3	-1.3	3.2	3.2
Long.	Q (SUS)	-14.2	14.2	20.9	-20.9	-14.2	14.2	20.9	-20.9
Shear	Pm (SUS)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Shear	Pl (SUS)	0.7	0.7	-0.7	-0.7	-0.7	-0.7	0.7	0.7
Shear	Q (SUS)	2.4	2.4	2.4	2.4	2.4	2.4	2.4	2.4
Pm (SUS)		46.1	46.1	46.1	46.1	46.1	46.1	46.1	46.1
Pm+Pl (SUS)		45.0	45.0	56.8	56.8	45.0	45.0	56.8	56.8
Pm+Pl+Q (Total)		38.0	90.9	127.2	42.3	37.6	90.7	127.4	42.6

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	46.12	118.00	Passed
Pm+Pl (SUS)	56.79	177.00	Passed
Pm+Pl+Q (TOTAL)	127.36	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Input, Nozzle Desc: K1a 2"

From: 80

Pressure for Reinforcement Calculations	P	0.4927	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	1226.1924	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		250.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		40	

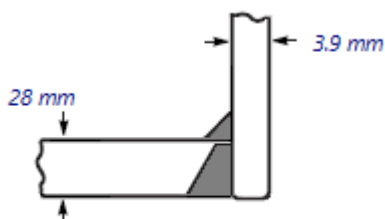
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K1a 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.49 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.49)$
= 14.0398 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.49 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.49)$
= 0.1257 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3250	mm
Parallel to Vessel Wall	Rn+tn+t	57.1625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1a 2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 15.0398	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 15.0398	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(15.0398, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 3.423, tr = 0.126, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: C

Govrn. thk, tg = 3.423, tr = 0.126, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve C	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.49/5.10 = 0.097

Weld Size Calculations, Description: K1a 2"

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.1355 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 57.16 mm

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Input, Nozzle Desc: K2a 2"

From: 80

Pressure for Reinforcement Calculations	P	0.4927	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	1226.1924	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		250.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		40	

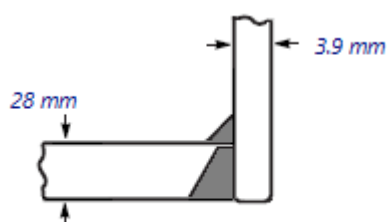
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K2a 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.49 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.49)$
= 14.0398 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.49 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.49)$
= 0.1257 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3250	mm
Parallel to Vessel Wall	Rn+tn+t	57.1625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2a 2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 15.0398	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 15.0398	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(15.0398, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.126, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.126, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.49/5.10 = 0.097

Weld Size Calculations, Description: K2a 2"

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.1355 mm

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Input, Nozzle Desc: K3a 2"

From: 80

Pressure for Reinforcement Calculations	P	0.4927	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	1226.1924	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		250.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		40	

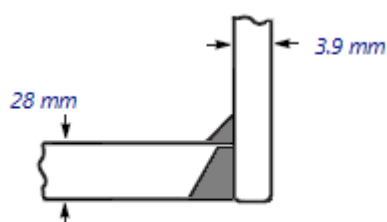
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K3a 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.49 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.49)$
= 14.0398 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.49 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.49)$
= 0.1257 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3250	mm
Parallel to Vessel Wall	Rn+tn+t	57.1625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3a 2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 15.0398	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 15.0398	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(15.0398, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.126, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.126, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.49/5.10 = 0.097

Weld Size Calculations, Description: K3a 2"

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.1355 mm

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Input, Nozzle Desc: K4a 2"

From: 80

Pressure for Reinforcement Calculations	P	0.4927	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	1226.1924	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		250.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		40	

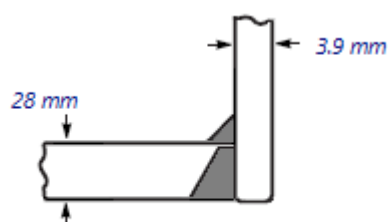
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K4a 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.49 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.49)$
= 14.0398 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.49 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.49)$
= 0.1257 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3250	mm
Parallel to Vessel Wall	Rn+tn+t	57.1625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K4a 2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 15.0398	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 15.0398	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[3.42, max(15.0398, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.126, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.126, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.49/5.10 = 0.097

Weld Size Calculations, Description: K4a 2"

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.1355 mm

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Input, Nozzle Desc: K1b 2"

From: 80

Pressure for Reinforcement Calculations	P	0.4927	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	5099.9990	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		2050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		40	

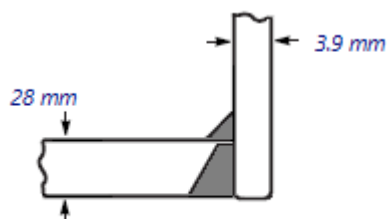
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K1b 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.49 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.49)$
= 14.0398 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.49 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.49)$
= 0.1257 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3250	mm
Parallel to Vessel Wall	Rn+tn+t	57.1625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K1b 2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 15.0398	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 15.0398	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(15.0398, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.126, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.126, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.49/5.10 = 0.097

Weld Size Calculations, Description: K1b 2"

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.1355 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 57.16 mm

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Input, Nozzle Desc: K2b 2"

From: 80

Pressure for Reinforcement Calculations	P	0.4927	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	5099.9990	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent 2050.00 mm

User Entered Minimum Design Metal Temperature -1.00 °C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		40	

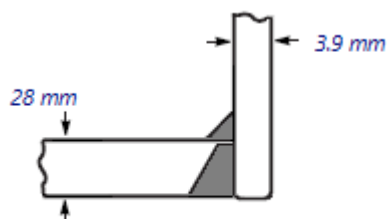
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K2b 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.49 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.49)$
= 14.0398 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.49 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.49)$
= 0.1257 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3250	mm
Parallel to Vessel Wall	Rn+tn+t	57.1625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K2b 2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 15.0398	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 15.0398	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[3.42, max(15.0398, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.126, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.126, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.49/5.10 = 0.097

Weld Size Calculations, Description: K2b 2"

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.1355 mm

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Input, Nozzle Desc: K3b 2"

From: 80

Pressure for Reinforcement Calculations	P	0.4927	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	5099.9990	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		2050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		40	

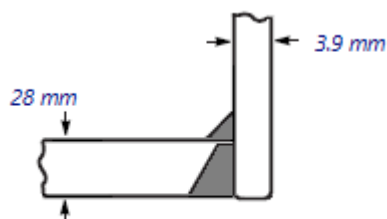
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K3b 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.49 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.49)$
= 14.0398 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.49 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.49)$
= 0.1257 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3250	mm
Parallel to Vessel Wall	Rn+tn+t	57.1625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K3b 2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 15.0398	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 15.0398	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(15.0398, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.126, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.126, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.49/5.10 = 0.097

Weld Size Calculations, Description: K3b 2"

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.1355 mm

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Input, Nozzle Desc: K4b 2"

From: 80

Pressure for Reinforcement Calculations	P	0.4927	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	5099.9990	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		2050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		40	

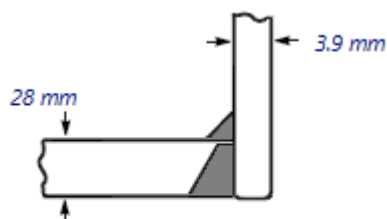
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K4b 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.49 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.49)$
= 14.0398 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.49 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.49)$
= 0.1257 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3250	mm
Parallel to Vessel Wall	Rn+tn+t	57.1625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K4b 2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 15.0398	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 15.0398	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[3.42, max(15.0398, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.126, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.126, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.49/5.10 = 0.097

Weld Size Calculations, Description: K4b 2"

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.1355 mm

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Input, Nozzle Desc: N10 2"

From: 80

Pressure for Reinforcement Calculations	P	0.4927	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	1226.1924	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		250.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		300.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		40	

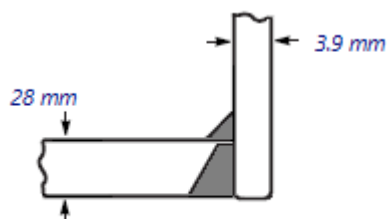
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: N10 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.49 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.49)$
= 14.0398 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.49 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.49)$
= 0.1257 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3250	mm
Parallel to Vessel Wall	Rn+tn+t	57.1625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N10 2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 15.0398	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 15.0398	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(15.0398, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.126, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.126, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.49/5.10 = 0.097

Weld Size Calculations, Description: N10 2"

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.1355 mm

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Input, Nozzle Desc: N14 4"

From: 80

Pressure for Reinforcement Calculations	P	0.4927	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	1226.1924	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		750.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		300.00	deg
Diameter		4.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

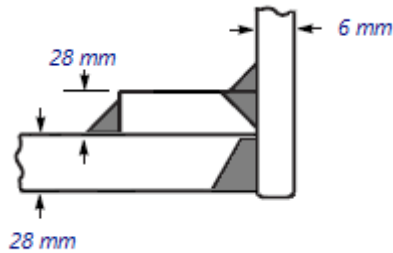
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	26.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	354.3000	mm
Thickness of Pad	te	28.0000	mm
Weld leg size between Pad and Shell	Wp	24.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm
Reinforcing Pad Width		120.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: N14 4"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.237 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.49 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.49)$
= 14.0398 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.49 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 0.49)$
= 0.2382 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.3417 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	204.5208 mm
Parallel to Vessel Wall, opening length	d	102.2604 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		43.0495 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 118.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 118.0)$
= 1.000

Weld Strength Reduction Factor [fr4]:
= $\min(1, S_p / S_v)$
= $\min(1, 138.0 / 118.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

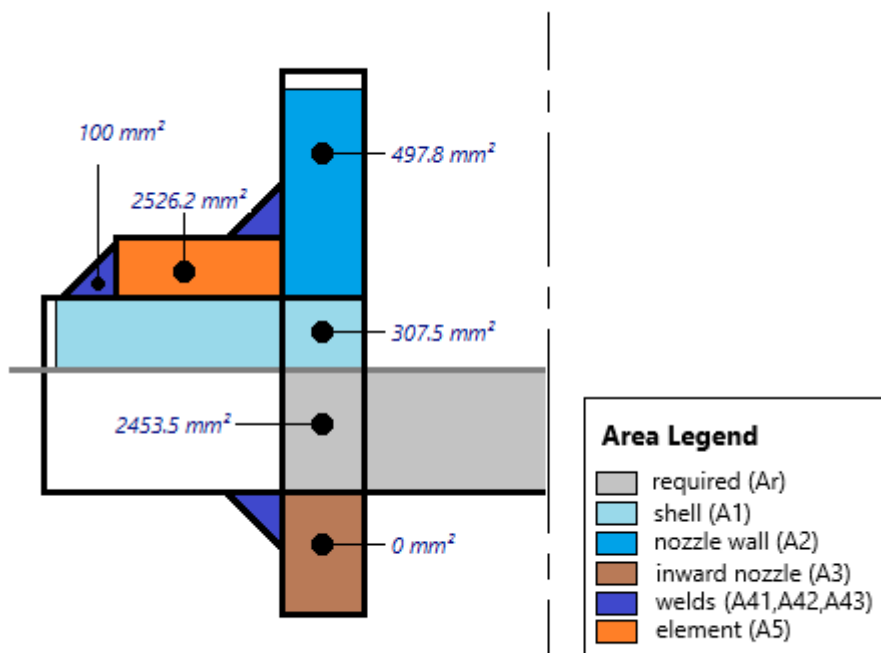


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	1435.717	403.538	NA
Area in Shell	A1	1325.314	1953.955	NA
Area in Nozzle Wall	A2	497.788	488.881	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	2526.182	2526.182	NA
TOTAL AREA AVAILABLE	Atot	4449.284	5069.018	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (102.2604 \cdot 14.0398 \cdot 1.0 + 2 \cdot 6.0198 \cdot 14.0398 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 1435.717 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 102.26(1.0 \cdot 27.0 - 1.0 \cdot 14.04) - 2 \cdot 6.02$$

$$(1.0 \cdot 27.0 - 1.0 \cdot 14.0398) \cdot (1 - 1.0)$$

$$= 1325.314 \text{ mm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 43.05) \cdot (6.02 - 0.24) \cdot 1.0$$

$$= 497.788 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0$$



= 100.000 mm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) • fr4
= (204.5208 - 114.3) • 28.0 • 1.0
= 2526.182 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 0.3417 mm
Wall Thickness per UG16(b), tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure trb1 = 15.0398 mm
Wall Thickness tb1 = max(trb1, tr16b) = 15.0398 mm
Wall Thickness, shell/head, external pressure trb2 = 2.3941 mm
Wall Thickness tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45 tb3 = 5.2578 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[5.258, max(15.0398, 2.5)]
= 5.2578 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(0.3417, 5.2578)
= 5.2578 mm

Available Nozzle Neck Thickness = 0.875 * 6.020 = 5.267 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	35.9,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	259.1 MPa	Passed
Occasional	:	2.0,	Allowable	:	156.9 MPa	Passed
Shear	:	27.5,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 5.267, tr = 0.238, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.045, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 5.267, tr = 0.238, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.045, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 5.267, tr = 0.238, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.045, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:C)

Govrn. thk, tg = 28.0, tr = 14.04, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.52, Temp. Reduction = 30 °C



Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -28 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 5.267, tr = 0.238, c = 0.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.045, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -28 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -28 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 0.49/5.10 = 0.097

Weld Size Calculations, Description: N14 4"

Intermediate Calc. for nozzle/shell Welds Tmin 6.0198 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.2139 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	16.9680 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$$

$$= \max(0, (1435.7168 - 1325.3141 + 2 \cdot 6.0198 \cdot 1.0 \cdot (1.0 \cdot 27.0 - 14.0398)) \cdot 118)$$

$$= 31436.48 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (497.7876 + 2526.1824 + 100.0 - 0.0 \cdot 1.0) \cdot 118$$

$$= 368590.47 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (497.7876 + 0.0 + 100.0 + (325.0692)) \cdot 118$$

$$= 108885.88 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (497.7876 + 0.0 + 100.0 + 2526.1824 + (325.0692)) \cdot 118$$

$$= 406944.66 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 103802. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 354.3 \cdot 24.0 \cdot 0.49 \cdot 118$$

$$= 772210. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 54.1401) \cdot (6.0198 - 0.0) \cdot 0.7 \cdot 118$$

$$= 84566. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 114.3 \cdot 28.0 \cdot 0.74 \cdot 138$$

$$= 513332. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 114.3 \cdot (26.0 - 0.0) \cdot 0.74 \cdot 118$$

$$= 407583. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (772210 + 84566) = 856776 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (103802 + 513332 + 407583 + 0) = 1024717 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (772210 + 407583 + 0) = 1179794 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 856776 N	, must exceed W = 31436 N	or W1 = 368590 N
Path 2-2 = 1024717 N	, must exceed W = 31436 N	or W2 = 108885 N
Path 3-3 = 1179793 N	, must exceed W = 31436 N	or W3 = 406944 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.4872 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.4870 mm

Input Echo, WRC297 Item 1, Description: N14 4" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	6706.000	mm
Shell Thickness	28.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 60	
Vessel Cold S.I. Allowable	Smc	118.00 MPa
Vessel Hot S.I. Allowable	Smh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	114.300	mm
Nozzle Thickness	6.0198	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa



See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad 354.300 mm
 Thickness of Reinforcing Pad 28.0000 mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	0.49	MPa
Radial Load	P	-7500.00	N
Circumferential Shear	Vc	5000.00	N
Longitudinal Shear	Vl	5000.00	N
Circumferential Moment	Mc	0.140E+07	N-mm
Longitudinal Moment	Ml	0.140E+07	N-mm
Torsional Moment	Mt	0.200E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, D_n/T_v , (4.082) is < 5
 check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{(3381.5(28.0 - 1.0))}$$

$$= 1111.427 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6763.000 mm
Nozzle Outside Diameter	(d) =	114.300 mm
Vessel Thickness used	(T) =	55.000 mm
Nozzle Thickness used	(t) =	6.020 mm
T / t	=	9.137
d / t	=	18.987

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.187$$

Nr/P	=	0.122
Mr/P	=	0.074
M0/P	=	0.397
N0/P	=	0.361
MrD/Mc	=	0.084
NrDL/Mc	=	0.087
M0d/Mc	=	0.334
N0DL/Mc	=	-0.155
MrD/Ml	=	0.085
NrDL/Ml	=	0.088
M0D/Ml	=	0.330
N0DL/Ml	=	-0.143

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	0	0	0	0
Outplane Bending (Ml)	-8	8	8	-8
Normal Pressure Stress	29	30	29	30
Outplane Stress Summary	29	33	44	16



Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-2	2	2	-2
Inplane Pressure Stress	14	14	14	14
Inplane Stress Summary	13	15	18	12

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	29	34	44	17

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	0	0	0	0
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-8	8	8	-8
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	14	14	14	14
Outplane Stress Summary	14	18	29	1

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	1	-1	1	-1
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-2	2	2	-2
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	29	30	29	30
Inplane Stress Summary	28	31	33	27

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0



Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	2	2
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	28	31	34	27

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	5	4	5
Hoop Stress Summary	6	6	5	5

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	3	3	3	3
Axial Bending (P)	16	-16	16	-16
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-26	-26	26	26
Axial Bending (ML)	-83	83	83	-83
Axial Pressure Stress	2	2	2	2
Axial Stress Summary	-87	46	131	-66

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	4	4	-4	-4
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	16	16	16	16
Shear Stress Summary	20	20	11	11

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	102	57	132	75

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	0	0	0	0
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	4	5	4	5



Hoop Stress Summary	6	6	5	5
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	3	3	3	3
Axial Bending (P)	16	-16	16	-16
Axial Membrane (Mc)	-26	-26	26	26
Axial Bending (Mc)	-82	82	82	-82
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	2	2	2	2

Axial Stress Summary	-86	45	130	-65
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-4	-4	4	4
Shear due to Torsion	16	16	16	16

Shear Stress Summary	11	11	20	20
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	95	48	134	82
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 114.3 + 2 \cdot 1.65 \cdot \sqrt{3367.5(28.0 - 1.0)}$$

$$= 1109.361 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6735.000 mm
Nozzle Outside Diameter	(d) =	354.300 mm
Vessel Thickness used	(T) =	27.000 mm
Nozzle Thickness used	(t) =	126.020 mm
T / t	=	0.214
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.831$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.117
Mr/P	=	0.254
M0/P	=	0.074
N0/P	=	0.128
MrD/Mc	=	0.314
NrDL/Mc	=	0.063
M0d/Mc	=	0.096
N0DL/Mc	=	0.036
MrD/Ml	=	0.219
NrDL/Ml	=	0.090
M0D/Ml	=	0.069
N0DL/Ml	=	0.098

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Stresses Normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-2	2	2	-2
Normal Pressure Stress	60	61	60	61
Outplane Stress Summary	64	59	69	56

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-7	7	7	-7
Inplane Pressure Stress	30	30	30	30
Inplane Stress Summary	39	22	54	9

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	64	59	69	56

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-3	3	3	-3
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	30	30	30	30
Outplane Stress Summary	33	30	39	24

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-10	10	10	-10

Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	60	61	60	61
Inplane Stress Summary	67	56	88	37

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	0	0

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	67	56	88	37

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	29	30	29	30	29	30	29	30
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	-3	3	13	-13	-1	1	3	-3
Long. Pm (SUS)	14	14	14	14	14	14	14	14
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	-1	1	3	-3	-3	3	13	-13
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	29.0	30.0	29.0	30.0	29.0	30.0	29.0	30.0
Pm+Pl (SUS)	29.0	30.0	29.0	30.0	29.0	30.0	29.0	30.0
Pm+Pl+Q (Total)	26.1	33.1	42.0	17.2	28.1	31.1	32.2	27.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	30.00	118.00	Passed
Pm+Pl (SUS)	30.00	177.00	Passed
Pm+Pl+Q (TOTAL)	42.04	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl

Circ. Pm (SUS)	60	61	60	61	60	61	60	61
Circ. Pl (SUS)	1	1	1	1	1	1	1	1
Circ. Q (SUS)	2	-2	6	-6	5	-5	25	-25
Long. Pm (SUS)	30	30	30	30	30	30	30	30
Long. Pl (SUS)	1	1	1	1	1	1	1	1
Long. Q (SUS)	8	-8	22	-22	1	-1	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	60.0	61.0	60.0	61.0	60.0	61.0	60.0	61.0
Pm+Pl (SUS)	61.0	62.0	61.0	62.0	61.0	62.0	61.0	62.0
Pm+Pl+Q (Total)	63.0	60.0	67.0	56.0	66.0	57.0	86.0	37.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	61.00	118.00	Passed
Pm+Pl (SUS)	62.00	177.00	Passed
Pm+Pl+Q (TOTAL)	86.00	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	4	5	4	5	4	5	4	5
Circ. Pl (SUS)	0	0	0	0	0	0	0	0
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	2	2	2	2	2	2	2	2
Long. Pl (SUS)	-23	-23	29	29	-23	-23	29	29
Long. Q (SUS)	-67	67	99	-99	-66	66	98	-98
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	4	4	-4	-4	-4	-4	4	4
Shear Q (SUS)	16	16	16	16	16	16	16	16
Pm (SUS)	4.0	5.0	4.0	5.0	4.0	5.0	4.0	5.0
Pm+Pl (SUS)	26.2	27.2	31.6	31.6	26.2	27.2	31.6	31.6
Pm+Pl+Q (Total)	100.3	57.3	131.1	76.8	94.1	48.3	132.1	82.4

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	5.00	118.00	Passed
Pm+Pl (SUS)	31.60	177.00	Passed
Pm+Pl+Q (TOTAL)	132.12	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N13A 8"

From: 80

Pressure for Reinforcement Calculations	P	0.4927	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	5099.9990	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		2550.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		240.00	deg
Diameter		8.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

Flange Material		SA-105	
Flange Type		Socket Weld	

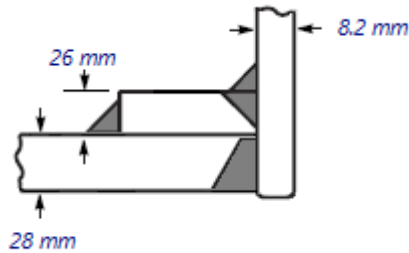
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	379.0750	mm
Thickness of Pad	te	26.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	26.0000	mm
Reinforcing Pad Width		80.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: N13A 8"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	8.625 in.
Actual Thickness Used in Calculation	0.322 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.49 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.49)$
 $= 14.0398 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.49 \cdot 109.5375 / (118 \cdot 1.0 + 0.4 \cdot 0.49)$
 $= 0.4566 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5022 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	405.4348 mm
Parallel to Vessel Wall, opening length	d	202.7174 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		46.4470 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

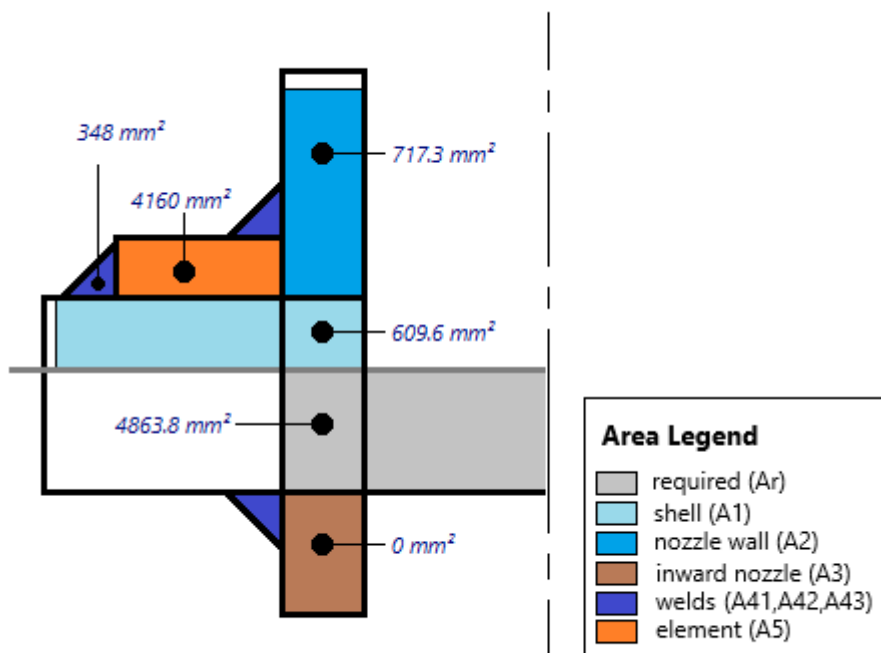


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	2846.114	1453.795	NA
Area in Shell	A1	2627.256	2565.779	NA
Area in Nozzle Wall	A2	717.346	713.113	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	348.047	348.047	NA
Area in Element	A5	4160.000	4160.000	NA
TOTAL AREA AVAILABLE	Atot	7852.649	7786.940	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (202.7174 \cdot 14.0398 \cdot 1.0 + 2 \cdot 8.1788 \cdot 14.0398 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 2846.114 \text{ mm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 202.717 (1.0 \cdot 27.0 - 1.0 \cdot 14.04) - 2 \cdot 8.179 \\
 &\quad (1.0 \cdot 27.0 - 1.0 \cdot 14.0398) \cdot (1 - 1.0) \\
 &= 2627.256 \text{ mm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 46.45) \cdot (8.18 - 0.46) \cdot 1.0 \\
 &= 717.346 \text{ mm}^2
 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned}
 &= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Trapfr4 \\
 &= (100.0) \cdot 1.0 + (0.0) \cdot 1.0 + 248.047^2 \cdot 1.0
 \end{aligned}$$



= 348.047 mm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) • fr4
= (379.075 - 219.075) • 26.0 • 1.0
= 4160.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 0.5022 mm
Wall Thickness per UG16(b), tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure trb1 = 15.0398 mm
Wall Thickness tb1 = max(trb1, tr16b) = 15.0398 mm
Wall Thickness, shell/head, external pressure trb2 = 2.3941 mm
Wall Thickness tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45 tb3 = 7.1600 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[7.16, max(15.0398, 2.5)]
= 7.1600 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(0.5022, 7.16)
= 7.1600 mm

Available Nozzle Neck Thickness = 0.875 * 8.179 = 7.156 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	28.9,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	266.1 MPa	Passed
Occasional	:	2.9,	Allowable	:	156.9 MPa	Passed
Shear	:	20.6,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 7.156, tr = 0.457, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.064, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 7.156, tr = 0.457, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.064, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 7.156, tr = 0.457, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.064, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:C)

Govrn. thk, tg = 26.0, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.52, Temp. Reduction = 30 °C



Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B 0 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -30 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 7.156, tr = 0.457, c = 0.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.064, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -30 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -30 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 0.49/5.10 = 0.097

Weld Size Calculations, Description: N13A 8"

Intermediate Calc. for nozzle/shell Welds Tmin 8.1788 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.7252 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	11.3120 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (2846.1143 - 2627.2556 + 2 \cdot 8.1788 \cdot 1.0 \cdot (1.0 \cdot 27.0 - 14.0398)) \cdot 118)$
 $= 50835.79 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (717.3456 + 4160.0005 + 348.047 - 0.0 \cdot 1.0) \cdot 118$
 $= 616532.81 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (717.3456 + 0.0 + 100.0 + (441.6552)) \cdot 118$
 $= 148546.80 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (717.3456 + 0.0 + 348.047 + 4160.0005 + (441.6552)) \cdot 118$
 $= 668642.81 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 219.075 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 198955. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 379.075 \cdot 16.0 \cdot 0.49 \cdot 118$



= 550806. N

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 105.4481) \cdot (8.1788 - 0.0) \cdot 0.7 \cdot 118$$

$$= 223780. N$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 219.075 \cdot 26.0 \cdot 0.74 \cdot 138$$

$$= 913608. N$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 219.075 \cdot (20.0 - 0.0) \cdot 0.74 \cdot 118$$

$$= 600924. N$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (550806 + 223780) = 774586 N$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (198955 + 913608 + 600924 + 0) = 1713486 N$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (550806 + 600924 + 0) = 1151730 N$$

Summary of Failure Path Calculations:

Path 1-1 = 774585 N	, must exceed W = 50835 N	or W1 = 616532 N
Path 2-2 = 1713486 N	, must exceed W = 50835 N	or W2 = 148546 N
Path 3-3 = 1151729 N	, must exceed W = 50835 N	or W3 = 668642 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 1.7898 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 229.7897 mm

Input Echo, WRC297 Item 1, Description: N13A 8" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	6706.000	mm
Shell Thickness	28.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516	60
Vessel Cold S.I. Allowable	Smc	118.00 MPa
Vessel Hot S.I. Allowable	Smh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	219.075	mm
Nozzle Thickness	8.1788	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106	B
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.



Diameter of Reinforcing Pad 379.075 mm
 Thickness of Reinforcing Pad 26.0000 mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	0.49	MPa
Radial Load	P	-14999.99	N
Circumferential Shear	Vc	9999.99	N
Longitudinal Shear	Vl	9999.99	N
Circumferential Moment	Mc	0.560E+07	N-mm
Longitudinal Moment	Ml	0.560E+07	N-mm
Torsional Moment	Mt	0.800E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(3380.5(28.0 - 1.0))}$
 $= 1216.055 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6761.000 mm
Nozzle Outside Diameter	(d) =	219.075 mm
Vessel Thickness used	(T) =	53.000 mm
Nozzle Thickness used	(t) =	8.179 mm
T / t	=	6.480
d / t	=	26.786

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.366$$

Nr/P	=	0.098
Mr/P	=	0.072
M0/P	=	0.254
N0/P	=	0.376
MrD/Mc	=	0.085
NrDL/Mc	=	0.098
M0d/Mc	=	0.324
N0DL/Mc	=	-0.106
MrD/Ml	=	0.086
NrDL/Ml	=	0.099
M0D/Ml	=	0.310
N0DL/Ml	=	0.112

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-1	-1	1	1
Outplane Bending (Ml)	-16	16	16	-16
Normal Pressure Stress	30	31	30	31
Outplane Stress Summary	23	41	58	9

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-4	4	4	-4
Inplane Pressure Stress	15	15	15	15
Inplane Stress Summary	12	17	23	9

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	23	41	59	11

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-17	17	17	-17
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	15	15	15	15
Outplane Stress Summary	8	27	42	-9

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-4	4	4	-4
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	30	31	30	31
Inplane Stress Summary	28	33	39	25

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1



Shear Stress Summary 1 1 2 2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	28	33	43	35

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1	-1	1	1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	7	7	7
Hoop Stress Summary	8	8	10	10

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	31	-31	31	-31
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-20	-20	20	20
Axial Bending (ML)	-84	84	84	-84
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-67	39	142	-88

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	12	12	12	12
Shear Stress Summary	16	16	9	9

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	82	46	142	101

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	7	7	7
Hoop Stress Summary	10	10	8	8



Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	31	-31	31	-31
Axial Membrane (Mc)	-20	-20	20	20
Axial Bending (Mc)	-82	82	82	-82
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-65	36	139	-86

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	12	12	12	12
Shear Stress Summary	9	9	16	16

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	77	39	141	100

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{3367.5(28.0 - 1.0)}$$

$$= 1214.136 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6735.000 mm
Nozzle Outside Diameter	(d) =	379.075 mm
Vessel Thickness used	(T) =	27.000 mm
Nozzle Thickness used	(t) =	88.179 mm
T / t	=	0.306
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 0.889$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.112
Mr/P	=	0.243
M0/P	=	0.071
N0/P	=	0.129
MrD/Mc	=	0.312
NrDL/Mc	=	0.073
M0d/Mc	=	0.096
N0DL/Mc	=	0.037
MrD/Ml	=	0.210
NrDL/Ml	=	0.091
M0D/Ml	=	0.067
N0DL/Ml	=	0.104

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-8	8	8	-8
Normal Pressure Stress	60	61	60	61

Outplane Stress Summary	62	61	82	49
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	29	-29	29	-29
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-25	25	25	-25
Inplane Pressure Stress	30	30	30	30

Inplane Stress Summary	35	26	90	-20
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	0	0
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	62	61	90	70

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-11	11	11	-11
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	30	30	30	30

Outplane Stress Summary	29	35	54	13
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	29	-29	29	-29
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-37	37	37	-37
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	60	61	60	61

Inplane Stress Summary 53 70 132 -2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		1	1	1	1

Shear Stress Summary 0 0 1 1

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress		53	70	132	16

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	30	31	30	31	30	31	30	31
Circ. Pl (SUS)	1	1	3	3	0	0	0	0
Circ. Q (SUS)	-8	8	24	-24	-2	2	6	-6
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	0	0	0	0	2	2	2	2
Long. Q (SUS)	-2	2	6	-6	-9	9	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	30.0	31.0	30.0	31.0	30.0	31.0	30.0	31.0
Pm+Pl (SUS)	31.0	32.0	33.0	34.0	30.0	31.0	30.0	31.0
Pm+Pl+Q (Total)	23.1	40.0	57.0	10.6	28.0	33.1	42.2	33.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	31.00	118.00	Passed
Pm+Pl (SUS)	34.00	177.00	Passed
Pm+Pl+Q (TOTAL)	57.03	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	60	61	60	61	60	61	60	61
Circ. Pl (SUS)	0	0	4	4	1	1	3	3
Circ. Q (SUS)	0	0	16	-16	-8	8	66	-66

Long. Pm (SUS)	30	30	30	30	30	30	30	30
Long. Pl (SUS)	1	1	3	3	2	2	2	2
Long. Q (SUS)	4	-4	54	-54	-3	3	19	-19
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	60.0	61.0	60.0	61.0	60.0	61.0	60.0	61.0
Pm+Pl (SUS)	60.0	61.0	64.0	65.0	61.0	62.0	63.0	64.0
Pm+Pl+Q (Total)	60.0	61.0	87.1	70.0	53.0	70.0	129.0	15.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	61.00	118.00	Passed
Pm+Pl (SUS)	65.00	177.00	Passed
Pm+Pl+Q (TOTAL)	129.01	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	7	7	7	7	7	7	7	7
Circ. Pl (SUS)	1	1	3	3	2	2	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-18	-18	22	22	-18	-18	22	22
Long. Q (SUS)	-53	53	115	-115	-51	51	113	-113
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0
Pm+Pl (SUS)	23.8	23.8	25.6	25.6	24.7	24.7	25.5	25.5
Pm+Pl+Q (Total)	81.7	44.2	140.6	101.6	77.1	38.7	139.7	101.5

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	118.00	Passed
Pm+Pl (SUS)	25.58	177.00	Passed
Pm+Pl+Q (TOTAL)	140.62	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: MH1 30"

From: 80

Pressure for Reinforcement Calculations	P	0.4927	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	1226.1924	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		850.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 60 +IT	
Material UNS Number		K02100	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		45.00	deg
Diameter		30.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	14.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

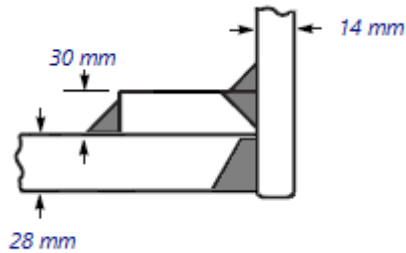
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 60 +IT	
Pad Allowable Stress at Temperature	Sp	118.00	MPa
Pad Allowable Stress At Ambient	Spa	118.00	MPa
Diameter of Pad along vessel surface	Dp	1522.0000	mm
Thickness of Pad	te	30.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	30.0000	mm
Reinforcing Pad Width		380.0000	mm
This is a Manway, Coupling or Access Opening.			

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: MH1 30"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	30.000 in.
Actual Thickness Used in Calculation	0.551 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

This nozzle looks to be 26 inch or greater OD and the flange series is set to None. Please set it to series A or B in the nozzle dialog or the flange weight will not be determined.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.49 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.49)$
= 14.0398 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.49 \cdot 381.0 / (118 \cdot 1.0 + 0.4 \cdot 0.49)$
= 1.5882 mm

Required Nozzle thickness under External Pressure per UG-28 : 1.3927 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	DI	1468.0000	mm
Parallel to Vessel Wall, opening length	d	734.0000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		65.0000	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $118.0 / 118.0$)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $118.0 / 118.0$)
= 1.000

Weld Strength Reduction Factor [fr4]:
= min(1, S_p / S_v)
= min(1, $118.0 / 118.0$)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000

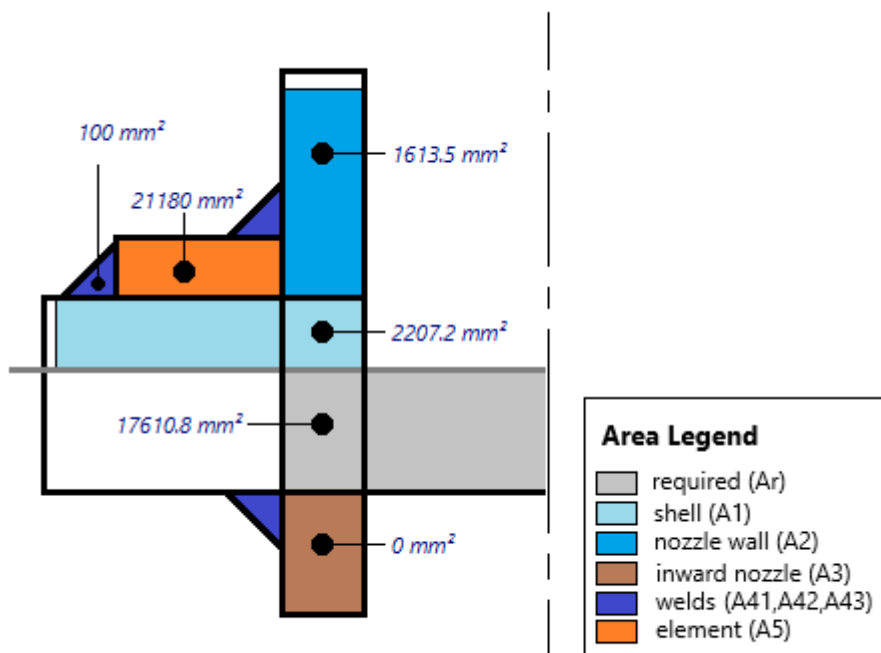


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	10305.223	2896.496	NA
Area in Shell	A1	9512.778	14025.010	NA
Area in Nozzle Wall	A2	1613.536	1638.946	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	21180.000	21180.000	NA
TOTAL AREA AVAILABLE	Atot	32406.314	36943.957	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (734.0 \cdot 14.0398 \cdot 1.0 + 2 \cdot 14.0 \cdot 14.0398 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 10305.223 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 734.0 (1.0 \cdot 27.0 - 1.0 \cdot 14.04) - 2 \cdot 14.0$$

$$(1.0 \cdot 27.0 - 1.0 \cdot 14.0398) \cdot (1 - 1.0)$$

$$= 9512.778 \text{ mm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 65.0) \cdot (14.0 - 1.59) \cdot 1.0$$

$$= 1613.536 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0$$



= 100.000 mm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) * fr4
= (1468.0 - 762.0) * 30.0 * 1.0
= 21180.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures ta = 1.5882 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= ta
= 1.5882 mm

Available Nozzle Neck Thickness = 14.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 14.0, tr = 1.588, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.113, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: C

Govrn. thk, tg = 28.0, tr = 14.04, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.52, Temp. Reduction = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve C	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: C

Govrn. thk, tg = 14.0, tr = 1.588, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.113, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve C	-36 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -48 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = 0.49/1.96 = 0.251

Weld Size Calculations, Description: MH1 30"

Intermediate Calc. for nozzle/shell Welds	Tmin	14.0000	mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000	mm

Results Per UW-16.1:



	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	11.3120 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (10305.2227 - 9512.7783 + 2 \cdot 14.0 \cdot 1.0 \cdot (1.0 \cdot 27.0 - 14.0398)) \cdot 118)$$

$$= 136314.86 \text{ N}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$$

$$= (1613.536 + 21180.0 + 100.0 - 0.0 \cdot 1.0) \cdot 118$$

$$= 2701158.50 \text{ N}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$$

$$= (1613.536 + 0.0 + 100.0 + (756.0)) \cdot 118$$

$$= 291375.19 \text{ N}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$$

$$= (1613.536 + 0.0 + 100.0 + 21180.0 + (756.0)) \cdot 118$$

$$= 2790357.50 \text{ N}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 762.0 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 692016. \text{ N}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 1522.0 \cdot 16.0 \cdot 0.49 \cdot 118$$

$$= 2211504. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 374.0) \cdot (14.0 - 0.0) \cdot 0.7 \cdot 118$$

$$= 1358603. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 762.0 \cdot 30.0 \cdot 0.74 \cdot 118$$

$$= 3135255. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 762.0 \cdot (20.0 - 0.0) \cdot 0.74 \cdot 118$$

$$= 2090170. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (2211505 + 1358603) = 3570108 \text{ N}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (692016 + 3135255 + 2090170 + 0) = 5917441 \text{ N}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (2211505 + 2090170 + 0) = 4301675 \text{ N}$$



Summary of Failure Path Calculations:

Path 1-1 = 3570107 N , must exceed W = 136314 N or W1 = 2701158 N
Path 2-2 = 5917440 N , must exceed W = 136314 N or W2 = 291375 N
Path 3-3 = 4301674 N , must exceed W = 136314 N or W3 = 2790357 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 21.7169 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 449.7167 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.872 %

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Input, Nozzle Desc: N12 2"

From: 80

Pressure for Reinforcement Calculations	P	0.4927	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	1226.1924	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		250.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		330.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		40	

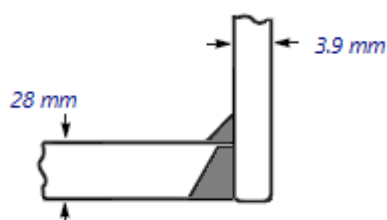
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: N12 2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.49 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.49)$
= 14.0398 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.49 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.49)$
= 0.1257 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	114.3250	mm
Parallel to Vessel Wall	Rn+tn+t	57.1625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: N12 2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 15.0398	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 15.0398	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(15.0398, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.126, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.126, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.037$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.49/5.10 = 0.097

Weld Size Calculations, Description: N12 2"

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 228.1355 mm

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Input, Nozzle Desc: N13B 8"

From: 80

Pressure for Reinforcement Calculations	P	0.4927	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	5099.9990	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		2550.00	mm
-----------------------------------	--	---------	----

User Entered Minimum Design Metal Temperature		-1.00	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		8.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

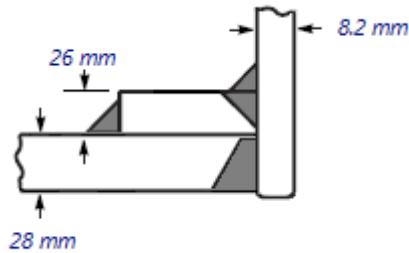
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	379.0750	mm
Thickness of Pad	te	26.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	26.0000	mm
Reinforcing Pad Width		80.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: N13B 8"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	8.625 in.
Actual Thickness Used in Calculation	0.322 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.49 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.49)$
 $= 14.0398 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.49 \cdot 109.5375 / (118 \cdot 1.0 + 0.4 \cdot 0.49)$
 $= 0.4566 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5022 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	405.4348 mm
Parallel to Vessel Wall, opening length	d	202.7174 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		46.4470 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

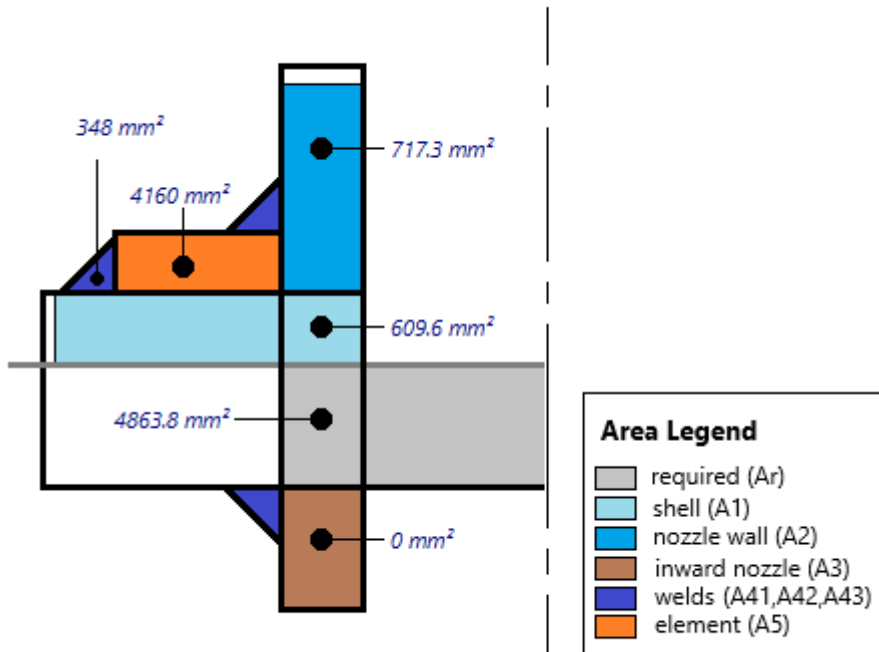


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	2846.114	1453.795	NA
Area in Shell	A1	2627.256	2565.779	NA
Area in Nozzle Wall	A2	717.346	713.113	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	348.047	348.047	NA
Area in Element	A5	4160.000	4160.000	NA
TOTAL AREA AVAILABLE	Atot	7852.649	7786.940	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (202.7174 \cdot 14.0398 \cdot 1.0 + 2 \cdot 8.1788 \cdot 14.0398 \cdot 1.0 \cdot (1 - 1.0))$
 $= 2846.114 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 202.717 (1.0 \cdot 27.0 - 1.0 \cdot 14.04) - 2 \cdot 8.179$
 $(1.0 \cdot 27.0 - 1.0 \cdot 14.0398) \cdot (1 - 1.0)$
 $= 2627.256 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 46.45) \cdot (8.18 - 0.46) \cdot 1.0$
 $= 717.346 \text{ mm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Trapfr4$
 $= (100.0) \cdot 1.0 + (0.0) \cdot 1.0 + 248.047^2 \cdot 1.0$



= 348.047 mm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) • fr4
= (379.075 - 219.075) • 26.0 • 1.0
= 4160.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 0.5022 mm
Wall Thickness per UG16(b), tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure trb1 = 15.0398 mm
Wall Thickness tb1 = max(trb1, tr16b) = 15.0398 mm
Wall Thickness, shell/head, external pressure trb2 = 2.3941 mm
Wall Thickness tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45 tb3 = 7.1600 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[7.16, max(15.0398, 2.5)]
= 7.1600 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(0.5022, 7.16)
= 7.1600 mm

Available Nozzle Neck Thickness = 0.875 * 8.179 = 7.156 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	28.9,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	266.1 MPa	Passed
Occasional	:	2.9,	Allowable	:	156.9 MPa	Passed
Shear	:	20.6,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 7.156, tr = 0.457, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.064, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 7.156, tr = 0.457, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.064, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 7.156, tr = 0.457, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.064, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:C)

Govrn. thk, tg = 26.0, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.52, Temp. Reduction = 30 °C



Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B 0 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -30 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 7.156, tr = 0.457, c = 0.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.064, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -30 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -30 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 0.49/5.10 = 0.097

Weld Size Calculations, Description: N13B 8"

Intermediate Calc. for nozzle/shell Welds Tmin 8.1788 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.7252 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	11.3120 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (2846.1143 - 2627.2556 + 2 \cdot 8.1788 \cdot 1.0 \cdot (1.0 \cdot 27.0 - 14.0398)) \cdot 118)$
 $= 50835.79 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (717.3456 + 4160.0005 + 348.047 - 0.0 \cdot 1.0) \cdot 118$
 $= 616532.81 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (717.3456 + 0.0 + 100.0 + (441.6552)) \cdot 118$
 $= 148546.80 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (717.3456 + 0.0 + 348.047 + 4160.0005 + (441.6552)) \cdot 118$
 $= 668642.81 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 219.075 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 198955. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 379.075 \cdot 16.0 \cdot 0.49 \cdot 118$



= 550806. N

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 105.4481) \cdot (8.1788 - 0.0) \cdot 0.7 \cdot 118$$

$$= 223780. N$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 219.075 \cdot 26.0 \cdot 0.74 \cdot 138$$

$$= 913608. N$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 219.075 \cdot (20.0 - 0.0) \cdot 0.74 \cdot 118$$

$$= 600924. N$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (550806 + 223780) = 774586 N$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (198955 + 913608 + 600924 + 0) = 1713486 N$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (550806 + 600924 + 0) = 1151730 N$$

Summary of Failure Path Calculations:

Path 1-1 = 774585 N	, must exceed W = 50835 N	or W1 = 616532 N
Path 2-2 = 1713486 N	, must exceed W = 50835 N	or W2 = 148546 N
Path 3-3 = 1151729 N	, must exceed W = 50835 N	or W3 = 668642 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 1.7898 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 229.7897 mm

Input Echo, WRC297 Item 1, Description: N13B 8" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	6706.000	mm
Shell Thickness	28.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 60	
Vessel Cold S.I. Allowable	Smc	118.00 MPa
Vessel Hot S.I. Allowable	Smh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	219.075	mm
Nozzle Thickness	8.1788	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.



Diameter of Reinforcing Pad 379.075 mm
 Thickness of Reinforcing Pad 26.0000 mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	0.49	MPa
Radial Load	P	-14999.99	N
Circumferential Shear	Vc	9999.99	N
Longitudinal Shear	Vl	9999.99	N
Circumferential Moment	Mc	0.560E+07	N-mm
Longitudinal Moment	Ml	0.560E+07	N-mm
Torsional Moment	Mt	0.800E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - c_a))}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(3380.5(28.0 - 1.0))}$
 $= 1216.055 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6761.000 mm
Nozzle Outside Diameter	(d) =	219.075 mm
Vessel Thickness used	(T) =	53.000 mm
Nozzle Thickness used	(t) =	8.179 mm
T / t	=	6.480
d / t	=	26.786

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.366$$

Nr/P	=	0.098
Mr/P	=	0.072
M0/P	=	0.254
N0/P	=	0.376
MrD/Mc	=	0.085
NrDL/Mc	=	0.098
M0d/Mc	=	0.324
N0DL/Mc	=	-0.106
MrD/Ml	=	0.086
NrDL/Ml	=	0.099
M0D/Ml	=	0.310
N0DL/Ml	=	0.112

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-1	-1	1	1
Outplane Bending (Ml)	-16	16	16	-16
Normal Pressure Stress	30	31	30	31
Outplane Stress Summary	23	41	58	9

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
--------------------	----	----	----	----



(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-4	4	4	-4
Inplane Pressure Stress	15	15	15	15
Inplane Stress Summary	12	17	23	9

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	23	41	59	11

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-17	17	17	-17
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	15	15	15	15
Outplane Stress Summary	8	27	42	-9

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-4	4	4	-4
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	30	31	30	31
Inplane Stress Summary	28	33	39	25

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1



Shear Stress Summary 1 1 2 2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	28	33	43	35

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1	-1	1	1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	7	7	7
Hoop Stress Summary	8	8	10	10

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	31	-31	31	-31
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-20	-20	20	20
Axial Bending (ML)	-84	84	84	-84
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-67	39	142	-88

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	12	12	12	12
Shear Stress Summary	16	16	9	9

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	82	46	142	101

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	7	7	7
Hoop Stress Summary	10	10	8	8



Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	31	-31	31	-31
Axial Membrane (Mc)	-20	-20	20	20
Axial Bending (Mc)	-82	82	82	-82
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-65	36	139	-86

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	12	12	12	12
Shear Stress Summary	9	9	16	16

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	77	39	141	100

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{3367.5(28.0 - 1.0)}$
 $= 1214.136 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 6735.000 mm
 Nozzle Outside Diameter (d) = 379.075 mm
 Vessel Thickness used (T) = 27.000 mm
 Nozzle Thickness used (t) = 88.179 mm
 $T / t = 0.306$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 0.889$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.112
 Mr/P = 0.243
 M0/P = 0.071
 N0/P = 0.129
 MrD/Mc = 0.312
 NrDL/Mc = 0.073
 M0d/Mc = 0.096
 N0DL/Mc = 0.037
 MrD/Ml = 0.210
 NrDL/Ml = 0.091
 M0D/Ml = 0.067
 N0DL/Ml = 0.104

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-8	8	8	-8
Normal Pressure Stress	60	61	60	61

Outplane Stress Summary	62	61	82	49
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	29	-29	29	-29
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-25	25	25	-25
Inplane Pressure Stress	30	30	30	30

Inplane Stress Summary	35	26	90	-20
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	0	0
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	62	61	90	70

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-11	11	11	-11
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	30	30	30	30

Outplane Stress Summary	29	35	54	13
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	29	-29	29	-29
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-37	37	37	-37
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	60	61	60	61

Inplane Stress Summary 53 70 132 -2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to
 circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary 0 0 1 1

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stress Intensities)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	53	70	132	16

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	30	31	30	31	30	31	30	31
Circ. Pl (SUS)	1	1	3	3	0	0	0	0
Circ. Q (SUS)	-8	8	24	-24	-2	2	6	-6
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	0	0	0	0	2	2	2	2
Long. Q (SUS)	-2	2	6	-6	-9	9	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	30.0	31.0	30.0	31.0	30.0	31.0	30.0	31.0
Pm+Pl (SUS)	31.0	32.0	33.0	34.0	30.0	31.0	30.0	31.0
Pm+Pl+Q (Total)	23.1	40.0	57.0	10.6	28.0	33.1	42.2	33.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	31.00	118.00	Passed
Pm+Pl (SUS)	34.00	177.00	Passed
Pm+Pl+Q (TOTAL)	57.03	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	60	61	60	61	60	61	60	61
Circ. Pl (SUS)	0	0	4	4	1	1	3	3
Circ. Q (SUS)	0	0	16	-16	-8	8	66	-66

Long. Pm (SUS)	30	30	30	30	30	30	30	30
Long. Pl (SUS)	1	1	3	3	2	2	2	2
Long. Q (SUS)	4	-4	54	-54	-3	3	19	-19
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	60.0	61.0	60.0	61.0	60.0	61.0	60.0	61.0
Pm+Pl (SUS)	60.0	61.0	64.0	65.0	61.0	62.0	63.0	64.0
Pm+Pl+Q (Total)	60.0	61.0	87.1	70.0	53.0	70.0	129.0	15.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	61.00	118.00	Passed
Pm+Pl (SUS)	65.00	177.00	Passed
Pm+Pl+Q (TOTAL)	129.01	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	7	7	7	7	7	7	7	7
Circ. Pl (SUS)	1	1	3	3	2	2	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-18	-18	22	22	-18	-18	22	22
Long. Q (SUS)	-53	53	115	-115	-51	51	113	-113
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0
Pm+Pl (SUS)	23.8	23.8	25.6	25.6	24.7	24.7	25.5	25.5
Pm+Pl+Q (Total)	81.7	44.2	140.6	101.6	77.1	38.7	139.7	101.5

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	118.00	Passed
Pm+Pl (SUS)	25.58	177.00	Passed
Pm+Pl+Q (TOTAL)	140.62	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N13C 8"

From: 80

Pressure for Reinforcement Calculations	P	0.4927	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	5099.9990	mm
Shell Finished (Minimum) Thickness	t	28.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		2550.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		300.00	deg
Diameter		8.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

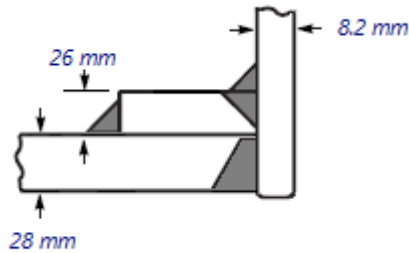
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	379.0750	mm
Thickness of Pad	te	26.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	26.0000	mm
Reinforcing Pad Width		80.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: N13C 8"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	8.625 in.
Actual Thickness Used in Calculation	0.322 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.49 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.49)$
 $= 14.0398 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.49 \cdot 109.5375 / (118 \cdot 1.0 + 0.4 \cdot 0.49)$
 $= 0.4566 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.5022 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	405.4348 mm
Parallel to Vessel Wall, opening length	d	202.7174 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		46.4470 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

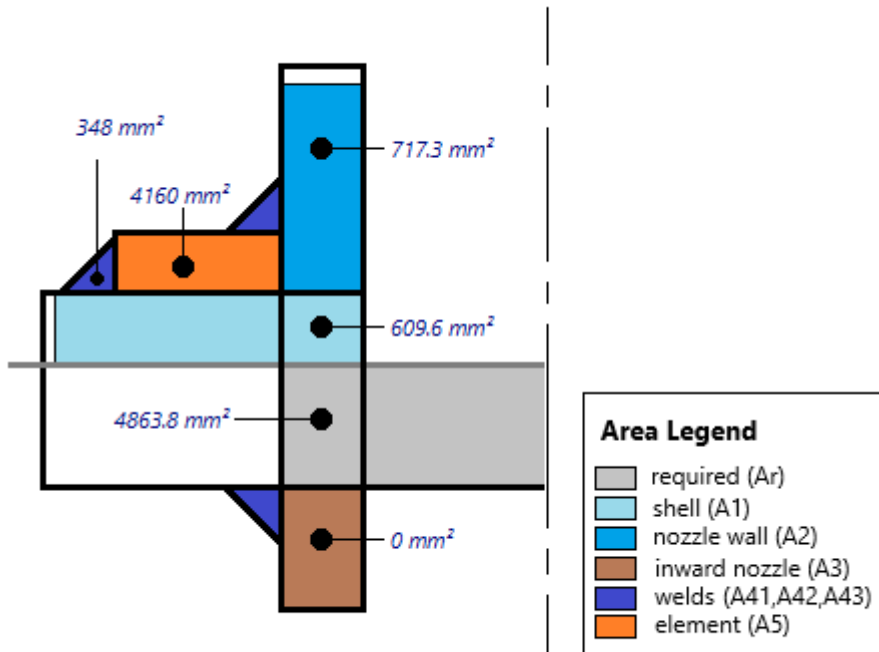


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	2846.114	1453.795	NA
Area in Shell	A1	2627.256	2565.779	NA
Area in Nozzle Wall	A2	717.346	713.113	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	348.047	348.047	NA
Area in Element	A5	4160.000	4160.000	NA
TOTAL AREA AVAILABLE	Atot	7852.649	7786.940	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (202.7174 \cdot 14.0398 \cdot 1.0 + 2 \cdot 8.1788 \cdot 14.0398 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 2846.114 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 202.717 (1.0 \cdot 27.0 - 1.0 \cdot 14.04) - 2 \cdot 8.179$$

$$(1.0 \cdot 27.0 - 1.0 \cdot 14.0398) \cdot (1 - 1.0)$$

$$= 2627.256 \text{ mm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 46.45) \cdot (8.18 - 0.46) \cdot 1.0$$

$$= 717.346 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Trapfr4$$

$$= (100.0) \cdot 1.0 + (0.0) \cdot 1.0 + 248.047^2 \cdot 1.0$$



= 348.047 mm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) • fr4
= (379.075 - 219.075) • 26.0 • 1.0
= 4160.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 0.5022 mm
Wall Thickness per UG16(b), tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure trb1 = 15.0398 mm
Wall Thickness tb1 = max(trb1, tr16b) = 15.0398 mm
Wall Thickness, shell/head, external pressure trb2 = 2.3941 mm
Wall Thickness tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45 tb3 = 7.1600 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[7.16, max(15.0398, 2.5)]
= 7.1600 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(0.5022, 7.16)
= 7.1600 mm

Available Nozzle Neck Thickness = 0.875 * 8.179 = 7.156 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	28.9,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	266.1 MPa	Passed
Occasional	:	2.9,	Allowable	:	156.9 MPa	Passed
Shear	:	20.6,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 7.156, tr = 0.457, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.064, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 7.156, tr = 0.457, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.064, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 7.156, tr = 0.457, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.064, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:C)

Govrn. thk, tg = 26.0, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.52, Temp. Reduction = 30 °C



Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B 0 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -30 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 7.156, tr = 0.457, c = 0.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.064, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -30 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -30 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 0.49/5.10 = 0.097

Weld Size Calculations, Description: N13C 8"

Intermediate Calc. for nozzle/shell Welds Tmin 8.1788 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	5.7252 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	11.3120 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (2846.1143 - 2627.2556 + 2 \cdot 8.1788 \cdot 1.0 \cdot (1.0 \cdot 27.0 - 14.0398)) \cdot 118)$
 $= 50835.79 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (717.3456 + 4160.0005 + 348.047 - 0.0 \cdot 1.0) \cdot 118$
 $= 616532.81 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (717.3456 + 0.0 + 100.0 + (441.6552)) \cdot 118$
 $= 148546.80 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (717.3456 + 0.0 + 348.047 + 4160.0005 + (441.6552)) \cdot 118$
 $= 668642.81 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 219.075 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 198955. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 379.075 \cdot 16.0 \cdot 0.49 \cdot 118$



= 550806. N

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 105.4481) \cdot (8.1788 - 0.0) \cdot 0.7 \cdot 118$$

$$= 223780. N$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 219.075 \cdot 26.0 \cdot 0.74 \cdot 138$$

$$= 913608. N$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 219.075 \cdot (20.0 - 0.0) \cdot 0.74 \cdot 118$$

$$= 600924. N$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (550806 + 223780) = 774586 N$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (198955 + 913608 + 600924 + 0) = 1713486 N$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (550806 + 600924 + 0) = 1151730 N$$

Summary of Failure Path Calculations:

Path 1-1 = 774585 N	, must exceed W = 50835 N	or W1 = 616532 N
Path 2-2 = 1713486 N	, must exceed W = 50835 N	or W2 = 148546 N
Path 3-3 = 1151729 N	, must exceed W = 50835 N	or W3 = 668642 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 1.7898 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 229.7897 mm

Input Echo, WRC297 Item 1, Description: N13C 8" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	6706.000	mm
Shell Thickness	28.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516	60
Vessel Cold S.I. Allowable	Smc	118.00 MPa
Vessel Hot S.I. Allowable	Smh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	219.075	mm
Nozzle Thickness	8.1788	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106	B
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.



Diameter of Reinforcing Pad 379.075 mm
 Thickness of Reinforcing Pad 26.0000 mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.*

Design Internal Pressure	Dp	0.49	MPa
Radial Load	P	-14999.99	N
Circumferential Shear	Vc	9999.99	N
Longitudinal Shear	Vl	9999.99	N
Circumferential Moment	Mc	0.560E+07	N-mm
Longitudinal Moment	Ml	0.560E+07	N-mm
Torsional Moment	Mt	0.800E+07	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 219.075 + 2 \cdot 1.65 \cdot \sqrt{(3380.5(28.0 - 1.0))}$
 $= 1216.055 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6761.000 mm
Nozzle Outside Diameter	(d) =	219.075 mm
Vessel Thickness used	(T) =	53.000 mm
Nozzle Thickness used	(t) =	8.179 mm
T / t	=	6.480
d / t	=	26.786

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.366$$

Nr/P	=	0.098
Mr/P	=	0.072
M0/P	=	0.254
N0/P	=	0.376
MrD/Mc	=	0.085
NrDL/Mc	=	0.098
M0d/Mc	=	0.324
N0DL/Mc	=	-0.106
MrD/Ml	=	0.086
NrDL/Ml	=	0.099
M0D/Ml	=	0.310
N0DL/Ml	=	0.112

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-1	-1	1	1
Outplane Bending (Ml)	-16	16	16	-16
Normal Pressure Stress	30	31	30	31
Outplane Stress Summary	23	41	58	9

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-4	4	4	-4
Inplane Pressure Stress	15	15	15	15
Inplane Stress Summary	12	17	23	9

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	23	41	59	11

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-17	17	17	-17
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	15	15	15	15
Outplane Stress Summary	8	27	42	-9

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	-4	4	4	-4
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	30	31	30	31
Inplane Stress Summary	28	33	39	25

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1



Shear Stress Summary 1 1 2 2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	28	33	43	35

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-1	-1	1	1
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	7	7	7
Hoop Stress Summary	8	8	10	10

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	31	-31	31	-31
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-20	-20	20	20
Axial Bending (ML)	-84	84	84	-84
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-67	39	142	-88

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	12	12	12	12
Shear Stress Summary	16	16	9	9

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	82	46	142	101

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	7	7	7	7
Hoop Stress Summary	10	10	8	8



Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	31	-31	31	-31
Axial Membrane (Mc)	-20	-20	20	20
Axial Bending (Mc)	-82	82	82	-82
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	3	3	3	3
Axial Stress Summary	-65	36	139	-86

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	12	12	12	12
Shear Stress Summary	9	9	16	16

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	77	39	141	100

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 219.075 + 2 \cdot 1.65 \cdot \sqrt{3367.5(28.0 - 1.0)}$$

$$= 1214.136 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 6735.000 mm
 Nozzle Outside Diameter (d) = 379.075 mm
 Vessel Thickness used (T) = 27.000 mm
 Nozzle Thickness used (t) = 88.179 mm
 $T / t = 0.306$
 $d / t = 10.000$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.889$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.112
 Mr/P = 0.243
 M0/P = 0.071
 N0/P = 0.129
 MrD/Mc = 0.312
 NrDL/Mc = 0.073
 M0d/Mc = 0.096
 N0DL/Mc = 0.037
 MrD/Ml = 0.210
 NrDL/Ml = 0.091
 M0D/Ml = 0.067
 N0DL/Ml = 0.104

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-8	8	8	-8
Normal Pressure Stress	60	61	60	61

Outplane Stress Summary	62	61	82	49
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	29	-29	29	-29
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-25	25	25	-25
Inplane Pressure Stress	30	30	30	30

Inplane Stress Summary	35	26	90	-20
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1

Shear Stress Summary	1	1	0	0
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	62	61	90	70

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	8	-8	8	-8
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	-11	11	11	-11
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	30	30	30	30

Outplane Stress Summary	29	35	54	13
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	29	-29	29	-29
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-37	37	37	-37
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	60	61	60	61

Inplane Stress Summary 53 70 132 -2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		1	1	1	1

Shear Stress Summary 0 0 1 1

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress		53	70	132	16

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	30	31	30	31	30	31	30	31
Circ. Pl (SUS)	1	1	3	3	0	0	0	0
Circ. Q (SUS)	-8	8	24	-24	-2	2	6	-6
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	0	0	0	0	2	2	2	2
Long. Q (SUS)	-2	2	6	-6	-9	9	25	-25
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	30.0	31.0	30.0	31.0	30.0	31.0	30.0	31.0
Pm+Pl (SUS)	31.0	32.0	33.0	34.0	30.0	31.0	30.0	31.0
Pm+Pl+Q (Total)	23.1	40.0	57.0	10.6	28.0	33.1	42.2	33.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	31.00	118.00	Passed
Pm+Pl (SUS)	34.00	177.00	Passed
Pm+Pl+Q (TOTAL)	57.03	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	60	61	60	61	60	61	60	61
Circ. Pl (SUS)	0	0	4	4	1	1	3	3
Circ. Q (SUS)	0	0	16	-16	-8	8	66	-66

Long. Pm (SUS)	30	30	30	30	30	30	30	30
Long. Pl (SUS)	1	1	3	3	2	2	2	2
Long. Q (SUS)	4	-4	54	-54	-3	3	19	-19
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	60.0	61.0	60.0	61.0	60.0	61.0	60.0	61.0
Pm+Pl (SUS)	60.0	61.0	64.0	65.0	61.0	62.0	63.0	64.0
Pm+Pl+Q (Total)	60.0	61.0	87.1	70.0	53.0	70.0	129.0	15.1

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	61.00	118.00	Passed
Pm+Pl (SUS)	65.00	177.00	Passed
Pm+Pl+Q (TOTAL)	129.01	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	7	7	7	7	7	7	7	7
Circ. Pl (SUS)	1	1	3	3	2	2	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	3	3	3	3	3	3	3	3
Long. Pl (SUS)	-18	-18	22	22	-18	-18	22	22
Long. Q (SUS)	-53	53	115	-115	-51	51	113	-113
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	7.0	7.0	7.0	7.0	7.0	7.0	7.0	7.0
Pm+Pl (SUS)	23.8	23.8	25.6	25.6	24.7	24.7	25.5	25.5
Pm+Pl+Q (Total)	81.7	44.2	140.6	101.6	77.1	38.7	139.7	101.5

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	7.00	118.00	Passed
Pm+Pl (SUS)	25.58	177.00	Passed
Pm+Pl+Q (TOTAL)	140.62	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N2A 84"

From: 90

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	5099.9990	mm
Shell Finished (Minimum) Thickness	t	40.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Note : User defined Limit(s) of Reinforcement specified below:

Physical Maximum for Diameter Limit	Dmax	5000.0000	mm
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Distance from Bottom/Left Tangent		4050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		200.00	deg
Diameter		84.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm

Flange Material		SA-105	
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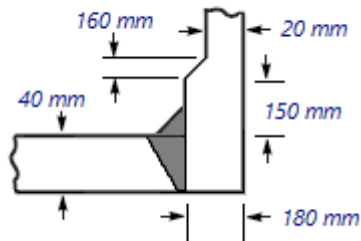
Hub Height of Integral Nozzle	h	150.0000	mm
Height of Beveled Transition	L'	160.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		180.0000	mm

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	800.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: N2A 84"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	84.000 in.
Actual Thickness Used in Calculation	0.787 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
 $= 13.5004 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.47 \cdot 1066.7999 / (138 \cdot 1.0 + 0.4 \cdot 0.47)$
 $= 3.6578 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 3.3853 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 160.0 / 0.5773$
 $= 277.1281 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

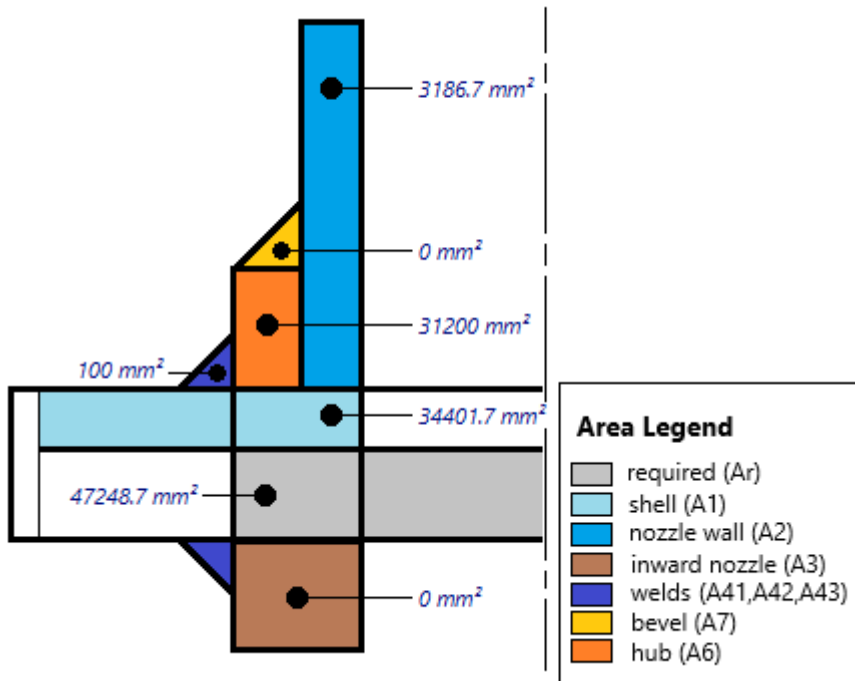
UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	4187.1997 mm
Parallel to Vessel Wall, opening length	d	2093.5999 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	97.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	28264.361	15008.950	NA
Area in Shell	A1	53386.031	51632.492	NA
Area in Nozzle Wall	A2	3186.734	3239.860	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	31199.996	31199.996	NA
TOTAL AREA AVAILABLE	Atot	87872.766	86172.352	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (2093.5999 \cdot 13.5004 \cdot 1.0 + 2 \cdot 180.0 \cdot 13.5004 \cdot 1.0 \cdot (1 - 1.0))$
 $= 28264.361 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 2093.6(1.0 \cdot 39.0 - 1.0 \cdot 13.5) - 2 \cdot 180.0$
 $(1.0 \cdot 39.0 - 1.0 \cdot 13.5004) \cdot (1 - 1.0)$
 $= 53386.031 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 97.5)(20.0 - 3.66)1.0$
 $= 3186.734 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubtk} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(97.5, 800.0, 150.0)) \cdot (180.0 - 20.0) \cdot 1.0 \\ &= 31199.996 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.6578 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 14.5004 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 14.5004 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.3941 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[8.331, \max(14.5004, 2.5)] \\ &= 8.3312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(3.6578, 8.3312) \\ &= 8.3312 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 23.1,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 321.9 MPa	Passed
Occasional	: 12.3,	Allowable	: 183.5 MPa	Passed
Shear	: 8.4,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 20.0, tr = 3.658, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.183, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 40.0, tr = 13.5, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.346, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/1.96 = 0.242

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:



$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 160.0/0.5773$$

$$= 277.1281 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 3140.3997 mm
 Parallel to Vessel Wall, opening length d 1570.1998 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 97.5000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (mm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	18843.850	18843.850
Area Available in Shell	A1	26693.014	26693.014
Area Available in Nozzle Wall	A2	3186.734	3186.734
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	100.000	100.000
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	31199.996	31199.996
TOTAL AREA AVAILABLE	Atot	61179.742	61179.742

The area available without a pad is Sufficient.

The following Calculation is performed using the internal design pressure (P) in the corroded condition.

Warning: The flange could not be considered in the moment of inertia calculations.

Area/Inertia Contributions from Fig.1-7-2 (for Bending stress only):

	Area	Distance	Inertia	d	I+Ad ²
Nozzle	7180.0	179.5	0.7711E+08	89.243	0.1343E+09
Hub	36800.0	119.6	0.2007E+09	29.380	0.2325E+09
Shell	24336.0	19.5	0.3085E+07	70.757	0.1249E+09
Sum	68316.0				0.4917E+09

Membrane stress per Appendix 1-7 (1 or 2) Sm	50.56	MPa
Allowable Membrane Stress S	118.00	MPa
Bending stress at the inside of the Nozzle Sb	38.24	MPa
Combined Membrane plus Bending Stress Sm + Sb	88.79	MPa
Allowable Membrane + Bending Stress 1.5 * S	177.00	MPa
M.A.W.P for the Appendix 1-7 Stress calc	0.94	MPa

Area contributions from Fig. 1-7-1 for Membrane stress per 1-7(b)(4):

Area from Nozzle = 3687.645 mm²
 Area from Shell = 14146.116 mm²
 Total Area, As = 46743.852 mm²

Membrane Stress Calc for Large Nozzles per 1-7(b)(4) [Sm]:

$$= P [R (Rn + tn + \sqrt{rm \cdot t}) + Rn (t + \sqrt{Rnm \cdot tn})] / As$$

$$= 0.474 [3354.0 (1046.8 + 20.0 + 362.721) + 1046.8 (184.382)] / 46743.852$$

$$= 50.557 \text{ MPa}$$

The area As was computed per figure 1-7-1 for the Sm calculation.

Moment [M]:

$$= (Rn^3/6 + R \cdot Rn \cdot e) P$$

$$= (1046.8^3/6 + 3354.0 \cdot 1046.8 \cdot 70.757) 0.474$$

$$= 0.20836E+09 \text{ N-mm}$$

Bending Stress Calc for Large Nozzles per 1-7(b)(4)(5) [Sb]:

$$= (M (e + t/2)) / I$$

$$= (0.20836E+09 (70.757 + 39.0/2)) / .49168E+09$$

$$= 38.236 \text{ MPa}$$



Weld Size Calculations, Description: N2A 84"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (28264.3613 - 53386.0312 + 2 • 180.0 • 1.0 •
(1.0 • 39.0 - 13.5004))118)
= max(0, -1880939.00) N

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (34386.7266 + 100.0 - 0.0 • 1.0) • 118
= 4069014.50 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (34386.7266 + 0.0 + 100.0 + (14040.0)) • 118
= 5725564.00 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (34386.7266 + 0.0 + 100.0 + 0.0 + (14040.0)) • 118
= 5725564.00 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 2133.5999 • 10.0 • 0.49 • 118
= 1937609. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 1056.8) • (180.0 - 0.0) • 0.7 • 138
= 57723860. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • Wgnvi • 0.74 • Sng
= (3.1416/2.0) • 2133.5999 • (20.0 - 0.0) • 0.74 • 138
= 6844422. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (1937609 + 57723860) = 59661472 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (1937609 + 0 + 6844422 + 0) = 8782031 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (1937609 + 6844422 + 0) = 8782031 N

Summary of Failure Path Calculations:

Path 1-1 = 59661472N	, must exceed W = 0 N	or W1 = 4069014 N
Path 2-2 = 8782031 N	, must exceed W = 0 N	or W2 = 5725564 N



Path 3-3 = 8782031 N , must exceed W = 0 N or W3 = 5725564 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 174.2351 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1014.2350 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N2A 84" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	6706.000	mm
Shell Thickness	40.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 60	
Vessel Cold S.I. Allowable	Smc	118.00 MPa
Vessel Hot S.I. Allowable	Smh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	2453.600	mm
Nozzle Thickness	180.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	0.47	MPa
Radial Load	P	-157500.11	N
Circumferential Shear	Vc	104999.93	N
Longitudinal Shear	Vl	104999.93	N
Circumferential Moment	Mc	0.588E+09	N-mm
Longitudinal Moment	Ml	0.588E+09	N-mm
Torsional Moment	Mt	0.840E+09	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 2453.6 + 2 \cdot 1.65 \cdot \sqrt{(3373.5(40.0 - 1.0))} \\
 &= 3650.579 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 6747.000 mm



Nozzle Outside Diameter (d) = 2453.600 mm
 Vessel Thickness used (T) = 39.000 mm
 Nozzle Thickness used (t) = 180.000 mm
 T / t = 0.217
 d / t = 13.631

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 4.783$$

Nr/P = 0.043
 Mr/P = 0.076
 M0/P = 0.022
 N0/P = 0.056
 MrD/Mc = 0.246
 NrDL/Mc = 0.079
 M0d/Mc = 0.074
 N0DL/Mc = 0.085
 MrD/Ml = 0.059
 NrDL/Ml = 0.037
 M0D/Ml = 0.018
 N0DL/Ml = 0.050

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	40	40	40	40
Outplane Stress Summary	34	42	85	23

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-55	55	55	-55
Inplane Pressure Stress	20	20	20	20
Inplane Stress Summary	10	27	133	-72

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	2	2	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	35	43	133	96

Vessel Stresses (MPa):



CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-13	-13	13	13
Outplane Bending (Mc)	-70	70	70	-70
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	20	20	20	20
Outplane Stress Summary	-43	69	123	-44

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-232	232	232	-232
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	40	40	40	40
Inplane Stress Summary	-152	217	336	-221

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	152	217	336	221

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-7	-7	7	7
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	3	3	3
Hoop Stress Summary	0	1	16	17

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0



Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	0	0	5	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	1	1	16	18
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-13	-13	13	13
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	3	3	3

Hoop Stress Summary	-4	-3	22	22
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-9	9	9	-9
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-6	8	13	-8
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	6	12	22	31
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Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	40	40	40	40	40	40	40	40
Circ. Pl (SUS)	-2	-2	12	12	-8	-8	16	16
Circ. Q (SUS)	-4	4	30	-30	-185	185	279	-279
Long. Pm (SUS)	20	20	20	20	20	20	20	20
Long. Pl (SUS)	-1	-1	9	9	-8	-8	18	18
Long. Q (SUS)	-8	8	102	-102	-57	57	83	-83
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	40.0	40.0	40.0	40.0	40.0	40.0	40.0	40.0
Pm+Pl (SUS)	38.0	38.0	52.0	52.0	32.0	32.0	56.0	56.0
Pm+Pl+Q (Total)	34.2	42.3	131.1	95.1	153.0	217.0	335.0	223.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	40.00	118.00	Passed
Pm+Pl (SUS)	56.00	177.00	Passed
Pm+Pl+Q (TOTAL)	335.02	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	3	3	3	3	3	3	3
Circ. Pl (SUS)	-2	-2	12	12	-8	-8	18	18
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	2	-2	-8	8	10	-10
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Pm+Pl (SUS)	1.0	1.0	15.0	15.0	6.0	6.0	21.0	21.0
Pm+Pl+Q (Total)	1.0	1.0	15.0	16.0	7.0	14.0	21.0	30.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	138.00	Passed
Pm+Pl (SUS)	21.00	207.00	Passed

PV Elite 28 U2 Licensee: SL User
FileName : C-1401_Ope_Rev 0
Nozzle Calcs.: N2A 84"

Page 320 of 564
Noz1: 129 4:01pm Jul 21,2026



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Pm+Pl+Q (TOTAL)	30.00	414.00	Passed
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*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: MH2 30"

From: 90

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	5099.9990	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		5850.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 70 +IT	
Material UNS Number		K02700	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		45.00	deg
Diameter		30.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

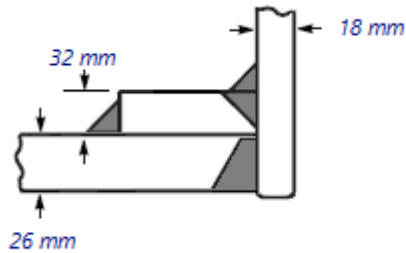
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1462.0000	mm
Thickness of Pad	te	32.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	32.0000	mm
Reinforcing Pad Width		350.0000	mm
This is a Manway, Coupling or Access Opening.			

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: MH2 30"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	30.000 in.
Actual Thickness Used in Calculation	0.709 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

This nozzle looks to be 26 inch or greater OD and the flange series is set to None. Please set it to series A or B in the nozzle dialog or the flange weight will not be determined.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 381.0 / (138 \cdot 1.0 + 0.4 \cdot 0.47)$
= 1.3063 mm

Required Nozzle thickness under External Pressure per UG-28 : 1.3927 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1452.0000	mm
Parallel to Vessel Wall, opening length	d	726.0000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		62.5000	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
= min(1, S_n / S_v)
= min(1, $138.0 / 118.0$)
= 1.000

Weld Strength Reduction Factor [fr2]:
= min(1, S_n / S_v)
= min(1, $138.0 / 118.0$)
= 1.000

Weld Strength Reduction Factor [fr4]:
= min(1, S_p / S_v)
= min(1, $138.0 / 118.0$)
= 1.000

Weld Strength Reduction Factor [fr3]:
= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000

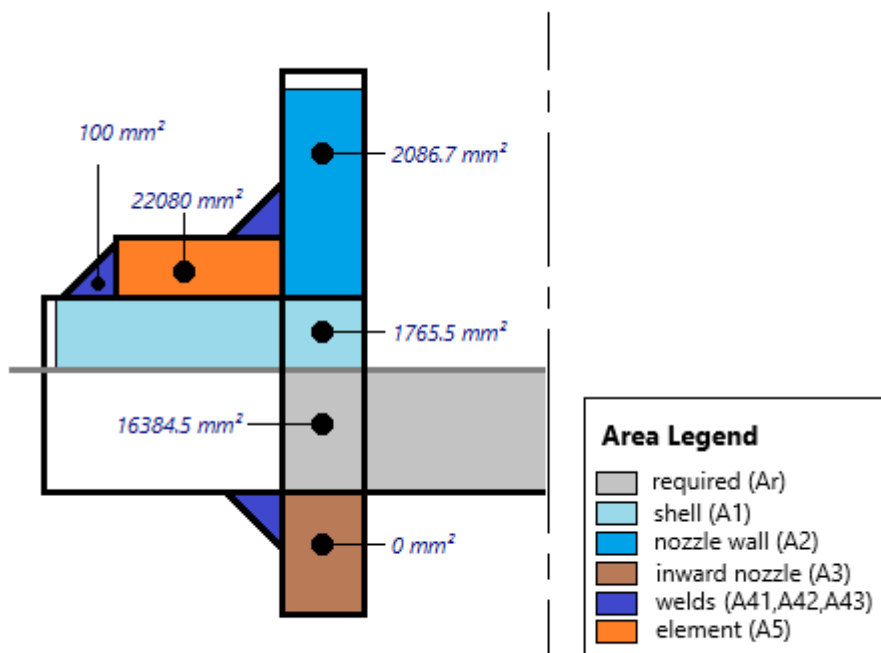


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	9801.265	5204.670	NA
Area in Shell	A1	8348.735	7740.658	NA
Area in Nozzle Wall	A2	2086.707	2075.911	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		100.000	100.000	NA
Area in Element	A5	22080.002	22080.002	NA
TOTAL AREA AVAILABLE	Atot	32615.443	31996.572	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (726.0 \cdot 13.5004 \cdot 1.0 + 2 \cdot 18.0 \cdot 13.5004 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 9801.265 \text{ mm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 726.0 (1.0 \cdot 25.0 - 1.0 \cdot 13.5) - 2 \cdot 18.0 \\
 &\quad (1.0 \cdot 25.0 - 1.0 \cdot 13.5004) \cdot (1 - 1.0) \\
 &= 8348.735 \text{ mm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 62.5) \cdot (18.0 - 1.31) \cdot 1.0 \\
 &= 2086.707 \text{ mm}^2
 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned}
 &= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\
 &= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0
 \end{aligned}$$



= 100.000 mm²

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot f_{r4} \\ &= (1452.0 - 762.0) \cdot 32.0 \cdot 1.0 \\ &= 22080.002 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 1.3927 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= t_a \\ &= 1.3927 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 18.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, $t_g = 18.0$, $t_r = 1.306$, $c = 0.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.073$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 18.0$, $t_r = 1.306$, $c = 0.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.073$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:C)

Govrn. thk, $t_g = 26.0$, $t_r = 13.5$, $c = 1.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.54$, Temp. Reduction = 28 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	0 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-28 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: C

Govrn. thk, $t_g = 18.0$, $t_r = 1.306$, $c = 0.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.073$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve C	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -28 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -28 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $0.47/1.96 = 0.242$

Weld Size Calculations, Description: MH2 30"



Intermediate Calc. for nozzle/shell Welds Tmin 18.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	11.3120 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (9801.2646 - 8348.7354 + 2 \cdot 18.0 \cdot 1.0 \cdot (1.0 \cdot 25.0 - 13.5004)) \cdot 118)$
 $= 220226.19 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (2086.7065 + 22080.002 + 100.0 - 0.0 \cdot 1.0) \cdot 118$
 $= 2863176.25 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (2086.7065 + 0.0 + 100.0 + (900.0)) \cdot 118$
 $= 364193.81 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (2086.7065 + 0.0 + 100.0 + 22080.002 + (900.0)) \cdot 118$
 $= 2969365.50 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 762.0 \cdot 10.0 \cdot 0.49 \cdot 138$
 $= 809306. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 1462.0 \cdot 16.0 \cdot 0.49 \cdot 118$
 $= 2124323. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 372.0) \cdot (18.0 - 0.0) \cdot 0.7 \cdot 138$
 $= 2031915. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 762.0 \cdot 32.0 \cdot 0.74 \cdot 138$
 $= 3911098. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 762.0 \cdot (20.0 - 0.0) \cdot 0.74 \cdot 138$
 $= 2444436. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (SPEW + SNW) = (2124323 + 2031915) = 4156238 \text{ N}$

Failure Path [Path 2-2]:
 $= (Sonw + Tpgw + Tngw + Sinw)$
 $= (809307 + 3911098 + 2444436 + 0) = 7164841 \text{ N}$



Failure Path [Path 3-3]:

= (Spew + Tngw + Sinw)

= (2124323 + 2444436 + 0) = 4568759 N

Summary of Failure Path Calculations:

Path 1-1 = 4156237 N , must exceed W = 220226 N or W1 = 2863176 N

Path 2-2 = 7164840 N , must exceed W = 220226 N or W2 = 364193 N

Path 3-3 = 4568759 N , must exceed W = 220226 N or W3 = 2969365 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 21.7169 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 447.7169 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 (50*tnom/Rf*(1-Rf/Ro)) 2.419 %

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Input, Nozzle Desc: K9a 2"

From: 90

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	5099.9990	mm
Shell Finished (Minimum) Thickness	t	26.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		5850.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-105 +IT	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

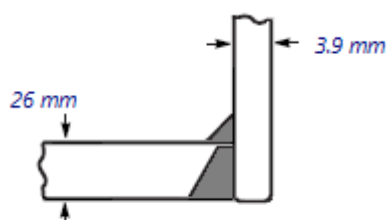
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K9a 2"



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (138 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1034 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	110.3250	mm
Parallel to Vessel Wall	Rn+tn+t	55.1625	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K9a 2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.5004	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.5004	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[3.42, max(14.5004, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 3.423, tr = 0.103, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.03$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: C

Govrn. thk, tg = 3.423, tr = 0.103, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.03$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve C	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K9a 2"

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 226.1356 mm

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Input, Nozzle Desc: N2B 84"

From: 90

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	5099.9990	mm
Shell Finished (Minimum) Thickness	t	40.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Note : User defined Limit(s) of Reinforcement specified below:

Physical Maximum for Diameter Limit	Dmax	5000.0000	mm
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Distance from Bottom/Left Tangent		4050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	MPa
Allowable Stress At Ambient	Sna	138.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		110.00	deg
Diameter		84.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm

Flange Material		SA-105	
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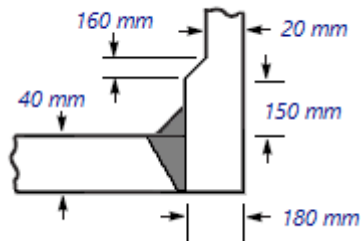
Hub Height of Integral Nozzle	h	150.0000	mm
Height of Beveled Transition	L'	160.0000	mm
Hub Thickness of Integral Nozzle (tn or x+tp)		180.0000	mm

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	800.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: N2B 84"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	84.000 in.
Actual Thickness Used in Calculation	0.787 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
 $= 13.5004 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.47 \cdot 1066.7999 / (138 \cdot 1.0 + 0.4 \cdot 0.47)$
 $= 3.6578 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 3.3853 mm

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:
 $= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$
 $= 160.0 / 0.5773$
 $= 277.1281 \text{ mm}$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

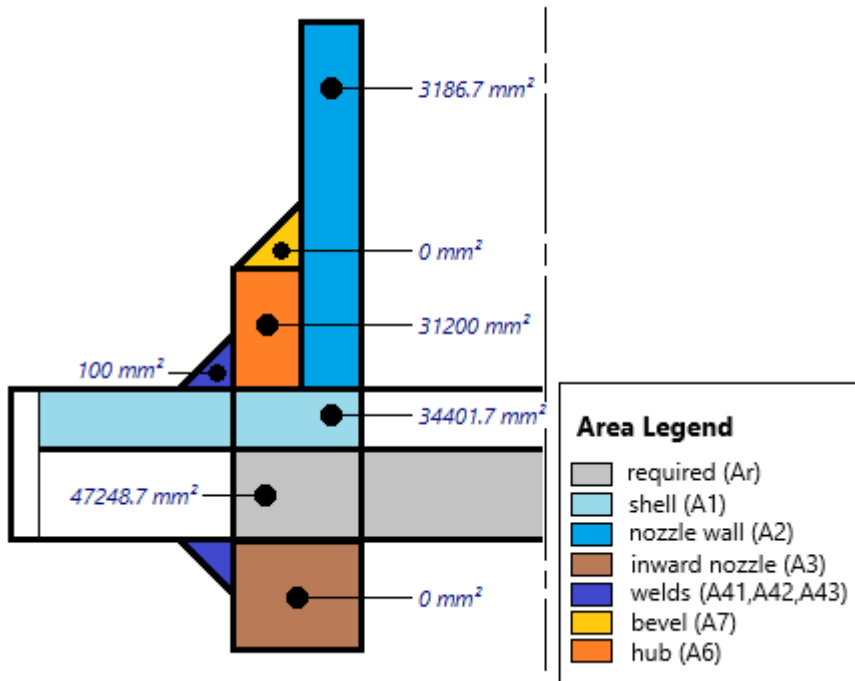
UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	4187.1997 mm
Parallel to Vessel Wall, opening length	d	2093.5999 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	97.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	28264.361	15008.950	NA
Area in Shell	A1	53386.031	51632.492	NA
Area in Nozzle Wall	A2	3186.734	3239.860	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	0.000	0.000	NA
Area in Hub	A6	31199.996	31199.996	NA
TOTAL AREA AVAILABLE	Atot	87872.766	86172.352	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (2093.5999 \cdot 13.5004 \cdot 1.0 + 2 \cdot 180.0 \cdot 13.5004 \cdot 1.0 \cdot (1 - 1.0))$
 $= 28264.361 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 2093.6(1.0 \cdot 39.0 - 1.0 \cdot 13.5) - 2 \cdot 180.0$
 $(1.0 \cdot 39.0 - 1.0 \cdot 13.5004) \cdot (1 - 1.0)$
 $= 53386.031 \text{ mm}^2$

Area Available in Nozzle Projecting Outward [A2]:
 $= (2 \cdot tlnp)(tn - trn)fr2$
 $= (2 \cdot 97.5)(20.0 - 3.66)1.0$
 $= 3186.734 \text{ mm}^2$

Area Available in Inward Weld + Outward Weld [A41 + A43]:
 $= Wo^2 \cdot fr2 + (Wi - can/0.707)^2 \cdot fr2$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0$
 $= 100.000 \text{ mm}^2$



Area Available in the Hub Section [A6]:

$$\begin{aligned} &= (2 \cdot \min(T_{lp}, h_o, H_{ubht})) \cdot (H_{ubt} - t_n) \cdot f_{r2} \\ &= (2 \cdot \min(97.5, 800.0, 150.0)) \cdot (180.0 - 20.0) \cdot 1.0 \\ &= 31199.996 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 3.6578 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 14.5004 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 14.5004 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.3941 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[8.331, \max(14.5004, 2.5)] \\ &= 8.3312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(3.6578, 8.3312) \\ &= 8.3312 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 23.1,	Allowable	: 138.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 321.9 MPa	Passed
Occasional	: 12.3,	Allowable	: 183.5 MPa	Passed
Shear	: 8.4,	Allowable	: 96.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 20.0, tr = 3.658, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.183, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 40.0, tr = 13.5, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.346, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	11 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/1.96 = 0.242

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:



$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 160.0/0.5773$$

$$= 277.1281 \text{ mm}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 3140.3997 mm
 Parallel to Vessel Wall, opening length d 1570.1998 mm
 Normal to Vessel Wall (Thickness Limit), no pad Tlnp 97.5000 mm

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (mm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure]	AR	18843.850	18843.850
Area Available in Shell	A1	26693.014	26693.014
Area Available in Nozzle Wall	A2	3186.734	3186.734
Area Available in Inward Nozzle	A3	0.000	0.000
Area Available in Welds	A4	100.000	100.000
Area Available in Pad	A5	0.000	0.000
Area Available in Hub	A6	31199.996	31199.996
TOTAL AREA AVAILABLE	Atot	61179.742	61179.742

The area available without a pad is Sufficient.

The following Calculation is performed using the internal design pressure (P) in the corroded condition.

Warning: The flange could not be considered in the moment of inertia calculations.

Area/Inertia Contributions from Fig.1-7-2 (for Bending stress only):

	Area	Distance	Inertia	d	I+Ad ²
Nozzle	7180.0	179.5	0.7711E+08	89.243	0.1343E+09
Hub	36800.0	119.6	0.2007E+09	29.380	0.2325E+09
Shell	24336.0	19.5	0.3085E+07	70.757	0.1249E+09
Sum	68316.0				0.4917E+09

Membrane stress per Appendix 1-7 (1 or 2) Sm	50.56	MPa
Allowable Membrane Stress S	118.00	MPa
Bending stress at the inside of the Nozzle Sb	38.24	MPa
Combined Membrane plus Bending Stress Sm + Sb	88.79	MPa
Allowable Membrane + Bending Stress 1.5 * S	177.00	MPa
M.A.W.P for the Appendix 1-7 Stress calc	0.94	MPa

Area contributions from Fig. 1-7-1 for Membrane stress per 1-7(b)(4):

Area from Nozzle = 3687.645 mm²
 Area from Shell = 14146.116 mm²
 Total Area, As = 46743.852 mm²

Membrane Stress Calc for Large Nozzles per 1-7(b)(4) [Sm]:

$$= P [R (Rn + tn + \sqrt{rm \cdot t}) + Rn (t + \sqrt{Rnm \cdot tn})] / As$$

$$= 0.474 [3354.0 (1046.8 + 20.0 + 362.721) + 1046.8 (184.382)] / 46743.852$$

$$= 50.557 \text{ MPa}$$

The area As was computed per figure 1-7-1 for the Sm calculation.

Moment [M]:

$$= (Rn^3/6 + R \cdot Rn \cdot e) P$$

$$= (1046.8^3/6 + 3354.0 \cdot 1046.8 \cdot 70.757) 0.474$$

$$= 0.20836E+09 \text{ N-mm}$$

Bending Stress Calc for Large Nozzles per 1-7(b)(4)(5) [Sb]:

$$= (M (e + t/2)) / I$$

$$= (0.20836E+09 (70.757 + 39.0/2)) / .49168E+09$$

$$= 38.236 \text{ MPa}$$



Weld Size Calculations, Description: N2B 84"

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•t_n•f_{r1}•(E1•t-tr))S_v)
= max(0, (28264.3613 - 53386.0312 + 2 • 180.0 • 1.0 •
(1.0 • 39.0 - 13.5004))118)
= max(0, -1880939.00) N

For hub type nozzles, A₂ includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A₂+A₄-(W_i-Can/.707)²•f_{r2})•S_v
= (34386.7266 + 100.0 - 0.0 • 1.0) • 118
= 4069014.50 N

Weld Load [W2]:
= (A₂ + A₃ + A₄ + (2 • t_n • t • f_{r1})) • S_v
= (34386.7266 + 0.0 + 100.0 + (14040.0)) • 118
= 5725564.00 N

Weld Load [W3]:
= (A₂+A₃+A₄+A₅+(2•t_n•t•f_{r1}))•S
= (34386.7266 + 0.0 + 100.0 + 0.0 + (14040.0)) • 118
= 5725564.00 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • D_{lo} • W_o • 0.49 • S_{nw}
= (3.1416/2.0) • 2133.5999 • 10.0 • 0.49 • 118
= 1937609. N

Shear, Nozzle Wall [Snw]:
= (π • (D_{lr} + D_{lo})/4) • (Thk - Can) • 0.7 • S_n
= (3.1416 • 1056.8) • (180.0 - 0.0) • 0.7 • 138
= 57723860. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • D_{lo} • W_{gnvi} • 0.74 • S_{ng}
= (3.1416/2.0) • 2133.5999 • (20.0 - 0.0) • 0.74 • 138
= 6844422. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (1937609 + 57723860) = 59661472 N

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (1937609 + 0 + 6844422 + 0) = 8782031 N

Failure Path [Path 3-3]:
= (Sonw + Tngw + Sinw)
= (1937609 + 6844422 + 0) = 8782031 N

Summary of Failure Path Calculations:

Path 1-1 = 59661472N	, must exceed W = 0 N	or W1 = 4069014 N
Path 2-2 = 8782031 N	, must exceed W = 0 N	or W2 = 5725564 N



Path 3-3 = 8782031 N , must exceed W = 0 N or W3 = 5725564 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 174.2351 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 1014.2350 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N2B 84" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	6706.000	mm
Shell Thickness	40.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 60	
Vessel Cold S.I. Allowable	Smc	118.00 MPa
Vessel Hot S.I. Allowable	Smh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	2453.600	mm
Nozzle Thickness	180.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc	138.00 MPa
Nozzle Hot S.I. Allowable	SNmh	138.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	0.47	MPa
Radial Load	P	-157500.11	N
Circumferential Shear	Vc	104999.93	N
Longitudinal Shear	Vl	104999.93	N
Circumferential Moment	Mc	0.588E+09	N-mm
Longitudinal Moment	Ml	0.588E+09	N-mm
Torsional Moment	Mt	0.840E+09	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))} \\
 &= 2453.6 + 2 \cdot 1.65 \cdot \sqrt{(3373.5(40.0 - 1.0))} \\
 &= 3650.579 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 6747.000 mm



Nozzle Outside Diameter (d) = 2453.600 mm
 Vessel Thickness used (T) = 39.000 mm
 Nozzle Thickness used (t) = 180.000 mm
 T / t = 0.217
 d / t = 13.631

$$\lambda = [(d/D) \cdot \sqrt{D/T}] = 4.783$$

Nr/P = 0.043
 Mr/P = 0.076
 M0/P = 0.022
 N0/P = 0.056
 MrD/Mc = 0.246
 NrDL/Mc = 0.079
 M0d/Mc = 0.074
 N0DL/Mc = 0.085
 MrD/Ml = 0.059
 NrDL/Ml = 0.037
 M0D/Ml = 0.018
 N0DL/Ml = 0.050

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7
Outplane Bending (ML)	-17	17	17	-17
Normal Pressure Stress	40	40	40	40
Outplane Stress Summary	34	42	85	23

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-55	55	55	-55
Inplane Pressure Stress	20	20	20	20
Inplane Stress Summary	10	27	133	-72

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	2	2	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	35	43	133	96

Vessel Stresses (MPa):



CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	13	-13	13	-13
Outplane Membrane (Mc)	-13	-13	13	13
Outplane Bending (Mc)	-70	70	70	-70
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	20	20	20	20
Outplane Stress Summary	-43	69	123	-44

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	-12	-12	12	12
Inplane Bending (Mc)	-232	232	232	-232
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	40	40	40	40
Inplane Stress Summary	-152	217	336	-221

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	2	2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	152	217	336	221

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-7	-7	7	7
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	3	3	3
Hoop Stress Summary	0	1	16	17

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0



Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	-1	1	1	-1
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	0	0	5	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	1	1	16	18
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-13	-13	13	13
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	3	3	3	3

Hoop Stress Summary	-4	-3	22	22
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	0	0	0	0
Axial Bending (P)	1	-1	1	-1
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	-9	9	9	-9
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	1	1	1	1

Axial Stress Summary	-6	8	13	-8
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0

Shear Stress Summary	0	0	0	0
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	6	12	22	31
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Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	40	40	40	40	40	40	40	40
Circ. Pl (SUS)	-2	-2	12	12	-8	-8	16	16
Circ. Q (SUS)	-4	4	30	-30	-185	185	279	-279
Long. Pm (SUS)	20	20	20	20	20	20	20	20
Long. Pl (SUS)	-1	-1	9	9	-8	-8	18	18
Long. Q (SUS)	-8	8	102	-102	-57	57	83	-83
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	40.0	40.0	40.0	40.0	40.0	40.0	40.0	40.0
Pm+Pl (SUS)	38.0	38.0	52.0	52.0	32.0	32.0	56.0	56.0
Pm+Pl+Q (Total)	34.2	42.3	131.1	95.1	153.0	217.0	335.0	223.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	40.00	118.00	Passed
Pm+Pl (SUS)	56.00	177.00	Passed
Pm+Pl+Q (TOTAL)	335.02	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	3	3	3	3	3	3	3	3
Circ. Pl (SUS)	-2	-2	12	12	-8	-8	18	18
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	1	1	1	1	1	1	1	1
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	0	0	2	-2	-8	8	10	-10
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	3.0	3.0	3.0	3.0	3.0	3.0	3.0	3.0
Pm+Pl (SUS)	1.0	1.0	15.0	15.0	6.0	6.0	21.0	21.0
Pm+Pl+Q (Total)	1.0	1.0	15.0	16.0	7.0	14.0	21.0	30.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	3.00	138.00	Passed
Pm+Pl (SUS)	21.00	207.00	Passed

PV Elite 28 U2 Licensee: SL User
FileName : C-1401_Ope_Rev 0
Nozzle Calcs.: N2B 84"

Page 341 of 564
Nozl: 132 4:01pm Jul 21, 2026



HEXAGON

Pm+Pl+Q (TOTAL)	30.00	414.00	Passed
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*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: K9b 2"

From: 130

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2000.0040	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		17650.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

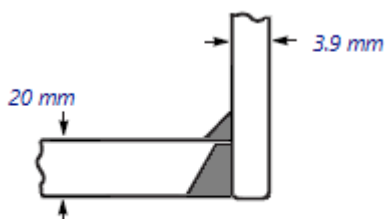
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	20.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K9b 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1209 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.0036	mm
Parallel to Vessel Wall, opening length	d	52.5018	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K9b 2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.5004	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.5004	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(14.5004, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K9b 2"

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.1356 mm

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Input, Nozzle Desc: MH3 30"

From: 140

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2000.0040	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		18150.01	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 60 +IT	
Material UNS Number		K02100	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		45.00	deg
Diameter		30.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

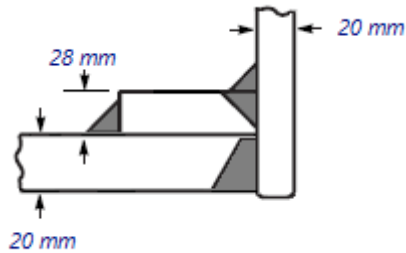
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 60 +IT	
Pad Allowable Stress at Temperature	Sp	118.00	MPa
Pad Allowable Stress At Ambient	Spa	118.00	MPa
Diameter of Pad along vessel surface	Dp	1422.0000	mm
Thickness of Pad	te	28.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm
Reinforcing Pad Width		330.0000	mm
This is a Manway, Coupling or Access Opening.			

Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: MH3 30"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	30.000 in.
Actual Thickness Used in Calculation	0.787 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
 $= 13.5004 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.47 \cdot 381.0 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
 $= 1.5274 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3927 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1444.0000 mm
Parallel to Vessel Wall, opening length	d	722.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		47.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

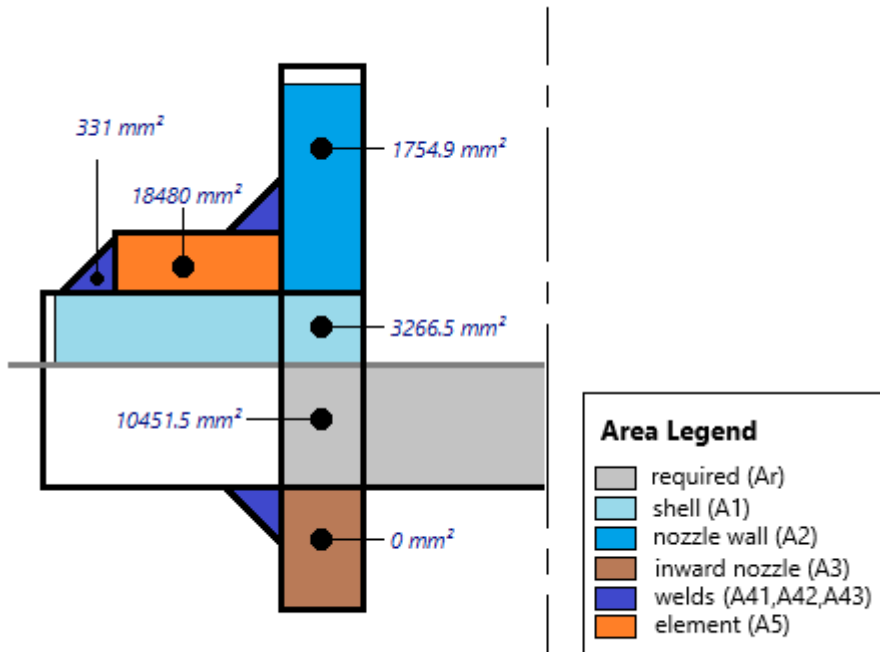


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	9747.263	3484.215	NA
Area in Shell	A1	3970.737	6749.571	NA
Area in Nozzle Wall	A2	1754.896	1767.692	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	331.000	331.000	NA
Area in Element	A5	18480.002	18480.002	NA
TOTAL AREA AVAILABLE	Atot	24536.635	27328.264	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 893.8082 28.0000 mm
 Based on given Pad Diameter: 1422.0000 5.5919 mm
 Based on Shell or Nozzle Thickness: 946.5316 20.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (722.0 \cdot 13.5004 \cdot 1.0 + 2 \cdot 20.0 \cdot 13.5004 \cdot 1.0 \cdot (1 - 1.0))$
 $= 9747.263 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 722.0(1.0 \cdot 19.0 - 1.0 \cdot 13.5) - 2 \cdot 20.0$
 $(1.0 \cdot 19.0 - 1.0 \cdot 13.5004) \cdot (1 - 1.0)$
 $= 3970.737 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 47.5) \cdot (20.0 - 1.53) \cdot 1.0$



= 1754.896 mm²

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (100.0) • 1.0 + (0.0) • 1.0 + 231.0² • 1.0
= 331.000 mm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (1422.0 - 762.0) • 28.0 • 1.0
= 18480.002 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures ta = 1.5274 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= ta
= 1.5274 mm

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 20.0, tr = 1.527, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.076, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: C

Govrn. thk, tg = 20.0, tr = 13.5, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.711, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve C	-27 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: C

Govrn. thk, tg = 20.0, tr = 13.5, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.711, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve C	-27 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-43 °C
Gov. MDMT of the Reinforcement Pad	:	-43 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/1.96 = 0.242



Weld Size Calculations, Description: MH3 30"

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm
Intermediate Calc. for pad/shell Welds T_{minPad} 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm
Pad Weld	9.5000 = 0.5*T _{minPad}	11.3120 = 0.7 * W _p mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•t_n•f_{r1}•(E1•t-tr))•S_v)
= max(0, (9747.2627 - 3970.7368 + 2 • 20.0 • 1.0 •
(1.0 • 19.0 - 13.5004))118)
= 707515.44 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(W_i-Can/.707)²•f_{r2})•S_v
= (1754.8962 + 18480.002 + 331.0002 - 0.0 • 1.0) • 118
= 2426525.75 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • t_n • t • f_{r1})) • S_v
= (1754.8962 + 0.0 + 100.0 + (760.0)) • 118
= 308525.94 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•t_n•t•f_{r1}))•S
= (1754.8962 + 0.0 + 331.0002 + 18480.002 + (760.0)) • 118
= 2516196.50 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [S_{onw}]:
= (π/2) • D_{lo} • W_o • 0.49 • S_{nw}
= (3.1416/2.0) • 762.0 • 10.0 • 0.49 • 118
= 692016. N

Shear, Pad Element Weld [S_{pew}]:
= (π/2) • D_p • W_p • 0.49 • S_{EW}
= (3.1416/2.0) • 1422.0 • 16.0 • 0.49 • 118
= 2066202. N

Shear, Nozzle Wall [S_{nw}]:
= (π • (D_{lr} + D_{lo})/4) • (Thk - Can) • 0.7 • S_n
= (3.1416 • 371.0) • (20.0 - 0.0) • 0.7 • 118
= 1925293. N

Tension, Pad Groove Weld [T_{pgw}]:
= (π/2) • D_{lo} • W_{gpn} • 0.74 • S_{eg}
= (3.1416/2) • 762.0 • 28.0 • 0.74 • 118
= 2926238. N

Tension, Shell Groove Weld [T_{ngw}]:
= (π/2) • D_{lo} • W_{gnvi} • 0.74 • S_{ng}
= (3.1416/2.0) • 762.0 • (18.0 - 0.0) • 0.74 • 118
= 1881153. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SPEW + SNW) = (2066202 + 1925293) = 3991495 N



Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (692016 + 2926238 + 1881153 + 0) = 5499407 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (2066202 + 1881153 + 0) = 3947355 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 3991495 N	, must exceed W = 707515 N	or W1 = 2426525 N
Path 2-2 = 5499407 N	, must exceed W = 707515 N	or W2 = 308525 N
Path 3-3 = 3947355 N	, must exceed W = 707515 N	or W3 = 2516196 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 21.7169 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 441.7168 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 2.695 %

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Input, Nozzle Desc: N4 20"

From: 140

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2000.0040	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		18150.01	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 60 +IT	
Material UNS Number		K02100	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		200.00	deg
Diameter		20.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

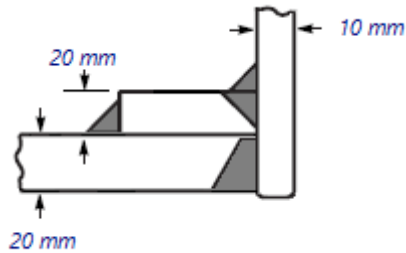
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 60 +IT	
Pad Allowable Stress at Temperature	Sp	118.00	MPa
Pad Allowable Stress At Ambient	Spa	118.00	MPa
Diameter of Pad along vessel surface	Dp	1148.0000	mm
Thickness of Pad	te	20.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm
Reinforcing Pad Width		320.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: N4 20"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	20.000 in.
Actual Thickness Used in Calculation	0.394 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 254.0 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 1.0183 mm

Required Nozzle thickness under External Pressure per UG-28 : 1.0963 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	976.0000 mm
Parallel to Vessel Wall, opening length	d	488.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		45.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 118.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 118.0)$
= 1.000

Weld Strength Reduction Factor [fr4]:
= $\min(1, S_p / S_v)$
= $\min(1, 118.0 / 118.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

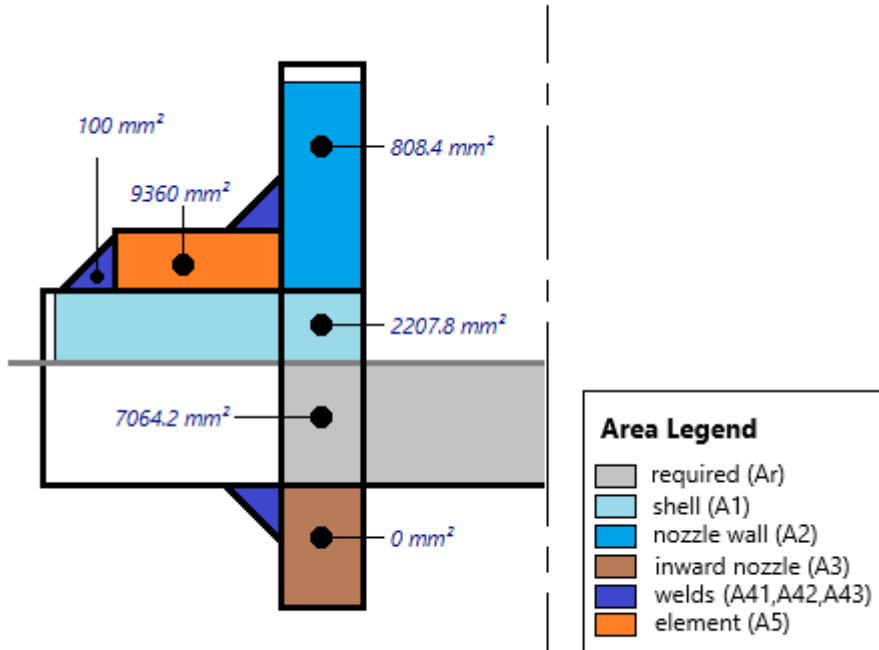


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	6588.178	2354.981	NA
Area in Shell	A1	2683.822	4562.037	NA
Area in Nozzle Wall	A2	808.356	801.330	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	9360.000	9360.000	NA
TOTAL AREA AVAILABLE	Atot	12952.178	14823.366	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 657.8000 20.0000 mm
 Based on given Pad Diameter: 1148.0000 6.4017 mm
 Based on Shell or Nozzle Thickness: 807.6000 10.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (488.0 \cdot 13.5004 \cdot 1.0 + 2 \cdot 10.0 \cdot 13.5004 \cdot 1.0 \cdot (1 - 1.0))$
 $= 6588.178 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 488.0(1.0 \cdot 19.0 - 1.0 \cdot 13.5) - 2 \cdot 10.0$
 $(1.0 \cdot 19.0 - 1.0 \cdot 13.5004) \cdot (1 - 1.0)$
 $= 2683.822 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 45.0) \cdot (10.0 - 1.02) \cdot 1.0$



= 808.356 mm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
= $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0$
= 100.000 mm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
= $(976.0 - 508.0) \cdot 20.0 \cdot 1.0$
= 9360.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.0963 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 14.5004 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 14.5004 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.3941 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[8.331, max(14.5004, 2.5)]
= 8.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.0963, 8.3312)
= 8.3312 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 29.2,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 265.8 MPa	Passed
Occasional	: 5.7,	Allowable	: 156.9 MPa	Passed
Shear	: 18.0,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govern. thk, tg = 10.0, tr = 1.018, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.102$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: C

Govern. thk, tg = 20.0, tr = 13.5, c = 1.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.711$, Temp. Reduction = 16 °C



Min Metal Temp. w/o impact per UCS-66, Curve C	-27 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: C

Govrn. thk, tg = 10.0, tr = 1.018, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.102$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve C	-44 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -43 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.47/1.96 = 0.242$

Weld Size Calculations, Description: N4 20"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o$ mm
Pad Weld	9.5000 = $0.5 \cdot T_{minPad}$	11.3120 = $0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (6588.1777 - 2683.822 + 2 \cdot 10.0 \cdot 1.0 \cdot (1.0 \cdot 19.0 - 13.5004)) \cdot 118)$
 $= 473644.31 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (808.3555 + 9360.0 + 100.0 - 0.0 \cdot 1.0) \cdot 118$
 $= 1211541.12 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (808.3555 + 0.0 + 100.0 + (380.0)) \cdot 118$
 $= 152010.28 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (808.3555 + 0.0 + 100.0 + 9360.0 + (380.0)) \cdot 118$
 $= 1256376.38 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 508.0 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 461344. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 1148.0 \cdot 16.0 \cdot 0.49 \cdot 118$
 $= 1668073. \text{ N}$



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 249.0) \cdot (10.0 - 0.0) \cdot 0.7 \cdot 118$$

$$= 646089. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 508.0 \cdot 20.0 \cdot 0.74 \cdot 118$$

$$= 1393447. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 508.0 \cdot (18.0 - 0.0) \cdot 0.74 \cdot 118$$

$$= 1254102. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (1668073 + 646089) = 2314162 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (461344 + 1393447 + 1254102 + 0) = 3108893 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (1668073 + 1254102 + 0) = 2922175 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 2314162 N , must exceed W = 473644 N or W1 = 1211541 N
 Path 2-2 = 3108892 N , must exceed W = 473644 N or W2 = 152010 N
 Path 3-3 = 2922175 N , must exceed W = 473644 N or W3 = 1256376 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 9.6347 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 429.6346 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.008 %

Input Echo, WRC297 Item 1, Description: N4 20" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	6706.000	mm
Shell Thickness	20.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 60	
Vessel Cold S.I. Allowable	Smc 118.00	MPa
Vessel Hot S.I. Allowable	Smh 118.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	508.000	mm
Nozzle Thickness	10.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 60	



Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	1148.000	mm
Thickness of Reinforcing Pad	20.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	0.47	MPa
Radial Load	P	-37500.02	N
Circumferential Shear	Vc	24999.98	N
Longitudinal Shear	Vl	24999.98	N
Circumferential Moment	Mc	0.350E+08	N-mm
Longitudinal Moment	Ml	0.350E+08	N-mm
Torsional Moment	Mt	0.500E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{(3373.5(20.0 - 1.0))}$$

$$= 1343.470 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6747.000	mm
Nozzle Outside Diameter	(d) =	508.000	mm
Vessel Thickness used	(T) =	39.000	mm
Nozzle Thickness used	(t) =	10.000	mm
T / t	=	3.900	
d / t	=	50.800	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.990$$

Nr/P	=	0.059
Mr/P	=	0.083
M0/P	=	0.106
N0/P	=	0.285
MrD/Mc	=	0.136
NrDL/Mc	=	0.091
M0d/Mc	=	0.224
N0DL/Mc	=	0.213
MrD/Ml	=	0.113
NrDL/Ml	=	0.080
M0D/Ml	=	0.165
N0DL/Ml	=	0.410

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-18	-18	18	18
Outplane Bending (ML)	-44	44	44	-44
Normal Pressure Stress	40	40	40	40



Outplane Stress Summary 0 58 126 6

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stresses parallel to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-30	30	30	-30
Inplane Pressure Stress	20	20	20	20

Inplane Stress Summary 0 36 68 -17

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary 3 3 2 2

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	7	59	126	24
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-60	60	60	-60
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	20	20	20	20

Outplane Stress Summary -27 62 113 -39

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	12	-12	12	-12
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-36	36	36	-36
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	40	40	40	40

Inplane Stress Summary 13 62 95 -2

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------



Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	3	3

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	40	65	114	40

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-18	-18	18	18
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	14	13	14
Hoop Stress Summary	2	2	39	39

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	120	-120	120	-120
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-18	-18	18	18
Axial Bending (ML)	-304	304	304	-304
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-192	174	452	-397

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	12	12	12	12
Shear Stress Summary	15	15	9	9

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	196	175	452	437

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0



Hoop Pressure Stress	13	14	13	14
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Hoop Stress Summary	10	11	30	30
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	120	-120	120	-120
Axial Membrane (Mc)	-18	-18	18	18
Axial Bending (Mc)	-373	373	373	-373
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-261	243	521	-466
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	12	12	12	12

Shear Stress Summary	9	9	15	15
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	273	243	522	498
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{3363.5(20.0 - 1.0)}$$

$$= 1342.231 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 6727.000 mm

Nozzle Outside Diameter (d) = 1148.000 mm

Vessel Thickness used (T) = 19.000 mm

Nozzle Thickness used (t) = 330.000 mm

T / t = 0.058

d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.211$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.063

Mr/P = 0.102

M0/P = 0.029

N0/P = 0.031

MrD/Mc = 0.265

NrDL/Mc = 0.117

M0d/Mc = 0.079

N0DL/Mc = 0.052

MrD/Ml = 0.073

NrDL/Ml = 0.084

M0D/Ml = 0.022

N0DL/Ml = 0.040

Vessel Stresses (MPa):



LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-11	11	11	-11
Normal Pressure Stress	83	83	83	83
Outplane Stress Summary	90	76	119	61

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	63	-63	63	-63
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-36	36	36	-36
Inplane Pressure Stress	41	41	41	41
Inplane Stress Summary	67	14	155	-45

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	90	76	155	106

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	17	-17	17	-17
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-40	40	40	-40
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	41	41	41	41
Outplane Stress Summary	18	62	107	-8

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	63	-63	63	-63
Inplane Membrane (Mc)	-9	-9	9	9

Inplane Bending (Mc)	-134	134	134	-134
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	83	83	83	83

Inplane Stress Summary	9	151	297	-97
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	1	1
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	18	151	297	97

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	40	40	40	40	40	40	40	40
Circ. Pl (SUS)	-11	-11	25	25	-3	-3	5	5
Circ. Q (SUS)	-29	29	59	-59	-24	24	48	-48
Long. Pm (SUS)	20	20	20	20	20	20	20	20
Long. Pl (SUS)	-2	-2	4	4	-2	-2	16	16
Long. Q (SUS)	-18	18	42	-42	-45	45	75	-75
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	40.0	40.0	40.0	40.0	40.0	40.0	40.0	40.0
Pm+Pl (SUS)	29.0	29.0	65.0	65.0	37.0	37.0	45.0	45.0
Pm+Pl+Q (Total)	6.0	58.4	124.2	24.7	40.4	65.2	111.5	39.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	40.00	118.00	Passed
Pm+Pl (SUS)	65.00	177.00	Passed
Pm+Pl+Q (TOTAL)	124.15	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl



Circ. Pm (SUS)	83	83	83	83	83	83	83	83
Circ. Pl (SUS)	0	0	6	6	-3	-3	15	15
Circ. Q (SUS)	6	-6	28	-28	-71	71	197	-197
Long. Pm (SUS)	41	41	41	41	41	41	41	41
Long. Pl (SUS)	-1	-1	13	13	-1	-1	7	7
Long. Q (SUS)	27	-27	99	-99	-23	23	57	-57
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	83.0	83.0	83.0	83.0	83.0	83.0	83.0	83.0
Pm+Pl (SUS)	83.0	83.0	89.0	89.0	80.0	80.0	98.0	98.0
Pm+Pl+Q (Total)	89.0	77.0	153.0	106.0	17.1	151.0	295.0	99.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	83.00	118.00	Passed
Pm+Pl (SUS)	98.00	177.00	Passed
Pm+Pl+Q (TOTAL)	295.01	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	14	13	14	13	14	13	14
Circ. Pl (SUS)	-11	-11	25	25	-2	-2	16	16
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-16	-16	20	20	-16	-16	20	20
Long. Q (SUS)	-184	184	424	-424	-253	253	493	-493
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	13.0	14.0	13.0	14.0	13.0	14.0	13.0	14.0
Pm+Pl (SUS)	13.4	14.3	38.7	39.7	21.8	22.8	30.9	31.6
Pm+Pl+Q (Total)	198.3	175.3	450.2	437.4	274.6	243.4	519.5	497.9

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.00	118.00	Passed
Pm+Pl (SUS)	39.66	177.00	Passed
Pm+Pl+Q (TOTAL)	519.46	354.00	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N3a 24"

From: 140

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2999.9988	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		19870.01	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 60 +IT	
Material UNS Number		K02100	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	12.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

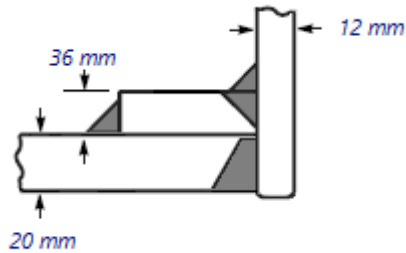
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	14.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 60 +IT	
Pad Allowable Stress at Temperature	Sp	118.00	MPa
Pad Allowable Stress At Ambient	Spa	118.00	MPa
Diameter of Pad along vessel surface	Dp	1563.6000	mm
Thickness of Pad	te	36.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	36.0000	mm
Reinforcing Pad Width		477.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: N3a 24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.472 in.

Note:

Post Weld Heat Treatment is required for this nozzle and it was not specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 304.8 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 1.2219 mm

Required Nozzle thickness under External Pressure per UG-28 : 1.2254 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1171.2000 mm
Parallel to Vessel Wall, opening length	d	585.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		47.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:

= min(1, S_n / S_v)
= min(1, $118.0 / 118.0$)
= 1.000

Weld Strength Reduction Factor [fr2]:

= min(1, S_n / S_v)
= min(1, $118.0 / 118.0$)
= 1.000

Weld Strength Reduction Factor [fr4]:

= min(1, S_p / S_v)
= min(1, $118.0 / 118.0$)
= 1.000

Weld Strength Reduction Factor [fr3]:

= min(fr2, fr4)
= min(1.0, 1.0)
= 1.000

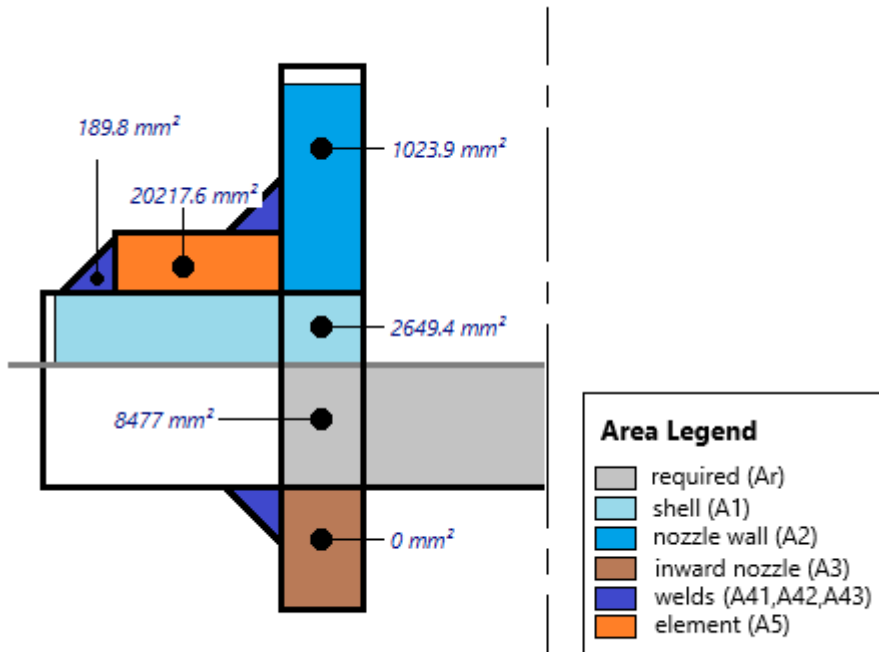


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	7905.813	3374.140	NA
Area in Shell	A1	3220.586	4378.120	NA
Area in Nozzle Wall	A2	1023.917	1023.588	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	189.750	189.750	NA
Area in Element	A5	20217.602	20217.602	NA
TOTAL AREA AVAILABLE	Atot	24651.855	25809.059	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 706.0322 36.0000 mm
 Based on given Pad Diameter: 1563.6000 6.1816 mm
 Based on Shell or Nozzle Thickness: 898.8965 12.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (585.6 \cdot 13.5004 \cdot 1.0 + 2 \cdot 12.0 \cdot 13.5004 \cdot 1.0 \cdot (1 - 1.0))$
 $= 7905.813 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 585.6(1.0 \cdot 19.0 - 1.0 \cdot 13.5) - 2 \cdot 12.0$
 $(1.0 \cdot 19.0 - 1.0 \cdot 13.5004) \cdot (1 - 1.0)$
 $= 3220.586 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 47.5) \cdot (12.0 - 1.22) \cdot 1.0$



= 1023.917 mm²

Area Available in Welds [A41 + A42 + A43]:

= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Wp²•fr4
= (189.75) • 1.0 + (0.0) • 1.0 + 0.0² • 1.0
= 189.750 mm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (1171.2 - 609.6) • 36.0 • 1.0
= 20217.602 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.2254 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 14.5004 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 14.5004 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.3941 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[8.331, max(14.5004, 2.5)]
= 8.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(1.2254, 8.3312)
= 8.3312 mm

Available Nozzle Neck Thickness = 12.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 25.3,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 269.7 MPa	Passed
Occasional	: 5.7,	Allowable	: 156.9 MPa	Passed
Shear	: 15.0,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 12.0, tr = 1.222, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.102, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: C

Govrn. thk, tg = 20.0, tr = 13.5, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.711, Temp. Reduction = 16 °C



Min Metal Temp. w/o impact per UCS-66, Curve C	-27 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: C

Govrn. thk, tg = 12.0, tr = 1.222, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.102$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve C	-39 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -43 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.47/1.96 = 0.242$

Weld Size Calculations, Description: N3a 24"

Intermediate Calc. for nozzle/shell Welds	Tmin	12.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	9.8980 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	11.3120 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (7905.8135 - 3220.5864 + 2 \cdot 12.0 \cdot 1.0 \cdot (1.0 \cdot 19.0 - 13.5004)) \cdot 118)$
 $= 568373.19 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (1023.9171 + 20217.6016 + 189.75 - 0.0 \cdot 1.0) \cdot 118$
 $= 2528629.00 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (1023.9171 + 0.0 + 196.0 + (456.0)) \cdot 118$
 $= 197737.84 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (1023.9171 + 0.0 + 189.75 + 20217.6016 + (456.0)) \cdot 118$
 $= 2582431.50 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 609.6 \cdot 14.0 \cdot 0.49 \cdot 118$
 $= 775058. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 1563.6 \cdot 16.0 \cdot 0.49 \cdot 118$
 $= 2271950. \text{ N}$



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 298.8) \cdot (12.0 - 0.0) \cdot 0.7 \cdot 118$$

$$= 930368. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 609.6 \cdot 36.0 \cdot 0.74 \cdot 118$$

$$= 3009845. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 609.6 \cdot (18.0 - 0.0) \cdot 0.74 \cdot 118$$

$$= 1504922. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (2271951 + 930368) = 3202319 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (775058 + 3009845 + 1504923 + 0) = 5289825 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (2271951 + 1504923 + 0) = 3776873 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 3202318 N	, must exceed W = 568373 N	or W1 = 2528629 N
Path 2-2 = 5289825 N	, must exceed W = 568373 N	or W2 = 197737 N
Path 3-3 = 3776873 N	, must exceed W = 568373 N	or W3 = 2582431 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 13.8825 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 433.8824 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.008 %

Input Echo, WRC297 Item 1, Description: N3a 24" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	6706.000	mm
Shell Thickness	20.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 60	
Vessel Cold S.I. Allowable	Smc	118.00 MPa
Vessel Hot S.I. Allowable	Smh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	609.600	mm
Nozzle Thickness	12.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 60	



Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	1563.600	mm
Thickness of Reinforcing Pad	36.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	0.47	MPa
Radial Load	P	-44999.97	N
Circumferential Shear	Vc	29999.98	N
Longitudinal Shear	Vl	29999.98	N
Circumferential Moment	Mc	0.504E+08	N-mm
Longitudinal Moment	Ml	0.504E+08	N-mm
Torsional Moment	Mt	0.720E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 609.6 + 2 \cdot 1.65 \cdot \sqrt{3381.5(20.0 - 1.0)}$$

$$= 1446.060 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6763.000	mm
Nozzle Outside Diameter	(d) =	609.600	mm
Vessel Thickness used	(T) =	55.000	mm
Nozzle Thickness used	(t) =	12.000	mm
T / t	=	4.583	
d / T	=	50.800	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.000$$

Nr/P	=	0.052
Mr/P	=	0.059
M0/P	=	0.112
N0/P	=	0.312
MrD/Mc	=	0.105
NrDL/Mc	=	0.086
M0d/Mc	=	0.252
N0DL/Mc	=	0.245
MrD/Ml	=	0.089
NrDL/Ml	=	0.076
M0D/Ml	=	0.185
N0DL/Ml	=	0.472

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au	Al	Bu	B1
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-12	-12	12	12
Outplane Bending (ML)	-30	30	30	-30
Normal Pressure Stress	28	29	28	29



Outplane Stress Summary 0 41 86 6

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stresses parallel to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	5	-5	5	-5
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-14	14	14	-14
Inplane Pressure Stress	14	14	14	14

Inplane Stress Summary 3 22 37 -2

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary 2 2 1 1

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress 6 41 86 9

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-41	41	41	-41
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	14	14	14	14

Outplane Stress Summary -18 43 76 -25

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	5	-5	5	-5
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-17	17	17	-17
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	28	29	28	29

Inplane Stress Summary 15 39 54 9

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------



Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	2	2
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	34	44	77	35
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-12	-12	12	12
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	14	13	14

Hoop Stress Summary	5	5	31	31
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	61	-61	61	-61
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-15	-15	15	15
Axial Bending (ML)	-177	177	177	-177
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-122	109	263	-215
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	10	10	10	10

Shear Stress Summary	12	12	7	7
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	129	110	263	247
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-6	-6	6	6
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0



Hoop Pressure Stress	13	14	13	14
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Hoop Stress Summary	11	12	24	25
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	61	-61	61	-61
Axial Membrane (Mc)	-15	-15	15	15
Axial Bending (Mc)	-211	211	211	-211
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-156	143	297	-249
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	10	10	10	10

Shear Stress Summary	7	7	12	12
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	168	143	298	276
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 609.6 + 2 \cdot 1.65 \cdot \sqrt{3363.5(20.0 - 1.0)}$
 $= 1443.831 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6727.000 mm
Nozzle Outside Diameter	(d) =	1563.600 mm
Vessel Thickness used	(T) =	19.000 mm
Nozzle Thickness used	(t) =	489.000 mm
T / t	=	0.039
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.374$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.051
Mr/P	=	0.078
M0/P	=	0.021
N0/P	=	0.022
MrD/Mc	=	0.246
NrDL/Mc	=	0.109
M0d/Mc	=	0.073
N0DL/Mc	=	0.045
MrD/Ml	=	0.053
NrDL/Ml	=	0.061
M0D/Ml	=	0.016
N0DL/Ml	=	0.025

Vessel Stresses (MPa):



LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-8	8	8	-8
Normal Pressure Stress	83	83	83	83
Outplane Stress Summary	91	77	112	64

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-28	28	28	-28
Inplane Pressure Stress	41	41	41	41
Inplane Stress Summary	72	12	140	-33

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	91	77	140	97

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-39	39	39	-39
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	41	41	41	41
Outplane Stress Summary	16	64	103	-6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	58	-58	58	-58
Inplane Membrane (Mc)	-9	-9	9	9

Inplane Bending (Mc)	-131	131	131	-131
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	83	83	83	83

Inplane Stress Summary	6	154	289	-90
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	0	0	1	1
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	16	154	289	90

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	28	29	28	29	28	29	28	29
Circ. Pl (SUS)	-8	-8	16	16	-2	-2	2	2
Circ. Q (SUS)	-20	20	40	-40	-12	12	22	-22
Long. Pm (SUS)	14	14	14	14	14	14	14	14
Long. Pl (SUS)	-2	-2	2	2	-2	-2	10	10
Long. Q (SUS)	-9	9	19	-19	-31	31	51	-51
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	28.0	29.0	28.0	29.0	28.0	29.0	28.0	29.0
Pm+Pl (SUS)	20.0	21.0	44.0	45.0	26.0	27.0	30.0	31.0
Pm+Pl+Q (Total)	5.0	41.2	84.1	8.9	33.2	43.8	75.2	36.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	29.00	118.00	Passed
Pm+Pl (SUS)	45.00	177.00	Passed
Pm+Pl+Q (TOTAL)	84.08	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl

Circ. Pm (SUS)	83	83	83	83	83	83	83	83
Circ. Pl (SUS)	0	0	4	4	-3	-3	15	15
Circ. Q (SUS)	7	-7	23	-23	-73	73	189	-189
Long. Pm (SUS)	41	41	41	41	41	41	41	41
Long. Pl (SUS)	1	1	11	11	-1	-1	5	5
Long. Q (SUS)	30	-30	86	-86	-24	24	54	-54
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	83.0	83.0	83.0	83.0	83.0	83.0	83.0	83.0
Pm+Pl (SUS)	83.0	83.0	87.0	87.0	80.0	80.0	98.0	98.0
Pm+Pl+Q (Total)	90.0	76.0	138.0	98.0	16.0	153.0	287.0	91.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	83.00	118.00	Passed
Pm+Pl (SUS)	98.00	177.00	Passed
Pm+Pl+Q (TOTAL)	287.00	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	14	13	14	13	14	13	14
Circ. Pl (SUS)	-8	-8	16	16	-2	-2	10	10
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-14	-14	16	16	-14	-14	16	16
Long. Q (SUS)	-116	116	238	-238	-150	150	272	-272
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	10	10	10	10	10	10	10	10
Pm (SUS)	13.0	14.0	13.0	14.0	13.0	14.0	13.0	14.0
Pm+Pl (SUS)	13.6	14.6	29.5	30.5	19.4	20.4	24.6	25.2
Pm+Pl+Q (Total)	131.2	109.4	260.3	246.5	169.8	142.5	294.5	275.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	14.00	118.00	Passed
Pm+Pl (SUS)	30.47	177.00	Passed
Pm+Pl+Q (TOTAL)	294.53	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: N3B 24"

From: 140

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2999.9988	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		19870.01	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 60 +IT	
Material UNS Number		K02100	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		270.00	deg
Diameter		24.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

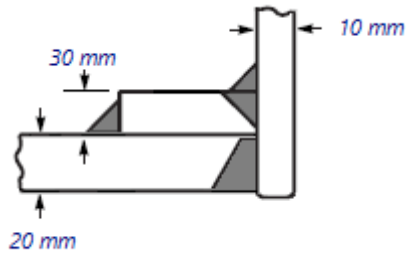
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 60 +IT	
Pad Allowable Stress at Temperature	Sp	118.00	MPa
Pad Allowable Stress At Ambient	Spa	118.00	MPa
Diameter of Pad along vessel surface	Dp	1509.6000	mm
Thickness of Pad	te	30.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	30.0000	mm
Reinforcing Pad Width		450.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: N3B 24"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	24.000 in.
Actual Thickness Used in Calculation	0.394 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 304.8 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 1.2219 mm

Required Nozzle thickness under External Pressure per UG-28 : 1.2254 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1179.2000 mm
Parallel to Vessel Wall, opening length	d	589.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		47.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 118.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 118.0)$
= 1.000

Weld Strength Reduction Factor [fr4]:
= $\min(1, S_p / S_v)$
= $\min(1, 118.0 / 118.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

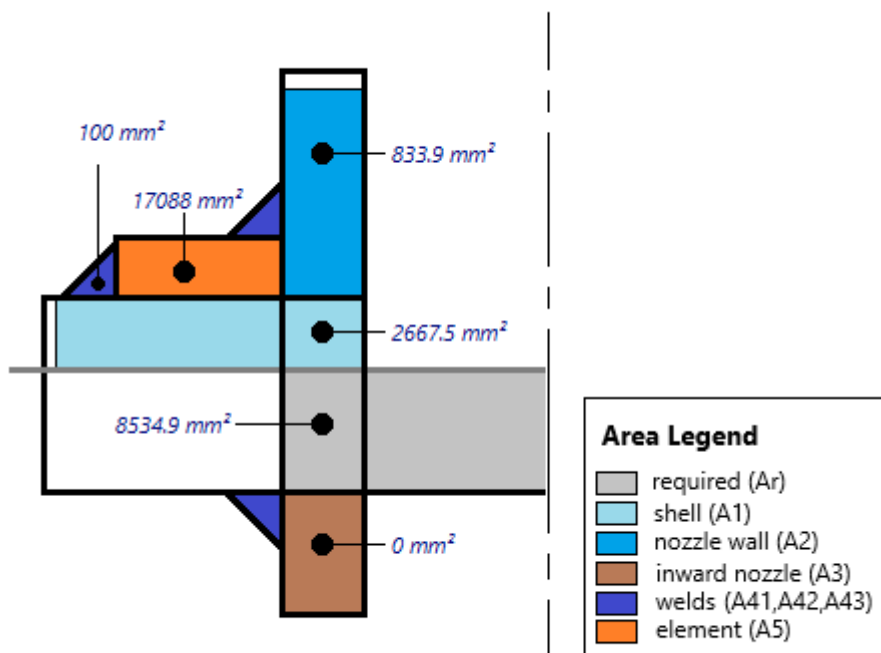


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	7959.815	3397.187	NA
Area in Shell	A1	3242.585	4408.025	NA
Area in Nozzle Wall	A2	833.917	833.589	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	17088.000	17088.000	NA
TOTAL AREA AVAILABLE	Atot	21264.502	22429.613	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 735.7104 30.0000 mm
 Based on given Pad Diameter: 1509.6000 6.6421 mm
 Based on Shell or Nozzle Thickness: 987.9313 10.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (589.6 \cdot 13.5004 \cdot 1.0 + 2 \cdot 10.0 \cdot 13.5004 \cdot 1.0 \cdot (1 - 1.0))$
 $= 7959.815 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 589.6(1.0 \cdot 19.0 - 1.0 \cdot 13.5) - 2 \cdot 10.0$
 $(1.0 \cdot 19.0 - 1.0 \cdot 13.5004) \cdot (1 - 1.0)$
 $= 3242.585 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 47.5) \cdot (10.0 - 1.22) \cdot 1.0$



= 833.917 mm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
= $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0$
= 100.000 mm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
= $(1179.2 - 609.6) \cdot 30.0 \cdot 1.0$
= 17088.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.2254 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 14.5004 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 14.5004 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.3941 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[8.331, max(14.5004, 2.5)]
= 8.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.2254, 8.3312)
= 8.3312 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 30.1,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 264.9 MPa	Passed
Occasional	: 6.9,	Allowable	: 156.9 MPa	Passed
Shear	: 17.8,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 10.0, tr = 1.222, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E*/(tg - c) = 0.122$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: C

Govrn. thk, tg = 20.0, tr = 13.5, c = 1.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E*/(tg - c) = 0.711$, Temp. Reduction = 16 °C



Min Metal Temp. w/o impact per UCS-66, Curve C	-27 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: C

Govrn. thk, tg = 10.0, tr = 1.222, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.122$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve C	-44 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -43 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $0.47/1.96 = 0.242$

Weld Size Calculations, Description: N3B 24"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o$ mm
Pad Weld	9.5000 = $0.5 \cdot T_{minPad}$	11.3120 = $0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A_1 + 2 \cdot t_n \cdot f_{r1} \cdot (E_1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (7959.8149 - 3242.5852 + 2 \cdot 10.0 \cdot 1.0 \cdot (1.0 \cdot 19.0 - 13.5004)) \cdot 118)$
 $= 569553.50 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A_2 + A_5 + A_4 - (W_i - Can / .707)^2 \cdot f_{r2}) \cdot S_v$
 $= (833.9169 + 17088.0 + 100.0 - 0.0 \cdot 1.0) \cdot 118$
 $= 2126367.00 \text{ N}$

Weld Load [W2]:
 $= (A_2 + A_3 + A_4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (833.9169 + 0.0 + 100.0 + (380.0)) \cdot 118$
 $= 155026.22 \text{ N}$

Weld Load [W3]:
 $= (A_2 + A_3 + A_4 + A_5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (833.9169 + 0.0 + 100.0 + 17088.0 + (380.0)) \cdot 118$
 $= 2171202.25 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 609.6 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 553613. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 1509.6 \cdot 16.0 \cdot 0.49 \cdot 118$
 $= 2193487. \text{ N}$



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 299.8) \cdot (10.0 - 0.0) \cdot 0.7 \cdot 118$$

$$= 777902. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 609.6 \cdot 30.0 \cdot 0.74 \cdot 118$$

$$= 2508204. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 609.6 \cdot (18.0 - 0.0) \cdot 0.74 \cdot 118$$

$$= 1504922. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (2193487 + 777902) = 2971389 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (553613 + 2508204 + 1504923 + 0) = 4566740 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (2193487 + 1504923 + 0) = 3698410 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 2971388 N , must exceed W = 569553 N or W1 = 2126367 N
 Path 2-2 = 4566739 N , must exceed W = 569553 N or W2 = 155026 N
 Path 3-3 = 3698409 N , must exceed W = 569553 N or W3 = 2171202 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 13.8825 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 433.8824 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.668 %

Input Echo, WRC297 Item 1, Description: N3B 24" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	6706.000	mm
Shell Thickness	20.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 60	
Vessel Cold S.I. Allowable	Smc	118.00 MPa
Vessel Hot S.I. Allowable	Smh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	609.600	mm
Nozzle Thickness	10.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 60	



Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	1509.600	mm
Thickness of Reinforcing Pad	30.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	0.47	MPa
Radial Load	P	-44999.97	N
Circumferential Shear	Vc	29999.98	N
Longitudinal Shear	Vl	29999.98	N
Circumferential Moment	Mc	0.504E+08	N-mm
Longitudinal Moment	Ml	0.504E+08	N-mm
Torsional Moment	Mt	0.720E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(\text{t} - \text{ca}))}$$

$$= 609.6 + 2 \cdot 1.65 \cdot \sqrt{(3378.5(20.0 - 1.0))}$$

$$= 1445.689 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6757.000	mm
Nozzle Outside Diameter	(d) =	609.600	mm
Vessel Thickness used	(T) =	49.000	mm
Nozzle Thickness used	(t) =	10.000	mm
T / t	=	4.900	
d / t	=	60.960	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.059$$

Nr/P	=	0.046
Mr/P	=	0.047
M0/P	=	0.111
N0/P	=	0.320
MrD/Mc	=	0.090
NrDL/Mc	=	0.080
M0d/Mc	=	0.259
N0DL/Mc	=	0.278
MrD/Ml	=	0.076
NrDL/Ml	=	0.068
M0D/Ml	=	0.184
N0DL/Ml	=	0.523

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-18	-18	18	18
Outplane Bending (Ml)	-38	38	38	-38
Normal Pressure Stress	32	32	32	32



Outplane Stress Summary -5 46 106 6

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stresses parallel to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	5	-5	5	-5
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-2	-2	2	2
Inplane Bending (ML)	-15	15	15	-15
Inplane Pressure Stress	16	16	16	16

Inplane Stress Summary 4 25 40 -1

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary 3 3 1 1

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 11 46 106 8

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-53	53	53	-53
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	16	16	16	16

Outplane Stress Summary -28 53 97 -34

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	5	-5	5	-5
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-18	18	18	-18
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	32	32	32	32

Inplane Stress Summary 17 44 59 12

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Two * Max Shear Stress	45	53	97	47
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-18	-18	18	18
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	17	16	17

Hoop Stress Summary	4	5	40	41
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Axial Membrane (P)	2	2	2	2
Axial Bending (P)	64	-64	64	-64
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-18	-18	18	18
Axial Bending (ML)	-209	209	209	-209
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	-151	136	302	-245
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	12	12	12	12

Shear Stress Summary	15	15	9	9
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
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Two * Max Shear Stress	159	138	303	286
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-9	-9	9	9
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0



Hoop Pressure Stress	16	17	16	17
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Hoop Stress Summary	12	13	32	32
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Axial Membrane (P)	2	2	2	2
Axial Bending (P)	64	-64	64	-64
Axial Membrane (Mc)	-18	-18	18	18
Axial Bending (Mc)	-246	246	246	-246
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	-189	174	340	-283
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	12	12	12	12

Shear Stress Summary	9	9	15	15
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Two * Max Shear Stress	203	174	341	317
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 609.6 + 2 \cdot 1.65 \cdot \sqrt{3363.5(20.0 - 1.0)}$$

$$= 1443.831 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 6727.000 mm

Nozzle Outside Diameter (d) = 1509.600 mm

Vessel Thickness used (T) = 19.000 mm

Nozzle Thickness used (t) = 460.000 mm

T / t = 0.041

d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.223$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.052

Mr/P = 0.080

M0/P = 0.022

N0/P = 0.023

MrD/Mc = 0.248

NrDL/Mc = 0.110

M0d/Mc = 0.074

N0DL/Mc = 0.046

MrD/Ml = 0.055

NrDL/Ml = 0.063

M0D/Ml = 0.017

N0DL/Ml = 0.027

Vessel Stresses (MPa):



LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-9	9	9	-9
Normal Pressure Stress	83	83	83	83
Outplane Stress Summary	90	77	114	63

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	60	-60	60	-60
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-30	30	30	-30
Inplane Pressure Stress	41	41	41	41
Inplane Stress Summary	72	12	144	-36

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	91	77	144	100

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-41	41	41	-41
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	41	41	41	41
Outplane Stress Summary	15	65	106	-8

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	60	-60	60	-60
Inplane Membrane (Mc)	-10	-10	10	10

Inplane Bending (Mc)	-137	137	137	-137
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	83	83	83	83

Inplane Stress Summary	2	157	298	-97
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	1	1
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	15	157	298	97

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	32	32	32	32	32	32	32	32
Circ. Pl (SUS)	-13	-13	23	23	-2	-2	2	2
Circ. Q (SUS)	-26	26	50	-50	-13	13	23	-23
Long. Pm (SUS)	16	16	16	16	16	16	16	16
Long. Pl (SUS)	-2	-2	2	2	-4	-4	14	14
Long. Q (SUS)	-10	10	20	-20	-41	41	65	-65
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	32.0	32.0	32.0	32.0	32.0	32.0	32.0	32.0
Pm+Pl (SUS)	19.0	19.0	55.0	55.0	30.0	30.0	34.0	34.0
Pm+Pl+Q (Total)	11.7	45.2	105.1	8.1	46.2	53.4	95.1	46.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	32.00	118.00	Passed
Pm+Pl (SUS)	55.00	177.00	Passed
Pm+Pl+Q (TOTAL)	105.06	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl

Circ. Pm (SUS)	83	83	83	83	83	83	83	83
Circ. Pl (SUS)	0	0	4	4	-4	-4	16	16
Circ. Q (SUS)	7	-7	25	-25	-77	77	197	-197
Long. Pm (SUS)	41	41	41	41	41	41	41	41
Long. Pl (SUS)	1	1	11	11	-2	-2	6	6
Long. Q (SUS)	30	-30	90	-90	-25	25	57	-57
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	83.0	83.0	83.0	83.0	83.0	83.0	83.0	83.0
Pm+Pl (SUS)	83.0	83.0	87.0	87.0	79.0	79.0	99.0	99.0
Pm+Pl+Q (Total)	90.1	76.0	142.0	100.0	14.1	156.0	296.0	98.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	83.00	118.00	Passed
Pm+Pl (SUS)	99.00	177.00	Passed
Pm+Pl+Q (TOTAL)	296.01	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	16	17	16	17	16	17	16	17
Circ. Pl (SUS)	-13	-13	23	23	-4	-4	14	14
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	-16	-16	20	20	-16	-16	20	20
Long. Q (SUS)	-145	145	273	-273	-182	182	310	-310
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	16.0	17.0	16.0	17.0	16.0	17.0	16.0	17.0
Pm+Pl (SUS)	12.5	13.4	39.8	40.7	20.9	21.8	32.2	32.9
Pm+Pl+Q (Total)	158.9	138.7	301.3	285.6	202.8	174.5	338.7	314.4

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	17.00	118.00	Passed
Pm+Pl (SUS)	40.71	177.00	Passed
Pm+Pl+Q (TOTAL)	338.73	354.00	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: K5a 2"

From: 140

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2999.9988	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		20070.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		200.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

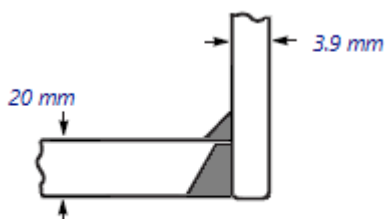
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K5a 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1209 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.0036	mm
Parallel to Vessel Wall, opening length	d	52.5018	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K5a 2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370 mm
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.5004 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.5004 mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm
Wall Thickness per table UG-45	tb3	= 3.4200 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(14.5004, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K5a 2"

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.1356 mm

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Input, Nozzle Desc: K6a 2"

From: 140

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2999.9988	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		20070.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

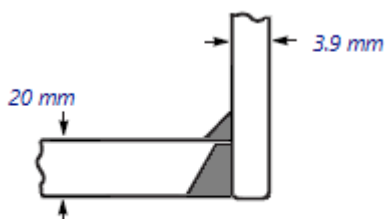
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K6a 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1209 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.0036	mm
Parallel to Vessel Wall, opening length	d	52.5018	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K6a 2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.5004	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.5004	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(14.5004, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: C

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve C	-48 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K6a 2"

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.1356 mm

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Input, Nozzle Desc: K14a 2"

From: 140

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2999.9988	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		20070.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		220.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

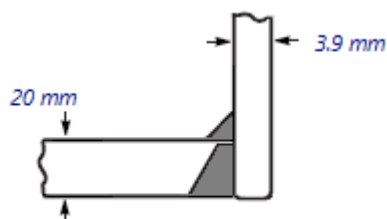
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K14a 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1209 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.0036	mm
Parallel to Vessel Wall, opening length	d	52.5018	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K14a 2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370 mm
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.5004 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.5004 mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm
Wall Thickness per table UG-45	tb3	= 3.4200 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(14.5004, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K14a 2"

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.1356 mm

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Input, Nozzle Desc: K14b 2"

From: 140

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2999.9988	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		20900.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		220.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

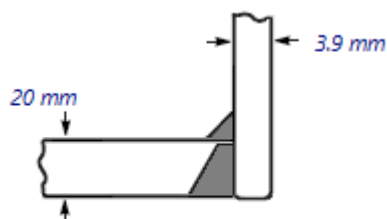
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K14b 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1209 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.0036	mm
Parallel to Vessel Wall, opening length	d	52.5018	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K14b 2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.5004	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.5004	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(14.5004, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K14b 2"

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.1356 mm

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Input, Nozzle Desc: K6b

From: 140

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2999.9988	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		20900.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

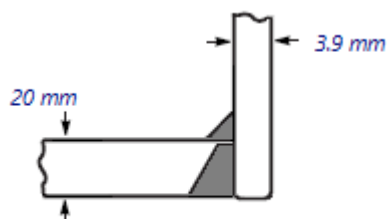
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K6b

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1209 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.0036	mm
Parallel to Vessel Wall, opening length	d	52.5018	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K6b.
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.5004	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.5004	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(14.5004, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K6b

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.1356 mm

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Input, Nozzle Desc: K5b 2"

From: 140

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2999.9988	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		20900.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		200.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

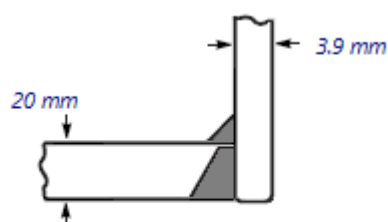
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K5b 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1209 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.0036	mm
Parallel to Vessel Wall, opening length	d	52.5018	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K5b 2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.5004	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.5004	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(14.5004, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K5b 2"

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.1356 mm

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Input, Nozzle Desc: MH4 30"

From: 140

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2999.9988	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		20550.01	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 60 +IT	
Material UNS Number		K02100	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		200.00	deg
Diameter		30.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

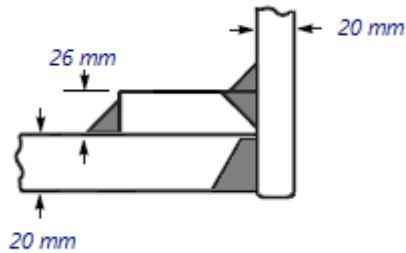
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 60 +IT	
Pad Allowable Stress at Temperature	Sp	118.00	MPa
Pad Allowable Stress At Ambient	Spa	118.00	MPa
Diameter of Pad along vessel surface	Dp	1422.0000	mm
Thickness of Pad	te	26.0000	mm
Weld leg size between Pad and Shell	Wp	20.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	26.0000	mm
Reinforcing Pad Width		330.0000	mm
This is a Manway, Coupling or Access Opening.			

Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: MH4 30"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	30.000 in.
Actual Thickness Used in Calculation	0.787 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
 $= 13.5004 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.47 \cdot 381.0 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
 $= 1.5274 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3927 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1444.0000 mm
Parallel to Vessel Wall, opening length	d	722.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		47.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

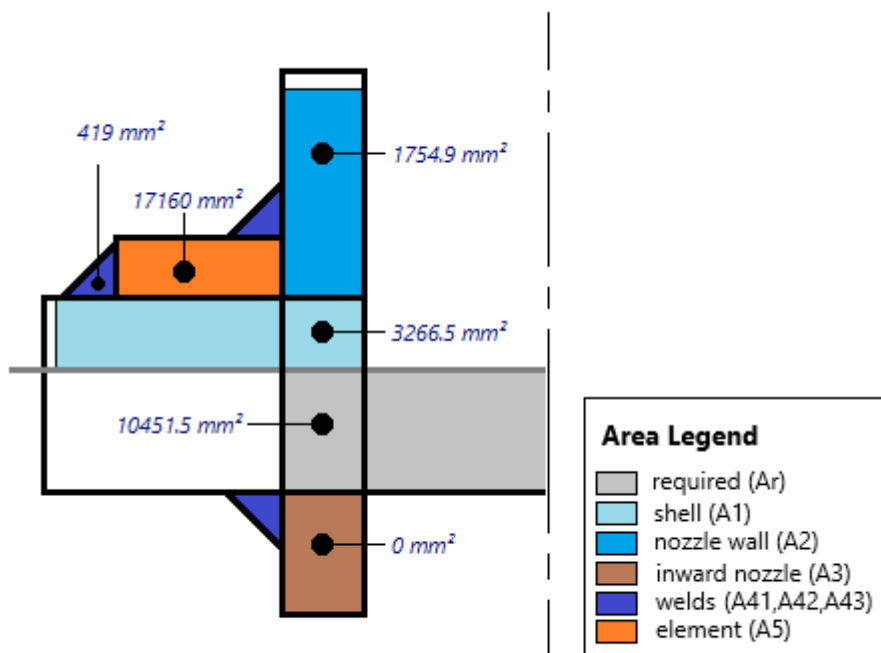


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	9747.263	4160.056	NA
Area in Shell	A1	3970.737	5397.887	NA
Area in Nozzle Wall	A2	1754.896	1767.692	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	419.000	419.000	NA
Area in Element	A5	17160.000	17160.000	NA
TOTAL AREA AVAILABLE	Atot	23304.633	24744.578	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 900.5627 26.0000 mm
 Based on given Pad Diameter: 1422.0000 5.4585 mm
 Based on Shell or Nozzle Thickness: 942.1315 20.0000 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (722.0 \cdot 13.5004 \cdot 1.0 + 2 \cdot 20.0 \cdot 13.5004 \cdot 1.0 \cdot (1 - 1.0))$$

$$= 9747.263 \text{ mm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 722.0(1.0 \cdot 19.0 - 1.0 \cdot 13.5) - 2 \cdot 20.0$$

$$(1.0 \cdot 19.0 - 1.0 \cdot 13.5004) \cdot (1 - 1.0)$$

$$= 3970.737 \text{ mm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$$

$$= (2 \cdot 47.5) \cdot (20.0 - 1.53) \cdot 1.0$$



$$= 1754.896 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (100.0) • 1.0 + (0.0) • 1.0 + 319.0003² • 1.0
= 419.000 mm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (1422.0 - 762.0) • 26.0 • 1.0
= 17160.000 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures ta = 1.5274 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$
$$= 1.5274 \text{ mm}$$

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 20.0, tr = 1.527, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.076, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: C

Govrn. thk, tg = 20.0, tr = 13.5, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.711, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve C	-27 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: C

Govrn. thk, tg = 20.0, tr = 13.5, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.711, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve C	-27 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-43 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-43 °C
Gov. MDMT of the Reinforcement Pad	:	-43 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/1.96 = 0.242



Weld Size Calculations, Description: MH4 30"

Intermediate Calc. for nozzle/shell Welds T_{min} 19.0000 mm
 Intermediate Calc. for pad/shell Welds T_{min}Pad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * W _o mm
Pad Weld	9.5000 = 0.5*T _{min} Pad	14.1400 = 0.7 * W _p mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (9747.2627 - 3970.7368 + 2 \cdot 20.0 \cdot 1.0 \cdot (1.0 \cdot 19.0 - 13.5004)) \cdot 118)$
 $= 707515.44 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1754.8962 + 17160.0 + 419.0003 - 0.0 \cdot 1.0) \cdot 118$
 $= 2281164.50 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1754.8962 + 0.0 + 100.0 + (760.0)) \cdot 118$
 $= 308525.94 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1754.8962 + 0.0 + 419.0003 + 17160.0 + (760.0)) \cdot 118$
 $= 2370835.50 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [S_{onw}]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 762.0 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 692016. \text{ N}$

Shear, Pad Element Weld [S_{pew}]:
 $= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$
 $= (3.1416/2.0) \cdot 1422.0 \cdot 20.0 \cdot 0.49 \cdot 118$
 $= 2582752. \text{ N}$

Shear, Nozzle Wall [S_{nw}]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.1416 \cdot 371.0) \cdot (20.0 - 0.0) \cdot 0.7 \cdot 118$
 $= 1925293. \text{ N}$

Tension, Pad Groove Weld [T_{pgw}]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$
 $= (3.1416/2) \cdot 762.0 \cdot 26.0 \cdot 0.74 \cdot 118$
 $= 2717221. \text{ N}$

Tension, Shell Groove Weld [T_{ngw}]:
 $= (\pi/2) \cdot D_{lo} \cdot W_{gnvi} \cdot 0.74 \cdot S_{ng}$
 $= (3.1416/2.0) \cdot 762.0 \cdot (18.0 - 0.0) \cdot 0.74 \cdot 118$
 $= 1881153. \text{ N}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (S_{PEW} + S_{SNW}) = (2582753 + 1925293) = 4508046 \text{ N}$



Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (692016 + 2717221 + 1881153 + 0) = 5290390 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (2582753 + 1881153 + 0) = 4463906 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 4508045 N	, must exceed W = 707515 N	or W1 = 2281164 N
Path 2-2 = 5290390 N	, must exceed W = 707515 N	or W2 = 308525 N
Path 3-3 = 4463905 N	, must exceed W = 707515 N	or W3 = 2370835 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 21.7169 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 441.7168 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 $(50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o))$ 2.695 %

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Input, Nozzle Desc: N15 38"

From: 140

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2999.9988	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		19550.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 60 +IT	
Material UNS Number		K02100	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		245.00	deg
Diameter		38.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

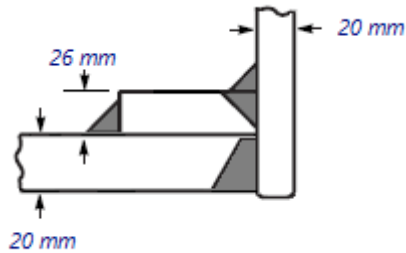
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 60 +IT	
Pad Allowable Stress at Temperature	Sp	118.00	MPa
Pad Allowable Stress At Ambient	Spa	118.00	MPa
Diameter of Pad along vessel surface	Dp	1665.1998	mm
Thickness of Pad	te	26.0000	mm
Weld leg size between Pad and Shell	Wp	20.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	24.0000	mm
Reinforcing Pad Width		349.9999	mm

Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: N15 38"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	38.000 in.
Actual Thickness Used in Calculation	0.787 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
 $= 13.5004 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.47 \cdot 482.6 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
 $= 1.9347 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.6032 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1850.4000 mm
Parallel to Vessel Wall, opening length	d	925.2000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		47.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

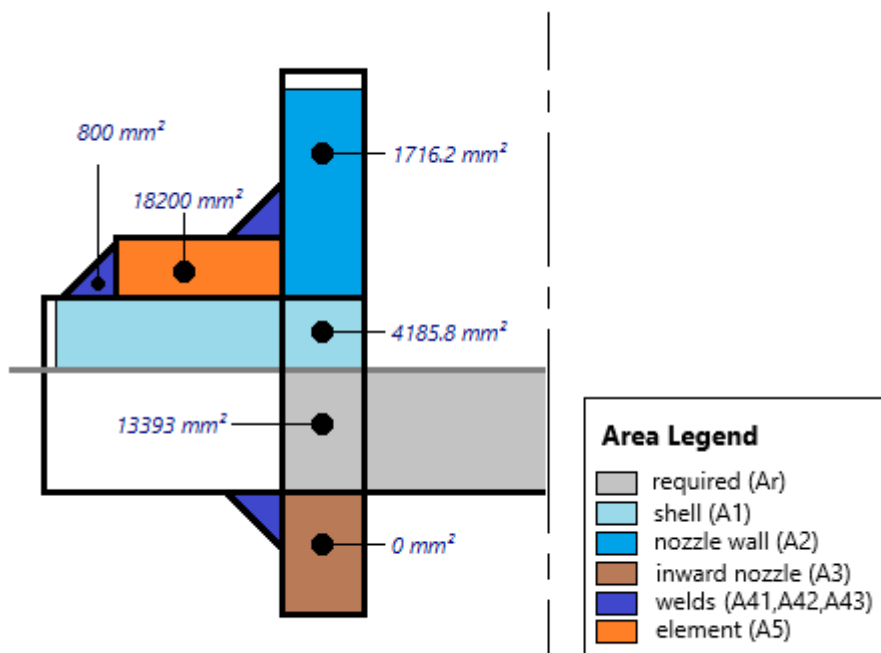


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	12490.537	5330.864	NA
Area in Shell	A1	5088.263	6917.071	NA
Area in Nozzle Wall	A2	1716.202	1747.692	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		800.000	800.000	NA
Area in Element	A5	18199.996	18199.996	NA
TOTAL AREA AVAILABLE	Atot	25804.463	27664.760	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	1153.1259	26.0000 mm
Based on given Pad Diameter:	1665.1998	6.9801 mm
Based on Shell or Nozzle Thickness:	1209.5035	20.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (925.2 \cdot 13.5004 \cdot 1.0 + 2 \cdot 20.0 \cdot 13.5004 \cdot 1.0 \cdot (1 - 1.0))$
 $= 12490.537 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 925.2(1.0 \cdot 19.0 - 1.0 \cdot 13.5) - 2 \cdot 20.0$
 $(1.0 \cdot 19.0 - 1.0 \cdot 13.5004) \cdot (1 - 1.0)$
 $= 5088.263 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 47.5) \cdot (20.0 - 1.93) \cdot 1.0$



= 1716.202 mm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$
= $20.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 20.0^2 \cdot 1.0$
= 800.000 mm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr_4$
= $(1665.1998 - 965.2) \cdot 26.0 \cdot 1.0$
= 18199.996 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.9347 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 14.5004 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 14.5004 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.3941 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[8.331, max(14.5004, 2.5)]
= 8.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.9347, 8.3312)
= 8.3312 mm

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	13.4,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	281.6 MPa	Passed
Occasional	:	5.4,	Allowable	:	156.9 MPa	Passed
Shear	:	9.7,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 20.0, tr = 1.935, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.097$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: C

Govrn. thk, tg = 20.0, tr = 13.5, c = 1.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.711$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve C	-27 °C
--	--------



Min Metal Temp. at Required thickness (UCS 66.1) -43 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: C

Govrn. thk, tg = 20.0, tr = 13.5, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.711$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -27 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -43 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -43 °C
 Gov. MDMT of the Reinforcement Pad : -43 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -43 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.47/1.96 = 0.242$

Weld Size Calculations, Description: N15 38"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$13.3000 = 0.7 \cdot t_{min}$	$14.1400 = 0.7 \cdot W_o$ mm
Pad Weld	$9.5000 = 0.5 \cdot T_{minPad}$	$14.1400 = 0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (12490.5371 - 5088.2627 + 2 \cdot 20.0 \cdot 1.0 \cdot (1.0 \cdot 19.0 - 13.5004)) \cdot 118)$
 $= 899333.94 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i-Can/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (1716.2019 + 18199.9961 + 800.0 - 0.0 \cdot 1.0) \cdot 118$
 $= 2444259.25 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (1716.2019 + 0.0 + 400.0 + (760.0)) \cdot 118$
 $= 339356.81 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (1716.2019 + 0.0 + 800.0 + 18199.9961 + (760.0)) \cdot 118$
 $= 2533930.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 965.2 \cdot 20.0 \cdot 0.49 \cdot 118$
 $= 1753107. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 1665.1998 \cdot 20.0 \cdot 0.49 \cdot 118$
 $= 3024472. \text{ N}$



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 472.6) \cdot (20.0 - 0.0) \cdot 0.7 \cdot 118$$

$$= 2452544. N$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 965.2 \cdot 24.0 \cdot 0.74 \cdot 118$$

$$= 3177059. N$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 965.2 \cdot (18.0 - 0.0) \cdot 0.74 \cdot 118$$

$$= 2382794. N$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (3024472 + 2452544) = 5477016 N$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (1753107 + 3177059 + 2382794 + 0) = 7312959 N$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (3024472 + 2382794 + 0) = 5407266 N$$

Summary of Failure Path Calculations:

Path 1-1 = 5477016 N , must exceed W = 899333 N or W1 = 2444259 N
 Path 2-2 = 7312959 N , must exceed W = 899333 N or W2 = 339356 N
 Path 3-3 = 5407265 N , must exceed W = 899333 N or W3 = 2533930 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 34.9122 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 454.9121 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.116 %

Input Echo, WRC297 Item 1, Description: N15 38" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	6706.000	mm
Shell Thickness	20.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 60	
Vessel Cold S.I. Allowable	Smc 118.00	MPa
Vessel Hot S.I. Allowable	Smh 118.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	965.200	mm
Nozzle Thickness	20.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 60	



Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	1665.200	mm
Thickness of Reinforcing Pad	26.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	0.47	MPa
Radial Load	P	-71249.84	N
Circumferential Shear	Vc	47500.19	N
Longitudinal Shear	Vl	47500.19	N
Circumferential Moment	Mc	0.126E+09	N-mm
Longitudinal Moment	Ml	0.126E+08	N-mm
Torsional Moment	Mt	0.199E+09	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(\text{t} - \text{ca}))}$$

$$= 965.2 + 2 \cdot 1.65 \cdot \sqrt{(3376.5(20.0 - 1.0))}$$

$$= 1801.042 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6753.000	mm
Nozzle Outside Diameter	(d) =	965.200	mm
Vessel Thickness used	(T) =	45.000	mm
Nozzle Thickness used	(t) =	20.000	mm
T / t	=	2.250	
d / t	=	48.260	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.751$$

Nr/P	=	0.053
Mr/P	=	0.092
M0/P	=	0.056
N0/P	=	0.168
MrD/Mc	=	0.184
NrDL/Mc	=	0.093
M0d/Mc	=	0.132
N0DL/Mc	=	0.213
MrD/Ml	=	0.110
NrDL/Ml	=	0.062
M0D/Ml	=	0.068
N0DL/Ml	=	0.320

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-2	-2	2	2
Outplane Bending (Ml)	-2	2	2	-2
Normal Pressure Stress	35	35	35	35



Outplane Stress Summary 48 30 57 29

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stresses parallel to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	19	-19	19	-19
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	-4	4	4	-4
Inplane Pressure Stress	17	17	17	17

Inplane Stress Summary 34 3 43 -3

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary 3 3 2 2

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress 49 30 57 33

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	-13	-13	13	13
Outplane Bending (Mc)	-51	51	51	-51
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	17	17	17	17

Outplane Stress Summary -29 49 100 -25

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	19	-19	19	-19
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-71	71	71	-71
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	35	35	35	35

Inplane Stress Summary -21 83 133 -47

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	3	3

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	30	83	134	48

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-2	-2	2	2
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	13	12	13
Hoop Stress Summary	16	17	20	21

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	69	-69	69	-69
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	-15	15	15	-15
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	60	-47	93	-76

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6
Shear Stress Summary	8	8	5	5

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	62	66	94	98

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	5	5	5	5
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-13	-13	13	13
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0



Hoop Pressure Stress	12	13	12	13
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Hoop Stress Summary	5	5	32	33
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	1	1	1	1
Axial Bending (P)	69	-69	69	-69
Axial Membrane (Mc)	-9	-9	9	9
Axial Bending (Mc)	-270	270	270	-270
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-202	199	357	-323
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6

Shear Stress Summary	5	5	8	8
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
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Two * Max Shear Stress	207	199	357	356
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 965.2 + 2 \cdot 1.65 \cdot \sqrt{3363.5(20.0 - 1.0)}$$

$$= 1799.431 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 6727.000 mm

Nozzle Outside Diameter (d) = 1665.200 mm

Vessel Thickness used (T) = 19.000 mm

Nozzle Thickness used (t) = 370.000 mm

T / t = 0.051

d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.658$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.048

Mr/P = 0.073

M0/P = 0.020

N0/P = 0.023

MrD/Mc = 0.243

NrDL/Mc = 0.105

M0d/Mc = 0.072

N0DL/Mc = 0.047

MrD/Ml = 0.051

NrDL/Ml = 0.055

M0D/Ml = 0.015

N0DL/Ml = 0.025

Vessel Stresses (MPa):



LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	23	-23	23	-23
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	-1	1	1	-1
Normal Pressure Stress	83	83	83	83
Outplane Stress Summary	109	66	114	63

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	9	9	9	9
Inplane Bending (P)	86	-86	86	-86
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-6	6	6	-6
Inplane Pressure Stress	41	41	41	41
Inplane Stress Summary	130	-30	145	-40

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	130	96	145	103

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	23	-23	23	-23
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-91	91	91	-91
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	41	41	41	41
Outplane Stress Summary	-31	104	171	-58

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	9	9	9	9
Inplane Bending (P)	86	-86	86	-86
Inplane Membrane (Mc)	-22	-22	22	22

Inplane Bending (Mc)	-306	306	306	-306
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	83	83	83	83

Inplane Stress Summary	-148	290	507	-277
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to	Left	Left	Right	Right
circumferential plane)	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	148	290	507	277

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	35	35	35	35	35	35	35	35
Circ. Pl (SUS)	3	3	7	7	-5	-5	7	7
Circ. Q (SUS)	9	-9	13	-13	-52	52	90	-90
Long. Pm (SUS)	17	17	17	17	17	17	17	17
Long. Pl (SUS)	1	1	1	1	-8	-8	18	18
Long. Q (SUS)	15	-15	23	-23	-40	40	62	-62
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	35.0	35.0	35.0	35.0	35.0	35.0	35.0	35.0
Pm+Pl (SUS)	38.0	38.0	42.0	42.0	30.0	30.0	42.0	42.0
Pm+Pl+Q (Total)	47.6	29.3	55.6	34.5	31.9	82.3	132.3	48.4

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	35.00	118.00	Passed
Pm+Pl (SUS)	42.00	177.00	Passed
Pm+Pl+Q (TOTAL)	132.26	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl

Circ. Pm (SUS)	83	83	83	83	83	83	83	83
Circ. Pl (SUS)	4	4	4	4	-13	-13	31	31
Circ. Q (SUS)	22	-22	24	-24	-220	220	392	-392
Long. Pm (SUS)	41	41	41	41	41	41	41	41
Long. Pl (SUS)	8	8	10	10	-5	-5	13	13
Long. Q (SUS)	80	-80	92	-92	-68	68	114	-114
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	83.0	83.0	83.0	83.0	83.0	83.0	83.0	83.0
Pm+Pl (SUS)	87.0	87.0	87.0	87.0	70.0	70.0	114.0	114.0
Pm+Pl+Q (Total)	129.2	96.1	143.1	104.1	150.0	290.0	506.0	278.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	83.00	118.00	Passed
Pm+Pl (SUS)	114.00	177.00	Passed
Pm+Pl+Q (TOTAL)	506.01	353.99	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	12	13	12	13	12	13	12	13
Circ. Pl (SUS)	3	3	7	7	-8	-8	18	18
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	1	1	1	1	-8	-8	10	10
Long. Q (SUS)	54	-54	84	-84	-201	201	339	-339
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	12.0	13.0	12.0	13.0	12.0	13.0	12.0	13.0
Pm+Pl (SUS)	15.1	16.1	19.1	20.1	6.3	7.3	30.1	31.1
Pm+Pl+Q (Total)	62.0	64.5	91.3	97.5	207.2	199.1	355.2	354.3

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	13.00	118.00	Passed
Pm+Pl (SUS)	31.07	177.00	Passed
Pm+Pl+Q (TOTAL)	355.15	354.00	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Input, Nozzle Desc: K10B

From: 180

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2000.0029	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		30300.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		200.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

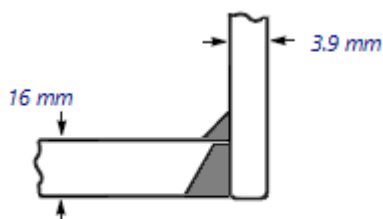
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K10B

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1209 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.0036	mm
Parallel to Vessel Wall, opening length	d	52.5018	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K10B.
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.5004	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.5004	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(14.5004, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K10B

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 216.1358 mm

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Input, Nozzle Desc: N5 14"

From: 180

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2000.0006	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		32000.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		250.00	deg
Diameter		14.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

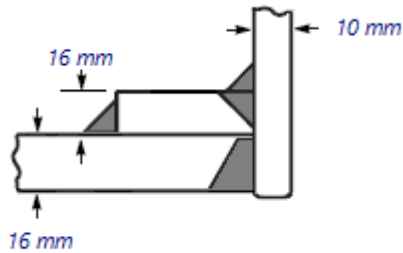
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	855.6000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		250.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: N5 14"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	14.000 in.
Actual Thickness Used in Calculation	0.394 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 177.8 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.7128 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.8913 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	671.2000 mm
Parallel to Vessel Wall, opening length	d	335.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 118.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 118.0)$
= 1.000

Weld Strength Reduction Factor [fr4]:
= $\min(1, S_p / S_v)$
= $\min(1, 138.0 / 118.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

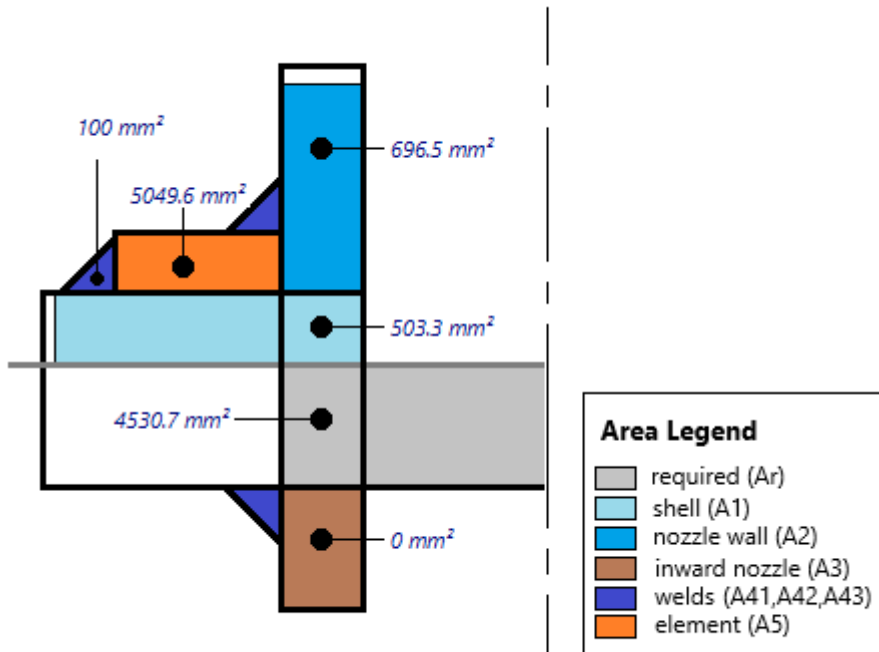


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	4530.722	1618.440	NA
Area in Shell	A1	503.278	1797.120	NA
Area in Nozzle Wall	A2	696.541	683.152	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	5049.600	5049.600	NA
TOTAL AREA AVAILABLE	Atot	6349.419	7629.873	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 557.5315 16.0000 mm
 Based on given Pad Diameter: 855.6000 10.2373 mm
 Based on the Estimated Diameter Limit: 669.6125 10.2891 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (335.6 \cdot 13.5004 \cdot 1.0 + 2 \cdot 10.0 \cdot 13.5004 \cdot 1.0 \cdot (1 - 1.0))$
 $= 4530.722 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 335.6(1.0 \cdot 15.0 - 1.0 \cdot 13.5) - 2 \cdot 10.0$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.5004) \cdot (1 - 1.0)$
 $= 503.278 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 37.5) \cdot (10.0 - 0.71) \cdot 1.0$



= 696.541 mm²

Area Available in Welds [A41 + A42 + A43]:
 = $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
 = $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0$
 = 100.000 mm²

Area Available in Element [A5]:
 = $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
 = $(671.2 - 355.6) \cdot 16.0 \cdot 1.0$
 = 5049.600 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 0.8913 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 14.5004 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 14.5004 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.3941 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[8.331, max(14.5004, 2.5)]
 = 8.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(0.8913, 8.3312)
 = 8.3312 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	28.0,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	267.0 MPa	Passed
Occasional	:	3.9,	Allowable	:	156.9 MPa	Passed
Shear	:	18.5,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 10.0, tr = 0.713, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.071$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 10.0, tr = 0.713, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.071$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 10.0, tr = 0.713, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.071$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:C)

Govrn. thk, tg = 16.0, tr = 13.5, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.9$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-20 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 10.0, tr = 0.713, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.071$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $0.47/1.96 = 0.242$

Weld Size Calculations, Description: N5 14"

Intermediate Calc. for nozzle/shell Welds	Tmin	10.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = $0.7 \cdot W_o$ mm
Pad Weld	7.5000 = $0.5 \cdot T_{minPad}$	11.3120 = $0.7 \cdot W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (4530.7222 - 503.2778 + 2 \cdot 10.0 \cdot 1.0 \cdot (1.0 \cdot 15.0 - 13.5004)) \cdot 118)$
 $= 478728.22 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - Can / 707)^2 \cdot f_{r2}) \cdot S_v$
 $= (696.5407 + 5049.6001 + 100.0 - 0.0 \cdot 1.0) \cdot 118$
 $= 689773.50 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (696.5407 + 0.0 + 100.0 + (300.0)) \cdot 118$
 $= 129378.46 \text{ N}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (696.5407 + 0.0 + 100.0 + 5049.6001 + (300.0)) \cdot 118$
 $= 725169.88 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 355.6 \cdot 10.0 \cdot 0.49 \cdot 118$



= 322941. N

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 855.6 \cdot 16.0 \cdot 0.49 \cdot 118$$

$$= 1243209. N$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 172.8) \cdot (10.0 - 0.0) \cdot 0.7 \cdot 118$$

$$= 448370. N$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 355.6 \cdot 16.0 \cdot 0.74 \cdot 138$$

$$= 912590. N$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 355.6 \cdot (16.0 - 1.0) \cdot 0.74 \cdot 118$$

$$= 731560. N$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (1243209 + 448370) = 1691579 N$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (322941 + 912590 + 731560 + 0) = 1967090 N$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (1243209 + 731560 + 0) = 1974768 N$$

Summary of Failure Path Calculations:

Path 1-1 = 1691578 N , must exceed W = 478728 N or W1 = 689773 N
 Path 2-2 = 1967089 N , must exceed W = 478728 N or W2 = 129378 N
 Path 3-3 = 1974768 N , must exceed W = 478728 N or W3 = 725169 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 4.7175 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 420.7177 mm

Input Echo, WRC297 Item 1, Description: N5 14" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	6706.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 60	
Vessel Cold S.I. Allowable	Smc	118.00 MPa
Vessel Hot S.I. Allowable	Smh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	355.600	mm



Nozzle Thickness		10.0000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	855.600	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.*

Design Internal Pressure	Dp	0.47	MPa
Radial Load	P	-26250.00	N
Circumferential Shear	Vc	17499.99	N
Longitudinal Shear	Vl	17499.99	N
Circumferential Moment	Mc	0.172E+08	N-mm
Longitudinal Moment	Ml	0.172E+08	N-mm
Torsional Moment	Mt	0.245E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 355.6 + 2 \cdot 1.65 \cdot \sqrt{(3369.5(16.0 - 1.0))}$$

$$= 1097.495 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6739.000	mm
Nozzle Outside Diameter	(d) =	355.600	mm
Vessel Thickness used	(T) =	31.000	mm
Nozzle Thickness used	(t) =	10.000	mm
T / t	=	3.100	
d / t	=	35.560	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.778$$

Nr/P	=	0.087
Mr/P	=	0.123
M0/P	=	0.113
N0/P	=	0.261
MrD/Mc	=	0.177
NrDL/Mc	=	0.115
M0d/Mc	=	0.203
N0DL/Mc	=	0.126
MrD/Ml	=	0.155
NrDL/Ml	=	0.106
M0D/Ml	=	0.164
N0DL/Ml	=	0.261

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-13	-13	13	13



Outplane Bending (ML)	-49	49	49	-49
Normal Pressure Stress	51	51	51	51

Outplane Stress Summary	14	76	139	3
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	20	-20	20	-20
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-46	46	46	-46
Inplane Pressure Stress	25	25	25	25
Inplane Stress Summary	-4	49	100	-33

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	20	77	139	37

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	7	7	7	7
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-61	61	61	-61
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	25	25	25	25
Outplane Stress Summary	-16	68	118	-40

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	20	-20	20	-20
Inplane Membrane (Mc)	-5	-5	5	5
Inplane Bending (Mc)	-53	53	53	-53
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	51	51	51	51
Inplane Stress Summary	14	81	132	-13

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	31	81	134	41

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-13	-13	13	13
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	9	9	9
Hoop Stress Summary	3	3	29	30

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	124	-124	124	-124
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-18	-18	18	18
Axial Bending (ML)	-295	295	295	-295
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	-182	158	446	-394

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	12	12	12	12
Shear Stress Summary	15	15	9	9

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	188	160	446	424

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	7	7	7	7
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-6	-6	6	6



Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	9	9	9

Hoop Stress Summary	10	10	22	23
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	124	-124	124	-124
Axial Membrane (Mc)	-18	-18	18	18
Axial Bending (Mc)	-344	344	344	-344
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4

Axial Stress Summary	-231	207	495	-443
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	12	12	12	12

Shear Stress Summary	9	9	15	15
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	242	208	495	467
------------------------	-----	-----	-----	-----

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 355.6 + 2 \cdot 1.65 \cdot \sqrt{(3361.5(16.0 - 1.0))}$
 $= 1096.614 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6723.000 mm
Nozzle Outside Diameter	(d) =	855.600 mm
Vessel Thickness used	(T) =	15.000 mm
Nozzle Thickness used	(t) =	260.000 mm
T / t	=	0.058
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.694$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.072
Mr/P	=	0.117
M0/P	=	0.034
N0/P	=	0.035
MrD/Mc	=	0.277
NrDL/Mc	=	0.121
M0d/Mc	=	0.083
N0DL/Mc	=	0.054
MrD/Ml	=	0.086
NrDL/Ml	=	0.102
M0D/Ml	=	0.026

NØDL/M1 = 0.048

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	23	-23	23	-23
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-13	13	13	-13
Normal Pressure Stress	105	106	105	106
Outplane Stress Summary	115	95	151	77

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	82	-82	82	-82
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-9	-9	9	9
Inplane Bending (ML)	-45	45	45	-45
Inplane Pressure Stress	52	52	52	52
Inplane Stress Summary	88	15	198	-57

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	115	95	198	134

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	23	-23	23	-23
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-44	44	44	-44
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	52	52	52	52
Outplane Stress Summary	31	72	129	-6

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------



Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	82	-82	82	-82
Inplane Membrane (Mc)	-10	-10	10	10
Inplane Bending (Mc)	-148	148	148	-148
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	105	106	105	106

Inplane Stress Summary	37	169	355	-104
------------------------	----	-----	-----	------

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	2	2
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	37	169	355	104

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	51	51	51	51	51	51	51	51
Circ. Pl (SUS)	-6	-6	20	20	-3	-3	7	7
Circ. Q (SUS)	-31	31	67	-67	-33	33	73	-73
Long. Pm (SUS)	25	25	25	25	25	25	25	25
Long. Pl (SUS)	-3	-3	7	7	1	1	13	13
Long. Q (SUS)	-26	26	66	-66	-43	43	79	-79
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	51.0	51.0	51.0	51.0	51.0	51.0	51.0	51.0
Pm+Pl (SUS)	45.0	45.0	71.0	71.0	48.0	48.0	58.0	58.0
Pm+Pl+Q (Total)	19.7	76.6	138.1	38.2	32.2	81.3	132.1	41.6

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	51.00	118.00	Passed
Pm+Pl (SUS)	71.03	177.00	Passed
Pm+Pl+Q (TOTAL)	138.10	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):



Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	105	106	105	106	105	106	105	106
Circ. Pl (SUS)	0	0	8	8	-2	-2	18	18
Circ. Q (SUS)	10	-10	36	-36	-66	66	230	-230
Long. Pm (SUS)	52	52	52	52	52	52	52	52
Long. Pl (SUS)	-1	-1	17	17	0	0	8	8
Long. Q (SUS)	37	-37	127	-127	-21	21	67	-67
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	105.0	106.0	105.0	106.0	105.0	106.0	105.0	106.0
Pm+Pl (SUS)	105.0	106.0	113.0	114.0	103.0	104.0	123.0	124.0
Pm+Pl+Q (Total)	115.0	96.0	196.0	136.0	37.2	170.0	353.0	106.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	106.00	118.00	Passed
Pm+Pl (SUS)	124.00	177.00	Passed
Pm+Pl+Q (TOTAL)	353.00	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9	9	9	9	9	9	9	9
Circ. Pl (SUS)	-6	-6	20	20	1	1	13	13
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-16	-16	20	20	-16	-16	20	20
Long. Q (SUS)	-171	171	419	-419	-220	220	468	-468
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	9.0	9.0	9.0	9.0	9.0	9.0	9.0	9.0
Pm+Pl (SUS)	16.2	16.2	30.4	30.4	22.8	22.8	26.2	26.2
Pm+Pl+Q (Total)	188.4	160.4	443.2	424.4	242.7	208.4	492.5	467.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	9.00	118.00	Passed
Pm+Pl (SUS)	30.41	177.00	Passed
Pm+Pl+Q (TOTAL)	492.48	354.00	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite 28 U2 Licensee: SL User
FileName : C-1401_Ope_Rev 0
Nozzle Calcs.: N5 14"

Nozl: 147 4:01pm Jul 21,2026

Page 443 of 564



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Input, Nozzle Desc: MH5 30"

From: 180

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2000.0006	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		32000.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		45.00	deg
Diameter		30.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

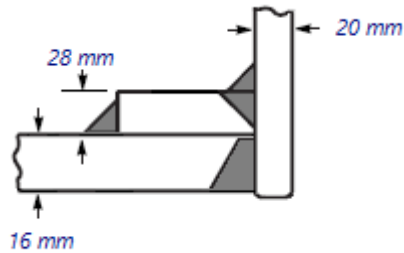
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1422.0000	mm
Thickness of Pad	te	28.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm
Reinforcing Pad Width		330.0000	mm
This is a Manway, Coupling or Access Opening.			

Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: MH5 30"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	30.000 in.
Actual Thickness Used in Calculation	0.787 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
 $= 13.5004 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.47 \cdot 381.0 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
 $= 1.5274 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3927 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1444.0000 mm
Parallel to Vessel Wall, opening length	d	722.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

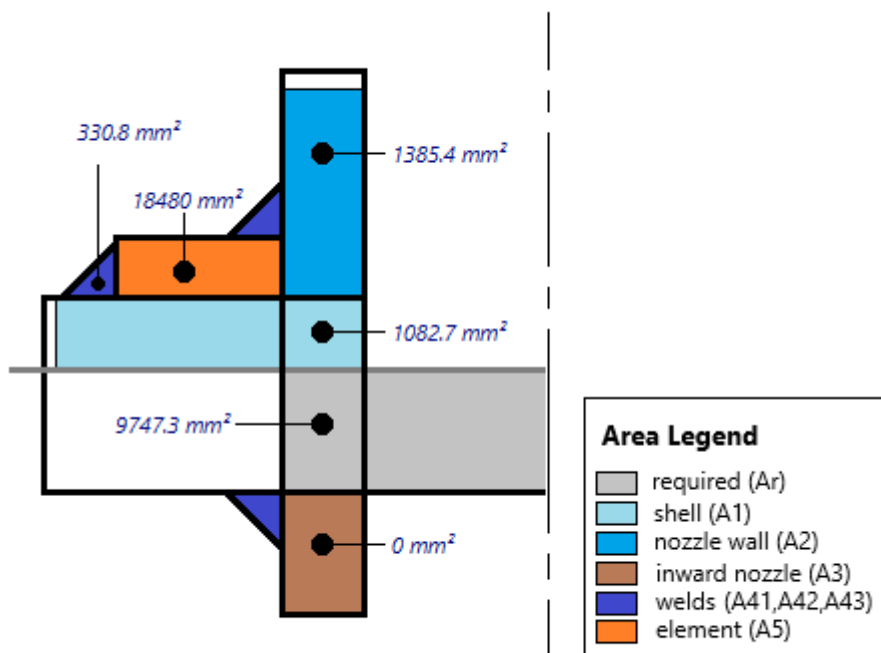


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	9747.263	3481.864	NA
Area in Shell	A1	1082.737	3866.272	NA
Area in Nozzle Wall	A2	1385.444	1395.547	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	330.750	330.750	NA
Area in Element	A5	18480.002	18480.002	NA
TOTAL AREA AVAILABLE	Atot	21278.934	24072.570	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	1010.1546	28.0000 mm
Based on given Pad Diameter:	1422.0000	10.5278 mm
Based on Shell or Nozzle Thickness:	1196.2706	16.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (722.0 \cdot 13.5004 \cdot 1.0 + 2 \cdot 20.0 \cdot 13.5004 \cdot 1.0 \cdot (1 - 1.0))$
 $= 9747.263 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 722.0(1.0 \cdot 15.0 - 1.0 \cdot 13.5) - 2 \cdot 20.0$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.5004) \cdot (1 - 1.0)$
 $= 1082.737 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 37.5) \cdot (20.0 - 1.53) \cdot 1.0$



= 1385.444 mm²

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (99.75) • 1.0 + (0.0) • 1.0 + 231.0² • 1.0
= 330.750 mm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (1422.0 - 762.0) • 28.0 • 1.0
= 18480.002 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures ta = 1.5274 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= ta
= 1.5274 mm

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 20.0, tr = 1.527, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.076, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 20.0, tr = 1.527, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.076, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 20.0, tr = 1.527, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.076, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:C)

Govrn. thk, tg = 16.0, tr = 13.5, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.9, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-20 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 16.0, tr = 13.5, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.9, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-20 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-29 °C
Gov. MDMT of the Reinforcement Pad	:	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-29 °C



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 0.47/1.96 = 0.242

Weld Size Calculations, Description: MH5 30"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	7.5000 = 0.5*TminPad	11.3120 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (9747.2627 - 1082.7372 + 2 \cdot 20.0 \cdot 1.0 \cdot (1.0 \cdot 15.0 - 13.5004)) \cdot 118)$
 $= 1029386.19 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (1385.4443 + 18480.002 + 330.7501 - 0.0 \cdot 1.0) \cdot 118$
 $= 2382905.25 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (1385.4443 + 0.0 + 100.0 + (600.0)) \cdot 118$
 $= 246057.06 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (1385.4443 + 0.0 + 330.7501 + 18480.002 + (600.0)) \cdot 118$
 $= 2453698.00 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 762.0 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 692016. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 1422.0 \cdot 16.0 \cdot 0.49 \cdot 118$
 $= 2066202. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 371.0) \cdot (20.0 - 0.0) \cdot 0.7 \cdot 118$
 $= 1925293. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 762.0 \cdot 28.0 \cdot 0.74 \cdot 138$
 $= 3422211. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 762.0 \cdot (16.0 - 1.0) \cdot 0.74 \cdot 118$
 $= 1567628. \text{ N}$



See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (2066202 + 1925293) = 3991495 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (692016 + 3422211 + 1567628 + 0) = 5681854 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (2066202 + 1567628 + 0) = 3633830 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 3991495 N , must exceed W = 1029386 N or W1 = 2382905 N

Path 2-2 = 5681854 N , must exceed W = 1029386 N or W2 = 246057 N

Path 3-3 = 3633829 N , must exceed W = 1029386 N or W3 = 2453698 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 21.7169 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 437.7171 mm

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Input, Nozzle Desc: N6a 18"

From: 180

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2000.0006	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		32070.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-516 60 +IT	
Material UNS Number		K02100	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		18.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

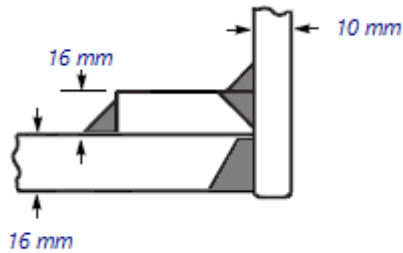
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1257.2000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		400.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: N6a 18"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	18.000 in.
Actual Thickness Used in Calculation	0.394 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
 $= 13.5004 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.47 \cdot 228.6 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
 $= 0.9164 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0297 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	874.4000 mm
Parallel to Vessel Wall, opening length	d	437.2000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

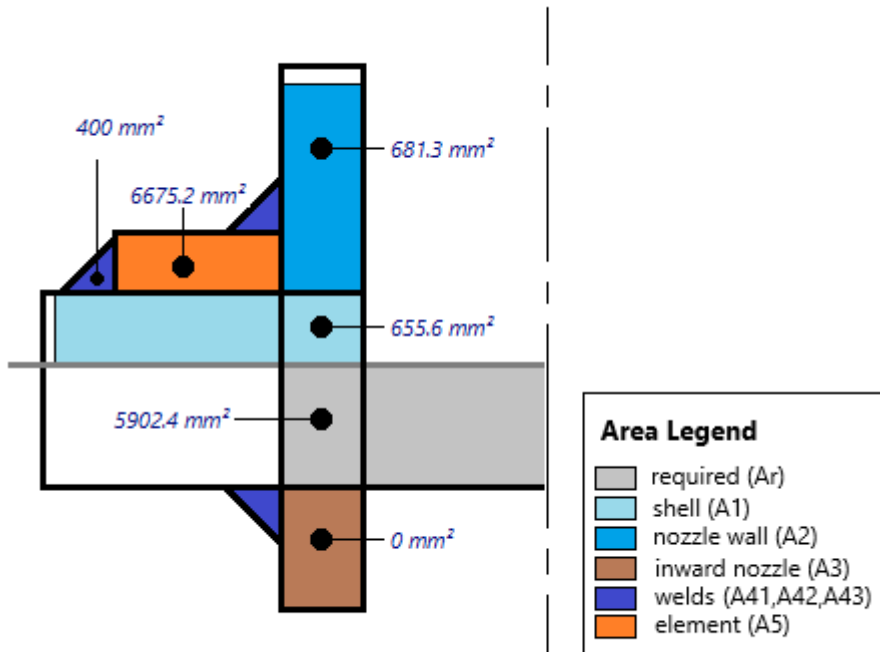


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	5902.359	2108.408	NA
Area in Shell	A1	655.641	2341.183	NA
Area in Nozzle Wall	A2	681.267	672.770	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	400.000	400.000	NA
Area in Element	A5	6675.200	6675.200	NA
TOTAL AREA AVAILABLE	Atot	8412.107	10089.153	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 717.5408 16.0000 mm
 Based on given Pad Diameter: 1257.2001 9.9843 mm
 Based on Shell or Nozzle Thickness: 873.7452 10.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (437.2 \cdot 13.5004 \cdot 1.0 + 2 \cdot 10.0 \cdot 13.5004 \cdot 1.0 \cdot (1 - 1.0))$
 $= 5902.359 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 437.2(1.0 \cdot 15.0 - 1.0 \cdot 13.5) - 2 \cdot 10.0$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.5004) \cdot (1 - 1.0)$
 $= 655.641 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 37.5) \cdot (10.0 - 0.92) \cdot 1.0$



= 681.267 mm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
= $20.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0$
= 400.000 mm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
= $(874.4 - 457.2) \cdot 16.0 \cdot 1.0$
= 6675.200 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.0297 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 14.5004 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 14.5004 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.3941 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[8.331, max(14.5004, 2.5)]
= 8.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.0297, 8.3312)
= 8.3312 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 28.7,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 266.3 MPa	Passed
Occasional	: 5.1,	Allowable	: 156.9 MPa	Passed
Shear	: 18.1,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 10.0, tr = 0.916, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.092$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 10.0, tr = 0.916, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.092$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:C)



Govrn. thk, tg = 16.0, tr = 13.5, c = 1.0 mm, E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.9$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
Min Metal Temp. at Required thickness (UCS 66.1) -20 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: C

Govrn. thk, tg = 10.0, tr = 0.916, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr * E* / (tg - c) = 0.092$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -44 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $0.47/1.96 = 0.242$

Weld Size Calculations, Description: N6a 18"

Intermediate Calc. for nozzle/shell Welds Tmin 10.0000 mm
Intermediate Calc. for pad/shell Welds TminPad 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$14.1400 = 0.7 * W_o \text{ mm}$
Pad Weld	$7.5000 = 0.5 * \text{TminPad}$	$11.3120 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (5902.3594 - 655.6408 + 2 \cdot 10.0 \cdot 1.0 \cdot (1.0 \cdot 15.0 - 13.5004)) \cdot 118)$
 $= 622587.81 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (681.2666 + 6675.2002 + 400.0 - 0.0 \cdot 1.0) \cdot 118$
 $= 915168.69 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (681.2666 + 0.0 + 400.0 + (300.0)) \cdot 118$
 $= 162972.66 \text{ N}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (681.2666 + 0.0 + 400.0 + 6675.2002 + (300.0)) \cdot 118$
 $= 950565.06 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot W_o \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 457.2 \cdot 20.0 \cdot 0.49 \cdot 118$
 $= 830419. \text{ N}$

Shear, Pad Element Weld [Spew]:



$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 1257.2 \cdot 16.0 \cdot 0.49 \cdot 118$$

$$= 1826744. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 223.6) \cdot (10.0 - 0.0) \cdot 0.7 \cdot 118$$

$$= 580183. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 457.2 \cdot 16.0 \cdot 0.74 \cdot 138$$

$$= 1173329. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 457.2 \cdot (16.0 - 1.0) \cdot 0.74 \cdot 118$$

$$= 940576. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (1826744 + 580183) = 2406926 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (830419 + 1173329 + 940577 + 0) = 2944325 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (1826744 + 940577 + 0) = 2767320 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 2406926 N	, must exceed W = 622587 N	or W1 = 915168 N
Path 2-2 = 2944324 N	, must exceed W = 622587 N	or W2 = 162972 N
Path 3-3 = 2767320 N	, must exceed W = 622587 N	or W3 = 950565 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 7.8018 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 423.8020 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.236 %

Input Echo, WRC297 Item 1, Description: N6a 18" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	6706.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 60	
Vessel Cold S.I. Allowable	Smc	118.00 MPa
Vessel Hot S.I. Allowable	Smh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	457.200	mm



Nozzle Thickness		10.0000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-516 60	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	1257.200	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	0.47	MPa
Radial Load	P	-33750.00	N
Circumferential Shear	Vc	22499.99	N
Longitudinal Shear	Vl	22499.99	N
Circumferential Moment	Mc	0.284E+08	N-mm
Longitudinal Moment	Ml	0.284E+08	N-mm
Torsional Moment	Mt	0.405E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 457.2 + 2 \cdot 1.65 \cdot \sqrt{(3369.5(16.0 - 1.0))}$
 $= 1199.095 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6739.000	mm
Nozzle Outside Diameter	(d) =	457.200	mm
Vessel Thickness used	(T) =	31.000	mm
Nozzle Thickness used	(t) =	10.000	mm
T / t	=	3.100	
d / t	=	45.720	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.000$$

Nr/P	=	0.068
Mr/P	=	0.108
M0/P	=	0.096
N0/P	=	0.246
MrD/Mc	=	0.172
NrDL/Mc	=	0.102
M0d/Mc	=	0.191
N0DL/Mc	=	0.176
MrD/Ml	=	0.140
NrDL/Ml	=	0.088
M0D/Ml	=	0.138
N0DL/Ml	=	0.338

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	8	8	8	8
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-21	-21	21	21



Outplane Bending (ML)	-53	53	53	-53
Normal Pressure Stress	51	51	51	51

Outplane Stress Summary	4	71	155	8
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	22	-22	22	-22
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-54	54	54	-54
Inplane Pressure Stress	25	25	25	25
Inplane Stress Summary	-9	53	110	-43

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	16	72	155	51

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	8	8	8	8
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-74	74	74	-74
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	25	25	25	25
Outplane Stress Summary	-30	76	139	-48

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	22	-22	22	-22
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-66	66	66	-66
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	51	51	51	51
Inplane Stress Summary	3	91	149	-28

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	34	91	151	49

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	8	8	8	8
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-21	-21	21	21
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	12	12	12
Hoop Stress Summary	0	0	42	43

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	149	-149	149	-149
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-18	-18	18	18
Axial Bending (ML)	-356	356	356	-356
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-217	197	532	-478

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	12	12	12	12
Shear Stress Summary	15	15	9	9

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	218	199	532	522

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	8	8	8	8
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-11	-11	11	11



Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	12	12	12

Hoop Stress Summary	9	9	32	32
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to pipe cross-section)	Left Outside	Left Inside	Right Outside	Right Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	149	-149	149	-149
Axial Membrane (Mc)	-18	-18	18	18
Axial Bending (Mc)	-449	449	449	-449
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-310	290	625	-571
----------------------	------	-----	-----	------

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	12	12	12	12

Shear Stress Summary	9	9	15	15
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Outside	Inside	Outside	Inside

Two * Max Shear Stress	320	290	625	605
------------------------	-----	-----	-----	-----

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 457.2 + 2 \cdot 1.65 \cdot \sqrt{(3361.5(16.0 - 1.0))}$
 $= 1198.214 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D)	=	6723.000 mm
Nozzle Outside Diameter (d)	=	1257.200 mm
Vessel Thickness used (T)	=	15.000 mm
Nozzle Thickness used (t)	=	410.000 mm
T / t	=	0.037
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.959$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.054
Mr/P	=	0.085
M0/P	=	0.023
N0/P	=	0.023
MrD/Mc	=	0.252
NrDL/Mc	=	0.114
M0d/Mc	=	0.075
N0DL/Mc	=	0.046
MrD/Ml	=	0.058
NrDL/Ml	=	0.069
M0D/Ml	=	0.018

NØDL/M1 = 0.028

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-10	10	10	-10
Normal Pressure Stress	105	106	105	106
Outplane Stress Summary	116	96	143	80

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	76	-76	76	-76
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-34	34	34	-34
Inplane Pressure Stress	52	52	52	52
Inplane Stress Summary	96	12	179	-44

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	116	96	179	124

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-45	45	45	-45
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	52	52	52	52
Outplane Stress Summary	27	76	127	-5

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------



Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	76	-76	76	-76
Inplane Membrane (Mc)	-11	-11	11	11
Inplane Bending (Mc)	-151	151	151	-151
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	105	106	105	106

Inplane Stress Summary	27	177	353	-102
------------------------	----	-----	-----	------

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	1	1
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	27	177	353	102

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	51	51	51	51	51	51	51	51
Circ. Pl (SUS)	-13	-13	29	29	-4	-4	8	8
Circ. Q (SUS)	-33	33	73	-73	-44	44	88	-88
Long. Pm (SUS)	25	25	25	25	25	25	25	25
Long. Pl (SUS)	-3	-3	7	7	-3	-3	19	19
Long. Q (SUS)	-32	32	76	-76	-54	54	94	-94
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	51.0	51.0	51.0	51.0	51.0	51.0	51.0	51.0
Pm+Pl (SUS)	38.1	38.1	80.0	80.0	47.0	47.0	59.1	59.1
Pm+Pl+Q (Total)	17.0	71.9	153.1	51.2	35.2	91.3	148.5	50.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	51.00	118.00	Passed
Pm+Pl (SUS)	80.02	177.00	Passed
Pm+Pl+Q (TOTAL)	153.09	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	105	106	105	106	105	106	105	106
Circ. Pl (SUS)	1	1	5	5	-3	-3	19	19
Circ. Q (SUS)	10	-10	30	-30	-75	75	227	-227
Long. Pm (SUS)	52	52	52	52	52	52	52	52
Long. Pl (SUS)	2	2	14	14	-1	-1	7	7
Long. Q (SUS)	42	-42	110	-110	-25	25	65	-65
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	105.0	106.0	105.0	106.0	105.0	106.0	105.0	106.0
Pm+Pl (SUS)	106.0	107.0	110.0	111.0	102.0	103.0	124.0	125.0
Pm+Pl+Q (Total)	116.0	97.0	176.0	125.0	27.6	178.0	351.0	102.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	106.00	118.00	Passed
Pm+Pl (SUS)	125.00	177.00	Passed
Pm+Pl+Q (TOTAL)	351.00	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	12	12	12	12	12	12	12	12
Circ. Pl (SUS)	-13	-13	29	29	-3	-3	19	19
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-16	-16	20	20	-16	-16	20	20
Long. Q (SUS)	-207	207	505	-505	-300	300	598	-598
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0
Pm+Pl (SUS)	10.9	10.9	41.6	41.6	19.9	19.9	32.4	32.4
Pm+Pl+Q (Total)	218.1	200.3	531.2	520.3	319.5	290.3	624.4	603.7

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12.00	118.00	Passed
Pm+Pl (SUS)	41.58	177.00	Passed
Pm+Pl+Q (TOTAL)	624.38	354.00	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite 28 U2 Licensee: SL User
FileName : C-1401_Ope_Rev 0
Nozzle Calcs.: N6a 18"

Nozl: 149 4:01pm Jul 21,2026

Page 463 of 564



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Input, Nozzle Desc: N6b 18"

From: 180

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2000.0006	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		32070.00	mm
-----------------------------------	--	----------	----

User Entered Minimum Design Metal Temperature		-1.00	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-516 60 +IT	
Material UNS Number		K02100	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		210.00	deg
Diameter		18.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

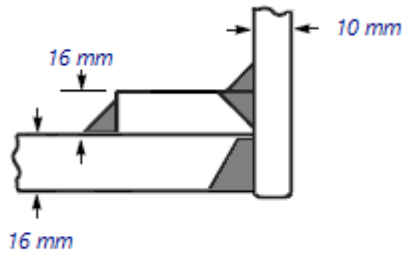
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1257.2000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		400.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: N6b 18"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	18.000 in.
Actual Thickness Used in Calculation	0.394 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
 $= 13.5004 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.47 \cdot 228.6 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
 $= 0.9164 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0297 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	874.4000 mm
Parallel to Vessel Wall, opening length	d	437.2000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

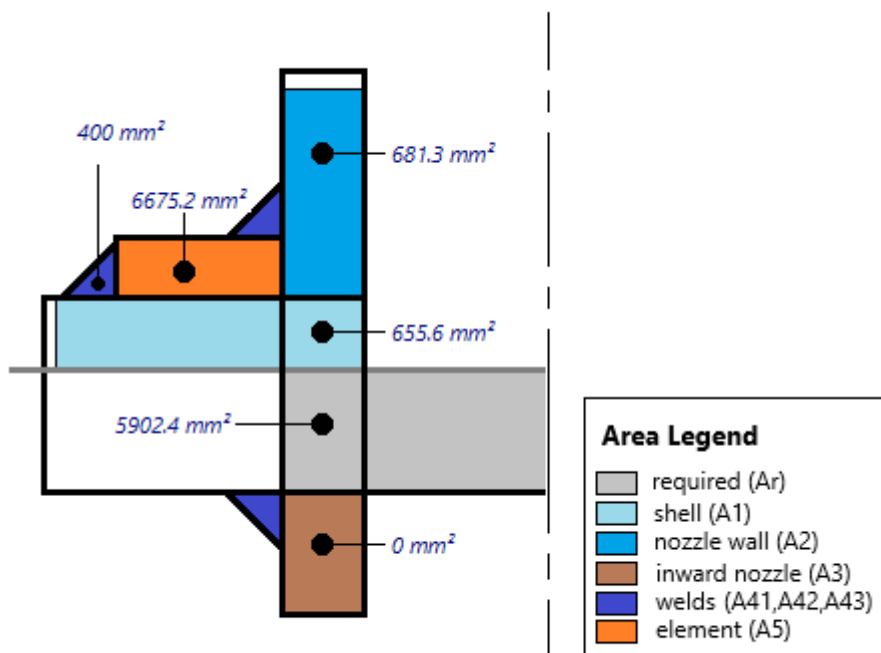


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	5902.359	2108.408	NA
Area in Shell	A1	655.641	2341.183	NA
Area in Nozzle Wall	A2	681.267	672.770	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	400.000	400.000	NA
Area in Element	A5	6675.200	6675.200	NA
TOTAL AREA AVAILABLE	Atot	8412.107	10089.153	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 717.5408 16.0000 mm
 Based on given Pad Diameter: 1257.2001 9.9843 mm
 Based on Shell or Nozzle Thickness: 873.7452 10.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (437.2 \cdot 13.5004 \cdot 1.0 + 2 \cdot 10.0 \cdot 13.5004 \cdot 1.0 \cdot (1 - 1.0))$
 $= 5902.359 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 437.2(1.0 \cdot 15.0 - 1.0 \cdot 13.5) - 2 \cdot 10.0$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.5004) \cdot (1 - 1.0)$
 $= 655.641 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 37.5) \cdot (10.0 - 0.92) \cdot 1.0$



= 681.267 mm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
= $20.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0$
= 400.000 mm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
= $(874.4 - 457.2) \cdot 16.0 \cdot 1.0$
= 6675.200 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.0297 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 14.5004 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 14.5004 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.3941 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[8.331, max(14.5004, 2.5)]
= 8.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.0297, 8.3312)
= 8.3312 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 28.7,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 266.3 MPa	Passed
Occasional	: 5.1,	Allowable	: 156.9 MPa	Passed
Shear	: 18.1,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 10.0, tr = 0.916, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.092$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 10.0, tr = 0.916, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.092$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:C)



Govrn. thk, tg = 16.0, tr = 13.5, c = 1.0 mm, E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.9$, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
Min Metal Temp. at Required thickness (UCS 66.1) -20 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: C

Govrn. thk, tg = 10.0, tr = 0.916, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.092$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -44 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
Gov. MDMT of the Reinforcement Pad : -29 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $0.47/1.96 = 0.242$

Weld Size Calculations, Description: N6b 18"

Intermediate Calc. for nozzle/shell Welds Tmin 10.0000 mm
Intermediate Calc. for pad/shell Welds TminPad 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$14.1400 = 0.7 * W_o \text{ mm}$
Pad Weld	$7.5000 = 0.5 * \text{TminPad}$	$11.3120 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot f r1 \cdot (E1 \cdot t - tr)) \cdot S_v)$
 $= \max(0, (5902.3594 - 655.6408 + 2 \cdot 10.0 \cdot 1.0 \cdot (1.0 \cdot 15.0 - 13.5004)) \cdot 118)$
 $= 622587.81 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (W_i - \text{Can} / .707)^2 \cdot f r2) \cdot S_v$
 $= (681.2666 + 6675.2002 + 400.0 - 0.0 \cdot 1.0) \cdot 118$
 $= 915168.69 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r1)) \cdot S_v$
 $= (681.2666 + 0.0 + 400.0 + (300.0)) \cdot 118$
 $= 162972.66 \text{ N}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f r1)) \cdot S$
 $= (681.2666 + 0.0 + 400.0 + 6675.2002 + (300.0)) \cdot 118$
 $= 950565.06 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.1416/2.0) \cdot 457.2 \cdot 20.0 \cdot 0.49 \cdot 118$
 $= 830419. \text{ N}$

Shear, Pad Element Weld [Spew]:



$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 1257.2 \cdot 16.0 \cdot 0.49 \cdot 118$$

$$= 1826744. \text{ N}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 223.6) \cdot (10.0 - 0.0) \cdot 0.7 \cdot 118$$

$$= 580183. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 457.2 \cdot 16.0 \cdot 0.74 \cdot 138$$

$$= 1173329. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 457.2 \cdot (16.0 - 1.0) \cdot 0.74 \cdot 118$$

$$= 940576. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (1826744 + 580183) = 2406926 \text{ N}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (830419 + 1173329 + 940577 + 0) = 2944325 \text{ N}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (1826744 + 940577 + 0) = 2767320 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 2406926 N	, must exceed W = 622587 N	or W1 = 915168 N
Path 2-2 = 2944324 N	, must exceed W = 622587 N	or W2 = 162972 N
Path 3-3 = 2767320 N	, must exceed W = 622587 N	or W3 = 950565 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 7.8018 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 423.8020 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.236 %

Input Echo, WRC297 Item 1, Description: N6b 18" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	6706.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 60	
Vessel Cold S.I. Allowable	Smc 118.00	MPa
Vessel Hot S.I. Allowable	Smh 118.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	457.200	mm



Nozzle Thickness		10.0000	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-516 60	
Nozzle Cold S.I. Allowable	SNmc	118.00	MPa
Nozzle Hot S.I. Allowable	SNmh	118.00	MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad	1257.200	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	0.47	MPa
Radial Load	P	-33750.00	N
Circumferential Shear	Vc	22499.99	N
Longitudinal Shear	Vl	22499.99	N
Circumferential Moment	Mc	0.284E+08	N-mm
Longitudinal Moment	Ml	0.284E+08	N-mm
Torsional Moment	Mt	0.405E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 457.2 + 2 \cdot 1.65 \cdot \sqrt{(3369.5(16.0 - 1.0))}$
 $= 1199.095 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6739.000	mm
Nozzle Outside Diameter	(d) =	457.200	mm
Vessel Thickness used	(T) =	31.000	mm
Nozzle Thickness used	(t) =	10.000	mm
T / t	=	3.100	
d / t	=	45.720	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.000$$

Nr/P	=	0.068
Mr/P	=	0.108
M0/P	=	0.096
N0/P	=	0.246
MrD/Mc	=	0.172
NrDL/Mc	=	0.102
M0d/Mc	=	0.191
N0DL/Mc	=	0.176
MrD/Ml	=	0.140
NrDL/Ml	=	0.088
M0D/Ml	=	0.138
N0DL/Ml	=	0.338

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	8	8	8	8
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-21	-21	21	21



Outplane Bending (ML)	-53	53	53	-53
Normal Pressure Stress	51	51	51	51

Outplane Stress Summary	4	71	155	8
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	22	-22	22	-22
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-54	54	54	-54
Inplane Pressure Stress	25	25	25	25
Inplane Stress Summary	-9	53	110	-43

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	16	72	155	51

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	8	8	8	8
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	-11	-11	11	11
Outplane Bending (Mc)	-74	74	74	-74
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	25	25	25	25
Outplane Stress Summary	-30	76	139	-48

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	22	-22	22	-22
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-66	66	66	-66
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	51	51	51	51
Inplane Stress Summary	3	91	149	-28

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
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(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	34	91	151	49

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	8	8	8	8
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-21	-21	21	21
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	12	12	12
Hoop Stress Summary	0	0	42	43

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	149	-149	149	-149
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-18	-18	18	18
Axial Bending (ML)	-356	356	356	-356
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-217	197	532	-478

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	12	12	12	12
Shear Stress Summary	15	15	9	9

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	218	199	532	522

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	8	8	8	8
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-11	-11	11	11



Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	12	12	12	12

Hoop Stress Summary	9	9	32	32
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	149	-149	149	-149
Axial Membrane (Mc)	-18	-18	18	18
Axial Bending (Mc)	-449	449	449	-449
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6

Axial Stress Summary	-310	290	625	-571
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	12	12	12	12

Shear Stress Summary	9	9	15	15
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	320	290	625	605
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 457.2 + 2 \cdot 1.65 \cdot \sqrt{(3361.5(16.0 - 1.0))}$
 $= 1198.214 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6723.000 mm
Nozzle Outside Diameter	(d) =	1257.200 mm
Vessel Thickness used	(T) =	15.000 mm
Nozzle Thickness used	(t) =	410.000 mm
T / t	=	0.037
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.959$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.054
Mr/P	=	0.085
M0/P	=	0.023
N0/P	=	0.023
MrD/Mc	=	0.252
NrDL/Mc	=	0.114
M0d/Mc	=	0.075
N0DL/Mc	=	0.046
MrD/M1	=	0.058
NrDL/M1	=	0.069
M0D/M1	=	0.018



NØDL/M1 = 0.028

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-10	10	10	-10
Normal Pressure Stress	105	106	105	106
Outplane Stress Summary	116	96	143	80

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	76	-76	76	-76
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-34	34	34	-34
Inplane Pressure Stress	52	52	52	52
Inplane Stress Summary	96	12	179	-44

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	116	96	179	124

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	20	-20	20	-20
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-45	45	45	-45
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	52	52	52	52
Outplane Stress Summary	27	76	127	-5

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	76	-76	76	-76
Inplane Membrane (Mc)	-11	-11	11	11
Inplane Bending (Mc)	-151	151	151	-151
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	105	106	105	106

Inplane Stress Summary	27	177	353	-102
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1

Shear Stress Summary	0	0	1	1
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	27	177	353	102

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	51	51	51	51	51	51	51	51
Circ. Pl (SUS)	-13	-13	29	29	-4	-4	8	8
Circ. Q (SUS)	-33	33	73	-73	-44	44	88	-88
Long. Pm (SUS)	25	25	25	25	25	25	25	25
Long. Pl (SUS)	-3	-3	7	7	-3	-3	19	19
Long. Q (SUS)	-32	32	76	-76	-54	54	94	-94
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	51.0	51.0	51.0	51.0	51.0	51.0	51.0	51.0
Pm+Pl (SUS)	38.1	38.1	80.0	80.0	47.0	47.0	59.1	59.1
Pm+Pl+Q (Total)	17.0	71.9	153.1	51.2	35.2	91.3	148.5	50.7

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	51.00	118.00	Passed
Pm+Pl (SUS)	80.02	177.00	Passed
Pm+Pl+Q (TOTAL)	153.09	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):



Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	105	106	105	106	105	106	105	106
Circ. Pl (SUS)	1	1	5	5	-3	-3	19	19
Circ. Q (SUS)	10	-10	30	-30	-75	75	227	-227
Long. Pm (SUS)	52	52	52	52	52	52	52	52
Long. Pl (SUS)	2	2	14	14	-1	-1	7	7
Long. Q (SUS)	42	-42	110	-110	-25	25	65	-65
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	105.0	106.0	105.0	106.0	105.0	106.0	105.0	106.0
Pm+Pl (SUS)	106.0	107.0	110.0	111.0	102.0	103.0	124.0	125.0
Pm+Pl+Q (Total)	116.0	97.0	176.0	125.0	27.6	178.0	351.0	102.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	106.00	118.00	Passed
Pm+Pl (SUS)	125.00	177.00	Passed
Pm+Pl+Q (TOTAL)	351.00	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	12	12	12	12	12	12	12	12
Circ. Pl (SUS)	-13	-13	29	29	-3	-3	19	19
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-16	-16	20	20	-16	-16	20	20
Long. Q (SUS)	-207	207	505	-505	-300	300	598	-598
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	12.0	12.0	12.0	12.0	12.0	12.0	12.0	12.0
Pm+Pl (SUS)	10.9	10.9	41.6	41.6	19.9	19.9	32.4	32.4
Pm+Pl+Q (Total)	218.1	200.3	531.2	520.3	319.5	290.3	624.4	603.7

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12.00	118.00	Passed
Pm+Pl (SUS)	41.58	177.00	Passed
Pm+Pl+Q (TOTAL)	624.38	354.00	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

PV Elite 28 U2 Licensee: SL User
FileName : C-1401_Ope_Rev 0
Nozzle Calcs.: N6b 18"

Nozl: 150 4:01pm Jul 21,2026

Page 477 of 564



HEXAGON

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Input, Nozzle Desc: K7a

From: 180

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2000.0029	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		30050.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

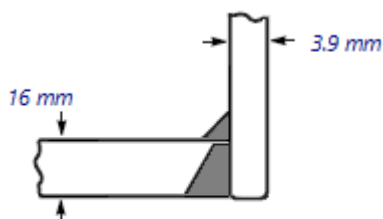
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K7a

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1209 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.0036	mm
Parallel to Vessel Wall, opening length	d	52.5018	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K7a.
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.5004	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.5004	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(14.5004, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K7a

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 216.1358 mm

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Input, Nozzle Desc: K8a

From: 180

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2000.0006	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		32870.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		30.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

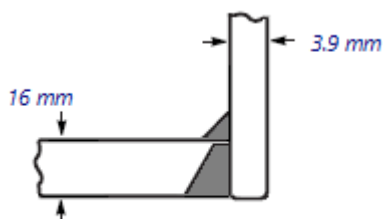
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K8a

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1209 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.0036	mm
Parallel to Vessel Wall, opening length	d	52.5018	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K8a.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.5004	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.5004	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(14.5004, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K8a

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 216.1358 mm

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Input, Nozzle Desc: K15a

From: 180

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2000.0006	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		32870.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		70.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

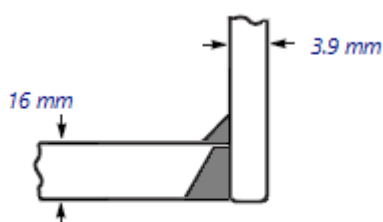
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K15a

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1209 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.0036	mm
Parallel to Vessel Wall, opening length	d	52.5018	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K15a.
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.5004	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.5004	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(14.5004, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K15a

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 216.1358 mm

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Input, Nozzle Desc: K7b

From: 190

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2000.0011	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		33800.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

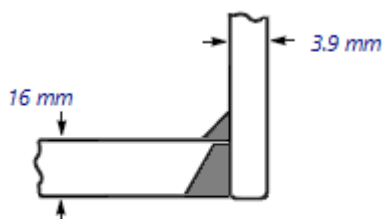
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K7b

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1209 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.0036	mm
Parallel to Vessel Wall, opening length	d	52.5018	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K7b.
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.5004	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.5004	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(14.5004, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K7b

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 216.1358 mm

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Input, Nozzle Desc: K8b

From: 190

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2000.0011	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		33800.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		30.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

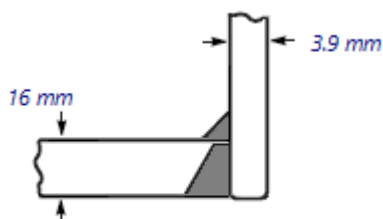
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K8b

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1209 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.0036	mm
Parallel to Vessel Wall, opening length	d	52.5018	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K8b.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.5004	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.5004	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(14.5004, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K8b

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 216.1358 mm

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Input, Nozzle Desc: K15b

From: 190

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2000.0011	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		33800.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		70.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

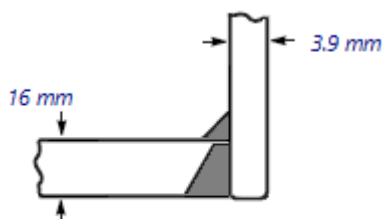
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K15b

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1209 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.0036	mm
Parallel to Vessel Wall, opening length	d	52.5018	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K15b.
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.5004	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.5004	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(14.5004, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K15b

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 216.1358 mm

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Input, Nozzle Desc: MH6 30"

From: 190

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2000.0011	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		34220.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		45.00	deg
Diameter		30.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

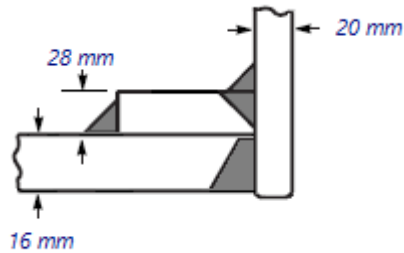
Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1422.0000	mm
Thickness of Pad	te	28.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm
Reinforcing Pad Width		330.0000	mm

This is a Manway, Coupling or Access Opening.

Flange Class		150
Flange Grade		GR 1.1
Flange Series		B

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: MH6 30"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	30.000 in.
Actual Thickness Used in Calculation	0.787 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
 $= 13.5004 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.47 \cdot 381.0 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
 $= 1.5274 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3927 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1444.0000 mm
Parallel to Vessel Wall, opening length	d	722.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

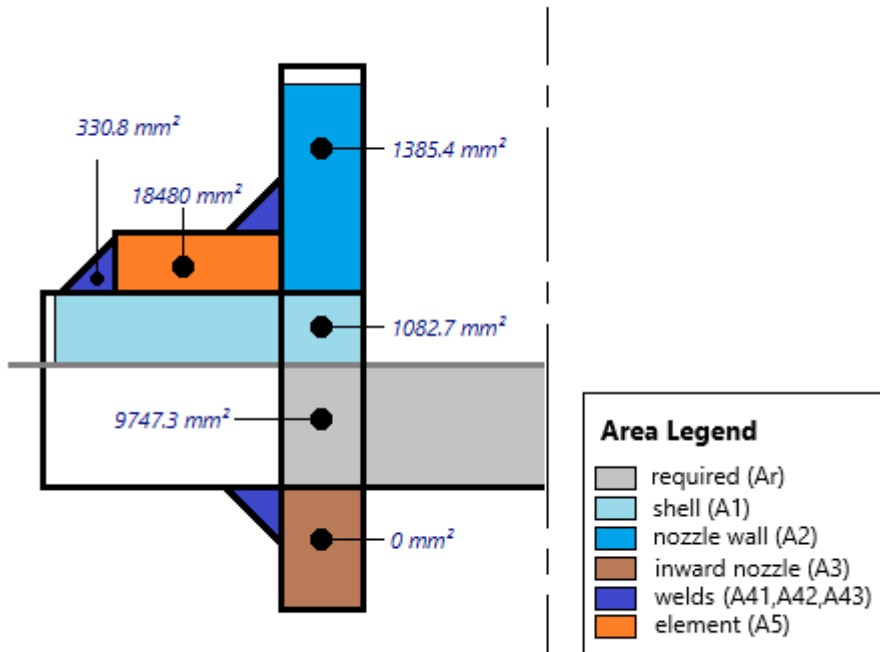


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	9747.263	3481.864	NA
Area in Shell	A1	1082.737	3866.271	NA
Area in Nozzle Wall	A2	1385.444	1395.547	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	330.750	330.750	NA
Area in Element	A5	18480.002	18480.002	NA
TOTAL AREA AVAILABLE	Atot	21278.934	24072.568	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS:	Diameter	Thickness
Based on given Pad Thickness:	1010.1546	28.0000 mm
Based on given Pad Diameter:	1422.0000	10.5278 mm
Based on Shell or Nozzle Thickness:	1196.2706	16.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (722.0 \cdot 13.5004 \cdot 1.0 + 2 \cdot 20.0 \cdot 13.5004 \cdot 1.0 \cdot (1 - 1.0))$
 $= 9747.263 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 722.0(1.0 \cdot 15.0 - 1.0 \cdot 13.5) - 2 \cdot 20.0$
 $(1.0 \cdot 15.0 - 1.0 \cdot 13.5004) \cdot (1 - 1.0)$
 $= 1082.737 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 37.5) \cdot (20.0 - 1.53) \cdot 1.0$



= 1385.444 mm²

Area Available in Welds [A41 + A42 + A43]:
= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Trapfr4
= (99.75) • 1.0 + (0.0) • 1.0 + 231.0² • 1.0
= 330.750 mm²

Area Available in Element [A5]:
= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
= (1422.0 - 762.0) • 28.0 • 1.0
= 18480.002 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures ta = 1.5274 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= ta
= 1.5274 mm

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 20.0, tr = 1.527, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.076, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 20.0, tr = 1.527, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.076, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 20.0, tr = 1.527, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.076, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:C)

Govrn. thk, tg = 16.0, tr = 13.5, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.9, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-20 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 16.0, tr = 13.5, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.9, Temp. Reduction = 6 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-20 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-29 °C
Gov. MDMT of the Reinforcement Pad	:	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-29 °C



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/1.96 = 0.242

Weld Size Calculations, Description: MH6 30"

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm
Intermediate Calc. for pad/shell Welds TminPad 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	7.5000 = 0.5*TminPad	11.3120 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (9747.2627 - 1082.7372 + 2 • 20.0 • 1.0 •
(1.0 • 15.0 - 13.5004))118)
= 1029386.19 N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)²•fr2)•Sv
= (1385.4443 + 18480.002 + 330.7501 - 0.0 • 1.0) • 118
= 2382905.25 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (1385.4443 + 0.0 + 100.0 + (600.0)) • 118
= 246057.06 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (1385.4443 + 0.0 + 330.7501 + 18480.002 + (600.0)) • 118
= 2453698.00 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.1416/2.0) • 762.0 • 10.0 • 0.49 • 118
= 692016. N

Shear, Pad Element Weld [Spew]:
= (π/2) • DP • WP • 0.49 • SEW
= (3.1416/2.0) • 1422.0 • 16.0 • 0.49 • 118
= 2066202. N

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.1416 • 371.0) • (20.0 - 0.0) • 0.7 • 118
= 1925293. N

Tension, Pad Groove Weld [Tpgw]:
= (π/2) • Dlo • Wgpn • 0.74 • Seg
= (3.1416/2) • 762.0 • 28.0 • 0.74 • 138
= 3422211. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.1416/2.0) • 762.0 • (16.0 - 1.0) • 0.74 • 118
= 1567628. N



See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (2066202 + 1925293) = 3991495 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (692016 + 3422211 + 1567628 + 0) = 5681854 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (2066202 + 1567628 + 0) = 3633830 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 3991495 N , must exceed W = 1029386 N or W1 = 2382905 N
Path 2-2 = 5681854 N , must exceed W = 1029386 N or W2 = 246057 N
Path 3-3 = 3633829 N , must exceed W = 1029386 N or W3 = 2453698 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 21.7169 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 437.7171 mm

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Input, Nozzle Desc: K11a 2"

From: 190

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6706.00	mm
Design Length of Section	L	2000.0011	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		34220.00	mm
-----------------------------------	--	----------	----

User Entered Minimum Design Metal Temperature		-1.00	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		120.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

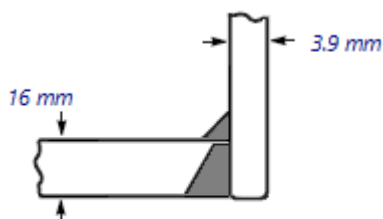
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel Wgnv		16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K11a 2"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3354.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.5004 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1209 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.0036	mm
Parallel to Vessel Wall, opening length	d	52.5018	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K11a 2".
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.5004	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.5004	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3941	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[3.42, max(14.5004, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K11a 2"

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 216.1358 mm

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Input, Nozzle Desc: N7 16"

From: 210

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6698.00	mm
Design Length of Section	L	1236.7529	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		39549.99	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-516 60 +IT	
Material UNS Number		K02100	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		240.00	deg
Diameter		16.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	10.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

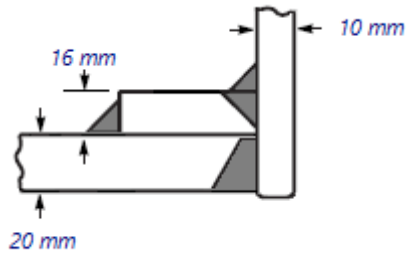
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	16.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 60 +IT	
Pad Allowable Stress at Temperature	Sp	118.00	MPa
Pad Allowable Stress At Ambient	Spa	118.00	MPa
Diameter of Pad along vessel surface	Dp	1066.4000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		330.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: N7 16"

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	16.000 in.
Actual Thickness Used in Calculation	0.394 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3350.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.4843 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 203.2 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.8146 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.9615 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	772.8000 mm
Parallel to Vessel Wall, opening length	d	386.4000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		41.0000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 118.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 118.0)$
= 1.000

Weld Strength Reduction Factor [fr4]:
= $\min(1, S_p / S_v)$
= $\min(1, 118.0 / 118.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

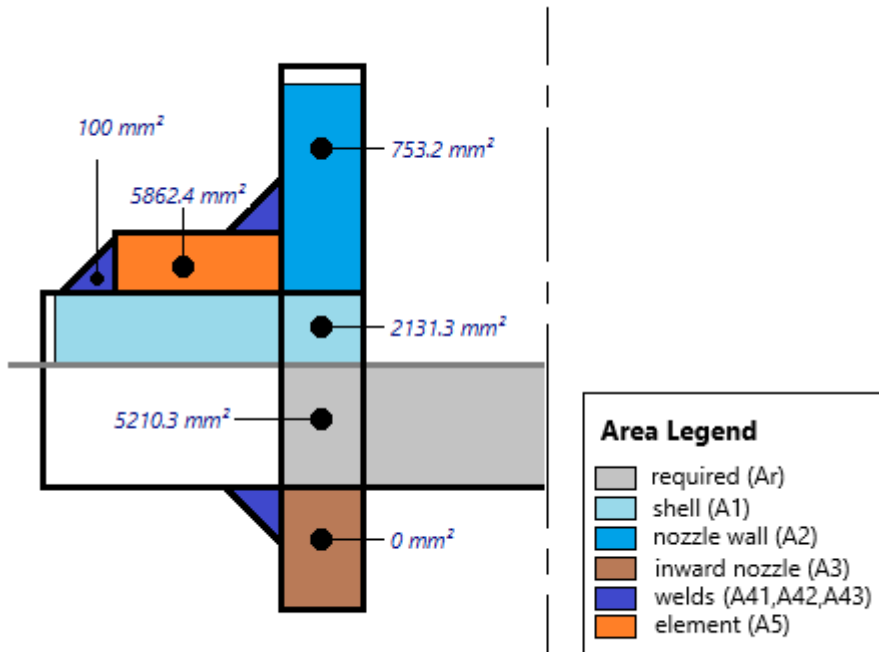


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	5210.319	1527.642	NA
Area in Shell	A1	2131.281	4286.316	NA
Area in Nozzle Wall	A2	753.201	741.161	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	5862.400	5862.400	NA
TOTAL AREA AVAILABLE	Atot	8846.882	10989.877	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 545.5148 16.0000 mm
 Based on given Pad Diameter: 1066.4001 6.0749 mm
 Based on Shell or Nozzle Thickness: 628.9838 10.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (386.4 \cdot 13.4843 \cdot 1.0 + 2 \cdot 10.0 \cdot 13.4843 \cdot 1.0 \cdot (1 - 1.0))$
 $= 5210.319 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 386.4(1.0 \cdot 19.0 - 1.0 \cdot 13.484) - 2 \cdot 10.0$
 $(1.0 \cdot 19.0 - 1.0 \cdot 13.4843) \cdot (1 - 1.0)$
 $= 2131.281 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 41.0) \cdot (10.0 - 0.81) \cdot 1.0$



= 753.201 mm²

Area Available in Welds [A41 + A42 + A43]:
= $W_o^2 \cdot fr3 + (W_i - can/0.707)^2 \cdot fr2 + W_p^2 \cdot fr4$
= $10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0$
= 100.000 mm²

Area Available in Element [A5]:
= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, Tlwp, te)) \cdot fr4$
= $(772.8 - 406.4) \cdot 16.0 \cdot 1.0$
= 5862.400 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 0.9615 mm
Wall Thickness per UG16(b),	tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 14.4843 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 14.4843 mm
Wall Thickness, shell/head, external pressure	trb2 = 2.3924 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[8.331, max(14.4843, 2.5)]
= 8.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.9615, 8.3312)
= 8.3312 mm

Available Nozzle Neck Thickness = 10.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 28.3,	Allowable	: 118.0 MPa	Passed
Expansion	: 0.0,	Allowable	: 266.7 MPa	Passed
Occasional	: 4.5,	Allowable	: 156.9 MPa	Passed
Shear	: 18.3,	Allowable	: 82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 10.0, tr = 0.815, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.081$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: C

Govrn. thk, tg = 16.0, c = 1.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.71$, Temp. Reduction = 16 °C
Attachment governing, Attachment stress = Shell stress per Code



Min Metal Temp. w/o impact per UCS-66, Curve C -32 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: C

Govrn. thk, tg = 10.0, tr = 0.815, c = 0.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.081, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -44 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 0.47/1.96 = 0.242

Weld Size Calculations, Description: N7 16"

Intermediate Calc. for nozzle/shell Welds Tmin 10.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 16.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	8.0000 = 0.5*TminPad	11.3120 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (5210.3193 - 2131.2805 + 2 \cdot 10.0 \cdot 1.0 \cdot (1.0 \cdot 19.0 - 13.4843)) \cdot 118)$
 $= 376304.94 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (753.2013 + 5862.3999 + 100.0 - 0.0 \cdot 1.0) \cdot 118$
 $= 792359.25 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (753.2013 + 0.0 + 100.0 + (380.0)) \cdot 118$
 $= 145502.75 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (753.2013 + 0.0 + 100.0 + 5862.3999 + (380.0)) \cdot 118$
 $= 837194.69 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 406.4 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 369075. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 1066.4 \cdot 16.0 \cdot 0.49 \cdot 118$
 $= 1549506. \text{ N}$



Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 198.2) \cdot (10.0 - 0.0) \cdot 0.7 \cdot 118$$

$$= 514276. \text{ N}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 406.4 \cdot 16.0 \cdot 0.74 \cdot 118$$

$$= 891806. \text{ N}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 406.4 \cdot (16.0 - 0.0) \cdot 0.74 \cdot 118$$

$$= 891806. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (1549506 + 514276) = 2063783 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$$

$$= (369075 + 891806 + 891806 + 0) = 2152687 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw})$$

$$= (1549506 + 891806 + 0) = 2441312 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 2063782 N , must exceed W = 376304 N or W1 = 792359 N
 Path 2-2 = 2152687 N , must exceed W = 376304 N or W2 = 145502 N
 Path 3-3 = 2441312 N , must exceed W = 376304 N or W3 = 837194 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 6.1702 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 426.1701 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.523 %

Input Echo, WRC297 Item 1, Description: N7 16" :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	6698.000	mm
Shell Thickness	20.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 60	
Vessel Cold S.I. Allowable	Smc	118.00 MPa
Vessel Hot S.I. Allowable	Smh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	406.400	mm
Nozzle Thickness	10.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 60	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa



See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter of Reinforcing Pad 1066.400 mm
 Thickness of Reinforcing Pad 16.0000 mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSTAINED loads.*

Design Internal Pressure	Dp	0.47	MPa
Radial Load	P	-29999.98	N
Circumferential Shear	Vc	19999.99	N
Longitudinal Shear	Vl	19999.99	N
Circumferential Moment	Mc	0.224E+08	N-mm
Longitudinal Moment	Ml	0.224E+08	N-mm
Torsional Moment	Mt	0.320E+08	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 406.4 + 2 \cdot 1.65 \cdot \sqrt{(3367.5(20.0 - 1.0))}$
 $= 1241.127 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6735.000 mm
Nozzle Outside Diameter	(d) =	406.400 mm
Vessel Thickness used	(T) =	35.000 mm
Nozzle Thickness used	(t) =	10.000 mm
T / t	=	3.500
d / t	=	40.640

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 0.837$$

Nr/P	=	0.076
Mr/P	=	0.105
M0/P	=	0.113
N0/P	=	0.275
MrD/Mc	=	0.158
NrDL/Mc	=	0.106
M0d/Mc	=	0.216
N0DL/Mc	=	0.153
MrD/Ml	=	0.137
NrDL/Ml	=	0.096
M0D/Ml	=	0.170
N0DL/Ml	=	0.316

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-14	-14	14	14
Outplane Bending (ML)	-45	45	45	-45
Normal Pressure Stress	45	45	45	45
Outplane Stress Summary	8	67	128	3



Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-36	36	36	-36
Inplane Pressure Stress	22	22	22	22
Inplane Stress Summary	-1	41	81	-23

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	13	68	128	28

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-58	58	58	-58
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	22	22	22	22
Outplane Stress Summary	-19	64	111	-38

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	-4	-4	4	4
Inplane Bending (Mc)	-42	42	42	-42
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	45	45	45	45
Inplane Stress Summary	15	69	109	-5

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0



Torsional Shear (Mt)	3	3	3	3
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Shear Stress Summary	2	2	4	4
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	34	70	114	39
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-14	-14	14	14
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	11	10	11

Hoop Stress Summary	3	3	31	32
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	121	-121	121	-121
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-18	-18	18	18
Axial Bending (ML)	-293	293	293	-293
Axial Pressure Stress	5	5	5	5

Axial Stress Summary	-183	161	441	-388
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	3	3	-3	-3
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	12	12	12	12

Shear Stress Summary	15	15	9	9
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Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	188	162	441	421
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Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-6	-6	6	6
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	10	11	10	11



Hoop Stress Summary 10 11 24 24

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to
 pipe cross-section)

	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	121	-121	121	-121
Axial Membrane (Mc)	-18	-18	18	18
Axial Bending (Mc)	-346	346	346	-346
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5

Axial Stress Summary -235 213 494 -441

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Shear stress)

	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	12	12	12	12

Shear Stress Summary 9 9 15 15

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE
 (Stress Intensities)

	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside

Two * Max Shear Stress	246	214	494	467
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 406.4 + 2 \cdot 1.65 \cdot \sqrt{(3359.5(20.0 - 1.0))}$$

$$= 1240.135 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	6719.000 mm
Nozzle Outside Diameter	(d) =	1066.400 mm
Vessel Thickness used	(T) =	19.000 mm
Nozzle Thickness used	(t) =	340.000 mm
T / t	=	0.056
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.985$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.066
Mr/P	=	0.108
M0/P	=	0.031
N0/P	=	0.032
MrD/Mc	=	0.270
NrDL/Mc	=	0.119
M0d/Mc	=	0.081
N0DL/Mc	=	0.053
MrD/Ml	=	0.078
NrDL/Ml	=	0.092
M0D/Ml	=	0.023
N0DL/Ml	=	0.043

Vessel Stresses (MPa):

LONGITUDINAL PLANE
 (Stresses Normal to

	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom



longitudinal plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-8	8	8	-8
Normal Pressure Stress	83	83	83	83
Outplane Stress Summary	90	76	112	65

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	54	-54	54	-54
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-27	27	27	-27
Inplane Pressure Stress	41	41	41	41
Inplane Stress Summary	68	15	133	-28

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	0	0

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	90	76	133	94

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	-3	-3	3	3
Outplane Bending (Mc)	-28	28	28	-28
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	41	41	41	41
Outplane Stress Summary	28	54	91	3

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	5	5	5	5
Inplane Bending (P)	54	-54	54	-54
Inplane Membrane (Mc)	-6	-6	6	6
Inplane Bending (Mc)	-94	94	94	-94
Inplane Membrane (ML)	0	0	0	0



Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	83	83	83	83
Inplane Stress Summary	41	122	243	-51

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	0	0	1	1

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	41	122	243	55

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	45	45	45	45	45	45	45	45
Circ. Pl (SUS)	-8	-8	20	20	-3	-3	5	5
Circ. Q (SUS)	-29	29	61	-61	-27	27	57	-57
Long. Pm (SUS)	22	22	22	22	22	22	22	22
Long. Pl (SUS)	-3	-3	5	5	0	0	12	12
Long. Q (SUS)	-21	21	51	-51	-42	42	74	-74
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	45.0	45.0	45.0	45.0	45.0	45.0	45.0	45.0
Pm+Pl (SUS)	37.0	37.0	65.0	65.0	42.0	42.0	50.0	50.0
Pm+Pl+Q (Total)	11.7	66.3	126.2	28.6	35.5	70.4	110.5	40.3

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	45.00	118.00	Passed
Pm+Pl (SUS)	65.00	177.00	Passed
Pm+Pl+Q (TOTAL)	126.19	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	83	83	83	83	83	83	83	83



Circ. P1 (SUS)	0	0	4	4	-1	-1	11	11
Circ. Q (SUS)	7	-7	23	-23	-40	40	148	-148
Long. Pm (SUS)	41	41	41	41	41	41	41	41
Long. P1 (SUS)	0	0	10	10	-1	-1	5	5
Long. Q (SUS)	27	-27	81	-81	-13	13	43	-43
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	83.0	83.0	83.0	83.0	83.0	83.0	83.0	83.0
Pm+P1 (SUS)	83.0	83.0	87.0	87.0	82.0	82.0	94.0	94.0
Pm+P1+Q (Total)	90.0	76.0	132.0	94.0	42.0	122.0	242.0	57.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	83.00	118.00	Passed
Pm+P1 (SUS)	94.00	177.00	Passed
Pm+P1+Q (TOTAL)	242.00	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10	11	10	11	10	11	10	11
Circ. P1 (SUS)	-8	-8	20	20	0	0	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. P1 (SUS)	-16	-16	20	20	-16	-16	20	20
Long. Q (SUS)	-172	172	414	-414	-225	225	467	-467
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	10.0	11.0	10.0	11.0	10.0	11.0	10.0	11.0
Pm+P1 (SUS)	14.3	15.2	31.4	32.2	21.8	22.8	26.9	27.2
Pm+P1+Q (Total)	187.4	162.4	439.2	420.4	246.7	214.4	492.5	466.0

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	11.00	118.00	Passed
Pm+P1 (SUS)	32.24	177.00	Passed
Pm+P1+Q (TOTAL)	492.48	354.00	Failed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



Input, Nozzle Desc: MH7

From: 210

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6698.00	mm
Design Length of Section	L	1236.7529	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		39870.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		45.00	deg
Diameter		30.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	20.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

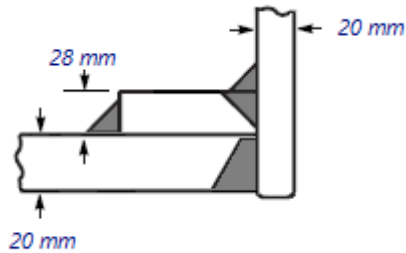
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	400.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	MPa
Pad Allowable Stress At Ambient	Spa	138.00	MPa
Diameter of Pad along vessel surface	Dp	1422.0000	mm
Thickness of Pad	te	28.0000	mm
Weld leg size between Pad and Shell	Wp	16.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm
Reinforcing Pad Width		330.0000	mm
This is a Manway, Coupling or Access Opening.			

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: MH7

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	30.000 in.
Actual Thickness Used in Calculation	0.787 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

This nozzle looks to be 26 inch or greater OD and the flange series is set to None. Please set it to series A or B in the nozzle dialog or the flange weight will not be determined.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.47 \cdot 3350.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
 $= 13.4843 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.47 \cdot 381.0 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
 $= 1.5274 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3927 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1444.0000 mm
Parallel to Vessel Wall, opening length	d	722.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		47.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

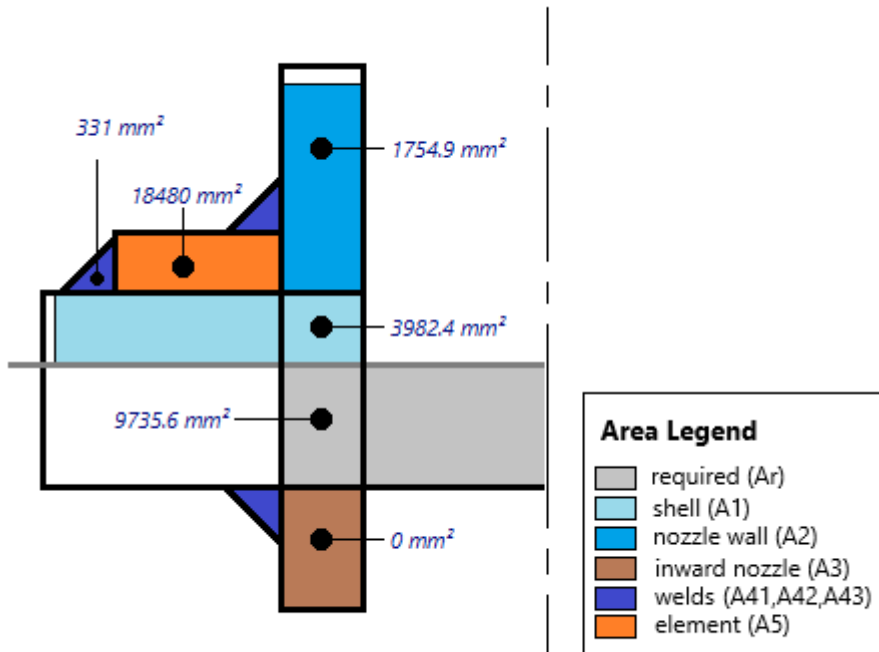


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	9735.639	2854.445	NA
Area in Shell	A1	3982.362	8009.111	NA
Area in Nozzle Wall	A2	1754.896	1767.692	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds A41+A42+A43		331.000	331.000	NA
Area in Element	A5	18480.002	18480.002	NA
TOTAL AREA AVAILABLE	Atot	24548.260	28587.805	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 892.9778 28.0000 mm
 Based on given Pad Diameter: 1422.0000 5.5566 mm
 Based on Shell or Nozzle Thickness: 945.3690 20.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (722.0 \cdot 13.4843 \cdot 1.0 + 2 \cdot 20.0 \cdot 13.4843 \cdot 1.0 \cdot (1 - 1.0))$
 $= 9735.639 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 722.0(1.0 \cdot 19.0 - 1.0 \cdot 13.484) - 2 \cdot 20.0$
 $(1.0 \cdot 19.0 - 1.0 \cdot 13.4843) \cdot (1 - 1.0)$
 $= 3982.362 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 47.5) \cdot (20.0 - 1.53) \cdot 1.0$



$$= 1754.896 \text{ mm}^2$$

Area Available in Welds [A41 + A42 + A43]:
$$= (W_o^2 - \text{Ar Lost}) \cdot \text{Fr3} + ((W_i - \text{can}/0.707)^2 - \text{Ar Lost}) \cdot \text{fr2} + \text{Trapfr4}$$
$$= (100.0) \cdot 1.0 + (0.0) \cdot 1.0 + 231.0^2 \cdot 1.0$$
$$= 331.000 \text{ mm}^2$$

Area Available in Element [A5]:
$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lwp}, t_e)) \cdot \text{fr4}$$
$$= (1422.0 - 762.0) \cdot 28.0 \cdot 1.0$$
$$= 18480.002 \text{ mm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement:

Wall Thickness for Internal/External pressures $t_a = 1.5274 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= t_a$$
$$= 1.5274 \text{ mm}$$

Available Nozzle Neck Thickness = 20.0000 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 20.0$, $t_r = 1.527$, $c = 0.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.076$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	12 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 20.0$, $t_r = 1.527$, $c = 0.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.076$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 20.0$, $t_r = 1.527$, $c = 0.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.076$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:C)

Govrn. thk, $t_g = 20.0$, $t_r = 13.484$, $c = 1.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.71$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, $t_g = 20.0$, $t_r = 13.484$, $c = 1.0 \text{ mm}$, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.71$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-29 °C
Gov. MDMT of the Reinforcement Pad	:	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-29 °C



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 0.47/1.96 = 0.242

Weld Size Calculations, Description: MH7

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	11.3120 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (9735.6387 - 3982.3616 + 2 \cdot 20.0 \cdot 1.0 \cdot (1.0 \cdot 19.0 - 13.4843)) \cdot 118)$
 $= 704848.19 \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (1754.8962 + 18480.002 + 331.0002 - 0.0 \cdot 1.0) \cdot 118$
 $= 2426525.75 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (1754.8962 + 0.0 + 100.0 + (760.0)) \cdot 118$
 $= 308525.94 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (1754.8962 + 0.0 + 331.0002 + 18480.002 + (760.0)) \cdot 118$
 $= 2516196.50 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 762.0 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 692016. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 1422.0 \cdot 16.0 \cdot 0.49 \cdot 118$
 $= 2066202. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 371.0) \cdot (20.0 - 0.0) \cdot 0.7 \cdot 118$
 $= 1925293. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 762.0 \cdot 28.0 \cdot 0.74 \cdot 138$
 $= 3422211. \text{ N}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 762.0 \cdot (18.0 - 0.0) \cdot 0.74 \cdot 118$
 $= 1881153. \text{ N}$



See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (\text{SPEW} + \text{SNW}) = (2066202 + 1925293) = 3991495 \text{ N}$$

Failure Path [Path 2-2]:

$$= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (692016 + 3422211 + 1881153 + 0) = 5995380 \text{ N}$$

Failure Path [Path 3-3]:

$$= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (2066202 + 1881153 + 0) = 3947355 \text{ N}$$

Summary of Failure Path Calculations:

Path 1-1 = 3991495 N	, must exceed W = 704848 N	or W1 = 2426525 N
Path 2-2 = 5995379 N	, must exceed W = 704848 N	or W2 = 308525 N
Path 3-3 = 3947355 N	, must exceed W = 704848 N	or W3 = 2516196 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 21.7429 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 441.7428 mm

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Input, Nozzle Desc: K16a

From: 210

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6698.00	mm
Design Length of Section	L	1236.7529	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		39870.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

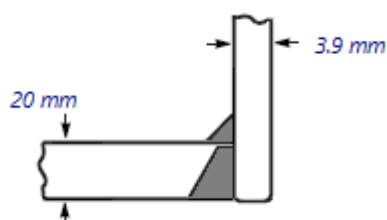
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K16a



ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3350.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.4843 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1209 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.0036	mm
Parallel to Vessel Wall, opening length	d	52.5018	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K16a.
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.4843	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.4843	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3924	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(14.4843, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K16a

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	2.7381 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.1356 mm

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Input, Nozzle Desc: K16b

From: 210

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6698.00	mm
Design Length of Section	L	1236.7529	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		39870.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

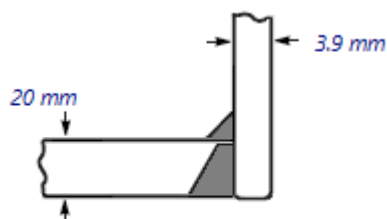
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K16b

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3350.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.4843 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1209 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.0036	mm
Parallel to Vessel Wall, opening length	d	52.5018	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K16b.
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.4843	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.4843	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3924	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1,tb2)]
= min[3.42, max(14.4843, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K16b

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.1356 mm

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Input, Nozzle Desc: K11b

From: 210

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6698.00	mm
Design Length of Section	L	1236.7529	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		39870.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		120.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

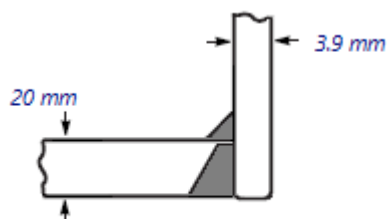
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K11b

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3350.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.4843 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1209 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.0036	mm
Parallel to Vessel Wall, opening length	d	52.5018	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K11b.
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370 mm
Wall Thickness per UG16(b),	tr16b	= 2.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.4843 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.4843 mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3924 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000 mm
Wall Thickness per table UG-45	tb3	= 3.4200 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(14.4843, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K11b

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.1356 mm

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Input, Nozzle Desc: K12

From: 210

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6698.00	mm
Design Length of Section	L	1236.7529	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		39870.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		150.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

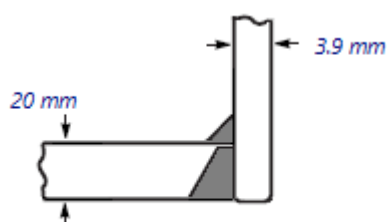
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K12

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3350.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.4843 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1209 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.0036	mm
Parallel to Vessel Wall, opening length	d	52.5018	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K12.
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.4843	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.4843	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3924	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(14.4843, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K12

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.1356 mm

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Input, Nozzle Desc: K13b

From: 210

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	6698.00	mm
Design Length of Section	L	1236.7529	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		39870.00	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		200.00	deg
Diameter		2.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

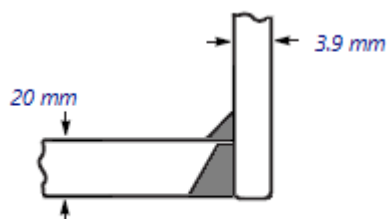
Flange Material		SA-105	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	200.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	18.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: K13b

ASME Code, Section VIII, Div. 1, UG-37 to UG-45



Actual Outside Diameter Used in Calculation 2.375 in.
Actual Thickness Used in Calculation 0.154 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 3350.0 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 13.4843 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 30.1625 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.1209 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.2370 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	105.0036	mm
Parallel to Vessel Wall, opening length	d	52.5018	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.7790	mm

*Taking a UG-36(c)(3)(a) exemption for nozzle: K13b.
This calculation is valid for nozzles that meet all the requirements of
paragraph UG-36. Please check the Code carefully, especially for nozzles
that are not isolated or do not meet Code spacing requirements. To force
the computation of areas for small nozzles go to Tools->Configuration
and check the box to force the UG-37 small nozzle area calculation.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.2370	mm
Wall Thickness per UG16(b),	tr16b	= 2.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 14.4843	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 14.4843	mm
Wall Thickness, shell/head, external pressure	trb2	= 2.3924	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 2.5000	mm
Wall Thickness per table UG-45	tb3	= 3.4200	mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.42, max(14.4843, 2.5)]
= 3.4200 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(0.237, 3.42)
= 3.4200 mm

Available Nozzle Neck Thickness = $0.875 \cdot 3.912 = 3.423$ mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, tg = 3.423, tr = 0.121, c = 0.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.035$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C



Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/5.10 = 0.093

Weld Size Calculations, Description: K13b

Intermediate Calc. for nozzle/shell Welds Tmin 3.9116 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.7381 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

The Drop for this Nozzle is : 0.1357 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 220.1356 mm

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Input, Nozzle Desc: P1(frm V-1403)

From: 250

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	3422.40	mm
Design Length of Section	L	3299.1997	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		43360.99	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B +IT	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		285.00	deg
Diameter		4.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

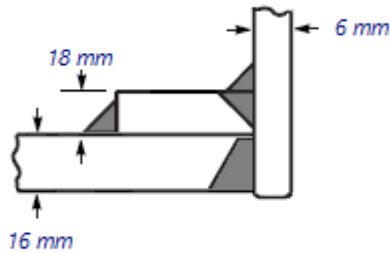
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	411.1000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137.90	MPa
Pad Allowable Stress At Ambient	Spa	137.90	MPa
Diameter of Pad along vessel surface	Dp	250.0000	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	14.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	18.0000	mm
Reinforcing Pad Width		67.8500	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: P1(frm V-1403)

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.237 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 1712.2 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 6.8919 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.2291 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4645 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	204.5208 mm
Parallel to Vessel Wall, opening length	d	102.2604 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		33.0495 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 118.0)$
= 1.000

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 118.0 / 118.0)$
= 1.000

Weld Strength Reduction Factor [fr4]:
= $\min(1, S_p / S_v)$
= $\min(1, 137.9 / 118.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(1.0, 1.0)$
= 1.000

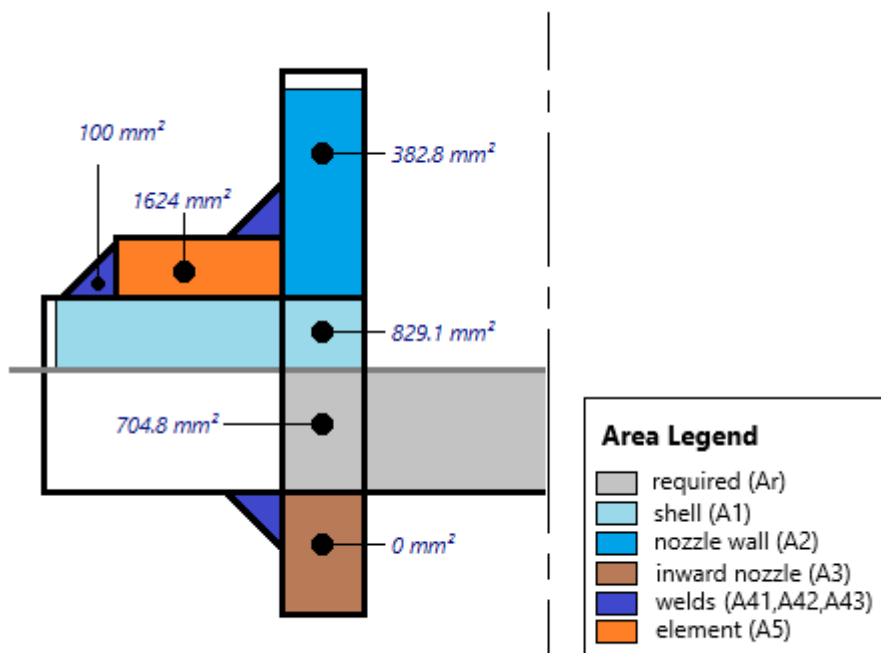


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	704.765	412.573	NA
Area in Shell	A1	829.141	708.761	NA
Area in Nozzle Wall	A2	382.759	367.203	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	100.000	100.000	NA
Area in Element	A5	1623.974	1623.974	NA
TOTAL AREA AVAILABLE	Atot	2935.874	2799.938	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (102.2604 \cdot 6.8919 \cdot 1.0 + 2 \cdot 6.0198 \cdot 6.8919 \cdot 1.0 \cdot (1 - 1.0))$
 $= 704.765 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 102.26(1.0 \cdot 15.0 - 1.0 \cdot 6.892) - 2 \cdot 6.02$
 $(1.0 \cdot 15.0 - 1.0 \cdot 6.8919) \cdot (1 - 1.0)$
 $= 829.141 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 33.05) \cdot (6.02 - 0.23) \cdot 1.0$
 $= 382.759 \text{ mm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi\text{-}can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0$



= 100.000 mm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) • fr4
= (204.5208 - 114.3) • 18.0 • 1.0
= 1623.974 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 0.4645 mm
Wall Thickness per UG16(b), tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure trb1 = 7.8919 mm
Wall Thickness tb1 = max(trb1, tr16b) = 7.8919 mm
Wall Thickness, shell/head, external pressure trb2 = 1.7117 mm
Wall Thickness tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45 tb3 = 5.2578 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[5.258, max(7.8919, 2.5)]
= 5.2578 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(0.4645, 5.2578)
= 5.2578 mm

Available Nozzle Neck Thickness = 0.875 * 6.020 = 5.267 mm -> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 5.267, tr = 0.229, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.043, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 5.267, tr = 0.229, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.043, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:C)

Govrn. thk, tg = 16.0, tr = 6.892, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.459, Temp. Reduction = 38 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: C

Govrn. thk, tg = 5.267, tr = 0.229, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.043, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -48 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C



Gov. MDMT of the Reinforcement Pad : -48 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 0.47/1.96 = 0.242

Weld Size Calculations, Description: P1(frm V-1403)

Intermediate Calc. for nozzle/shell Welds Tmin 6.0198 mm
Intermediate Calc. for pad/shell Welds TminPad 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.2139 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	7.5000 = 0.5*TminPad	9.8980 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)
= max(0, (704.765 - 829.141 + 2 * 6.0198 * 1.0 *
(1.0 * 15.0 - 6.8919))118)
= max(0, -3157.03) N

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)*fr2)*Sv
= (382.7588 + 1623.9744 + 100.0 - 0.0 * 1.0) * 118
= 248568.88 N

Weld Load [W2]:
= (A2 + A3 + A4 + (2 * tn * t * fr1)) * Sv
= (382.7588 + 0.0 + 100.0 + (180.594)) * 118
= 78267.56 N

Weld Load [W3]:
= (A2+A3+A4+A5+(2*tn*t*fr1))*S
= (382.7588 + 0.0 + 100.0 + 1623.9744 + (180.594)) * 118
= 269876.78 N

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) * Dlo * Wo * 0.49 * Snw
= (3.1416/2.0) * 114.3 * 10.0 * 0.49 * 118
= 103802. N

Shear, Pad Element Weld [Spew]:
= (π/2) * DP * WP * 0.49 * SEW
= (3.1416/2.0) * 250.0 * 14.0 * 0.49 * 118
= 317849. N

Shear, Nozzle Wall [Snw]:
= (π * (Dlr + Dlo)/4) * (Thk - Can) * 0.7 * Sn
= (3.1416 * 54.1401) * (6.0198 - 0.0) * 0.7 * 118
= 84566. N

Tension, Pad Groove Weld [Tpgw]:
= (π/2) * Dlo * Wgpn * 0.74 * Seg
= (3.1416/2) * 114.3 * 18.0 * 0.74 * 138
= 329760. N

Tension, Shell Groove Weld [Tngw]:
= (π/2) * Dlo * Wgnvi * 0.74 * Sng



$$= (3.1416/2.0) \cdot 114.3 \cdot (14.0 - 0.0) \cdot 0.74 \cdot 118 \\ = 219468. \text{ N}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNW}) = (317849 + 84566) = 402415 \text{ N}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw}) \\ = (103802 + 329760 + 219468 + 0) = 653030 \text{ N}$

Failure Path [Path 3-3]:
 $= (\text{Spew} + \text{Tngw} + \text{Sinw}) \\ = (317849 + 219468 + 0) = 537317 \text{ N}$

Summary of Failure Path Calculations:

Path 1-1 = 402415 N , must exceed W = 0 N	or W1 = 248568 N
Path 2-2 = 653030 N , must exceed W = 0 N	or W2 = 78267 N
Path 3-3 = 537317 N , must exceed W = 0 N	or W3 = 269876 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.9546 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 428.0546 mm

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Input, Nozzle Desc: P2(frm ME-1702)

From: 250

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	3422.40	mm
Design Length of Section	L	3299.1997	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		43360.99	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
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Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117.90	MPa
Allowable Stress At Ambient	Sna	117.90	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		245.00	deg
Diameter		4.0000	in.

Size and Thickness Basis		Nominal	
Nominal Thickness		STD	

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

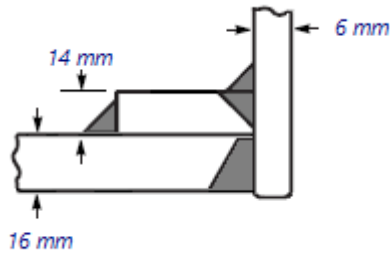
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	423.8000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	137.90	MPa
Pad Allowable Stress At Ambient	Spa	137.90	MPa
Diameter of Pad along vessel surface	Dp	250.0000	mm
Thickness of Pad	te	14.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	14.0000	mm
Reinforcing Pad Width		67.8500	mm
User Defined Nozzle/Shell Centerline Angle		90.0000	deg.

Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: P2(frm ME-1702)

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	4.500 in.
Actual Thickness Used in Calculation	0.237 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $0.47 \cdot 1712.2 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
= 6.8919 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $0.47 \cdot 57.15 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
= 0.2293 mm

Required Nozzle thickness under External Pressure per UG-28 : 0.4700 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	204.5208 mm
Parallel to Vessel Wall, opening length	d	102.2604 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		29.0495 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
= $\min(1, S_n / S_v)$
= $\min(1, 117.9 / 118.0)$
= 0.999

Weld Strength Reduction Factor [fr2]:
= $\min(1, S_n / S_v)$
= $\min(1, 117.9 / 118.0)$
= 0.999

Weld Strength Reduction Factor [fr4]:
= $\min(1, S_p / S_v)$
= $\min(1, 137.9 / 118.0)$
= 1.000

Weld Strength Reduction Factor [fr3]:
= $\min(fr2, fr4)$
= $\min(0.999, 1.0)$
= 0.999

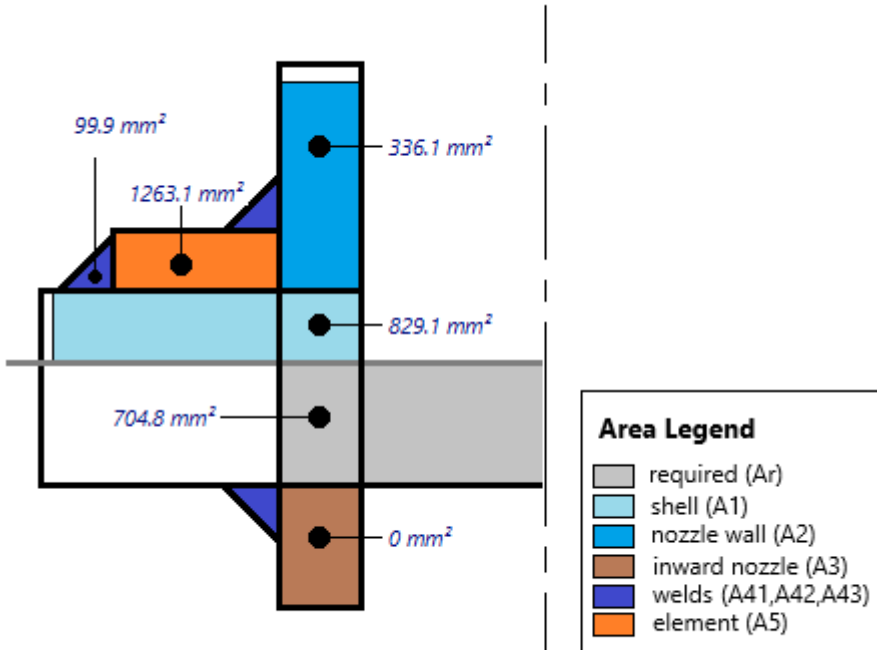


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	704.834	412.613	NA
Area in Shell	A1	829.060	708.691	NA
Area in Nozzle Wall	A2	336.143	322.169	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	99.917	99.917	NA
Area in Element	A5	1263.091	1263.091	NA
TOTAL AREA AVAILABLE	Atot	2528.211	2393.869	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (102.2604 \cdot 6.8919 \cdot 1.0 + 2 \cdot 6.0198 \cdot 6.8919 \cdot 1.0 \cdot (1 - 1.))$
 $= 704.834 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d (E1 \cdot t - F \cdot tr) - 2 \cdot tn (E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 102.26 (1.0 \cdot 15.0 - 1.0 \cdot 6.892) - 2 \cdot 6.02$
 $(1.0 \cdot 15.0 - 1.0 \cdot 6.8919) \cdot (1 - 0.999)$
 $= 829.060 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 29.05) \cdot (6.02 - 0.23) \cdot 0.9992$
 $= 336.143 \text{ mm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi \cdot can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10.0^2 \cdot 1. + (0.0)^2 \cdot 1. + 0.0^2 \cdot 1.0$



$$= 99.917 \text{ mm}^2$$

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lwp}, t_e)) \cdot f_r \\ &= (204.5208 - 114.3) \cdot 14.0 \cdot 1.0 \\ &= 1263.091 \text{ mm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 0.4700 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 2.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$trb1 = 7.8919 \text{ mm}$
Wall Thickness	$tb1 = \max(trb1, tr_{16b}) = 7.8919 \text{ mm}$
Wall Thickness, shell/head, external pressure	$trb2 = 1.7117 \text{ mm}$
Wall Thickness	$tb2 = \max(trb2, tr_{16b}) = 2.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb3 = 5.2578 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[5.258, \max(7.8919, 2.5)] \\ &= 5.2578 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(t_a, t_b) \\ &= \max(0.47, 5.2578) \\ &= 5.2578 \text{ mm} \end{aligned}$$

$$\text{Available Nozzle Neck Thickness} = 0.875 \cdot 6.020 = 5.267 \text{ mm} \rightarrow \text{OK}$$

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

$$\begin{aligned} \text{Govrn. thk, } t_g &= 5.267, tr = 0.229, c = 0.0 \text{ mm, } E^* = 1.0 \\ \text{Thickness Ratio} &= tr \cdot E^* / (t_g - c) = 0.044, \text{Temp. Reduction} = 78 \text{ }^\circ\text{C} \end{aligned}$$

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

$$\begin{aligned} \text{Govrn. thk, } t_g &= 5.267, tr = 0.229, c = 0.0 \text{ mm, } E^* = 1.0 \\ \text{Thickness Ratio} &= tr \cdot E^* / (t_g - c) = 0.044, \text{Temp. Reduction} = 78 \text{ }^\circ\text{C} \end{aligned}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

$$\begin{aligned} \text{Govrn. thk, } t_g &= 5.267, tr = 0.229, c = 0.0 \text{ mm, } E^* = 1.0 \\ \text{Thickness Ratio} &= tr \cdot E^* / (t_g - c) = 0.044, \text{Temp. Reduction} = 78 \text{ }^\circ\text{C} \end{aligned}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, min(Curve:B, Curve:C)

$$\begin{aligned} \text{Govrn. thk, } t_g &= 14.0, c = 1.0 \text{ mm, } E^* = 1.0 \\ \text{Thickness Ratio} &= tr \cdot E^* / (t_g - c) = 0.459, \text{Temp. Reduction} = 38 \text{ }^\circ\text{C} \\ \text{Attachment governing, Attachment stress} &= \text{Shell stress per Code} \end{aligned}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

$$\begin{aligned} \text{Govrn. thk, } t_g &= 5.267, tr = 0.229, c = 0.0 \text{ mm, } E^* = 1.0 \\ \text{Thickness Ratio} &= tr \cdot E^* / (t_g - c) = 0.044, \text{Temp. Reduction} = 78 \text{ }^\circ\text{C} \end{aligned}$$



Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -48 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -48 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 0.47/1.96 = 0.242

Weld Size Calculations, Description: P2(frm ME-1702)

Intermediate Calc. for nozzle/shell Welds Tmin 6.0198 mm
 Intermediate Calc. for pad/shell Welds TminPad 14.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	4.2139 = 0.7 * tmin.	7.0700 = 0.7 * Wo mm
Pad Weld	7.0000 = 0.5*TminPad	8.4840 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$
 $= \max(0, (704.8339 - 829.06 + 2 \cdot 6.0198 \cdot 0.9992 \cdot (1.0 \cdot 15.0 - 6.8919)) \cdot 118)$
 $= \max(0, -3148.91) \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (336.1429 + 1263.0912 + 99.9171 - 0.0 \cdot 1.) \cdot 118$
 $= 200479.16 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (336.1429 + 0.0 + 99.9171 + (180.4442)) \cdot 118$
 $= 72740.00 \text{ N}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (336.1429 + 0.0 + 99.9171 + 1263.0912 + (180.4442)) \cdot 118$
 $= 221769.39 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 114.3 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 103714. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 250.0 \cdot 12.0 \cdot 0.49 \cdot 118$
 $= 272442. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 54.1401) \cdot (6.0198 - 0.0) \cdot 0.7 \cdot 118$
 $= 84494. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:
 $= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$
 $= (3.1416/2) \cdot 114.3 \cdot 14.0 \cdot 0.74 \cdot 138$



= 256480. N

Tension, Shell Groove Weld [Tngw]:

= $(\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$

= $(3.1416/2.0) \cdot 114.3 \cdot (14.0 - 0.0) \cdot 0.74 \cdot 118$

= 219464. N

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

= (SPEW + SNW) = (272442 + 84494) = 356936 N

Failure Path [Path 2-2]:

= (Sonw + Tpgw + Tngw + Sinw)

= (103714 + 256480 + 219464 + 0) = 579658 N

Failure Path [Path 3-3]:

= (Spew + Tngw + Sinw)

= (272442 + 219464 + 0) = 491906 N

Summary of Failure Path Calculations:

Path 1-1 = 356936 N , must exceed W = 0 N or W1 = 200479 N

Path 2-2 = 579658 N , must exceed W = 0 N or W2 = 72740 N

Path 3-3 = 491906 N , must exceed W = 0 N or W3 = 221769 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 0.9546 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 440.7546 mm

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Input, Nozzle Desc: P3 (from C-1402)

From: 250

Pressure for Reinforcement Calculations	P	0.4738	MPa
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	0.05	MPa
Temperature for External Pressure	Tempex	150	°C

Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	MPa
Parent Allowable Stress At Ambient	Sva	118.00	MPa

Inside Diameter of Cylindrical Shell	D	3422.40	mm
Design Length of Section	L	3299.1997	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	cas	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Distance from Bottom/Left Tangent		44360.99	mm
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User Entered Minimum Design Metal Temperature		-1.00	°C
---	--	-------	----

Type of Element Connected to the Parent : Nozzle

Material		SA-516 60 +IT	
Material UNS Number		K02100	
Material Specification/Type		Plate	
Allowable Stress at Temperature	Sn	118.00	MPa
Allowable Stress At Ambient	Sna	118.00	MPa

Diameter Basis (for tr calc only)		Outside	
Layout Angle		45.00	deg
Diameter		32.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	18.0000	mm

Flange Material		SA-105	
Flange Type		Weld Neck Flange	

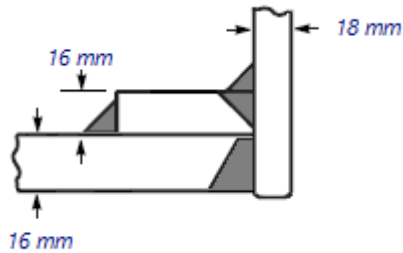
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	250.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 60 +IT	
Pad Allowable Stress at Temperature	Sp	118.00	MPa
Pad Allowable Stress At Ambient	Spa	118.00	MPa
Diameter of Pad along vessel surface	Dp	1212.8000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		200.0000	mm

Flange Class		150	
Flange Grade		GR 1.1	
Flange Series		A	

The Pressure Design option was MAWP + static head (bottom of element).



Reinforcement CALCULATION, Description: P3 (from C-1402)

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	32.000 in.
Actual Thickness Used in Calculation	0.709 in.

Messages regarding the Nozzle Input Data:

For this nozzle, the actual wall thickness was specified. In these cases, the weight of the attached nozzle flange cannot be determined. It is required to manually compute the weight of the assembly and enter it into the nozzle dialog's overriding weight field.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 0.47 \cdot 1712.2 / (118 \cdot 1.0 - 0.6 \cdot 0.47)$
 $= 6.8919 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 0.47 \cdot 406.4 / (118 \cdot 1.0 + 0.4 \cdot 0.47)$
 $= 1.6292 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.1813 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1553.6000 mm
Parallel to Vessel Wall, opening length	d	776.8000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

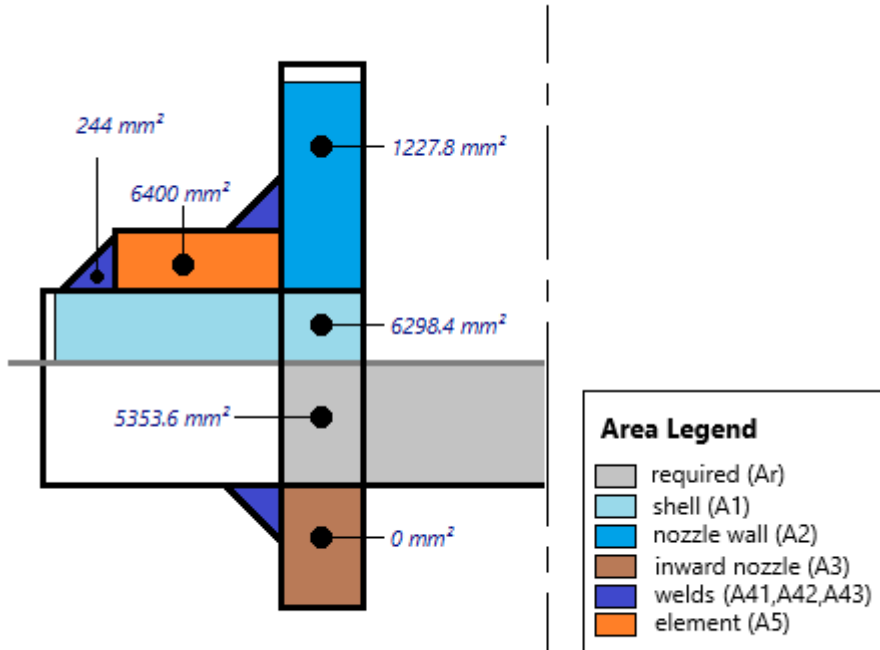


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (mm²)

AREA AVAILABLE, A1 to A5		MAWP	External	Mapnc
Area Required	Ar	5353.602	3134.023	NA
Area in Shell	A1	6298.398	5383.954	NA
Area in Nozzle Wall	A2	1227.807	1261.403	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	244.000	244.000	NA
Area in Element	A5	6400.001	6400.001	NA
TOTAL AREA AVAILABLE	Atot	14170.205	13289.356	NA

The MAWP Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.
 The area available with the given pad is Sufficient.

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (776.8 \cdot 6.8919 \cdot 1.0 + 2 \cdot 18.0 \cdot 6.8919 \cdot 1.0 \cdot (1 - 1.0))$
 $= 5353.602 \text{ mm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 776.8(1.0 \cdot 15.0 - 1.0 \cdot 6.892) - 2 \cdot 18.0$
 $(1.0 \cdot 15.0 - 1.0 \cdot 6.8919) \cdot (1 - 1.0)$
 $= 6298.398 \text{ mm}^2$

Area Available in Nozzle Wall Projecting Outward [A2]:
 $= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2$
 $= (2 \cdot 37.5) \cdot (18.0 - 1.63) \cdot 1.0$
 $= 1227.807 \text{ mm}^2$

Area Available in Welds [A41 + A42 + A43]:
 $= Wo^2 \cdot fr3 + (Wi \cdot can / 0.707)^2 \cdot fr2 + Wp^2 \cdot fr4$
 $= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 12.0^2 \cdot 1.0$



= 244.000 mm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))*(min(tp,Tlwp,te)) • fr4
= (1212.8 - 812.8) • 16.0 • 1.0
= 6400.001 mm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 1.6292 mm
Wall Thickness per UG16(b), tr16b = 2.5000 mm
Wall Thickness, shell/head, internal pressure trb1 = 7.8919 mm
Wall Thickness tb1 = max(trb1, tr16b) = 7.8919 mm
Wall Thickness, shell/head, external pressure trb2 = 1.7117 mm
Wall Thickness tb2 = max(trb2, tr16b) = 2.5000 mm
Wall Thickness per table UG-45 tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
= min[8.331, max(7.8919, 2.5)]
= 7.8919 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
= max(1.6292, 7.8919)
= 7.8919 mm

Available Nozzle Neck Thickness = 18.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	18.2,	Allowable	:	118.0 MPa	Passed
Expansion	:	0.0,	Allowable	:	276.8 MPa	Passed
Occasional	:	5.0,	Allowable	:	156.9 MPa	Passed
Shear	:	10.1,	Allowable	:	82.6 MPa	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 18.0, tr = 1.629, c = 0.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.091, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A 9 °C
Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for the Nozzle (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Nozzle Neck to Pad Weld for Reinf. pad (Impact tested) :

Note: This Element/Detail was specified as being Impact Tested.

Shell to Pad Weld Junction at Pad OD, Curve: C

Govrn. thk, tg = 16.0, tr = 6.892, c = 1.0 mm, E* = 1.0
Thickness Ratio = tr * E*/(tg - c) = 0.459, Temp. Reduction = 38 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -32 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), Curve: C

Govrn. thk, tg = 16.0, tr = 6.892, c = 1.0 mm, E* = 1.0



Thickness Ratio = $tr * E / (tg - c) = 0.459$, Temp. Reduction = 38 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -32 °C
Min Metal Temp. at Required thickness (UCS 66.1) -48 °C

Gov. MDMT of the Nozzle : -39 °C
Gov. MDMT of the Reinforcement Pad : -39 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -39 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = $0.47/1.96 = 0.242$

Weld Size Calculations, Description: P3 (from C-1402)

Intermediate Calc. for nozzle/shell Welds Tmin 16.0000 mm
Intermediate Calc. for pad/shell Welds TminPad 15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * Wo \text{ mm}$
Pad Weld	$7.5000 = 0.5 * TminPad$	$8.4840 = 0.7 * Wp \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (5353.6021 - 6298.3979 + 2 \cdot 18.0 \cdot 1.0 \cdot (1.0 \cdot 15.0 - 6.8919)) \cdot 118)$
 $= \max(0, -77034.62) \text{ N}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (1227.8074 + 6400.001 + 244.0 - 0.0 \cdot 1.0) \cdot 118$
 $= 928777.69 \text{ N}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$
 $= (1227.8074 + 0.0 + 100.0 + (540.0)) \cdot 118$
 $= 220378.53 \text{ N}$

Weld Load [W3]:
 $= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$
 $= (1227.8074 + 0.0 + 244.0 + 6400.001 + (540.0)) \cdot 118$
 $= 992491.12 \text{ N}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$
 $= (3.1416/2.0) \cdot 812.8 \cdot 10.0 \cdot 0.49 \cdot 118$
 $= 738150. \text{ N}$

Shear, Pad Element Weld [Spew]:
 $= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$
 $= (3.1416/2.0) \cdot 1212.8 \cdot 12.0 \cdot 0.49 \cdot 118$
 $= 1321672. \text{ N}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$
 $= (3.1416 \cdot 397.4) \cdot (18.0 - 0.0) \cdot 0.7 \cdot 118$
 $= 1856066. \text{ N}$

Tension, Pad Groove Weld [Tpgw]:



$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 812.8 \cdot 16.0 \cdot 0.74 \cdot 118$$

$$= 1783612. N$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 812.8 \cdot (14.0 - 0.0) \cdot 0.74 \cdot 118$$

$$= 1560661. N$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (1321672 + 1856066) = 3177738 N$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (738150 + 1783612 + 1560661 + 0) = 4082423 N$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (1321672 + 1560661 + 0) = 2882333 N$$

Summary of Failure Path Calculations:

Path 1-1 = 3177738 N , must exceed W = 0 N	or W1 = 928777 N
Path 2-2 = 4082422 N , must exceed W = 0 N	or W2 = 220378 N
Path 3-3 = 2882332 N , must exceed W = 0 N	or W3 = 992491 N

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1 MPa

Nozzle is O.K. for the External Pressure: 0 MPa

The Drop for this Nozzle is : 48.9592 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 314.9592 mm

Percent Elongation Calculations:

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 2.265 %

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 776.80 mm

Input Echo, WRC297 Item 1, Description: P3 (from C-140:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	3422.400	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 60	
Vessel Cold S.I. Allowable	Smc	118.00 MPa
Vessel Hot S.I. Allowable	Smh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	812.800	mm
Nozzle Thickness	18.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-516 60	
Nozzle Cold S.I. Allowable	SNmc	118.00 MPa
Nozzle Hot S.I. Allowable	SNmh	118.00 MPa

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.



Diameter of Reinforcing Pad 1212.800 mm
 Thickness of Reinforcing Pad 16.0000 mm

Note:

*External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUstained loads.*

Design Internal Pressure	Dp	0.47	MPa
Radial Load	P	-59999.96	N
Circumferential Shear	Vc	40000.02	N
Longitudinal Shear	Vl	40000.02	N
Circumferential Moment	Mc	0.896E+08	N-mm
Longitudinal Moment	Ml	0.896E+08	N-mm
Torsional Moment	Mt	0.128E+09	N-mm

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$
 $= 812.8 + 2 \cdot 1.65 \cdot \sqrt{(1727.7(16.0 - 1.0))}$
 $= 1344.044 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3455.400 mm
Nozzle Outside Diameter	(d) =	812.800 mm
Vessel Thickness used	(T) =	31.000 mm
Nozzle Thickness used	(t) =	18.000 mm
T / t	=	1.722
d / t	=	45.156

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.483$$

Nr/P	=	0.048
Mr/P	=	0.086
M0/P	=	0.040
N0/P	=	0.142
MrD/Mc	=	0.194
NrDL/Mc	=	0.085
M0d/Mc	=	0.104
N0DL/Mc	=	0.187
MrD/Ml	=	0.091
NrDL/Ml	=	0.046
M0D/Ml	=	0.042
N0DL/Ml	=	0.248

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	8	8	8	8
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-28	-28	28	28
Outplane Bending (Ml)	-28	28	28	-28
Normal Pressure Stress	25	26	25	26
Outplane Stress Summary	-7	20	107	19

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses parallel to	Au Top	Al Top	Bu Bottom	Bl Bottom
---	-----------	-----------	--------------	--------------



longitudinal plane)	Outside	Inside	Outside	Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	32	-32	32	-32
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-5	-5	5	5
Inplane Bending (ML)	-62	62	62	-62
Inplane Pressure Stress	12	12	12	12
Inplane Stress Summary	-19	40	115	-73

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	21	42	116	93

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	8	8	8	8
Outplane Bending (P)	15	-15	15	-15
Outplane Membrane (Mc)	-21	-21	21	21
Outplane Bending (Mc)	-71	71	71	-71
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	12	12	12	12
Outplane Stress Summary	-56	56	129	-43

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	32	-32	32	-32
Inplane Membrane (Mc)	-9	-9	9	9
Inplane Bending (Mc)	-133	133	133	-133
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	25	26	25	26
Inplane Stress Summary	-82	121	204	-126

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4



Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	82	121	204	126

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	8	8	8	8
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-28	-28	28	28
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	12	11	12
Hoop Stress Summary	-7	-7	49	49

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	68	-68	68	-68
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-10	-10	10	10
Axial Bending (ML)	-138	138	138	-138
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-72	66	224	-189

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6
Shear Stress Summary	8	8	5	5

Nozzle Stresses (MPa):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	73	75	224	239

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	8	8	8	8
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-21	-21	21	21
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	11	12	11	12
Hoop Stress Summary	0	0	42	42

Nozzle Stresses (MPa):



CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	68	-68	68	-68
Axial Membrane (Mc)	-10	-10	10	10
Axial Bending (Mc)	-309	309	309	-309
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	5	5	5	5
Axial Stress Summary	-244	238	395	-360

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6
Shear Stress Summary	5	5	8	8

Nozzle Stresses (MPa):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	244	238	396	403

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(R_{\text{mean}}(t - ca))}$
 $= 812.8 + 2 \cdot 1.65 \cdot \sqrt{(1719.7(16.0 - 1.0))}$
 $= 1342.812 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	3439.400 mm
Nozzle Outside Diameter	(d) =	1212.800 mm
Vessel Thickness used	(T) =	15.000 mm
Nozzle Thickness used	(t) =	218.000 mm
T / t	=	0.069
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.340$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.045
Mr/P	=	0.071
M0/P	=	0.019
N0/P	=	0.025
MrD/Mc	=	0.239
NrDL/Mc	=	0.099
M0d/Mc	=	0.070
N0DL/Mc	=	0.049
MrD/Ml	=	0.046
NrDL/Ml	=	0.045
M0D/Ml	=	0.014
N0DL/Ml	=	0.024

Vessel Stresses (MPa):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6



Outplane Bending (P)	30	-30	30	-30
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7
Outplane Bending (ML)	-27	27	27	-27
Normal Pressure Stress	53	54	53	54

Outplane Stress Summary	55	50	126	10
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses parallel to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Inplane Membrane (P)	12	12	12	12
Inplane Bending (P)	113	-113	113	-113
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-14	-14	14	14
Inplane Bending (ML)	-90	90	90	-90
Inplane Pressure Stress	26	26	26	26

Inplane Stress Summary	46	1	257	-150
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary	5	5	2	2
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Vessel Stresses (MPa):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	57	51	257	160
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	30	-30	30	-30
Outplane Membrane (Mc)	-16	-16	16	16
Outplane Bending (Mc)	-138	138	138	-138
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	26	26	26	26

Outplane Stress Summary	-90	125	219	-119
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Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	12	12	12	12
Inplane Bending (P)	113	-113	113	-113
Inplane Membrane (Mc)	-32	-32	32	32
Inplane Bending (Mc)	-469	469	469	-469
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	53	54	53	54



Inplane Stress Summary -323 390 681 -484

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		-1	-1	1	1
Torsional Shear (Mt)		3	3	3	3

Shear Stress Summary 2 2 5 5

Vessel Stresses (MPa):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities)		Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress		323	390	681	484

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	25	26	25	26	25	26	25	26
Circ. Pl (SUS)	-20	-20	36	36	-6	-6	12	12
Circ. Q (SUS)	-13	13	43	-43	-101	101	165	-165
Long. Pm (SUS)	12	12	12	12	12	12	12	12
Long. Pl (SUS)	-2	-2	8	8	-13	-13	29	29
Long. Q (SUS)	-30	30	94	-94	-56	56	86	-86
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	25.0	26.0	25.0	26.0	25.0	26.0	25.0	26.0
Pm+Pl (SUS)	10.2	10.2	61.0	62.0	20.1	21.1	41.2	41.3
Pm+Pl+Q (Total)	21.2	40.7	114.4	93.1	82.2	121.1	202.2	127.2

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	26.00	118.00	Passed
Pm+Pl (SUS)	62.02	177.00	Passed
Pm+Pl+Q (TOTAL)	202.21	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	53	54	53	54	53	54	53	54
Circ. Pl (SUS)	-1	-1	13	13	-20	-20	44	44
Circ. Q (SUS)	3	-3	57	-57	-356	356	582	-582

Long. Pm (SUS)	26	26	26	26	26	26	26	26
Long. Pl (SUS)	-2	-2	26	26	-10	-10	22	22
Long. Q (SUS)	23	-23	203	-203	-108	108	168	-168
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	53.0	54.0	53.0	54.0	53.0	54.0	53.0	54.0
Pm+Pl (SUS)	52.0	53.0	66.1	67.1	33.1	34.1	97.0	98.0
Pm+Pl+Q (Total)	56.7	50.3	255.0	161.0	323.0	390.0	679.0	484.0

Vessel Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	54.00	118.00	Passed
Pm+Pl (SUS)	98.02	177.00	Passed
Pm+Pl+Q (TOTAL)	679.03	353.99	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (MPa):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	11	12	11	12	11	12	11	12
Circ. Pl (SUS)	-20	-20	36	36	-13	-13	29	29
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-9	-9	11	11	-9	-9	11	11
Long. Q (SUS)	-70	70	206	-206	-241	241	377	-377
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	11.0	12.0	11.0	12.0	11.0	12.0	11.0	12.0
Pm+Pl (SUS)	9.2	8.2	47.0	48.0	4.4	4.3	40.0	41.0
Pm+Pl+Q (Total)	74.7	75.3	222.1	238.2	245.1	238.2	393.1	402.2

Nozzle Stress Summation Comparison (MPa):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	12.00	118.00	Passed
Pm+Pl (SUS)	48.03	177.00	Passed
Pm+Pl+Q (TOTAL)	402.24	354.00	Failed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: PWHT was required for at least 1 Nozzle/Pad in this vessel!**

**** Warning: One or more nozzles do not meet proper Spacing Requirements!**

Please review all reports carefully!

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